# Steeper Simplicity Priors in Solomonoff Function Induction:
## The Model–Randomness Split and the Teacher's Mistakes

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**Abstract**

Solomonoff function induction (SFI) predicts labels from inputs with a universal prior on random functions. We study a complexity penalty steeper than log-likelihood by a factor $\lambda > 1$, for private-randomness hypotheses, and give two reasons for it. In each class containing its own universal mixture that mixture, taken as one hypothesis, absorbs all structure; no private-randomness hypothesis is universal. The first and main reason is the split into model and noise. With i.i.d. inputs and a computable sampler with fresh coins per query, for every $1 \le \lambda_n = o(n)$ every $\lambda_n$-MAP is eventually a shortest program for the sampler; for fixed $\lambda > 1$, with probability $1 - \varepsilon$ at each $n$ beyond a bound depending only on $\lambda$, the program's length, a margin and $\varepsilon$. At $\lambda = 1$ no such bound exists, and in a parity class recovery is exponentially slower in that length. On one data set, a kink of the product structure function at a sufficient program makes every $\lambda$ between 1 and its steepness select it up to windows; per-query samples have one, and so, for large $d$, does the full table of a noisy random parity at each dyadic flip rate below $1/4$, but set models have no steep one. The second reason, the same geometry at a positive rate, is not learning a teacher's errors: an error rule is reused at many inputs, so $\lambda$ must grow linearly in the sample size. For a random parity on uniform inputs, with labels flipped at a rate below 0.05596 on an error set of product form incompressible relative to the parity, every near-optimal hypothesis of the population objective with $\lambda/n$ in an explicit window knows the parity and, for large error rules, gives the erroneous labels far less probability than a memoriser. These are toy cases of imitation and of learning from a fallible teacher; we offer tools, not solutions.

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# 1 Introduction

Solomonoff function induction (SFI) is Bayesian prediction of a label $y$ from an input $x$ with a universal prior over random partial functions (Hänni, 2026a). It is the supervised-learning counterpart of Solomonoff induction (SI), and like SI it learns every computable regularity of the labels. We study the following proposal, a stronger simplicity prior: penalise the length of a hypothesis more than its likelihood term, so that what is selected is a simple model with noise.[1] In the form studied here, a hypothesis is a program $\pi$ of length $\ell(\pi)$ with likelihood $P_\pi$, and one minimises $J_\lambda(\pi; D) = \lambda\,\ell(\pi) - \log P_\pi(D)$ with $\lambda > 1$, or uses the power prior $2^{-\lambda\ell(\pi)}$ (Definition 3.6), over probabilistic programs that use fresh coins at every input: private randomness.

We give two reasons for such a penalty. The first, and the main one, is to split data into a model and noise: labels drawn by a simple sampler are explained by the sampler and by *fill-ins*, deterministic programs that store its coin flips, and at $\lambda = 1$ the two get similar posterior weight on a data set, in SFI and in SI (Theorem 6.11 and Proposition D.7). Does a steeper prior select

[1] The proposal first appeared in the author's note (Hänni, 2026b) and in an online discussion (on LessWrong), where the absorption objection described below was also raised. So did the suggestion that private randomness at each input blocks this objection, the three-hypothesis calculation below (in both versions of Remark 8.1), the question whether its good model is a vertex of the convex hull, and an absolute complexity bound as an alternative. SFI itself first appeared in the author's working notes (Hänni, 2026c).

the sampler? One motivation is imitation: a learner trained on a person's answers should return a copy of the person, a sampler with its own randomness, rather than the sampler with some of its sampled bits stored, or a different system that predicts the person's answers better than the person does. If the person's coins are private this is the first reason; if the environment drives them and the input partly reveals them, it needs the schedule of the second reason below (Section 10). The second reason is not to learn a teacher's systematic mistakes, as a human learner grasps what a teacher means by a word without learning to predict what the teacher would say. With a 100-bit simple model and systematic errors on a $2^{-10}$ fraction of the inputs, whose set takes 10,000 bits to specify, the simple model with noise beats a model that learns the errors and a fair coin for a range of $\lambda/n$ (Remark 8.1); but three hypotheses are not all hypotheses. Both reasons rest on one geometry, a kink of the hull of the points (length, fit) with the discarded part to its right; noise is used at one input and an error rule at many, so noise needs only a fixed $\lambda > 1$ and errors need $\lambda$ proportional to the number $n$ of samples (Table 4).

A standard objection, the *absorption objection*, is that under a steeper penalty a hypothesis that is itself a universal mixture is short, penalises complexity inside itself only in the usual way, and so absorbs all structure, noise and errors included. It holds in SI and in SFI with shared randomness (Theorems 4.2 and 4.4); if each input samples its own random bits, no hypothesis is universal (Corollary 4.5).

**Conventions.** Logarithms are to base 2; $N = 2^d$, $c = \lambda/n$, and $\kappa$ is a constant depending only on the reference machine and fixed programs; numbers computed with $\kappa = 0$ are illustrations. A shared hypothesis $P$ (Definition 3.4) predicts $P(D + (x, y))/P(D)$ (the *semimeasure convention*, whose log loss telescopes in every order to at most $-\log P(D)$), or normalises by $\sum_{y'} P(D + (x, y')) \le P(D)$, which never increases losses. The data objects DS (one data set), FR (frozen labels, one per distinct input) and PQ (fresh labels at every query) are those of Definition 3.1.

**Main results.** **Theorem A** (informal; exact recovery). *Let the inputs be i.i.d. from any law $\mu$ on $\{0,1\}^*$ and each label drawn by a sampler $\nu$ with fresh coins at every query, so that a repeated input gets a fresh label. Let the hypotheses be the programs of a countable class with Kraft weights that contains a program for $\nu$, each a kernel with private randomness (all lower-semicomputable ones, or a time-bounded, circuit or finite subclass). For every schedule $1 \le \lambda_n = o(n)$, $\lambda = 1$ included, almost surely every $\lambda_n$-MAP is eventually a shortest program whose kernel equals $\nu$ $\mu$-almost everywhere. For fixed $\lambda > 1$, the $\lambda$-MAPs are exactly these programs with probability at least $1 - \varepsilon$ at every $n \ge \lambda(\ell_0 + \log(2/\varepsilon)) \ln 2/(2\min(1, \lambda - 1)\gamma)$, where $\ell_0$ is their length and $\gamma$ a margin, a lower bound on the average squared Hellinger distance to $\nu$ (the Hellinger gap of Definition 6.1) of the inequivalent programs up to a length fixed by $\lambda$, $\ell_0$ and $\varepsilon$; and at every $n$ the fill-ins have posterior mass exponentially small in the sampled randomness. At $\lambda = 1$ no recovery time depends only on $(\ell_0, \gamma, \varepsilon)$; in a class of parities with a uniform margin it is $\Theta(2^{\ell_0/2})$, against $O(\lambda\ell_0/(\lambda-1))$ for fixed $\lambda > 1$. With frozen labels no procedure recovers every computable sampler, and every $\lambda \ge 1$ eventually selects programs equivalent to $\nu$ on the run, with summable Hellinger gaps to $\nu$ along the queried inputs.* (Theorems 6.3 and 6.9 and Propositions 6.6 and 6.10.)

**Theorem B** (informal; a kink implies the split). *On one data set, plot each program at its length and its fit, minus its log-likelihood; the $\lambda$-MAPs lie on the lower convex hull where its slope crosses $-\lambda$. Let a program be $b$-sufficient (its two-part code length is within $b$ bits of $K(y_D \mid \operatorname{dom} D)$, the least possible up to a constant) and sit at a kink: every program more than $w$ bits shorter fits worse by at least $\sigma > 1$ bits per bit saved. Then for every $\lambda \in (1, \sigma)$ every $\lambda$-MAP is between $w$ bits shorter and $(b + \kappa)/(\lambda - 1)$ bits longer, fits at most $\lambda w$ bits worse and $\lambda(b + \kappa)/(\lambda - 1)$ bits*

*better, and is* $(b+(\lambda-1)w)$*-sufficient. Conversely, MAP lengths that stay in an interval for all* $\lambda$ *in an interval occur only at kinks.* (Lemma 3.8, Corollary 5.5, and Theorem 5.6.) With high probability a per-query sample has a kink of the same kind at the shortest equivalent program, with steepness linear in $n$ and the Ville deficiency in place of the sufficiency slack (Theorem 6.7). The full table of a noisy random parity has one at the parity, and for large $d$ and typical error sets ($\delta_E = O(d)$) every fixed $\lambda > 1$ selects "parity plus noise" at each dyadic flip rate below $1/4$; set models have no steep kink there (Propositions 7.7, 7.8 and 7.10). A universal member flattens every kink far beyond its own length (Proposition 5.7).

**Theorem C** (informal; the teacher's mistakes). *Let the inputs lie in* $\{0,1\}^d$*, let the simple model be a random parity* $f_a$*, and let a teacher flip its labels on the inputs whose first* $m$ *bits lie in an error rule* $S \subseteq \{0,1\}^m$ *of density* $p < 0.05596$*, so that the error set* $S \times \{0,1\}^{d-m}$ *has product form. Score each hypothesis* $\pi$ *with private randomness by the population objective* $c\,\ell(\pi) + R(\pi)$*, with* $R(\pi)$ *its expected log loss on the teacher's labels. If* $S$ *is incompressible relative to the parity, the inputs are uniform and* $\lambda = cn$ *with* $c$ *in an explicit window* $(2^{-m}, c_{\mathrm{hi}})$*, every near-optimal product hypothesis of the population objective knows the parity, is short, holds little information about* $S$ *and gives the erroneous labels average probability at most* $0.200$ $(\kappa = 0)$ *or* $0.274$ $(\kappa = 30)$ *at* $c = 10^{-4}$ *and the sizes given below, against* $1$ *for a memoriser. Below about* $2^{-m}$ *the memoriser beats every hypothesis that knows the parity and fits no better than noise.* (Theorem 8.7 and Proposition 8.6.)

The example's sizes are $d = 100$, $m = 20$, $p = 2^{-10}$ and 11,710 bits for $S$, with deficiency $\delta_S = 0$; there the window is $(9.54 \cdot 10^{-7}, 8.06 \cdot 10^{-3}]$ at $\kappa = 0$ and $(9.54 \cdot 10^{-7}, 3.65 \cdot 10^{-3}]$ at $\kappa = 30$ (Remark 8.8). The lower end $2^{-m}$ is the *reuse factor*: each bit of the error rule serves about $n2^{-m}$ of the $n$ samples, a bit of noise one, which is why errors need $\lambda \propto n$ and noise only a fixed $\lambda > 1$ (Lemma 8.2 and Proposition 8.3). Asymptotically in $m$, windows exist at every flip rate $p \leq 0.1233$ (Proposition 8.13).

**Contributions.**

- *Absorption* (Section 4). With a universal member every $\lambda$-MAP is shorter than a data-independent bound, a computable teacher's errors are eventually predicted for fixed $\lambda > 1$, and no sampler longer than that bound is selected (Theorems 4.2 and 4.4); no private-randomness hypothesis is universal, at a price on shared latent coins and up to labelled history (Corollary 4.5 and Propositions B.5 and B.6).
- *Kinks* (Section 5). One kink lemma for both reasons (Lemma 5.3) and Theorem B; a universal member flattens every kink far beyond its own length (Proposition 5.7).
- *Exact recovery, the first reason* (Section 6). Theorem A, with each ingredient of its rate necessary separately, up to logarithms (Theorem 6.3 and Propositions 6.6 and D.2).
- *The noisy random parity* (Section 7). The full-table kink, with certified $\lambda$-ranges (Propositions 7.7 and 7.8); no steep set kink, and product sophistication above set sophistication by $d - O(\log d)$ bits (Propositions 7.10 and E.5).
- *The teacher's mistakes, the second reason* (Section 8). Theorem C as the kink lemma with right slope $2^{-m}$ (Proposition 8.6 and Theorem 8.7); the reuse factor, the schedules that keep the identification, and plain Bayes, MDL and SI, which learn the errors (Lemma 8.2 and Propositions 8.3 and F.2); full tables for every $m$ (Corollary 8.11); flip rates below 0.1233 (Propositions 8.13 and F.8).

| Question | Answer | Where |
|---|---|---|
| Does absorption occur in the sequence case, and not if each input samples its own random bits? | Yes, in every class that contains its own mixture; no private-randomness hypothesis is universal. In part: absorption returns with labelled history, and randomness shared by several inputs costs about a factor $\min(\lambda, 2)$ more than in a shared class | Thm. 4.2, Cor. 4.5, and Props. B.5 and B.6 |
| Does a steeper prior separate a sampler from its fill-ins? | Yes per query, exactly, and for fixed $\lambda > 1$ at rates that depend on the class only through a margin. With frozen labels no procedure recovers every sampler; every $\lambda \geq 1$ eventually selects programs equivalent to it on the run (summable Hellinger gaps along the queried inputs) | Thms. 6.3 and 6.9 and Prop. 6.10 |
| Against hypothesis length, does the fit change slope at the model? | For per-query samples (with high probability) and for the full table of the noisy parity at every dyadic flip rate below $1/4$ (large $d$), yes, and then every $\lambda$ between the slopes selects the model, up to windows; not for set models | Cor. 5.5, Thm. 6.7, and Props. 7.8 and 7.10 |
| Can the good model of the example be selected for a range of $c$, or does another hypothesis do better? | For error rules incompressible relative to the model, every near-optimal product hypothesis knows the model and not the errors, though the good model itself need not be the minimiser on the whole window. Not for every rule with the same density and description length: for one, no hypothesis that fits no better than the good model is near-optimal on about a third ($\kappa = 0$) to a quarter ($\kappa = 30$) of the window's logarithmic width | Thm. 8.7, Rem. 8.10, and Prop. 9.1 |
| Must the length coefficient grow with the data? | For errors yes: $\lambda/n$ must stay above about $2^{-m}$, the reuse factor. For noise no: a fixed $\lambda > 1$ suffices | Lem. 8.2, Prop. 8.3, and Thm. 6.3 |
| Does an absolute bound on model complexity select the same hypotheses as the penalty? | In part: every penalised minimiser solves a length-bounded problem, but a length bound can also select points that no penalty selects | Lem. 3.8 and Rem. A.2 |
| Is a sampler recovered when other hypotheses predict better? | In part: near-copies for $\lambda/n$ in a window, if its coins are an incompressible function of an environment variable on many more cells than it has bits; not for cheap couplings, nor reliably by any procedure when it is entangled with them | Thm. 10.2 and Props. 10.3 and H.4 |
| Is some other pathology present? | Yes: rate blindness, error envelopes, guard erosion, non-identifiability of $\lambda$, input-law dependence, labelled history, possible confident wrong extrapolation off the training support | § 9, Prop. B.6 |

Table 1: Questions studied in this paper and where they are answered. Numbers with $\kappa = 0$ set the machine constant to 0 and are illustrations.

- *Limits and computable classes* (Sections 9 and 11). Structured error rules add a second kink, and cheap frequent rules defeat every $c$ (Propositions 9.1 and 9.2 and Lemma G.1); selection sorts components by rate, not validity (Proposition 9.3); both reasons survive resource bounds (Proposition 11.1).
- *Imitation* (Section 10). If a sampler's coins are an incompressible function of an environment variable on many more cells than the sampler has bits, other hypotheses can predict better, yet for $\lambda/n$ in a window every near-optimal private-randomness hypothesis, whatever its structure, is close to the sampler in behaviour (Theorem 10.2); imitation data identify only the data kernel, not the mechanism or the split into person and environment, cheap couplings defeat every $\lambda/n$, and no procedure reliably copies a sampler entangled with its coupling (Propositions 10.3 and H.4).

**Scope.** We do not claim to solve the problems that motivate the two reasons, imitating a person and learning what a teacher means rather than what the teacher says. We offer tools for them: the private-randomness class, the kink geometry, the reuse factor and a per-sample threshold on the rate at which a component buys fit. With them we solve toy cases, for idealised learners that minimise the objective exactly over a hypothesis class, in the main results all programs of a fixed reference machine: exact recovery of a computable sampler from per-query samples with i.i.d. inputs (Theorem 6.3); the full table of a noisy random parity with a dyadic flip rate below $\frac{1}{4}$, for large $d$ and typical error sets (Proposition 7.8); a teacher who flips a random parity on an incompressible error rule of product form, with uniform inputs and flip rates below 0.05596, or 0.1233 asymptotically (Theorem 8.7, Remark 8.9, and Proposition 8.13); and, up to near-copies and for $\lambda/n$ in a window, a sampler whose coins are an incompressible function of the environment, with far more entropy summed over the cells than the sampler has bits (Theorem 10.2); the last two for the population objective, and from samples only asymptotically or with large constants. For realistic error patterns we have counterexamples but no positive result, and conceptual problems remain open (Section 12), among them that imitation data identify behaviour, not mechanism, and settle neither which structure belongs to the person nor the complexity weight. Exact recovery holds per query; from labels of a computable function flipped at a computable rate below $\frac{1}{2}$ it recovers the function ($\mu$-a.e.) and the rate (Corollary 6.8). With $\lambda = cn$ the learner is a family of batch objectives (Remark 3.7).

**Reader's guide.** After related work and the setting (Sections 2 and 3) come absorption, the precondition of both reasons (Section 4), kinks (Section 5), the first reason (Section 6), the shared instance (Section 7), the second reason (Section 8), where it fails (Section 9), imitation (Section 10) and resource bounds (Section 11), each with an appendix of proofs; Section 12 lists open problems. Notation follows the companion paper.

## 2 Related work

**Generalised posteriors and two-part codes.** The tempered posterior $\Pi^{1/\lambda}$ is a generalised (Gibbs, fractional or safe) posterior with learning rate $1/\lambda$ (Zhang, 2006; Bissiri et al., 2016; Bhattacharya et al., 2019; Grünwald, 2012; Grünwald and van Ommen, 2017); its maximiser, the $\lambda$-MAP, is a two-part code with complexity weight $\lambda$ (Definition 3.6). Cold posteriors (Wenzel et al., 2020), which raise prior and likelihood to the same power, keep the ratio of $\lambda = 1$. With $\lambda = cn$ the prior changes with the sample size, as the Gaussian prior implicit in weight decay does (Remark 8.5); Zhu et al. (2026) likewise read a large complexity weight in a PAC-Bayes rule as a sample-size-dependent prior. For i.i.d. density estimation with a countable list of densities, two-part (MDL) estimators are studied by Barron and Cover (1991); Theorem 6.3(a),(b) is the supervised analogue, for sub-probability kernels, an arbitrary input law and every schedule $1 \leq \lambda_n = o(n)$, and it identifies the limit as the shortest $\mu$-almost everywhere equivalent programs. Its proof is the standard argument from Ville's inequality and the strong law, and we claim no novelty for it; what we add is the rate for fixed $\lambda > 1$, the contrast with $\lambda = 1$ and the necessity results (Theorem 6.3(c),(d), Propositions 6.6 and D.2).

**$\lambda$-weighted two-part MDL in noisy classification.** The closest prior work studies two-part codes in noisy binary classification, where a hypothesis is a classifier together with a code for the label noise, essentially a product hypothesis. Grünwald and Langford (2007) show that Bayes and MDL classifiers can behave suboptimally under misspecification. Zhu and Srebro (2025) quantify

overfitting along the regularisation path of two-part-code MDL in supervised classification, and Zhu et al. (2026) study overfitting and generalisation of (PAC-)Bayesian prediction in noisy binary classification. We do not claim as new the qualitative finding that a bounded $\lambda$ fits the noise and that $\lambda$ must grow with $n$. Their target is the data distribution; for the teacher's errors ours deliberately differs from it (Remark 8.4). An earlier report (Claude (Anthropic), 2026, Section 8) formalised the proposal for a general imitation channel; Appendix I.1 lists what we re-derive from it.

**Algorithmic statistics.** The structure function and sophistication split the description of a string into a model and a data-to-model code (Koppel, 1987; Gács et al., 2001; Vereshchagin and Vitányi, 2004; Vereshchagin and Shen, 2017); restricting the model family, for example to Hamming balls (Vereshchagin and Vitányi, 2010), changes the structure function. Product models, a parameter with private noise at each input, are such a restriction; Section 5 uses their structure function, and Section 7 compares the two splits for a noisy parity.

**Imitation and identifiability from behaviour.** Behavioural cloning treats imitation as supervised learning of a demonstrator's actions given observations (Bain and Sammut, 1999; Pomerleau, 1991; de Haan et al., 2019); because this ignores the causal structure of demonstrator and environment, more information can make the imitator worse (de Haan et al., 2019), and Zhang et al. (2020) give a graphical criterion, from a causal model, for when imitation is possible if the learner does not observe everything the demonstrator's decisions depend on. Inverse reinforcement learning infers a reward for which the observed behaviour is optimal, and many rewards qualify (Ng and Russell, 2000); a policy does not determine its decomposition into a planner and a reward, and even a simplicity prior does not single out the true decomposition (Armstrong and Mindermann, 2018), while demonstrations in sufficiently different environments identify an entropy-regularised reward up to a constant (Cao et al., 2021). Cohen et al. (2022) give a Bayesian imitation learner for environments that never reset. Our negative results are of the same kind, for one program selected from i.i.d. imitation data: $\lambda$ cannot be chosen from imitation data alone, since twin worlds with one teacher channel need different selections (Propositions 9.3 and G.3), and no procedure reliably copies the sampler in each of several worlds with one data law (Proposition 10.3(a)).

**Novelty.** Known are $\lambda$-weighted MDL in noisy classification, two-part MDL for density estimation, the structure function, sophistication and restricted model families, generalised and tempered posteriors, and the proof techniques we use (dominance bounds, Ville's inequality, the strong law, Kakutani's dichotomy, list decoding, Hoeffding's inequality). The hull–penalty relation itself is elementary convex geometry, and the method of Theorem 6.3(a),(b) is standard. To our knowledge the following are new: the contrast under a steeper prior between classes with a universal member and the private-randomness class; the exact-recovery rates for fixed $\lambda > 1$, with the contrast at $\lambda = 1$ and the necessity results; the kink vocabulary, the converse and the flattening by a universal member; the product kink of the noisy parity against set models; the teacher theorem over all hypotheses through list decoding, and its extension to flip rates below 0.1233; and the reuse factor.

# 3 Setting: data, hypothesis classes and steeper objectives

This section fixes the data, two kinds of hypotheses, classes of programs and the steeper objectives, whose selections lie on a convex hull in the plane of length and fit (proofs in Appendix A).

**Objects from the companion paper (Hänni, 2026a).** A *function machine* $T$ reads, on input $x$, a prefix of one random tape $\omega \sim \mathbb{U}$ shared by all inputs and outputs a label $f^T_\omega(x)$; its *machine law* is $\nu_T(D) = \mathbb{U}\{\omega : f^T_\omega \supseteq D\}$. Solomonoff function induction (SFI) is the law $M_F$ of the universal function machine, $M_F(D) = \sum_i \mathbf{m}(i)\,\nu_{T_i}(D)$ for $D \neq \emptyset$. $M_{\text{prod}}$ mixes the product laws of lower-semicomputable (lsc) conditional semimeasures, with fresh random bits at each input; $M_{\text{pq}}$ is the per-query mixture.

## 3.1 Data and the two kinds of hypotheses

**Definition 3.1** (Data)**.** The input set is $X = \{0,1\}^*$; statements about a fixed input length use $\{0,1\}^d \subseteq X$. The label set $Y$ is finite, $|Y| \geq 2$. We use three data objects.

(DS) A *data set* is a finite partial function $D : X \rightharpoonup Y$, with $\operatorname{dom} D = \{x_1, \ldots, x_n\}$ ($n$ distinct inputs) and $y_i = D(x_i)$. No probability model is assumed.

(FR) *Frozen labels.* Inputs $x_1, x_2, \ldots$ are chosen by an input rule that fixes each new input before its label is revealed (for example i.i.d. from a law $\mu$ on $X$); each distinct input carries one label, drawn once. After $n$ queries the data are the data set $D_n$ of the distinct pairs seen.

(PQ) *Per-query labels.* The pairs $(x_i, y_i)$ are i.i.d. from $\mu \otimes \nu$ for an input law $\mu$ and a probability kernel $\nu$: the label is drawn afresh at every query, also at a repeated input. The data are the sequence $h_n := ((x_1, y_1), \ldots, (x_n, y_n))$.

On DS and FR a hypothesis is scored by its likelihood of the data set, each distinct input counting once. On PQ a product hypothesis $\nu_\pi$ (Definition 3.3) is scored by $\nu_\pi(h_n) := \prod_{i \leq n} \nu_\pi(y_i \mid x_i)$.

*Remark* 3.2 (Repeated inputs). (a) On the countable $X$ i.i.d. inputs repeat almost surely, so FR and PQ differ (Section 6); (b) on distinct inputs they coincide. (c) Only product hypotheses are scored per query, by their terms in $M_{\text{pq}}$ (Hänni, 2026a, Proposition 5.7); SFI's expected loss would be infinite (Hänni, 2026a, Proposition 5.6). (d) FQ (Section 8) counts a frozen table per query.

**Definition 3.3** (Product hypothesis; private randomness)**.** A *product hypothesis* is an lsc conditional semimeasure $\nu(y \mid x)$ ($\sum_y \nu(y \mid x) \leq 1$): the law of a random partial function with independent values $F(x) \sim \nu(\cdot \mid x)$ at distinct $x$, as for a machine that uses fresh coins of its own at each input. Its likelihood of a data set is the *product law* $\nu(D) := \prod_{x \in \operatorname{dom} D} \nu(D(x) \mid x)$, and its prediction at a new input $x$ is $\nu(\cdot \mid x)$, whatever $D$ is. It is *deterministic* if it is a partial computable function $f$; then $\nu(D) = \mathbf{1}[f \supseteq D]$. On a data set the randomness is private to each distinct input; on per-query data it is private to each query, and the likelihood $\nu(h_n)$ is the hypothesis's term in $M_{\text{pq}}$. On distinct inputs the two readings coincide (Hänni, 2026a, Proposition 5.7(a)).

**Definition 3.4** (Shared hypothesis)**.** A *shared hypothesis* is an lsc weak function semimeasure $P$: $P(\emptyset) \leq 1$ and $\sum_y P(D + (x,y)) \leq P(D)$ for $x \notin \operatorname{dom} D$. Its prediction is $P(y \mid x, D) := P(D + (x,y))/P(D)$. Machine laws, Bayes mixtures $\sum_i w_i P_i$ with $\sum_i w_i \leq 1$, and product laws are shared hypotheses. By telescoping, the cumulative sequential log loss of $P$ on $D_n$ is $-\log P(D_n) + \log P(\emptyset) \leq -\log P(D_n)$ in every reveal order. Hypotheses of Solomonoff induction

(SI), semimeasures on an interleaved string $x_1y_1x_2y_2\cdots$, are not of this kind: they depend on the order and model the inputs.

### 3.2 Hypothesis classes

A steeper penalty acts on single hypotheses, so its effect depends on what counts as one hypothesis and on its length; we make the programs explicit.

**Definition 3.5** (Hypothesis class)**.** Fix a countable set $\mathcal{O}$ of *observations*: data sets for SFI, and per-query samples $h_n$ for product classes; binary strings for SI; input–label sequences for history-aware prediction. A *hypothesis class* is a c.e. prefix-free set $\mathcal{H} \subseteq \{0,1\}^*$ of *programs* $\pi$, of length $\ell(\pi) := |\pi|$, with maps $P_\pi : \mathcal{O} \to [0,1]$ that are lsc uniformly in $\pi$; for a product class, $P_\pi(h_n) := \nu_\pi(h_n)$. Two programs for the same law are different hypotheses. By Kraft's inequality, $\sum_\pi 2^{-\ell(\pi)} \leq 1$. In a *canonical class*, $\mathcal{H} := \operatorname{dom} V$ for the fixed universal prefix machine $V$, and $P_\pi$ is the member with index $V(\pi)$ of a fixed effective enumeration of a family, trimmed so that exactly the members of the family occur. The canonical classes, with their mixtures $\xi_1 := \sum_\pi 2^{-\ell(\pi)} P_\pi$ in brackets, are: $\mathcal{H}_\times$, lsc conditional semimeasures with their product laws $\nu_\pi$ ($\xi_\times = \sum_i \mathbf{m}(i)\nu_i = M_{\mathrm{prod}}$); $\mathcal{H}_F$, machine laws ($M_F$ on nonempty data sets); $\mathcal{H}_w$, lsc weak function semimeasures ($M^w$, a universal one); $\mathcal{H}_{\mathrm{SI}}$, lsc semimeasures on strings ($M$); and $\mathcal{H}_{\mathrm{chr}}$, history-oracle function machines with their chronological semimeasures ($M_H$, the law of $\mathrm{SFI}^{\mathrm{hist}}$, SFI whose hypotheses see the earlier inputs: an lsc chronological semimeasure that dominates every one (Hänni, 2026a, Corollary 6.17(a)); we use only this universality). Canonical classes are *closed under prefix-free simulation*: for every prefix machine $V'$ whose outputs are indices of the family there is a string $\sigma$ with $V(\sigma w) = V'(w)$ for all $w \in \operatorname{dom} V'$, so distinct $w$ give distinct programs $\sigma w$.

Throughout, $\kappa$ is a constant that depends only on the reference machine, the class and the programs built in the proofs, and may differ between occurrences. The standard facts (F1)–(F5) are in Appendix A.1; (F1) is $K(\pi) \leq \ell(\pi) + \kappa$ in every class.

### 3.3 Steeper objectives

**Definition 3.6** (Steeper objectives)**.** Fix a class, an observation $o$ (for a product class possibly a per-query sample $h_n$, with $P_\pi(h_n) = \nu_\pi(h_n)$) and a *complexity weight* $\lambda \geq 1$; we call $\lambda > 1$ *steeper*. The *fit* of $\pi$ is $L(\pi; o) := -\log P_\pi(o) \in [0, \infty]$, and the *two-part objective* is $J_\lambda(\pi; o) := \lambda\,\ell(\pi) + L(\pi; o)$; a minimiser is a *$\lambda$-MAP*. The *power-prior posterior* is $\Pi_\lambda(\pi \mid o) := 2^{-\lambda\ell(\pi)} P_\pi(o)/\xi_\lambda(o)$, with the *steeper mixture* $\xi_\lambda(o) := \sum_\pi 2^{-\lambda\ell(\pi)} P_\pi(o)$: Bayes' rule with the prior $2^{-\lambda\ell}$, whose Kraft sum is at most 1. We write $\xi_\lambda^\times$, $\xi_\lambda^F$ for $\mathcal{H}_\times$, $\mathcal{H}_F$, and $\xi_\mathcal{C} := \sum_{\pi\in\mathcal{C}} 2^{-\ell(\pi)} P_\pi$ for a class $\mathcal{C}$. The *tempered posterior* with exponent $\eta \in (0,1]$ is $\Pi^\eta(\pi \mid o) \propto 2^{-\ell(\pi)} P_\pi(o)^\eta$. The objective $J_\lambda$ and its minimisers are defined in the same way for every $\lambda > 0$.

For data-set classes $\xi_\lambda$ is a shared hypothesis. The tempered posterior $\Pi^{1/\lambda}$ is a generalised posterior with learning rate $1/\lambda$ (Section 2). The power-prior posterior $\Pi_\lambda$ has the same maximiser and is proportional to $(\Pi^{1/\lambda})^\lambda$. Our posterior results concern $\Pi_\lambda$ unless stated. If some program has finite $J_\lambda$, the $\lambda$-MAPs are finitely many and nonempty, they maximise both posteriors, and on deterministic hypotheses tempering changes nothing (Lemma A.1).

*Remark* 3.7 (Per-sample form; $\lambda = cn$ gives batch objectives). For a product class and $n$ data points put $c := \lambda/n$; then $J_\lambda/n = \hat{J}_c(\pi) := c\,\ell(\pi) + \frac{1}{n}\sum_{i\leq n} -\log \nu_\pi(y_i \mid x_i)$. For an input law $\mu$ and a fixed teacher $y^* : X \to Y$ the population versions are the risk $R(\pi) := \mathbb{E}_{x\sim\mu}[-\log \nu_\pi(y^*(x) \mid x)]$,

$J_c(\pi) := c\,\ell(\pi) + R(\pi)$ and $\Phi(c) := \inf_\pi J_c(\pi)$. Against a teacher's systematic errors the coefficient of the length term must grow with the size of the data set (Remark 8.4); $\lambda = cn$ with $c$ fixed is this choice. For fixed $\lambda$, $\xi_\lambda$ is one shared hypothesis and its sequential loss telescopes; with $\lambda = cn$ the prior changes with $n$, so statements with $\lambda = cn$ are *batch* statements at a given $n$, and in $\mathcal{H}_\times$ no fixed prior with full support gives the $cn$-MAPs on every sample (Proposition F.6).

Both objectives select points on the lower boundary of a convex hull, whose kinks are where both reasons for a steeper penalty take effect (Section 5).

**Lemma 3.8** (Hull lemma)**.** *Let $(\ell(\pi), L(\pi))_\pi$ be a countable family with $\ell(\pi) \in \{0, 1, \dots\}$, $L(\pi) \in [0, \infty]$, finitely many $\pi$ of each length and some $L(\pi) < \infty$; let $\mathcal{A}$ be the set of its finite points, $C := \mathrm{conv}(\mathcal{A}) + \mathbb{R}^2_{\geq 0}$, the* vertices *the extreme points of $C$, and $J_c := c\,\ell + L$ for $c > 0$. (*Data reading: *$L = L(\cdot; \bar{D})$, $c = \lambda$, and the minimisers are the $\lambda$-MAPs.* Population reading: *$L = R$, $c = \lambda/n$.)*

*(i) For $c > 0$, every minimiser of $J_c$ lies on the boundary of $C$, on a supporting line with normal $(c, 1)$.*

*(ii) If $c < c'$ and $\pi, \pi'$ minimise $J_c, J_{c'}$, then $\ell(\pi) \geq \ell(\pi')$ and $L(\pi) \leq L(\pi')$.*

*(iii) A point of $\mathcal{A}$ is the unique minimising point of $J_c$ for every $c$ in some nonempty open interval if and only if it is a vertex of $C$; the least-$\ell$ vertex and the least-$L$ vertex, if it exists, count. A program is then the unique minimiser if and only if no other program has the same point.*

*(iv) (Absolute length bound.) Every minimiser $\pi_c$ of $J_c$ solves $\min\{L(\pi) : \ell(\pi) \leq \ell(\pi_c)\}$. Conversely, a solution of $\min\{L(\pi) : \ell(\pi) \leq \bar{\ell}\}$ minimises some $J_c$, $c > 0$, if and only if its point lies on the boundary of $C$ but not on the open horizontal ray to the right of the least-$L$ vertex.*

*(v) (Vertices and minimisers.) The vertices are points $v_i = (\alpha_i, \rho_i)$, $i = 0, 1, \dots$ (finitely or infinitely many), $\alpha_0 < \alpha_1 < \cdots$, each the point of a program of length exactly $\alpha_i$, joined by edges of slopes $-\sigma_i$, $\sigma_1 > \sigma_2 > \cdots > 0$, with $\sigma_i \to 0$ if there are infinitely many; put $\sigma_0 := \infty$, and $\sigma_{q+1} := 0$ if $v_q$ is the last vertex. For $c > 0$ the minimisers of $J_c$ are finitely many: for $\sigma_{i+1} < c < \sigma_i$ the programs at $v_i$; for $c = \sigma_i$ ($i \geq 1$) those on the closed edge $[v_{i-1}, v_i]$, interior collinear points included.*

*(vi) (Duality.) $\Phi(c) := \min_\pi J_c(\pi) = \min_i(c\,\alpha_i + \rho_i)$ is concave and piecewise linear, with derivative $\alpha_i$ on $(\sigma_{i+1}, \sigma_i)$, and the lower boundary $\underline{C}(\alpha) := \min\{y : (\alpha, y) \in C\}$ is $\underline{C}(\alpha) = \sup_{c>0}(\Phi(c) - c\,\alpha)$ for $\alpha \geq \alpha_0$.*

*Proof idea.* Only finitely many points lie below a line of negative slope through a point of $\mathcal{A}$, so the lower boundary of $C$ is a polygon built by gift wrapping; (i)–(iv) are elementary convex geometry on it, (v) follows by minimising $J_c$ along it, and (vi) is the Legendre transform (Appendix A.3). So the $\lambda$-MAPs on a data set sit where the hull's slope crosses $-\lambda$. An absolute length bound, a natural alternative to the penalty, also selects points that no penalty selects (Remark A.2). The theorems on the teacher's errors (Section 8) ask not whether the good model is a vertex but whether every near-optimal hypothesis knows the model and not the errors (Remark 8.10).

The last lemma is the sufficiency line: one bit of description pays for at most one bit of fit.

**Lemma 3.9** (Coding bound)**.** *In every data-set class (product or shared hypotheses), for every program $\pi$ and data set $D$, $\ell(\pi) + L(\pi; D) \geq K(y_D \mid \mathrm{dom}\, D) - \kappa$, where $y_D$ lists the labels of $D$ in the canonical order of* $\mathrm{dom}\, D$.

*Proof idea.* Given $\pi$ and $\operatorname{dom} D$, the labelling $y \mapsto P_\pi(\operatorname{dom} D \mapsto y)$ is an lsc semimeasure on $Y^{\operatorname{dom} D}$, uniformly in $(\pi, \operatorname{dom} D)$; apply (F2), (F3) and (F1) (Appendix A.4). In the plane of Lemma 3.8 the bound is a line of slope $-1$ below all points. If $\pi_s$ is *b-sufficient*, $\ell(\pi_s) + L(\pi_s; D) \leq K(y_D \mid \operatorname{dom} D) + b$, then a program longer than $\pi_s$ by $x$ bits fits better by at most $x + b + \kappa$ bits: a right line of slope 1 with offset $b + \kappa$ at $\pi_s$ (Definition 5.2 and Corollary 5.5).

What separates the two kinds of class is whether one member attains the best two-part code length $\min_\pi[\ell(\pi) + L(\pi; D)]$ up to a constant on every data set: under (Dom) $U$ does, $\ell_U - \log U(D) \leq \ell(\pi) + L(\pi; D) + \ell_U + \kappa_0$ for every $\pi$ (Section 4); in the product class no member does (Corollary 4.5).

# 4 What counts as one hypothesis: absorption and private randomness

Under a steeper penalty, a hypothesis that is itself a universal mixture, and so penalises complexity only in the usual way inside itself, can absorb all the structure of the data. This is a standard point, familiar from algorithmic statistics, where the model classes are restricted, for example to finite sets (Gács et al., 2001; Vereshchagin and Shen, 2017). In function induction there is a way out: if each input samples its own random bits, a hypothesis pays its likelihood term separately at every input. Both reasons for a steeper penalty need the selected hypothesis to be a model, not a universal learner, so this section is their precondition. Proofs are in Appendix B.

## 4.1 Shared randomness: a universal hypothesis absorbs everything

A class satisfies (Dom) if it contains a program $\pi_U$, with $U := P_{\pi_U}$ and $\ell_U := \ell(\pi_U)$, such that

$$P_\pi(o) \;\leq\; 2^{\ell(\pi)+\kappa_0}\, U(o) \qquad \text{for all programs } \pi \text{ and observations } o. \tag{Dom}$$

*Remark* 4.1 (Classes with an absorbing member). (Dom) holds with $\kappa_0 = 0$, and $\xi_1 \leq U$, in every class that contains its own mixture as a member: in $\mathcal{H}_F$ with $U = M_F$, in $\mathcal{H}_w$ with $U = M^w$, in $\mathcal{H}_{\mathrm{SI}}$ with $U = M$, and in $\mathcal{H}_{\mathrm{chr}}$ with $U = M_H$ (proof in Appendix B.1).

Let $\ell^*_\lambda := (\lambda \ell_U + \kappa_0)/(\lambda - 1)$.

**Theorem 4.2** (Absorption)**.** *Assume (Dom) and let $\lambda > 1$.*

*(a)* MAP. *For every observation $o$ with $U(o) > 0$, every $\lambda$-MAP $\hat\pi$ has $\ell(\hat\pi) \leq \ell^*_\lambda$ and $L(\hat\pi; o) \leq -\log U(o) + \lambda\ell_U - \lambda\ell(\hat\pi)$. So if $\bar\ell > \ell_U$ and $\lambda > (\bar\ell + \kappa_0)/(\bar\ell - \ell_U)$, no hypothesis of length at least $\bar\ell$ is a $\lambda$-MAP for any such observation; this holds for fixed $\lambda$, and for $\lambda = cn$ once $cn$ exceeds the bound.*

*(b)* Power posterior. $2^{-\lambda\ell_U} U(o) \leq \xi_\lambda(o) \leq \xi_1(o)$. *If $\xi_1 \leq 2^{\kappa_1} U$, then $-\kappa_1 \leq -\log \xi_\lambda(o) + \log U(o) \leq \lambda\ell_U$ for every $o$.*

*(c)* Deterministic teacher, errors included (data-set classes). *Let $\lambda > 1$ be fixed. Let $t : X \to Y$ be total computable with a program $\pi_t$ of length $\ell_t$ in the class, let $x_1, x_2, \ldots$ be distinct and $D_n := \{(x_i, t(x_i)) : i \leq n\}$.*

*(c1)* $\sum_{i \leq n} -\log \xi_\lambda(t(x_i) \mid x_i, D_{i-1}) \leq \min\{\lambda\ell_t,\ \lambda\ell_U + \kappa_0 + \ell_t\}$ *for every $n$; so at most that many inputs get probability below $1/2$ for the teacher's label.*

*(c2) Every $\lambda$-MAP for $D_n$ has $L(\hat\pi; D_n) \leq \lambda\ell_U + \kappa_0 + \ell_t$.*

*(c3) There are $n_0$ and a finite set $\mathcal{H}_{\mathrm{fin}}$ such that every $\lambda$-MAP for $D_n$, $n \geq n_0$, lies in $\mathcal{H}_{\mathrm{fin}}$, and at only finitely many $n$ does a $\lambda$-MAP give $t(x_{n+1})$ probability below $1/2$.*

*With $\lambda = cn$, (c1) and (c3) are not available; (a) and (c2) hold at each $n$.*

*(d)* SI with random inputs. *Let the inputs lie in $\{0,1\}^d$, encode $(x_1, y_1, x_2, \dots)$ as a binary string with fixed-length blocks, let the $x_i$ be i.i.d. from a computable law and $y_i = t(x_i)$, and let $\mathbb{P}$ be the resulting computable measure, with a program $\pi_{\mathbb{P}} \in \mathcal{H}_{\mathrm{SI}}$. Then $\mathbb{E}_{\mathbb{P}} \sum_{i \leq n} -\log \xi_\lambda(y_i \mid x_1 y_1 \cdots x_i) \leq \lambda \ell_U + \kappa_0 + \ell(\pi_{\mathbb{P}})$.*

*Proof idea.* (a) A $\lambda$-MAP is at least as good as $\pi_U$, and (Dom) bounds its gain in fit by $\ell(\hat{\pi}) + \kappa_0$, so $(\lambda - 1)\ell(\hat{\pi}) \leq \lambda \ell_U + \kappa_0$. (b) Keep one term of the mixture. (c) Telescope $\xi_\lambda(D_n) \geq 2^{-\lambda \ell_U} U(D_n)$; for (c3), each of the finitely many short programs either has bounded loss along the sequence or is eventually beaten by $\pi_U$. (d) is Solomonoff's bound (Solomonoff, 1978) restricted to the label positions (Appendix B.2).

On one data set (Dom) also flattens the hull of Lemma 3.8: every left kink $(\sigma, w)$ (Definition 5.2) at a reference point $(\alpha_{\mathrm{r}}, L_{\mathrm{r}})$ with $\alpha_{\mathrm{r}} - w > \ell_U$ has $\sigma \leq 1 + (\ell_U + \kappa_0)/(\alpha_{\mathrm{r}} - \ell_U)$ (Proposition 5.7), so no steep change of slope far beyond $\ell_U$ can single out a model.

*Remark* 4.3 (Reading). Part (a) is the absorption objection made exact. The threshold $\ell^*_\lambda = \ell_U + (\ell_U + \kappa_0)/(\lambda - 1)$, that of the earlier report (Claude (Anthropic), 2026), tends to $\ell_U$, not 0, as $\lambda \to \infty$; a model of millions of bits is longer than $\ell_U$ for any reasonable reference machine. For adaptive or non-computable inputs, $\mathcal{H}_{\mathrm{chr}}$ replaces (d). Part (c3) has no explicit rate; explicit online bounds for two-part MDL are in general exponentially worse than for Bayes mixtures (Poland and Hutter, 2005).

### 4.2 No split with shared randomness

A *sampler* is a probability kernel $\nu$ (Definition 6.1); here it has a program $\pi_\nu$ of length $k$ in the class considered, so in the canonical classes it is a computable conditional measure. The first reason for a steeper penalty is to select $\pi_\nu$, model plus noise, not a *fill-in*, a deterministic program storing the sampled labels (Section 6). With shared randomness no long sampler is selected.

**Theorem 4.4** (SI, or SFI with shared randomness, $\lambda > 1$)**.** *Assume (Dom) and $\xi_1 \leq 2^{\kappa_1} U$, let $\lambda > 1$, and let $\pi_\nu$ be a program of length $k$ for a sampler in the class. For every observation $o$ with $U(o) > 0$:*

*(i) if $(\lambda - 1)k > \lambda \ell_U + \kappa_0$, then $J_\lambda(\pi_U; o) < J_\lambda(\pi_\nu; o)$: the sampler program is never a $\lambda$-MAP, and every $\lambda$-MAP has length at most $\ell^*_\lambda < k$;*

*(ii) for every nonempty set $A$ of programs, $\Pi_\lambda(A \mid o) \leq 2^{-(\lambda-1)\min_{\pi \in A} \ell(\pi) + \lambda \ell_U + \kappa_1}$; in particular the power posterior puts at most $2^{-(\lambda-1)\bar{\ell} + \lambda \ell_U + \kappa_1}$ on all programs of length at least $\bar{\ell}$.*

*(For SI with adaptive or non-computable inputs the sampler, which does not model the inputs, is not an SI hypothesis; use $\mathcal{H}_{\mathrm{chr}}$ with $U = M_H$. With i.i.d. computable inputs, $\mathcal{H}_{\mathrm{SI}}$ with $U = M$ works.)*

*Proof idea.* (i) is Theorem 4.2(a) plus one comparison: by (Dom), $\pi_U$ fits at most $k + \kappa_0$ bits worse than $\pi_\nu$ and is $k - \ell_U$ bits shorter. (ii) bounds the numerator by $2^{-(\lambda-1)\min_A \ell} \xi_1(o)$ and the normaliser by $2^{-\lambda \ell_U} U(o)$ (Appendix B.3).

So in a shared class the steeper posterior lives on short programs whatever the data, while in the product class it concentrates on programs equivalent to the sampler (Theorem 6.3(b), Theorem 6.9(iv)), whose length is unbounded over samplers. The difference is *what the selected*

*hypothesis is*: a model plus private noise, or a short universal learner that contains no model of $\nu$ (Example B.1). In SI's representation by deterministic programs no complexity weight separates a sampler from its fill-ins (Remark D.8).

### 4.3 Private randomness: no hypothesis absorbs anything

On distinct inputs a product hypothesis pays its likelihood term at every input: for a finite set $F$ of functions and any law $Q$ on $F$, some $f \in F$ costs it at least $\sum_i H(Q_{x_i})$ bits, with $Q_x$ the law of $f(x)$. On the parities of $\{0,1\}^d$, at nonzero inputs, it pays at least $n$ bits on some parity, where the shared Bayes mixture over parities pays at most $d$ (Proposition B.2 in Appendix B.4).

**Corollary 4.5** (No universal product hypothesis)**.** *For every product hypothesis $\nu_U$ and every $d \geq 1$ there are a data set $D$ with $\operatorname{dom} D = \{0,1\}^d \setminus \{0^d\}$ and a deterministic $f \supseteq D$ with a program of length at most $d + 2\log d + \kappa$ such that $\nu_U(D) \leq 2^{-(2^d-1)}$. Hence for every $C$ there are such $D$ and $f$ with $\nu_U(D) < 2^{-C} 2^{-\ell(\pi_f)} f(D)$, and (Dom) fails in $\mathcal{H}_\times$, and in every product class that contains, for every $d$, programs of length at most $d + 2\log d + \kappa$ for the parities on $\{0,1\}^d$.*

*Proof.* Take the parity given by Proposition B.2(ii) on which $\nu_U$ loses at least $2^d - 1$ bits; it is described by $d$ and $a \in \{0,1\}^d$, and (F5) gives the length bound. Choose $d$ with $2^d - 1 - d - 2\log d - \kappa > C$. □

The natural candidate for a universal product hypothesis, the universal lsc conditional semimeasure, pays a constant at every input and never learns (Remark B.3).

### 4.4 Why private randomness, and what it costs

Steeper SFI is therefore defined over $\mathcal{H}_\times$, whose $\lambda \to 1$ limit is $M_{\mathrm{prod}}$, not $M_F$ (Remark B.4 gives the reasons and the cost at $\lambda = 1$). At $\lambda > 1$ the price is a constant factor on randomness shared by several inputs. On data $D_\beta$ that show each of $s$ fair coins $\beta_i$ on two inputs, every product program has $J_\lambda(\pi; D_\beta) \geq \min(\lambda, 2)\,(K(\beta \mid s) - \kappa)$, while $-\log M_F(D_\beta) \leq s + \kappa$ (Proposition B.5). A product class pays for such a coin either as model, at $\lambda$ bits, or as noise at every input that shows it; a shared class pays one bit. Persistent latent causes, such as a per-batch coin, are of this kind.

The protection also depends on what an input is. If inputs contain the earlier labelled pairs of their block, an in-context Bayes mixture is one product hypothesis that pays its best member's log-weight once per block (Proposition B.6); if they contain only the earlier inputs, pricing stays per input (Proposition B.7); with shared randomness ($\mathrm{SFI}^{\mathrm{hist}}$) Theorem 4.2 applies with $U = M_H$.

### 4.5 Answer to the absorption objection

*Remark* 4.6 (Answer to the absorption objection)*.* (i) *The objection is correct wherever a universal hypothesis is one member of the class*: in SI, in SFI with shared randomness, in the weak class and in history-aware SFI. There every $\lambda$-MAP has length at most $\ell^*_\lambda$ and fits almost as well as $U$; for fixed $\lambda$ it reproduces a deterministic teacher's errors (Theorem 4.2(c)); no sampler longer than $\ell^*_\lambda$ is selected (Theorem 4.4); with $\lambda = cn$, $\ell^*_\lambda \to \ell_U$. (ii) *The construction is unavailable in $\mathcal{H}_\times$.* No product hypothesis is universal (Corollary 4.5), the universal conditional semimeasure never learns (Remark B.3), and seeing earlier inputs does not help while the randomness stays private (Proposition B.7). (iii) *The restriction has a price and a boundary*: randomness shared across inputs costs more, and the protection ends where hypotheses see labelled history (Remark B.4

and Propositions B.5 and B.6). (iv) *Algorithmic statistics*: the $d$-bit parameter of a noisy random parity is noise for set models, which share their randomness, and model for product models (Propositions 7.7, 7.10 and E.5). The restriction is a modelling choice, supported by two reasons: the product hypotheses are exactly the computably samplable predictors that use different random bits on different inputs, which SFI dominates (Hänni, 2026a, Theorem 5.1(a)); and in supervised learning the model is naturally the part of the description that is reused across inputs.

**Question 4.7** (shared hypotheses)**.** Is every lower-semicomputable law of a random partial function dominated by $M_F$ up to a constant factor? This question of the companion paper (Hänni, 2026a, Question 3.14) decides which laws with shared randomness the class $\mathcal{H}_F$ covers; Theorem 4.2 holds for $\mathcal{H}_w$ and for $\mathcal{H}_F$ either way.

# 5 Kinks: the model–randomness split on one data set

Besides a steeper penalty, one can look for the split between model and noise by plotting the entropy left given the hypothesis against the hypothesis length and looking for a change of slope. With shared randomness, as in the sequence case, one expects none where a random parameter of the model ends, because each of its bits pays for about one bit of data (Remark 7.11).[2] We make it precise as a *kink*: every complexity weight $\lambda$ between the two slopes of a kink selects the model there, up to explicit windows, and stable selection happens only at kinks. The geometry is common to both reasons for a steeper penalty: right of the kink lies sampled noise (Section 6) or a teacher's error rule (Section 8). Proofs are in Appendix C.

## 5.1 The product structure function

**Definition 5.1** (Structure functions and sufficiency)**.** A *point family* is a countable set $\mathcal{P}$ of programs $\pi$ with integer lengths $\ell(\pi) \geq 0$, finitely many of each length, and fits $L(\pi) \in [0, \infty]$, not all infinite; for example, a data-set class on one observation $o$ (Definitions 3.1 and 3.5) with $L(\pi) := L(\pi; o)$. Its *structure function* is $\rho(\alpha) := \min\{L(\pi) : \ell(\pi) \leq \alpha\}$ ($\min \emptyset := \infty$), and $\breve{\rho}$ is the greatest convex minorant of $\rho$. On a data set $D$ we write $\rho_D$, and $\rho_D^{\times}$ for $\mathcal{H}_{\times}$, the *product structure function*; in population form, $\rho_{\mathrm{pop}}(\alpha) := \inf\{R(\pi) : \ell(\pi) \leq \alpha\}$, with $R$ as in Remark 3.7. In a data-set class, $\pi$ is *$b$-sufficient* for $D$ if $\ell(\pi) + L(\pi; D) \leq K(y_D \mid \mathrm{dom}\, D) + b$; in $\mathcal{H}_{\times}$ we say *product-$b$-sufficient*, and $\mathrm{soph}_b^{\times}(D) := \min\{\ell(\pi) : \pi \in \mathcal{H}_{\times} \text{ is product-}b\text{-sufficient for } D\}$ is the *product sophistication*.

**The hull.** A *$\lambda$-MAP* minimises $J_\lambda := \lambda\ell + L$ ($\lambda > 0$). Let $C$ be the hull of Lemma 3.8 for the points $(\ell(\pi), L(\pi))$, with vertices $v_i = (\alpha_i, \rho_i)$ in order of length and edge slopes $-\sigma_i$, so that $\sigma_0 = \infty > \sigma_1 > \sigma_2 > \cdots > 0$ and $\sigma_{i+1} = 0$ at a last vertex $v_i$. By Lemma 3.8(v), for $\lambda$ in the *$\lambda$-interval* $(\sigma_{i+1}, \sigma_i)$ of $v_i$ the $\lambda$-MAPs are the programs at $v_i$. The lower boundary of $C$ is the graph of $\breve{\rho}$ (Appendix C.1), so the $\lambda$-MAPs lie where the slope of $\breve{\rho}$ crosses $-\lambda$.

## 5.2 Kinks, the kink lemma and the split

**Definition 5.2** (Kinks, right lines and secants)**.** Let $\mathcal{P}$ be a countable set of programs with lengths $\ell(\pi) \geq 0$ and two fits $0 \leq L'(\pi) \leq L(\pi) \leq \infty$ ($L' = L$ on a point family). A point

[2] The suggestion to look for this change of slope in function induction, and the reason why none is expected with shared randomness, first appeared in the author's note (Hänni, 2026b).

$(\alpha_{\mathrm{r}}, L_{\mathrm{r}}) \in \mathbb{R}^2$ is *bounded* if some program $\pi_{\mathrm{r}}$ has $\ell(\pi_{\mathrm{r}}) \le \alpha_{\mathrm{r}}$ and $L(\pi_{\mathrm{r}}) \le L_{\mathrm{r}}$. A bounded point is a *left kink of steepness* $\sigma \in (0,\infty]$ *and window* $w \ge 0$ if every $\pi$ with $\ell(\pi) < \alpha_{\mathrm{r}} - w$ has $L(\pi) \ge L_{\mathrm{r}} + \sigma(\alpha_{\mathrm{r}} - \ell(\pi))$, and a *right kink of flatness* $\tau \ge 0$ *and window* $w_+ \ge 0$ if every $\pi$ with $\ell(\pi) > \alpha_{\mathrm{r}} + w_+$ has $L(\pi) \ge L_{\mathrm{r}} - \tau(\ell(\pi) - \alpha_{\mathrm{r}})$; both together form a $(\sigma,\tau)$-*kink*. A set $\mathcal{M} \subseteq \mathcal{P}$ has a *right line* $(\tau, g)$, with $\tau \ge 0$ and $g \in \mathbb{R}$, and *length floor* $\ell_{\mathcal{M}} \le \alpha_{\mathrm{r}}$ if every $\pi \in \mathcal{M}$ has $\ell(\pi) \ge \ell_{\mathcal{M}}$ and $L'(\pi) \ge L_{\mathrm{r}} - g - \tau(\ell(\pi) - \alpha_{\mathrm{r}})$. The *right secant* on $\mathcal{M}$ is $s_+ := \max\{0, \sup\{(L_{\mathrm{r}} - L'(\pi))/(\ell(\pi) - \alpha_{\mathrm{r}}) : \pi \in \mathcal{M},\ \ell(\pi) > \alpha_{\mathrm{r}}\}\}$, and the *left steepness* is $s_- := \inf\{(L(\pi) - L_{\mathrm{r}})/(\alpha_{\mathrm{r}} - \ell(\pi)) : \ell(\pi) < \alpha_{\mathrm{r}}\}$ ($\inf \emptyset := \infty$).

For $\mathcal{M} = \mathcal{P}$ and $L' = L$, a bounded point is a zero-window $(\sigma,\tau)$-kink iff $s_- \ge \sigma$ and $s_+ \le \tau$; a right line $(\tau, g)$, $g \ge 0$, gives right kinks of flatness $\tau' > \tau$ and window $g/(\tau' - \tau)$ (Lemma C.1).

For $c > 0$ put $J_c := c\,\ell + L$, $J'_c := c\,\ell + L'$, $\Phi(c) := \inf_\pi J_c(\pi)$, the reference value $J_{\mathrm{r}}(c) := c\alpha_{\mathrm{r}} + L_{\mathrm{r}}$, and, for a partition $\mathcal{P} = \mathcal{M} \sqcup \mathcal{N}$, the *left margin* $\eta_{\mathcal{N}}(c) := \inf_{\pi \in \mathcal{N}} J'_c(\pi) - J_{\mathrm{r}}(c) \in (-\infty, \infty]$. In the *data reading* $L = L' = L(\cdot; o)$ and $c = \lambda$, in total bits; in the *population reading* $L = R$, $L' \le R$ (for example the truncated risk $R_B$ of Definition 7.1) and $c = \lambda/n$, per sample.

**Lemma 5.3** (The kink lemma)**.** *Let $(\alpha_{\mathrm{r}}, L_{\mathrm{r}})$ be bounded, $\mathcal{P} = \mathcal{M} \sqcup \mathcal{N}$, $c > 0$ and $\epsilon \ge 0$. In (i) and (ii) let $\mathcal{M}$ have a right line $(\tau, g)$ with length floor $\ell_{\mathcal{M}}$, let $c > \tau$ and put $\Delta(c) := g + (c - \tau)(\alpha_{\mathrm{r}} - \ell_{\mathcal{M}})$.*

*(i)* $\inf_\pi J'_c(\pi) \ge J_{\mathrm{r}}(c) + \min\{\eta_{\mathcal{N}}(c), -\Delta(c)\}$. *If a program of $\mathcal{M}$ bounds the point, then $g \ge 0$ and $\Delta(c) \ge 0$. If $\eta_{\mathcal{N}}(c) \ge 0$ and $\Delta(c) \ge 0$, then $J_{\mathrm{r}}(c) - \Delta(c) \le \Phi(c) \le J_{\mathrm{r}}(c)$.*

*(ii) If every $\pi \in \mathcal{N}$ has $J'_c(\pi) > J_{\mathrm{r}}(c) + \epsilon$ (for instance, if $\eta_{\mathcal{N}}(c) > \epsilon$), then every $\pi$ with $J'_c(\pi) \le J_{\mathrm{r}}(c) + \epsilon$, in particular every $\pi$ with $J_c(\pi) \le \Phi(c) + \epsilon$, lies in $\mathcal{M}$ and has $\ell(\pi) \le \alpha_{\mathrm{r}} + (g + \epsilon)/(c - \tau)$ and $L'(\pi) \ge L_{\mathrm{r}} - (cg + \tau\epsilon)/(c - \tau)$.*

*(iii) $\eta_{\mathcal{N}}$ is concave on $(0, \infty)$.*

*(iv)* No right line needed: *if every $\pi \in \mathcal{M}$ has $\ell(\pi) \ge \ell_{\mathcal{M}}$, every $\pi \in \mathcal{M}$ with $J'_c(\pi) \le J_{\mathrm{r}}(c) + \epsilon$ has $L'(\pi) \le L_{\mathrm{r}} + \epsilon + c(\alpha_{\mathrm{r}} - \ell(\pi)) \le L_{\mathrm{r}} + \epsilon + c(\alpha_{\mathrm{r}} - \ell_{\mathcal{M}})$.*

*(v)* Wedge criterion: *if $\sigma > 0$, $\vartheta \ge 0$ and every $\pi \in \mathcal{N}$ has $L'(\pi) \ge L_{\mathrm{r}} + \max\{\sigma(\alpha_{\mathrm{r}} - \ell(\pi)), \vartheta\}$, then $\eta_{\mathcal{N}}(c) \ge \vartheta(1 - c/\sigma) \ge 0$ for $c \in (0, \sigma]$.*

*Proof idea.* On $\mathcal{M}$ the line gives $J'_c \ge J_{\mathrm{r}}(c) - \Delta(c)$, as $c > \tau$. A program at most $\epsilon$ worse than the bounding one avoids $\mathcal{N}$ by the margin, and the line bounds its length (Appendix C.2).

**Corollary 5.4** (Kinks locate the $\lambda$-MAPs)**.** *Let $\mathcal{P}$ be a point family and $\lambda > 0$. (a) If $\mathcal{P}$ has a left kink $(\sigma, w)$ at $(\alpha_-, L_-)$ and $\lambda < \sigma$, every $\lambda$-MAP $\hat{\pi}$ has $\ell(\hat{\pi}) \ge \alpha_- - w$ and $L(\hat{\pi}) \le L_- + \lambda(\alpha_- - \ell(\hat{\pi})) \le L_- + \lambda w$; if also $\lambda \ge 1$, then $\ell(\hat{\pi}) + L(\hat{\pi}) \le \alpha_- + L_- + (\lambda - 1)w$. (b) If $\mathcal{P}$ has a right kink $(\tau, w_+)$ at $(\alpha_+, L_+)$ and $\lambda > \tau$, every $\lambda$-MAP has $\ell(\hat{\pi}) \le \alpha_+ + w_+$. (c) If $\mathcal{P}$ has a zero-window $(\sigma,\tau)$-kink at $(\alpha_{\mathrm{r}}, L_{\mathrm{r}})$ with $\tau < \sigma$, then $\rho(\alpha_{\mathrm{r}})$ is attained at length exactly $\alpha_{\mathrm{r}}$, $(\alpha_{\mathrm{r}}, \rho(\alpha_{\mathrm{r}}))$ is a vertex $v_i$ with $(\tau, \sigma) \subseteq (\sigma_{i+1}, \sigma_i)$, and for $\lambda \in (\tau, \sigma)$ the $\lambda$-MAPs are exactly the programs at $v_i$. Conversely, every vertex $v_i$ is a zero-window $(\sigma_i, \sigma_{i+1})$-kink, and $\sigma_{i+1} < \sigma_i$.*

*Proof idea.* A $\lambda$-MAP is no worse than the bounding program; beyond the left window one loses at least $\sigma - \lambda > 0$ per bit, beyond the right one gains at most $\tau < \lambda$ per bit. In (c) the $\lambda$-MAP point is fixed on $(\tau, \sigma)$, hence a vertex. No strict inequality can be dropped (Appendix C.3).

**Corollary 5.5** (The split on one data set)**.** *Let $D$ be a data set and $\mathcal{C}$ a hypothesis class of data-set hypotheses, with the constant $\kappa$ of the sufficiency line (Lemma 3.9); let $\pi_s \in \mathcal{C}$ be $b$-sufficient for $D$, and $(\alpha_{\mathrm{r}}, L_{\mathrm{r}}) := (\ell(\pi_s), L(\pi_s; D))$. (a) Every $\pi$ has $L(\pi; D) \ge L_{\mathrm{r}} - (b + \kappa) - (\ell(\pi) - \alpha_{\mathrm{r}})$: a right line $(1, b + \kappa)$ on all programs. So for every $\tau > 1$ there is a right kink of flatness $\tau$ and window*

$(b+\kappa)/(\tau-1)$ *at* $(\alpha_{\mathrm{r}}, L_{\mathrm{r}})$*, and for every* $\lambda > 1$ *every* $\lambda$*-MAP* $\hat{\pi}$ *has* $\ell(\hat{\pi}) \le \alpha_{\mathrm{r}} + (b+\kappa)/(\lambda-1)$ *and* $L(\hat{\pi}; D) \ge L_{\mathrm{r}} - \lambda(b+\kappa)/(\lambda-1)$*. (b) If moreover* $D$ *has a left kink* $(\sigma, w)$ *at* $(\alpha_{\mathrm{r}}, L_{\mathrm{r}})$*, then for every* $\lambda \in (1, \sigma)$ *every* $\lambda$*-MAP* $\hat{\pi}$ *is* $(b + (\lambda-1)w)$*-sufficient and satisfies*

$$\alpha_{\mathrm{r}} - w \le \ell(\hat{\pi}) \le \alpha_{\mathrm{r}} + \frac{b+\kappa}{\lambda-1}, \qquad L_{\mathrm{r}} - \frac{\lambda(b+\kappa)}{\lambda-1} \le L(\hat{\pi}; D) \le L_{\mathrm{r}} + \lambda w.$$

*Proof idea.* The right line is the coding bound rearranged around $\pi_s$, and (a) is Lemma 5.3(ii) with $\mathcal{N} = \emptyset$, $\tau = 1$ and $c = \lambda$; (b) adds Corollary 5.4(a) (Appendix C.3).

At a program attaining $\mathrm{soph}_b^{\times}(D)$, (a) gives every $\lambda$-MAP over $\mathcal{H}_{\times}$ length at most $\mathrm{soph}_b^{\times}(D) + (b+\kappa)/(\lambda-1)$ (Proposition C.2), a bound vacuous as $\lambda \to 1$ (Remark E.2). The content of (b) is a steepness $\sigma > 1$, which the coding bound alone does not give; per-query samples have one with high probability (Theorem 6.7, where the Ville line of Theorem 6.7(iii) replaces the coding bound), and so does the full table of a noisy random parity under the conditions of Proposition 7.7.

**Theorem 5.6** (Stable MAP lengths occur only at kinks)**.** *Let* $\mathcal{P}$ *be a point family,* $0 \le \tau < \sigma \le \infty$ *and* $\alpha' \le \alpha''$*. Every* $\lambda$*-MAP with* $\lambda \in (\tau, \sigma)$ *has length in* $[\alpha', \alpha'']$ *if and only if* $\mathcal{P}$ *has a zero-window left kink of steepness* $\sigma$ *at a point* $(\alpha_-, \rho(\alpha_-))$ *with* $\alpha_- \ge \alpha'$ *and a zero-window right kink of flatness* $\tau$ *at a point* $(\alpha_+, \rho(\alpha_+))$ *with* $\alpha_+ \le \alpha''$*. With the indices* $i, j$ *such that* $\sigma_{i+1} < \sigma \le \sigma_i$ *and* $\sigma_{j+1} \le \tau < \sigma_j$*, one can take* $\alpha_- = \alpha_i$*,* $\alpha_+ = \alpha_j$*, and the* $\lambda$*-MAP lengths over* $(\tau, \sigma)$ *have minimum* $\alpha_i$ *and maximum* $\alpha_j$*. The index* $j$ *exists unless* $\tau = 0$ *and there are infinitely many vertices; then both conditions fail for every finite* $\alpha''$*.*

*Proof idea.* One direction is Corollary 5.4(a),(b). Conversely, the $\lambda$-MAPs sit at $v_i$ for $\lambda$ just below $\sigma$ and at $v_j$ just above $\tau$, and by Corollary 5.4(c) these are zero-window kinks of steepness $\sigma_i \ge \sigma$ and flatness $\sigma_{j+1} \le \tau$ (Appendix C.3).

**Proposition 5.7** (A universal member flattens every kink far beyond its length)**.** *Let a point family contain* $\pi_U$*, with* $\ell_U := \ell(\pi_U)$ *and* $L(\pi_U) \le L(\pi) + \ell(\pi) + \kappa_0$ *for every* $\pi$*, which is* (Dom) *on one observation. (i) Every left kink* $(\sigma, w)$ *at* $(\alpha_{\mathrm{r}}, L_{\mathrm{r}})$ *with* $\alpha_{\mathrm{r}} - w > \ell_U$ *has* $\sigma \le 1 + (\ell_U + \kappa_0)/(\alpha_{\mathrm{r}} - \ell_U)$*; so for* $\epsilon > 0$ *every such kink steeper than* $1+\epsilon$ *has* $\alpha_{\mathrm{r}} \le \ell_U + (\ell_U + \kappa_0)/\epsilon$*. (ii) For every* $\lambda > 1$ *every* $\lambda$*-MAP has length at most* $\ell_U + (\ell_U + \kappa_0)/(\lambda-1)$*. For* $\epsilon > 0$ *this is at most* $\ell_U + (\ell_U + \kappa_0)/\epsilon$ *when* $\lambda \ge 1 + \epsilon$*, and, if* $\ell_U + \kappa_0 > 0$*, only then.*

*Proof idea.* (i) The hypothesis puts $L(\pi_U)$ at most $\alpha_{\mathrm{r}} + \kappa_0$ above $L_{\mathrm{r}}$, the kink at least $\sigma(\alpha_{\mathrm{r}} - \ell_U)$ above. (ii) Insert it into $J_\lambda(\hat{\pi}) \le J_\lambda(\pi_U)$ (Appendix C.3).

*Remark* 5.8 (Reading). The $\lambda$-MAP takes exactly the hull edges with $\sigma_i > \lambda$: the model keeps the blocks of program bits that explain more than $\lambda$ bits of data per bit, on average over an edge, not bit by bit. Several vertices with $\sigma_i > 1$ give different models for different $\lambda$ (Section 9). Under (Dom), which holds in $\mathcal{H}_F$ and fails in $\mathcal{H}_{\times}$ (Corollary 4.5), Proposition 5.7(ii) is Theorem 4.2(a).

## 6 Exact model recovery from samples

This section gives the first and main reason for a complexity penalty steeper than the likelihood penalty: it separates a model from its noise. Labels drawn from a sampler are explained by the sampler, with private coins, and by *fill-ins*, deterministic programs that store the sampled bits; on one data set, at $\lambda = 1$, the two get similar posterior weight (Theorem 6.11). Yet model-plus-noise splits work exactly per query: with i.i.d. inputs and a fresh label at every query, every $\lambda$-MAP is eventually a shortest program for the sampler, for every schedule $1 \le \lambda_n = o(n)$

(Theorem 6.3). A steeper prior adds a rate that depends only on $\lambda$, the program's length, a margin and the confidence, which $\lambda = 1$ lacks (Proposition 6.6); the sample has a kink at the model (Theorem 6.7). With frozen labels no procedure recovers every sampler, and every $\lambda \geq 1$ eventually selects programs that are $\nu$-equivalent on the run: their Hellinger gaps to $\nu$ along the queried inputs are summable (Theorem 6.9 and Proposition 6.10). Proofs are in Appendix D.

### 6.1 Samplers, equivalent programs and the Ville deficiency

**Definition 6.1** (Samplers, equivalent programs, margins and fill-ins)**.** Let $\mathcal{C}$ be a countable class of programs $\pi$ with lengths $\ell(\pi)$, $\sum_{\pi\in\mathcal{C}} 2^{-\ell(\pi)} \leq 1$, and kernels $\nu_\pi(y \mid x) \geq 0$, $\sum_y \nu_\pi(y \mid x) \leq 1$: for example $\mathcal{H}_\times$, a time-bounded or circuit subclass of it, or a finite class. A *sampler* is a probability kernel $\nu$. We write $o_n$ for the data after $n$ queries ($h_n$ per query, $D_n$ with frozen labels; Definition 3.1), $L_n(\pi) := L(\pi; o_n)$, and $L_n(\nu) := -\log \nu(o_n)$, the *sampled randomness. Per query* (PQ data, inputs i.i.d. from any law $\mu$): $\mathcal{E}$ is the set of programs whose kernel equals $\nu$ at $\mu$-almost every input, assumed nonempty; $\ell_0 := \min_{\mathcal{E}} \ell$, and $\mathcal{E}_0$ is the finite set of programs of $\mathcal{E}$ of length $\ell_0$. The *Hellinger gap* is $d^2_{\mathrm{H}\mu}(\pi) := \mathbb{E}_{x\sim\mu} d^2_{\mathrm{H}}(\nu(\cdot \mid x), \nu_\pi(\cdot \mid x))$, with $d^2_{\mathrm{H}}(P, Q) := 1 - \sum_y \sqrt{P(y)Q(y)}$, also for a sub-probability $Q$; it is positive iff $\pi \notin \mathcal{E}$. The *margins* $\gamma_{\bar\ell}$, $\gamma_-$, $\gamma_=(\ell)$ are the least gaps of programs outside $\mathcal{E}$ of length at most $\bar\ell$, below $\ell_0$, and exactly $\ell$ ($\min\emptyset := \infty$). *Any input rule:* the inputs are chosen by any rule that fixes $x_{n+1}$ before $y_{n+1}$ is drawn (adaptive, randomised or non-computable), the labels are frozen on distinct inputs or drawn per query, and $\pi_\nu \in \mathcal{C}$ is a program of length $k$ whose kernel equals $\nu$ at every input the rule can query; in $\mathcal{H}_\times$ such a $\nu$ is a computable conditional measure. With frozen labels a repeated query changes no data; where a process is needed, it is indexed by distinct inputs in order of first appearance, and frozen labels with i.i.d. inputs are an input rule of this kind. A *fill-in* is a deterministic program consistent with the data; in a data-set class their total weight is $W_{\mathrm{det}}(D) := \sum\{2^{-\ell(\pi)} : \pi \text{ deterministic}, \ \nu_\pi \supseteq D\}$.

**Lemma 6.2** (Ville deficiency)**.** *Let the inputs follow any rule as in Definition 6.1, and let $\xi$ be a shared hypothesis with $\xi(\emptyset) \leq 1$ (frozen labels) or the mixture $\xi_{\mathcal{C}}(o_n) := \sum_{\pi\in\mathcal{C}} 2^{-\ell(\pi)}\nu_\pi(o_n)$ (either kind of labels; $M_{\mathrm{prod}}$ for $\mathcal{H}_\times$ on data sets). The* current deficiency $\delta_n := \log(\xi(o_n)/\nu(o_n))$ *and the* Ville deficiency $\delta^* := \sup_n \delta_n$ *satisfy $\mathbb{P}[\delta^* \geq j] \leq 2^{-j}$ for every $j \geq 0$, whether or not $\nu$ has a program in $\mathcal{C}$. For $\xi = \xi_{\mathcal{C}}$, every $\pi \in \mathcal{C}$ and every $n$: $L_n(\pi) \geq L_n(\nu) - \ell(\pi) - \delta_n \geq L_n(\nu) - \ell(\pi) - \delta^*$.*

*Proof idea.* As $x_{n+1}$ is fixed before $y_{n+1}$, $\xi(o_n)/\nu(o_n)$ is a nonnegative supermartingale started at most at 1; apply Ville's inequality (Ville, 1939) and $\xi_{\mathcal{C}} \geq 2^{-\ell(\pi)}\nu_\pi$ (Appendix D.7).

### 6.2 Exact recovery

Let $H_\mu(\nu) := \mathbb{E}_{x\sim\mu} H(\nu(\cdot \mid x))$, $W_\lambda := \sum_{\pi\in\mathcal{E}} 2^{-\lambda\ell(\pi)}$, and $\mathsf{C}_s(\pi) := \mathbb{E}_{x\sim\mu} \sum_{y:\nu(y|x)>0} \nu(y \mid x)^{1-s}\nu_\pi(y \mid x)^s$, the *Chernoff coefficient* of $\pi$ at $s \in (0, 1]$.

**Theorem 6.3** (Exact model recovery)**.** *In the per-query setting of Definition 6.1 (PQ data, inputs i.i.d. from $\mu$, $\mathcal{E} \neq \emptyset$), with $\delta_n, \delta^*$ the deficiencies of $\xi_{\mathcal{C}}$ (Lemma 6.2):*

*(a)* The limit. *For every schedule $\lambda_n \geq 1$ with $\lambda_n/n \to 0$, fixed $\lambda \geq 1$ included, almost surely the set of $\lambda_n$-MAPs equals $\mathcal{E}_0$ for all large $n$.*

*(b)* The posteriors. *For fixed $\lambda \geq 1$ and $\eta \in (0, 1]$, almost surely $\Pi_\lambda(\mathcal{E} \mid h_n) \to 1$, with $\Pi_\lambda(\pi \mid h_n) \to 2^{-\lambda\ell(\pi)}/W_\lambda$ for $\pi \in \mathcal{E}$; and $\Pi^\eta(\mathcal{E} \mid h_n) \to 1$, with limit shares $2^{-\ell(\pi)}/W_1$.*

*(c)* Every $n$, fixed $\lambda > 1$. *Every $\lambda$-MAP has length at most $(\lambda\ell_0 + \delta_n)/(\lambda - 1) \leq \ell_0 + (\ell_0 + \delta^*)/(\lambda - 1)$. Every fill-in has length at least $L_n(\nu) - \delta^*$; the fill-ins' total mass is at most*

$2^{\lambda\ell_0+\lambda\delta^*-(\lambda-1)L_n(\nu)}$ *under* $\Pi_\lambda$ *and at most* $2^{\ell_0+\delta^*-(1-1/\lambda)L_n(\nu)}$ *under* $\Pi^{1/\lambda}$*; and* $L_n(\nu)/n \to H_\mu(\nu)$ *almost surely.*

*(d)* Each $n$. *For every* $n$ *and every* $\lambda > 0$,

$$\mathbb{P}[\textit{the set of } \lambda\textit{-MAPs} \neq \mathcal{E}_0] \le \sum\nolimits_{\pi\notin\mathcal{E}} \inf\nolimits_{0<s\le 1} 2^{-s\lambda(\ell(\pi)-\ell_0)}\mathsf{C}_s(\pi)^n.$$

*If* $\lambda > 1$, $\varepsilon \in (0,1)$, $\bar{\ell}_\varepsilon := \lceil(\lambda\ell_0 + \log(2/\varepsilon))/(\lambda-1)\rceil$ *and* $0 < \gamma \le \gamma_{\bar{\ell}_\varepsilon}$, *this probability is at most* $\varepsilon$ *for every* $n \ge \lceil\lambda(\ell_0 + \log(2/\varepsilon))\ln 2/(2\min(1,\lambda-1)\gamma)\rceil$.

*Proof idea.* A $\lambda_n$-MAP is no worse than $\mathcal{E}_0$, so, as $\lambda_n \ge 1$, selecting a program longer than $\bar{\ell}$ makes the prior-weighted likelihood ratios against $\nu$ of such programs sum to at least $2^{-\ell_0}$; this sum is a nonnegative supermartingale started at the prior's tail mass, so by Ville's inequality long programs are almost surely never selected. Among the rest, the strong law makes the likelihood ratio of every $\pi \notin \mathcal{E}$ decay exponentially, which beats a length advantage of $o(n)$. (b) is similar, (c) uses Lemma 6.2, and (d) is Markov's inequality for powers of the likelihood ratio (Appendix D.2).

*Remark* 6.4 (Concentration on $\mathcal{E}$, not on $\mathcal{E}_0$). The natural strengthening to concentration on $\mathcal{E}_0$ fails: with programs for $\nu$ of lengths 5 and 7, the limit share of $\mathcal{E}_0$ in (b) is $8/9$ at $\lambda = 1.5$ and $4/5$ for every tempered posterior, and below 1 in every canonical class. Only the MAP singles out $\mathcal{E}_0$.

*Remark* 6.5 (Method and literature). Parts (a) and (b) use the standard argument from Ville's inequality and the strong law, and we claim no novelty for them (Section 2); what needs $\lambda > 1$ is (c) and the margin form of (d).

### 6.3 What a steeper prior buys, and what is necessary

**Two parity classes.** Let the inputs be uniform on $\{0,1\}^d$, $d$ large, and $\nu$ the fair coin. In $\mathcal{C}_k$ the coin has length $k$, and for $1 \le r \le d$ and $\theta \in \{0,1\}^r$ the parity $x \mapsto \langle\theta, x_{1:r}\rangle \bmod 2$ has length $r + 2\lceil\log(r+1)\rceil + 1$; in $\mathcal{C}_{k,r_0}$ ($r_0 \le d$) the coin has length $k$ and the $2^{r_0}$ parities on the first $r_0$ bits have length $r_0 + 2$. In both, $\mathcal{E}_0 = \{\text{coin}\}$, $\ell_0 = k$, and every parity has gap $1 - 2^{-1/2} > 0.29$. The *recovery time* $n_{\mathrm{rec}}(\varepsilon)$ is the least $n$ with $\mathbb{P}[\text{the set of } \lambda\text{-MAPs} \neq \{\text{coin}\}] \le \varepsilon$ at all larger $n$.

**Proposition 6.6** ($\lambda = 1$ against $\lambda > 1$)**.** *(a) Let* $k \ge 8$, $r_0 \ge 6$ *and* $n := r_0 - 5$. *In* $\mathcal{C}_{k,r_0}$, *with probability at least* $31/32$ *every* 1*-MAP at* $n$ *is a parity. So for* $\varepsilon < 31/32$ *no function of* $(\ell_0, \gamma, \varepsilon)$ *bounds the recovery time at* $\lambda = 1$, *even with a uniform margin. (b) In* $\mathcal{C}_k$ *with* $k$ *even,* $d \ge 2^{(k-8)/2}$ *and* $\varepsilon < 31/32$, *at* $\lambda = 1$*:* $2^{(k-8)/2} - 6 < n_{\mathrm{rec}}(\varepsilon) \le \max(32, \lceil 2^{(k+1)/2}/\sqrt{\varepsilon}\,\rceil)$. *(c) For fixed* $\lambda > 1$, *in both classes, whatever* $r_0$ *and* $d$, $\mathbb{P}[\textit{the set of } \lambda\textit{-MAPs} \neq \{\textit{coin}\}] \le 2^{k-2-(1-1/\lambda)n}$ *for every* $n$*; so* $n_{\mathrm{rec}}(\varepsilon) \le \lceil\lambda(k + \log(1/(4\varepsilon)))/(\lambda-1)\rceil$.

*Proof idea.* A parity is consistent iff the labels lie in the column span of the input matrix over GF(2). In (a) and the lower end of (b) a parity on $n+5$ bits is consistent with probability at least $31/32$ and beats the coin; the upper end bounds the dimension of the parities that can tie the coin. (c) is Theorem 6.3(d) with $s = 1/\lambda$ (Appendix D.3).

So in $\mathcal{C}_k$ the recovery time is $\Theta(2^{\ell_0/2})$ at $\lambda = 1$ and $O(\lambda\ell_0/(\lambda-1))$ for fixed $\lambda > 1$, tight up to logarithms (Proposition D.2(c)). A direct computation (code in the accompanying repository), exact on every integer $n$, gives Table 2; at $\lambda = 1$ the times equal $2^{(k+4)/2} + 6$. In Theorem 6.3(d), for $\lambda > 1$, the steeper prior charges every program longer than $\bar{\ell}_\varepsilon$ by $2^{-\lambda(\ell(\pi)-\ell_0)}$, at most $\varepsilon/2$ in total in every class, and $s = 1/\lambda$ turns the prior factor of the shorter programs into $2^{\ell_0-\ell(\pi)}$, which is Kraft-summable; so the margin matters only up to length $\bar{\ell}_\varepsilon$. At $\lambda = 1$ one exponent

Table 2: Recovery times $n_{\mathrm{rec}}(0.01)$ in $\mathcal{C}_k$ for large $d$ (Proposition 6.6), computed exactly.

| $\ell_0 = k$ | $\lambda = 1$ | $\lambda = 1.1$ | $\lambda = 1.5$ | $\lambda = 2$ | $\lambda = 3$ |
|---|---|---|---|---|---|
| 16 | 1030 | 78 | 34 | 23 | 20 |
| 24 | 16390 | 144 | 52 | 38 | 29 |
| 32 | 262150 | 232 | 70 | 51 | 38 |
| 40 | 4194310 | 298 | 94 | 67 | 50 |

$s < 1$ for all programs leaves the factor $Z(s) := \sum_\pi 2^{-s\ell(\pi)}$, which depends on the class (it is finite in finite classes), and by (a) no bound in $(\ell_0, \gamma, \varepsilon)$ alone exists.

Each ingredient of Theorem 6.3(d) is needed (Proposition D.2): a margin; for $\lambda \geq 2$ the factor $1/\gamma$, up to a constant and the ratio of $\ell_0$ to the length a wrong program saves; and $1/(\lambda - 1)$, up to logarithms in the recovery time of $\mathcal{C}_k$ and up to $\log(1 + 2^{2-k})/(\lambda - 1)$ bits in the length bound of (c), in separate classes. In $\mathcal{H}_\times$, with $Y = \{0, 1\}$ and $\nu$ bounded away from 0 and 1, $\gamma_{\bar\ell}$ is eventually below every computable function of $\bar\ell$, so (d) gives no computable sample size.

### 6.4 The sample kink

Consider the point family $\{(\ell(\pi), L_n(\pi)) : \pi \in \mathcal{C}\}$ of a per-query sample (Section 5), with left steepness $s_-(n)$ and right secant $s_+(n)$ at $(\ell_0, L_n(\nu))$ (Definition 5.2, with $\mathcal{M} = \mathcal{C}$).

**Theorem 6.7** (A sample has a kink at the shortest equivalent length)**.** *In the per-query setting, fix $n$ and $\varepsilon \in (0, 1)$, and let $\delta^*$ be the Ville deficiency of $\xi_{\mathcal{C}}$.*

*(i) With probability at least $1 - \varepsilon$, simultaneously for all $\pi \in \mathcal{C}$: $L_n(\pi) - L_n(\nu) > 2\log e \cdot n\, d^2_{\mathrm{H}_\mu}(\pi) - 2\ell(\pi) - 2\log(1/\varepsilon)$.*

*(ii) With probability at least $1 - \varepsilon$, $(\ell_0, L_n(\nu))$ is a zero-window left kink of steepness $\sigma_n := \min_{\ell<\ell_0}(2\log e \cdot n\gamma_=(\ell) - 2\ell - 2\log(1/\varepsilon))/(\ell_0 - \ell)$ if $\sigma_n > 0$ ($\sigma_n := \infty$ if $\ell_0 = 0$); if $A := 2\log e \cdot n\gamma_- - 2\log(1/\varepsilon) \geq 2\ell_0$, then $\sigma_n \geq A/\ell_0$.*

*(iii) With probability at least $1 - \varepsilon$, for every $\tau > 1$, $(\ell_0, L_n(\nu))$ is a right kink of flatness $\tau$ and window $(\ell_0 + \log(1/\varepsilon))/(\tau - 1)$. For all $n$ at once, $L_n(\pi) \geq L_n(\nu) - (\ell_0 + \delta^*) - (\ell(\pi) - \ell_0)$ for every $\pi$: a right line of slope 1 and offset $\ell_0 + \delta^*$, giving the window $(\ell_0 + \delta^*)/(\tau - 1)$.*

*(iv) Let $\bar\ell > \ell_0$ and $n > (\bar\ell + \log(1/\varepsilon))/(\log e \cdot \gamma_{\bar\ell})$. With probability at least $1 - 2\varepsilon$: programs of length at most $\bar\ell$ that fit no worse than $\nu$ are in $\mathcal{E}$; every right secant from $(\ell_0, L_n(\nu))$ is below $\tau_{\bar\ell} := 1 + (\ell_0 + \log(1/\varepsilon))/(\bar\ell - \ell_0)$; and if $\sigma_n > \tau_{\bar\ell}$, the point is a vertex of the hull and for every $\lambda \in (\tau_{\bar\ell}, \sigma_n)$ the set of $\lambda$-MAPs is $\mathcal{E}_0$.*

*(v) For an infinite sequence of queries, almost surely $s_-(n) \geq n\,\underline{s} + o(n)$ with $\underline{s} := 2\log e \cdot \min_{\ell<\ell_0} \gamma_=(\ell)/(\ell_0 - \ell) > 0$, and $s_+(n) < 1 < s_-(n)$ for all large $n$: $\lambda = 1$ lies strictly inside the kink.*

*Proof idea.* (i) is Markov's inequality for $\sqrt{\nu_\pi(h_n)/\nu(h_n)}$, whose mean is $(1 - d^2_{\mathrm{H}_\mu}(\pi))^n \leq e^{-nd^2_{\mathrm{H}_\mu}(\pi)}$, with a union bound weighted by $2^{-\ell(\pi)}$; (ii) reads it below $\ell_0$; (iii) is Markov's inequality for $\xi_{\mathcal{C}}(h_n)/\nu(h_n)$, or Lemma 6.2; (iv) adds a flat part and applies Corollary 5.4(c); (v) takes $\varepsilon_n = n^{-2}$ in (i) and uses Theorem 6.3(a) at $\lambda = 1$ with Lemma 3.8(v) (Appendix D.4).

So the model is the hull vertex at length $\ell_0$: to its left a bit of model explains a number of bits of data linear in $n$, to its right a bit of program buys at most about one bit of noise beyond a window. The $\ell_0$ in the right window of (iii) cannot be dropped at finite $n$ (Remark D.3).

### 6.5 Noisy computable functions

**Corollary 6.8** (Noisy computable functions)**.** *Let $Y = \{0,1\}$, $f : X \to Y$ total computable, $p \in [0, \frac{1}{2}]$ computable, and $\nu_{f,p}(y \mid x) := 1 - p$ if $y = f(x)$, $p$ otherwise. Let $\mathcal{C}$ contain a program for $\nu_{f,p}$ ($\mu$-a.e.), with PQ data from i.i.d. inputs: a noisy oracle that redraws its flip at every query. (i) For $p < \frac{1}{2}$ a kernel in $\mathcal{E}$ determines $f$ $\mu$-a.e. (its more likely label) and $p$; so for every schedule $1 \le \lambda_n = o(n)$, almost surely every $\lambda_n$-MAP is eventually a shortest program in $\mathcal{C}$ whose kernel is $f$, up to $\mu$-null sets, with flips at rate exactly $p$. (ii) For fixed $\lambda > 1$ the fill-ins, "$f$ plus the list of flipped queries" among them, have total posterior mass at most $2^{\lambda\ell_0 + \lambda\delta^* - (\lambda-1)L_n(\nu_{f,p})}$, with $L_n(\nu_{f,p})/n \to H(p)$ almost surely. (iii) If $p = \frac{1}{2}$, $f$ is not identifiable; if $p = 0$, $\mathcal{E}$ contains fill-ins and (ii) gives no decay. In $\mathcal{H}_\times$, for $p > 0$, the margin form gives no computable sample size (Proposition D.2(e) with a rational $\beta \in (0, p]$).*

*Proof idea.* Theorem 6.3(a),(c), with $H(\nu_{f,p}(\cdot \mid x)) = H(p)$ for every $x$ (Appendix D.5).

The rate, a real number, is identified exactly, and the flips stay out of the model. With frozen labels on a finite support the data set is eventually the full table, so the selection depends on one table: no procedure identifies every such sampler (Proposition 6.10(a)), and in an explicit class of parities with eight noise rates the $\lambda$-MAP is then wrong with probability at least $(7/8)^{64}$ for every $\lambda$, while per query (i) applies (Remark D.4 in Appendix D.5).

### 6.6 Frozen labels, adaptive inputs and misspecification

Call $\pi$ *$\nu$-equivalent on the run* if $\sum_i d_{\mathrm{H}}^2(\nu(\cdot \mid x_i), \nu_\pi(\cdot \mid x_i)) < \infty$, the sum running over the queries (per query) or over the distinct inputs (frozen labels).

**Theorem 6.9** (Private randomness; any input rule)**.** *Let $\mathcal{C}$ be as in Definition 6.1, the inputs follow any rule, with frozen or per-query labels, and let $\pi_\nu \in \mathcal{C}$ have length $k$ and $\nu_{\pi_\nu}(o_n) = \nu(o_n)$ for all $n$ almost surely. Define $\delta^*$ from $\xi_{\mathcal{C}}$. For fixed $\lambda > 1$, simultaneously for all $n$: (i) every $\lambda$-MAP $\hat{\pi}$ has $\ell(\hat{\pi}) \le (\lambda k + \delta^*)/(\lambda - 1)$ and $L_n(\nu) - \ell(\hat{\pi}) - \delta^* \le L_n(\hat{\pi}) \le L_n(\nu) + \lambda k$; (ii) $\Pi_\lambda(\{\pi : \ell(\pi) > \bar{\ell}\} \mid o_n) \le 2^{\lambda k + \delta^* - (\lambda-1)\bar{\ell}}$ for every $\bar{\ell}$; (iii) every fill-in has length at least $L_n(\nu) - \delta^*$, $\Pi_\lambda(\text{fill-ins} \mid o_n) \le 2^{\lambda k + \lambda\delta^* - (\lambda-1)L_n(\nu)}$, and $\Pi^{1/\lambda}(\text{fill-ins} \mid o_n) \le 2^{k + \delta^* - (1-1/\lambda)L_n(\nu)}$. (iv)* Every $\lambda \ge 1$. *If infinitely many queries are made, then for fixed $\lambda \ge 1$, almost surely the $\lambda$-MAPs have bounded length, uniformly in $n$, from some $n$ on every $\lambda$-MAP is $\nu$-equivalent on the run, and $\Pi_\lambda(\nu\text{-equivalent on the run} \mid o_n) \to 1$.*

*Proof idea.* (i)–(iii) compare with $J_\lambda(\pi_\nu; o_n) = \lambda k + L_n(\nu)$ and use Lemma 6.2. In (iv) the tail supermartingale of Theorem 6.3(a), with $k$ for $\ell_0$, confines the $\lambda$-MAPs to a finite set, and the likelihood ratio of a program not $\nu$-equivalent on the run tends to 0 by the martingale argument behind Kakutani's dichotomy (Kakutani, 1948), valid for adaptive inputs (Appendix D.10).

Per query with i.i.d. inputs, $\nu$-equivalence on the run is almost surely membership in $\mathcal{E}$. With frozen labels and i.i.d. inputs from a $\mu$ of infinite support, any program of $\mathcal{E}_0$ can serve as $\pi_\nu$, and $\nu$-equivalence on the run means $\sum_{x \in \operatorname{supp}\mu} d_{\mathrm{H}}^2(\nu(\cdot \mid x), \nu_\pi(\cdot \mid x)) < \infty$ (Appendix D.6).

**Proposition 6.10** (Frozen labels)**.** *Take frozen labels with inputs i.i.d. from $\mu$. (a) Let $\nu, \nu'$ differ only at one input $x_0$ with $\mu(x_0) > 0$, where $\nu(\cdot \mid x_0)$ and $\nu'(\cdot \mid x_0)$ are distinct and mutually absolutely continuous. Every estimator with values in $\mathcal{C}$ that is almost surely eventually in $\mathcal{E}_\nu$ under $\nu$ is almost surely not eventually in $\mathcal{E}_{\nu'}$ under $\nu'$. Here $\mathcal{E}_\nu$ is the set of programs whose kernel equals $\nu$ at $\mu$-almost every input. In $\mathcal{H}_\times$ every computable $\nu$ with $\nu(y_0 \mid x_0) \in (0,1)$ at some $x_0$ of positive mass has such a computable $\nu'$: no procedure, the $\lambda$-MAP for any $\lambda$ included, recovers every computable sampler with noise on the support of $\mu$. (b) Let $\mathcal{E} \neq \emptyset$ and*

$\beta_x := 1 - \max_y \nu(y \mid x)$. *If* $\sum_{x \in \operatorname{supp}\mu} \beta_x = \infty$, *then for every* $\lambda \geq 1$ *the posterior mass of the deterministic programs tends to* 0 *almost surely. (c) Let* $\mu$ *be uniform on* $\{0, \ldots, 9\}$, *and let the class consist of a program of length* 5 *for* $\nu = \mathrm{Bern}(1/64)$ *and programs* $e_j$ *of length* 9 *equal to* $\nu$ *except* $e_j(1 \mid j) = 1$. *For every* $\lambda < 1.5$ *the* $\lambda$*-MAP is eventually some* $e_j$ *with probability exactly* $1 - (63/64)^{10}$.

*Proof idea.* (a) The two laws of the data process differ in one factor of a product. (b) A deterministic program has gap at least $\beta_x/2$ at $x$; apply Theorem 6.9(iv). (c) An $e_j$ with frozen label 1 at $j$ gains 6 bits of fit for $4\lambda$ bits of length; per query it fails at the first label 0 at $j$; and $1 - (63/64)^{10} = 0.1457$ to four digits (Appendix D.6).

Proposition D.5 gives further limits: with adaptive inputs a shorter program that differs from $\nu$ at every query but is $\nu$-equivalent on the run can win forever with positive probability; under misspecification fill-ins do not compete per query; and with $\lambda = cn$, $\mathcal{E}_0$ is recovered for $c$ below the truth's left steepness in population and not above, as for the teacher's errors (Section 8).

### 6.7 $\lambda = 1$ on one data set

On a data set fill-ins are cheap: a deterministic program can store a finite piece of each input's private random stream. Adapting the tape-length bound of Hänni (2026a, Proposition 4.6(a)) gives, in $\mathcal{H}_\times$, $W_{\det}(D) \geq 2^{-k-2\log(t_D+1)-\kappa}\nu(D)$ whenever $\beta_D := \min_{(x,y)\in D} \nu(y \mid x) > 0$, with $2\log(t_D + 1)$ doubly logarithmic in the input length and triply logarithmic in $|\operatorname{dom} D|$ and $-\log \nu(D)$ (Lemma D.6). For frozen labels on distinct inputs let $d_n$ be the largest input length in $D_n$, $\beta_n := \beta_{D_n}$, $i_n := \lceil \log(6|\operatorname{dom} D_n|/\beta_n) \rceil$, $t_n := \lceil \log(2d_n + 2 + 2\log(i_n + 1)) \rceil$, and $\Pi_1(\det \mid D_n)$ the posterior mass of all deterministic programs.

**Theorem 6.11** (SFI at $\lambda = 1$)**.** *Let the class be* $\mathcal{H}_\times$, *the labels frozen on distinct inputs chosen by any rule,* $\nu$ *a sampler with a program* $\pi_\nu$ *of length* $k$, *and* $\delta^*$ *the Ville deficiency of* $\xi_\times = M_{\mathrm{prod}}$. *Simultaneously for all* $n$: *(i)* $\Pi_1(\pi_\nu \mid D_n) \geq 2^{-k-\delta^*}$; *(ii)* $\Pi_1(\det \mid D_n) = W_{\det}(D_n)/\xi_\times(D_n) \geq 2^{-k-2\log(t_n+1)-\kappa-\delta^*}$; *(iii) the posterior odds of all fill-ins against the sampler program lie in* $[2^{-2\log(t_n+1)-\kappa}, 2^{k+\delta^*}]$, *with a lower end that depends on the sampled randomness only through* $t_n$.

*Proof.* $\Pi_1(\pi_\nu \mid D_n) = 2^{-k}\nu(D_n)/\xi_\times(D_n)$; bound $\xi_\times(D_n) \leq 2^{\delta^*}\nu(D_n)$ by Lemma 6.2, and $W_{\det}(D_n)$ below by Lemma D.6 and above by $\xi_\times(D_n)$. □

The lower ends cover non-dyadic samplers, such as the Bern(1/3) threshold hypotheses of Sam Eisenstat's coin example (personal communication; Hänni, 2026a, Section 7.3), and in SI's representation by deterministic programs no complexity weight separates a sampler from its fill-ins (Proposition D.7 and Remark D.8).

*Remark* 6.12 (What frozen labels give). With frozen labels and fixed $\lambda > 1$, no $\lambda$-MAP is a fill-in once $L_n(\nu) > \lambda(k + \delta^*)/(\lambda - 1)$, and the fill-ins' posterior is exponentially small in the sampled randomness, but positive (Theorem 6.9). The selected program need not be $\nu$: it may carry up to $(k + \delta^*)/(\lambda - 1)$ bits beyond it, such as noise memorised at a few inputs. In the limit every $\lambda \geq 1$ selects programs that are $\nu$-equivalent on the run, although at $\lambda = 1$ the posterior odds stay within the factors of Theorem 6.11. Equivalence on the run says nothing about unseen inputs and is vacuous on a finite domain; exact recovery needs per-query labels (Theorem 6.3 and Proposition 6.10).

**Question 6.13** (rates, frozen labels, misspecification)**.** (a) Is there a recovery-time bound in $\mathcal{H}_\times$ computable from $(\lambda, \ell_0, \varepsilon)$ and a margin for the programs far from $\nu$? (b) Can $1/\gamma_{\bar{\ell}}$ in Theorem 6.3(d) be replaced by a term of order $\log n$ for each family of near-equivalent refinements? (c) Is the joint factor $1/(\min(1, \lambda{-}1)\gamma)$ necessary in one class? (d) With frozen labels and repeated inputs, for which classes and input laws does the evidence $\sum_{x \in \operatorname{dom} D_n} d_{\mathrm{H}}^2(\nu(\cdot \mid x), \nu_\pi(\cdot \mid x))$ of every program shorter than $\ell_0$ grow without bound? On one frozen data set far from the full table, can single fill-ins in $\mathcal{H}_\times$ give a right secant of at least 1 at the shortest equivalent program? (e) If $\mathcal{E} = \emptyset$ in a countable class, is the $\lambda_n$-MAP eventually a shortest minimiser of the mean KL divergence for every schedule $\lambda_n = o(n)$ that grows fast enough?

# 7 The noisy random parity: one instance, two readings

Both reasons meet in one instance: a random parity on $\{0,1\}^d$ with labels flipped on an error set. If the error set is a random set of inputs, the full table's flips are noise and the coding bound is a right line of slope 1 at the good model (Section 5); if it is a random rule on $m < d$ input bits, they are a teacher's systematic errors, with slope $2^{-m}$ per sample (Section 8). We fix the instance and its lemmas, find a steep kink of the full table at the parity, selecting "parity plus noise" on a range of $\lambda$, and contrast set models, for which the parity is noise (proofs in Appendix E).

## 7.1 The instance

**Notation** for this section and Section 8:

| | | | |
|---|---|---|---|
| $d,\ m,\ p,\ s_0$ | lengths, flip rate, rule size | $a,\ f_a,\ S,\ E,\ y^*$ | parity, rule, error set, teacher |
| $L_S,\ B_S,\ \delta_S,\ I,\ e$ | codes, deficiency, information | $B,\ R,\ R_B,\ J_c,\ J_c^B,\ \Phi,\ \Phi^B$ | risks, objectives, infima |
| $u,\ \mathrm{lev},\ s(t),\ G_j,\ F_t^\sharp$ | agreement, level, floors | $\pi_{\mathrm{coin}},\ \pi_g,\ \pi_b,\ \kappa_g,\ \kappa_b,\ \Lambda_1$ | reference programs, overheads |
| $b_0,\ \Gamma_t,\ \Gamma_t^\circ$ | full table (Section 7.3) | $\Phi_2,\ J_g,\ J_b,\ \Delta,\ c_{\mathrm{hi}}$ | bounds of Section 8 |

**Definition 7.1** (The instance)**.** In this section and Sections 8 and 9, $Y = \{0,1\}$. Fix integers $2 \le m \le d$ and $1 \le s_0 \le 2^{m-2}$, and put $p := s_0/2^m \le \frac{1}{4}$.

(a) *Model, errors, teacher.* The *simple model* is the parity $f_a(x) = \langle a, x\rangle \bmod 2$ on $\{0,1\}^d$, for some $a \in \{0,1\}^d$ with $K(a) \ge d$. The *error rule* is a set $S \subseteq \{0,1\}^m$ with $|S| = s_0$, the *error set* is $E := S \times \{0,1\}^{d-m}$, and the *teacher* is $y^* := f_a \oplus \mathbf{1}_E$: a systematic rule (a random junta) for $m < d$, an idiosyncratic random set for $m = d$.

(b) *Description lengths.* $L_S := \log \binom{2^m}{s_0}$, $B_S := K(S \mid a, K(a))$ and the *deficiency* $\delta_S := L_S - B_S$. The *information* of a program about the rule is $I(\pi) := B_S - K(S \mid a, \pi)$, so $L_S - K(S \mid a, \pi) = \delta_S + I(\pi)$. One has $2^m H(p) - \log(2^m + 1) \le L_S \le 2^m H(p)$ (Cover and Thomas, 2006), $\delta_S \ge -2\log m - 2\log s_0 - \kappa$, and $\Pr_S[\delta_S > j] < 2^{-j}$ for a uniformly random rule.

(c) *Population quantities.* Inputs are uniform on $\{0,1\}^d$, so $\Pr[x \in E] = p$. For a product program $\pi$ and a truncation level $B \in [\log(1/p), \infty]$: $R(\pi) := \mathbb{E}_x[-\log \nu_\pi(y^*(x) \mid x)]$, $R_B(\pi) := \mathbb{E}_x \min\{B, -\log \nu_\pi(y^*(x) \mid x)\} \le R(\pi)$, $J_c := c\,\ell + R$, $J_c^B := c\,\ell + R_B$, $\Phi(c) := \inf_\pi J_c(\pi)$ and $\Phi^B(c) := \inf_\pi J_c^B(\pi) \le \Phi(c)$, the infima over the class; $2^{-\infty} := 0$. Results that use (D2) of Lemma 7.5 take $B$ to be $\infty$ or an integer.

(d) *Agreement, level, errors.* $u(\pi) := \mathbb{E}_x \nu_\pi(f_a(x) \mid x)$ and $e(\pi) := \mathbb{E}[\nu_\pi(y^*(x) \mid x) \mid x \in E]$. With $\theta_t := \frac{1}{2} + 2^{-t-1}$, the *level* is $\mathrm{lev}(\pi) := \min\{t \ge 1 : u(\pi) > \theta_t\}$ ($\infty$ if $u(\pi) \le \frac{1}{2}$); so $\mathrm{lev}(\pi) = 1$ if and only if $u(\pi) > \frac{3}{4}$, and $\mathrm{lev}(\pi) > t$ if and only if $u(\pi) \le \theta_t$. Put $s(t) := 2t + 2\log(t+1) \le 4t$ and $G_j := 1 - \log(1 + 2^{-j} + 2p + 2^{1-B})$ for $j \in [0, \infty]$, increasing and concave in $j$.

(e) *Reference programs* in $\mathcal{H}_\times$: the *coin* $\pi_{\text{coin}}$, with $\ell \le \kappa$ and $R = 1$; the *good model* $\pi_g$, which gives $f_a(x)$ probability $1-p$, with $\ell(\pi_g) \le d + \kappa_g$, $\kappa_g := 2\log d + K(p) + \kappa$, and $R(\pi_g) = R_B(\pi_g) = H(p)$; the *memoriser* $\pi_b$, the deterministic $y^*$, with $\ell(\pi_b) \le d + \lceil L_S \rceil + \kappa_b$, $\kappa_b := 2\log d + 2\log m + 2\log s_0 + \kappa$, and $R(\pi_b) = 0$.

(f) *Slack constant.* $\Lambda_1 := 2\log d + 2\log m + 2\log B \cdot \mathbf{1}[B < \infty] + 2^m \log(1 + 2^{1-B}) + \kappa$.

The *example's sizes* are $d = 100$, $m = 20$, $p = 2^{-10}$; then $s_0 = 1024$ and $L_S = 11{,}710.27$ bits.

For $p = 2^{-r}$, (F5) of Appendix A.1 gives $K(p) \le 2\log r + \kappa$, and numbers for such $p$ use $\kappa_g = 2\log d + 2\log r + \kappa$. Every slack term below contains $\delta_S$ and every number states the $\delta_S$ it assumes; numbers with $\delta_S = 0$ say nothing about rules compressible given the model (Proposition 9.1).

### 7.2 Four lemmas and a sharper floor

We use parities for list decoding: beating a coin noticeably on $f_a$ determines $a$ up to a few bits.

**Lemma 7.2** (List decoding for semimeasures)**.** *If* $\text{lev}(\pi) = t < \infty$*, then* $K(a \mid \pi) \le s(t) + 2\log d + \kappa$.

**Lemma 7.3** (Risk floor)**.** $R_B(\pi) \ge -\log\left(u(\pi) + p + 2^{-B}\right)$. *Hence* $\text{lev}(\pi) = t \ge 2$ *implies* $R_B(\pi) \ge G_{t-1}$*, and* $\text{lev}(\pi) = \infty$ *implies* $R_B(\pi) \ge G_\infty$.

**Lemma 7.4** (Coding the error rule with a hypothesis)**.** *For integer or infinite* $B$*,* $K(S \mid a, \pi) \le 2^m R_B(\pi) + 2^m \log(1 + 2^{1-B}) + 2\log m + 2\log B \cdot \mathbf{1}[B < \infty] + \kappa$.

**Lemma 7.5** (Two-part lower bounds)**.** *Let* $\text{lev}(\pi) = t < \infty$*. Then (D1)* $\ell(\pi) \ge d - s(t) - \Lambda_1$ *and, if* $B$ *is an integer or* $\infty$*, (D2)* $\ell(\pi) + 2^m R_B(\pi) \ge d + B_S - s(t) - \Lambda_1$.

Lemma 7.3 is tight in $u$ alone, but the additive $p$ goes if $S$ is random given the hypothesis.

**Lemma 7.6** (Randomness of the error rule sharpens the floor)**.** *For every program* $\pi$*, integer* $g \ge 1$ *and* $B \in [\log(1/p), \infty]$*,* $R_B(\pi) \ge -\log\left(p + (1-2p)\,u(\pi) + p(2\gamma(\pi) + 2^{1-g}) + 2^{-B}\right)$*, where* $\gamma(\pi) := \sqrt{j \ln 2/(2s_0)}$ *and* $j$ *is the least integer with* $j - 2\log(j+1) > \delta_S + I(\pi) + 1 + 2\log m + 2\log s_0 + g + 2\log g + \kappa$*. Always* $I(\pi) \le \ell(\pi) + \kappa$*, and the bound remains valid with any* $I \ge I(\pi)$ *in place of* $I(\pi)$.

*Proof ideas.* By Parseval, fewer than $2^{2t}$ parities agree with $\nu_\pi$ above $\theta_t$, a list enumerable from $\pi$ (Goldreich and Levin, 1989; Kushilevitz and Mansour, 1993). The floor is Jensen's inequality. The coding lemma makes the likelihoods per prefix a semimeasure on rules, of cost $2^m R_B$ as $E$ is a product. (D1) is $d \le K(a) \le \ell(\pi) + K(a \mid \pi) + \kappa$; (D2) adds the coding lemma. In the sharp floor, unusual mass on $E$ would put $S$ in a small enumerable set (Hoeffding, 1963) (Appendices E.1 and E.2).

For $t \in [1, \infty]$ and an information budget $I \ge 0$ the sharp floor at $B = \infty$ is

$$F_t^\sharp(I) := \sup_{g \ge 1} \Big[1 - \log\Big(1 + (1-2p)2^{-t} + 2p(2\gamma_g(I) + 2^{1-g})\Big)\Big],$$

with $\gamma_g(I)$ the $\gamma(\pi)$ of Lemma 7.6 with $I$ for $I(\pi)$. By that lemma, every $\pi$ with $\text{lev}(\pi) > t$ and $I(\pi) \le I$ has $R(\pi) \ge F_t^\sharp(I)$; $F_t^\sharp(I)$ is nondecreasing in $t$ and nonincreasing in $I$ (Appendix E.2).

### 7.3 The full table: a kink at the parity

Take $m = d$, $B = \infty$ and $p < 1/4$, so $E = S$, $N := 2^d$ and $|E| = pN$; write $L_E, B_E, \delta_E$ for $L_S, B_S, \delta_S$. Let $z := y^* \in \{0,1\}^N$ be the teacher's table and $D$ the full data set: frozen noise at rate $p$, conditioned on its number of flips. A product program has fit $L(\pi; D) = NR(\pi)$; here $G_t = 1 - \log(1 + 2^{-t} + 2p)$, $\Lambda_1 = 4\log d + \kappa$ and $\kappa_b = 4\log d + 2\log(pN) + \kappa$. The class is $\mathcal{H}_\times$, or any product hypothesis class that contains a good model $\pi_g$, with $\kappa_g := \ell(\pi_g) - d$ and $\kappa$ the class's constant (as in Proposition 11.1(a)). Put $\Gamma_t^\circ := \max\{G_t, F_t^\sharp(d+\kappa)\}$, $\mathrm{den}_t := \kappa_g + \min\{d, s(t+1) + \Lambda_1\}$ for $t \in [1, \infty]$, and, as $0 \le NH(p) - L_E \le \log(N+1)$,

$$b_0 := \kappa_g + (NH(p) - L_E) + \delta_E + 2\log(d+1) + \kappa \;=\; \delta_E + O(d).$$

**Proposition 7.7** (A kink at the parity)**.** *Let $t_0 \ge 1$ be an integer and $\sigma^{(t_0)} := \inf_{t \ge t_0} N(\Gamma_t^\circ - H(p))/\mathrm{den}_t$. (i) $\pi_g$ is $b_0$-sufficient for $D$. (ii) If $\sigma^{(t_0)} > 0$, then $D$ has a left kink of steepness $\sigma^{(t_0)}$ and window $w := \ell(\pi_g) - (d - s(t_0) - \Lambda_1) \in [0, \kappa_g + s(t_0) + \Lambda_1]$ at $(\ell(\pi_g), NH(p))$. (iii) Hence, if $\sigma^{(t_0)} > 0$, Corollary 5.5(b) applies to $\pi_s = \pi_g$ and $b = b_0$ for every $\lambda \in (1, \sigma^{(t_0)})$.*

*Proof idea.* (i): minimum-distance decoding recovers $(a, E)$ from $z$, as two parities differ on $N/2 > 2pN$ inputs. (ii): by (D1), a program shorter than $d - s(t_0) - \Lambda_1$ has level $t + 1 > t_0$, so it saves at most $\mathrm{den}_t$ bits against $\pi_g$ and pays at least $N\Gamma_t^\circ$ in fit (Appendix E.3).

As $\Gamma_t^\circ$ is nondecreasing in $t$, $\sigma^{(t_0)} \ge N(\Gamma_{t_0}^\circ - H(p))/(\kappa_g + d)$, exponential in $d$ once $\Gamma_{t_0}^\circ > H(p)$; beyond $\pi_g$ the coding bound allows at most about one bit of fit per bit. For $\lambda > 1$ let

$$\bar{\ell}_\lambda := d + \kappa_g + \frac{b_0 + \kappa}{\lambda - 1}, \qquad \Gamma_t(\lambda) := \max\{G_t, F_t^\sharp(\bar{\ell}_\lambda + \kappa)\},$$

with $\kappa$ the constant of Lemma 3.9, let $T$ be the least integer $t \ge 1$ with $s(t+1) + \Lambda_1 \ge d$, and for an integer $t_0 \ge 1$ let $C_{t_0}(\lambda)$ be the condition $\lambda\,\mathrm{den}_t < N(\Gamma_t(\lambda) - H(p))$ for $t = t_0, \dots, \max(t_0, T)$.

**Proposition 7.8** (The $\lambda$-range of "parity plus noise")**.** *(i) Let $\lambda > 1$ satisfy $C_{t_0}(\lambda)$. Every $\lambda$-MAP $\hat{\pi}$ on $D$ satisfies: (a) $\mathrm{lev}(\hat{\pi}) \le t_0$, so $u(\hat{\pi}) > \theta_{t_0}$ and $K(a \mid \hat{\pi}) \le s(t_0) + 2\log d + \kappa$: it knows the parity; (b) $d - s(t_0) - \Lambda_1 \le \ell(\hat{\pi}) \le \bar{\ell}_\lambda$; (c) $NH(p) - \lambda(b_0 + \kappa)/(\lambda - 1) \le L(\hat{\pi}; D) \le NH(p) + \lambda(\kappa_g + s(t_0) + \Lambda_1)$: it fits like the noise; (d) $\hat{\pi}$ is $(b_0 + (\lambda - 1)(\kappa_g + s(t_0) + \Lambda_1))$-sufficient; (e) $I(\hat{\pi}) \le I_e := \bar{\ell}_\lambda - d + s(t_0) + 2\log d + \kappa$; (f) $e(\hat{\pi}) \le 1 - u(\hat{\pi}) + \gamma_g(I_e) + 2^{-g}$ for every integer $g \ge 1$; (g) with $R_{\max} := H(p) + \lambda(\kappa_g + s(t_0) + \Lambda_1)/N$, $u(\hat{\pi}) \ge \max\{\theta_{t_0}, 2^{-R_{\max}} - p, \max_{g \ge 1}(2^{-R_{\max}} - p - p(2\gamma_g(I_e) + 2^{1-g}))/(1 - 2p)\}$.*

*(ii)* Interval form. *If $1 < \lambda_{\mathrm{lo}} \le \lambda_{\mathrm{hi}}$ and $\lambda_{\mathrm{hi}}\,\mathrm{den}_t < N(\Gamma_t(\lambda_{\mathrm{lo}}) - H(p))$ for $t = t_0, \dots, \max(t_0, T)$, then $C_{t_0}(\lambda)$, and with it (a)–(g), holds for every $\lambda \in [\lambda_{\mathrm{lo}}, \lambda_{\mathrm{hi}}]$.*

*(iii)* Every flip rate below 1/4. *In $\mathcal{H}_\times$, fix a dyadic rational $p \in (0, \frac{1}{4})$ and $\lambda > 1$, and let $m = d \to \infty$ with $\delta_E = O(d)$. For all large $d$, $C_2(\lambda)$ holds; if moreover $1 - \log(\frac{3}{2} - p) > H(p)$, then $C_1(\lambda)$ holds.*

*Proof idea.* The upper length bound is Corollary 5.5(a) at $\pi_g$. A $\lambda$-MAP of level $t + 1 > t_0$ is at least $(d - s(t+1) - \Lambda_1)^+$ long and has budget at most $\bar{\ell}_\lambda + \kappa$, so the floors charge it $N\Gamma_t(\lambda)$, and $C_{t_0}(\lambda)$ puts it above $\pi_g$; the rest is (D1), the comparison with $\pi_g$ and the sharp floor's proof. (ii): $\Gamma_t(\lambda)$ is nondecreasing in $\lambda$. (iii): the budget is $O(d)$ while $s_0 = pN$ (Appendix E.3).

With $f_t(p) := 1 - \log(1 + (1-2p)2^{-t})$, the crossing condition of (iii) is $f_1(p) > H(p)$, which holds exactly for $p < p_1$, the unique root in $(0, \frac{1}{4})$; likewise $G_1 > H(p)$ exactly for $p < p_0$. A direct computation (code in the accompanying repository) gives $p_1 = 0.1233072\ldots$ and $p_0 =$

$0.0559643\ldots$, so the conditions hold for $p \le 0.1233$ and $p < 0.05596$ respectively; it also certifies by interval arithmetic that $f_2 - H \ge 0.0187$ on $(0, \frac{1}{4}]$, which the proof of (iii) uses.

By (ii), finitely many cells certify $C_{t_0}(\lambda)$ on an interval of $\lambda$. A direct computation of such covers (code in the accompanying repository; Appendix E.3) shows that for $p = 2^{-10}$ the range reaches $\lambda \to 1$; for $p \ge 1/16$, at the levels $t_0$ of Table 5, it starts above 1, for example $[1.00059, 6529.7]$ at $p = 1/8$, $d = 20$, with the machine constant set to 0 (an illustration).

### 7.4 Set models: the parity is noise

**Definition 7.9** (Set models)**.** A *set model* of a string $z$ is a finite set $A \ni z$, with two-part length $K(A) + \log|A|$. The *structure function* is $h_z(\alpha) := \min\{\log|A| : A \ni z,\ K(A) \le \alpha\}$, and the *sufficiency line* is $K(A) + \log|A| \ge K(z) - \kappa$, valid since $K(z \mid A) \le \log|A| + \kappa$. $A$ is *b-sufficient* if $K(A) + \log|A| \le K(z) + b$; a *minimal sufficient statistic* minimises $K(A)$ among these; the *set sophistication* is $\mathrm{soph}_b^{\mathrm{set}}(z) := \min\{K(A) : A \ni z \text{ } b\text{-sufficient}\}$. Conditional versions replace $K$ by $K(\cdot \mid w)$.

These notions are standard (Gács et al., 2001; Vereshchagin and Vitányi, 2004; Vereshchagin and Shen, 2017); sophistication goes back to Koppel (1987); Koppel and Atlan (1991) (see also Vitányi, 2006; Antunes and Fortnow, 2009). The set models of $z$ form a point family (Definition 5.1) with lengths $K(A)$, fits $\log|A|$ and structure function $h_z$. Steeper two-part codes over sets select an approximately minimal sufficient statistic only in a limited sense, and thresholding a computable product hypothesis gives a set model at most logarithmically longer (Appendix E.4).

Let $c^* := 2\log d + K(p) + \kappa$ and $A_0 := \{f_{a'} \oplus \mathbf{1}_{E'} : a' \in \{0,1\}^d,\ |E'| = pN\}$, a $(\delta_E + c^*)$-sufficient set model with $K(A_0) \le c^*$. For sets the parity is noise: every $\lambda$-optimal set model holds at most $O(\log d) + (\delta_E + O(\log d))/(\lambda - 1)$ bits about $a$. For products it is model: for large $d$ and $\delta_E = O(d)$, product sophistication is $d \pm O(\log d)$ on a slack range of order $N$ (Proposition E.5).

**Proposition 7.10** (Set models have no steep kink at the parity)**.** *In the setting of Section 7.3, with $c^*$ as above, every left kink $(\sigma, w)$ of the set models of $z$ at $(\alpha_{\mathrm{r}}, L_{\mathrm{r}})$ with $\alpha_{\mathrm{r}} - w > c^*$ has $\sigma \le 1 + (c^* + \delta_E + 2\kappa)/(\alpha_{\mathrm{r}} - c^*)$.*

*Proof idea.* The kink point lies above the sufficiency line and $A_0$, shorter than $\alpha_{\mathrm{r}} - w$, at most $\delta_E + c^*$ above it, so between them the fit falls about one bit per bit (Appendix E.5).

So at $\alpha_{\mathrm{r}} \approx d$ the set steepness is at most $1 + O((\log d + \delta_E)/d)$, and the product steepness at least $N(\Gamma_1^\circ - H(p))/(\kappa_g + d)$, exponential in $d$ when $\Gamma_1^\circ > H(p)$: for $p < 0.05596$ always, for $p \le 0.1233$ once $d$ is large if $\delta_E = O(d)$. $A_0$ acts as the universal member of Proposition 5.7. For $\lambda - 1$ of order $(\log d + \delta_E)/d$ nothing is claimed, and the code's $O(\log d)$ terms decide (Appendix E.5).

*Remark* 7.11 (Reading: one bit for one bit)*.* Knowing a random $d$-bit parameter saves sets one bit per bit, products about a bit *per input*. This is why steeper simplicity is useless over set models in this example once $\lambda - 1 \gg (\log d + \delta_E)/d$ (Propositions 7.10 and E.5; closer to 1 the code's $O(\log d)$ terms decide), and decisive over product models (Corollary 5.5 and Proposition 7.8).

*Remark* 7.12 (Restricted model families)*.* Restricting the model family changes the structure function; Hamming-ball models (Vereshchagin and Vitányi, 2010) are the closest case, and the Hamming ball of radius $pN$ around $f_a$ is a set model of complexity about $d$ and log-size about $NH(p)$, the parity plus independent flips as a set (see also Milovanov, 2016).

**Question 7.13** (set kinks)**.** Let $\pi_s = q\theta \in \mathcal{H}_\times$ be product-$b$-sufficient for a data set $D$, with a $k$-bit parameter $\theta$ such that $K(\theta \mid q) \ge k$, and let $\rho_D^\times$ have a left kink of steepness $\sigma > 1$ at the point

of $\pi_s$. Is there a function $\epsilon(k) = o(k)$, not depending on $D$, such that there is always a set model $A \ni y_D$ with $K(A \mid \mathrm{dom}\, D) \le |q| + \epsilon(k)$ and $K(A \mid \mathrm{dom}\, D) + \log|A| \le K(y_D \mid \mathrm{dom}\, D) + b + \epsilon(k)$? By the argument of Proposition 7.10, such a set caps the steepness of the set models' left kinks beyond $K(A \mid \mathrm{dom}\, D)$ at $1 + (b + \epsilon(k) + 2\kappa)/(\alpha_{\mathrm{r}} - K(A \mid \mathrm{dom}\, D))$. For the noisy parity, $A_0$ has complexity $O(\log d)$ and is $(\delta_E + O(\log d))$-sufficient (Section 7.4).

# 8 The teacher's mistakes: the split at a positive rate

This section gives the second reason for a complexity penalty steeper than the likelihood penalty: a learner that grasps what a teacher means, rather than predicting what the teacher would say, should not reproduce the teacher's systematic mistakes. It is the split of Section 5 at a positive rate: a bit of noise enters one likelihood term, and a fixed $\lambda > 1$ discards it (Section 6); a bit of an error rule enters the term of every input it governs, so $\lambda$ must grow linearly in $n$. We prove this for the instance of Definition 7.1, over all product hypotheses; Section 9 shows where it stops. Proofs are in Appendix F.

## 8.1 From noise to errors

*Remark* 8.1 (The three-hypothesis calculation). (1) A simple model of 100 bits predicts 100 bits of labels per bit of description, and a systematic error model of 100 further bits predicts 10 bits per bit. The coin, the simple model and the simple model with its errors lie on a hull with slopes 100 and 10, so the simple model alone is selected exactly for $\lambda \in (10, 100)$. (2) Per sample, let the errors cover a $2^{-10}$ fraction of the inputs and take 10,000 bits. Of the good model (flip rate $p = 2^{-10}$, 110 bits), the memoriser (10,100 bits, risk 0) and the coin, the good model minimises $c\,\ell + R$ exactly for $H(p)/9990 < c < (1 - H(p))/110$; a direct computation (code in the accompanying repository) gives the window $(1.1185 \cdot 10^{-6}, 8.989 \cdot 10^{-3})$, and plain MDL ($\lambda = 1$) switches to the memoriser at $n \approx 8.94 \cdot 10^5$. Other hypotheses might do better.

The *reuse factor* of a unit of randomness (one sampled label, or one cell $z \in \{0,1\}^m$ of an error rule) is the number of likelihood terms it determines. A label of noise, drawn per query or frozen per distinct input (PQ, FR of Definition 3.1), enters one term; a cell of the rule enters the term of every input in it, $2^{d-m}$ on the full table and about $n2^{-m}$ among $n$ samples. Repetition scales every threshold:

**Lemma 8.2** (Duplication)**.** *If every term of a data set $D$ is counted $r \ge 1$ times, so that each program has fit $rL(\pi; D)$, then $J_\lambda = r\, J_{\lambda/r}(\cdot\,; D)$ and the $\lambda$-MAPs are the $(\lambda/r)$-MAPs of $D$. With every reference point $(\alpha_{\mathrm{r}}, L_{\mathrm{r}})$ moved to $(\alpha_{\mathrm{r}}, rL_{\mathrm{r}})$, every $\lambda$-threshold of $D$ (the hull's edge slopes and vertex $\lambda$-intervals, left steepness, the slopes of right lines and right secants) is multiplied by $r$, and so are the wedge margins and the offsets of right lines.*

*Proof.* $\lambda\ell + rL = r\big((\lambda/r)\ell + L\big)$, and the conditions of Definition 5.2 and Lemma 5.3 are homogeneous in fits, offsets and slopes. □

Frozen noise counted once (FR) is like fresh noise: for fixed $\lambda > 1$ every $\lambda$-MAP is at most $(k + \delta^*)/(\lambda - 1)$ bits longer than the sampler's program, at every $n$. Counted per query (*FQ*: a fixed table $F$ queried at i.i.d. inputs) it is reused at every repeat, like an error rule with $m = d$: if the class can memorise finitely many inputs, every $\lambda_n = o(n)$ eventually rejects every program for the sampler (Proposition F.1). Plain Bayes, MDL and SI learn the errors, as does every schedule with $(\lambda_n/n)(d + L_S) \to 0$ along a sequence of instances (Proposition F.2).

**Proposition 8.3** (Which schedules keep the identification)**.** *In the instance of Definition 7.1, let the data be PQ with $\nu = \delta_{y^*}$ (FQ with $F = y^*$), the inputs uniform on $\{0,1\}^d$, and $\lambda_n > 0$.*

*(a) If $\lambda_n/n \to 0$, almost surely for all large $n$ every $\lambda_n$-MAP has $R = 0$: it gives $y^*(x)$ probability 1 at every $x \in \{0,1\}^d$, so $e = 1$.*

*(b) If $\lambda_n/n \to \infty$, almost surely for all large $n$ every $\lambda_n$-MAP has length at most $\ell_{\min} := \min\{\ell(\pi) : R(\pi) < \infty\}$. In $\mathcal{H}_\times$, $\ell_{\min} \le \kappa$, so once $d - 4 - \Lambda_1 > \kappa$ these $\lambda_n$-MAPs have level at least 2.*

*(c) Let $B < \infty$ be an integer in $[\log(1/p), \infty)$, $c_{\mathrm{hi}}$ computed at this $B$, $2^{-m} < c_1 \le c_2 < c_{\mathrm{hi}}$, $\delta \in (0,1)$, $\epsilon_1 := \log(1/p)\sqrt{\ln(2/\delta)/(2n)}$, $\bar{L} := (J_g(c_2) + \epsilon_1)/c_1$ and $\bar{\epsilon}_n := \epsilon_1 + B\sqrt{(\bar{L}\ln 2 + \ln(2/\delta))/(2n)}$. For each $n$, with these data or with $n \le 2^d$ inputs drawn without replacement, and each $\lambda_n$ with $\lambda_n/n \in [c_1, c_2]$: with probability at least $1 - \delta$, every $\lambda_n$-MAP satisfies Theorem 8.7(ii) with $c = \lambda_n/n$ and $\epsilon = \bar{\epsilon}_n$, whenever $J_g(c) + \bar{\epsilon}_n < \Phi_2(c)$ on $[c_1, c_2]$, which holds for all large $n$. For PQ data with $\delta = n^{-2}$, almost surely this holds for all large $n$.*

*Proof idea.* (a) The memoriser has fit 0, which leaves finitely many candidates, and the strong law eliminates those of positive risk. (b) One bit of length eventually outweighs the fit of a shortest finite-risk program. (c) is Lemma F.3 with a length budget uniform over $[c_1, c_2]$ (Appendix F.1).

So a ratio $\lambda_n/n$ that stays in a compact subset of the window $(2^{-m}, c_{\mathrm{hi}})$ of Theorem 8.7 keeps the identification; $\lambda = cn$ is the simplest such schedule.

*Remark* 8.4 (Fixed $\lambda$ against $\lambda \propto n$). For noise a fixed $\lambda > 1$ suffices (Theorems 6.3 and 6.9); for errors $\lambda/n$ must stay above about $2^{-m}$, the reuse factor per sample (Theorem 8.7(iii),(iv)). With $\lambda = cn$ the learner discards every component whose rate is below $c$, valid or not (Proposition 9.3): it is deliberately inconsistent for the teacher's label distribution, at the price of a batch excess: $-\log \xi_\lambda(D_n) \le n\,(c\,\ell(\pi) + \hat{R}(\pi))$ for every $\pi$, with $\hat{R}$ the empirical risk, so $c\,\ell(\pi)$ bits per sample over $\pi$'s own loss (not a sequential regret, Remark 3.7; literature: Section 2).

*Remark* 8.5 ($\lambda = cn$ and weight decay). With $\lambda = cn$ the length is charged once per sample: $2^{-cn\ell(\pi)}\nu_\pi(h_n) = \prod_{i \le n} 2^{-c\ell(\pi)}\nu_\pi(y_i \mid x_i)$. A fixed $\lambda$ is Bayes' rule with the fixed prior $2^{-\lambda\ell}$, but no fixed prior with full support gives the $cn$-MAPs on every sample: the $cn$-MAPs depend on the sample only through its empirical distribution, whereas under such a prior a program that fits a sample worse than another stops being a MAP once the sample is repeated often enough, and in $\mathcal{H}_\times$, for every $c > 0$, some samples have only such $cn$-MAPs. Per query, for $c$ above the truth's left steepness $c^*$ (Proposition D.5) the $cn$-MAPs end at shorter programs that predict worse, whereas the MAPs of every summable prior that charges $\mathcal{E}$ end in $\mathcal{E}$ (Proposition F.6). The window of Theorem 8.7, outside a thin band just above $2^{-m}$, is an instance: there the $cn$-MAPs eventually do not reproduce the teacher's labels, while every summable prior that charges a program reproducing them eventually selects such a program.

Weight decay (Krogh and Hertz, 1992) has the same form. With the loss averaged over $n$ samples, gradient descent with weight-decay coefficient $\omega$ descends $\hat{L}_n(\theta) + \frac{\omega}{2}\|\theta\|^2$, and, for the log loss, $n$ times this is, up to a constant, the negative log posterior under a Gaussian prior (MacKay, 1992) with precision $n\omega$, which grows linearly in $n$ like the exponent of $2^{-cn\ell}$; a fixed prior needs $\omega \propto 1/n$. In the limit, for a differentiable population loss $L$, the minimiser of $L(\theta) + \frac{\omega}{2}\|\theta\|^2$ minimises $L$ only if $\theta = 0$ does, and it can classify worse (Example F.7). In both cases the penalty per sample does not vanish with the data. Discarding a component whose risk reduction per unit of complexity is small requires this (Proposition 9.3): every $\lambda_n = o(n)$ eventually learns the errors (Proposition 8.3(a)), and the same per-sample threshold keeps the near-copies of Theorem 10.2.

The analogy is one of objectives. Unlike a vertex of the hull, a nonzero minimiser of $L+\frac{\omega}{2}\|\theta\|^2$ with $L$ differentiable changes with every $\omega$, so the analogue of a window is a range of $\omega$ with one classifier, not with one hypothesis (Appendix F.7). $\|\theta\|^2$ is not a description length: the information in a weight depends on its precision, which Hinton and van Camp (1993) control by adding Gaussian noise, and with normalisation layers the norm mainly sets the effective learning rate (van Laarhoven, 2017). Stochastic gradient iterates are not minimisers (Mandt et al., 2017); decoupled weight decay in adaptive optimisers is not an $L_2$ penalty (Loshchilov and Hutter, 2019; Xie and Li, 2024); and the decay acts per step, so before convergence its effect depends on the learning rate and the number of steps. Grokking (Power et al., 2022; Liu et al., 2022) has been related to the weight norm and to weight decay (Liu et al., 2023), and analysed as memorisation followed by a cleanup that removes memorising components (Nanda et al., 2023).

## 8.2 The two kink bounds and the theorem

The objective is the population objective $J_c$ of Definition 7.1, in the population reading of Lemma 5.3; samples are treated in Section 8.3. Define

$$t^* := \tfrac{d-\Lambda_1}{4}, \qquad \Phi_2(c) := \begin{cases} \min\{c(d-8-\Lambda_1)+G_1,\ G_{t^*-1}\} & t^* \geq 2, \\ G_1 & t^* < 2, \end{cases} \qquad \Lambda_3 := 4 + 2\log d + \kappa,$$

$$J_g(c) := c(d+\kappa_g) + H(p) \geq J_c(\pi_g), \qquad J_b(c) := c(d + \lceil L_S \rceil + \kappa_b) \geq J_c(\pi_b),$$

$$\Lambda_2 := 2^m H(p) - B_S + 4 + \Lambda_1 + \kappa_g \leq \log(2^m+1) + \delta_S + 4 + \Lambda_1 + \kappa_g,$$

$$\Delta(c) := \max\{0,\ c(4+\Lambda_1+\kappa_g) + H(p) - B_S/2^m\}.$$

For $c \geq 2^{-m}$ and $\Lambda_2 \geq 0$, $\Delta(c) = \Lambda_2/2^m + (c-2^{-m})(4+\Lambda_1+\kappa_g)$. $J_g$ minus each piece of $\Phi_2$ is affine and increasing in $c$, so $\{c>0 : J_g(c) \leq \Phi_2(c)\} = (0, c_{\mathrm{hi}}]$ (empty if $c_{\mathrm{hi}} \leq 0$) with

$$c_{\mathrm{hi}} := \min\Big\{\frac{G_1 - H(p)}{8+\Lambda_1+\kappa_g},\ \frac{G_{t^*-1} - H(p)}{d+\kappa_g}\Big\} \ \ (t^* \geq 2), \qquad c_{\mathrm{hi}} := \frac{G_1 - H(p)}{d+\kappa_g} \ \ (t^* < 2).$$

$c_{\mathrm{hi}} > 0$ requires $G_1 > H(p)$; at $B = \infty$ this holds exactly for $p < p_0 = 0.05596\ldots$ (Section 7.3).

**Proposition 8.6** (The two kink bounds)**.** *Let $B = \infty$ or let $B \geq \log(1/p)$ be an integer. Take the reference point $(d+\kappa_g, H(p))$, bounded by $\pi_g$, the fits $R \geq R_B$, the set $\mathcal{N}$ of programs of level at least 2 ($\infty$ included) and the set $\mathcal{M}$ of programs of level 1. (a)* Left. *Every $\pi \in \mathcal{N}$ has $J_c^B(\pi) \geq \Phi_2(c)$ for every $c > 0$, so $\eta_{\mathcal{N}}(c) \geq \Phi_2(c) - J_g(c)$, which is nonnegative on $(0, c_{\mathrm{hi}}]$ and positive on $(0, c_{\mathrm{hi}})$. If $c_{\mathrm{hi}} > 0$, every $\pi \in \mathcal{N}$ lies above the wedge with apex $(d+\kappa_g, H(p))$, steepness $c_{\mathrm{hi}}$ and margin $G_1 - H(p)$: $R_B(\pi) \geq H(p) + \max\{c_{\mathrm{hi}}(d+\kappa_g-\ell(\pi)),\ G_1 - H(p)\}$. (b)* Right. *Every $\pi \in \mathcal{M}$ has $\ell(\pi) \geq d-4-\Lambda_1$ and $R_B(\pi) \geq H(p) - \Lambda_2/2^m - 2^{-m}(\ell(\pi) - d - \kappa_g)$: $\mathcal{M}$ has the right line of slope $2^{-m}$ and offset $\Lambda_2/2^m$, with length floor $d-4-\Lambda_1$. If $p < \frac{1}{4}$, then $\Lambda_2 \geq 0$.*

*Proof idea.* (a): by (D1) a program of level $t \geq 2$ is at least $d - s(t) - \Lambda_1$ long or pays the risk floor $G_{t-1}$, and the concavity of $G$ leaves two cases. (b) is (D1) and (D2) at level 1 (Appendix F.2).

Not knowing the parity costs a positive amount per sample; knowing the rule buys only $2^{-m}$ of risk per bit. Between the two slopes lies the window:

**Theorem 8.7** (The teacher's-mistakes theorem)**.** *In the setting of Definition 7.1, with $B = \infty$ or an integer $B \geq \log(1/p)$, the following hold for every program $\pi$ of $\mathcal{H}_\times$, and of every product class covered by Proposition 11.1.*

*(o)* Level bounds. *If* $\mathrm{lev}(\pi) = 1$, *then* $J_c^B(\pi) \geq c(d - 4 - \Lambda_1) + \max\{R_B(\pi),\, cB_S - (c2^m - 1)R_B(\pi)\}$. *If* $\mathrm{lev}(\pi) \geq 2$, *including* $\infty$, *then* $J_c^B(\pi) \geq \Phi_2(c)$.

*(i)* Window. *If* $c\,2^m > 1$ *and* $c \leq c_{\mathrm{hi}}$, *then* $\Phi(c) \geq \Phi^B(c) \geq \min\{c(d-4-\Lambda_1) + B_S/2^m,\ \Phi_2(c)\}$ *and* $J_c(\pi_g) \leq \Phi(c) + \Delta(c)$, *where* $\Delta(c) \leq c(4 + \Lambda_1 + \kappa_g) + (\log(2^m + 1) + \delta_S)/2^m$.

*(ii)* Identification. *Let* $c\,2^m > 1$, $\epsilon \geq 0$ *and* $J_g(c) + \epsilon < \Phi_2(c)$, *and let* $J_c^B(\pi) \leq J_g(c) + \epsilon$; *this includes every* $\pi$ *with* $J_c(\pi) \leq \Phi(c) + \epsilon$. *Then* $\mathrm{lev}(\pi) = 1$. *Put*

$$\Delta_\ell := \kappa_g + \frac{\epsilon}{c} + \frac{\Lambda_2 + \epsilon/c}{c2^m - 1}, \qquad R_{\max} := J_g(c) + \epsilon - c\,(d - 4 - \Lambda_1)^+.$$

*(a)* It knows the simple model: $u(\pi) \geq \max\{\frac{3}{4},\ 2^{-R_{\max}} - p - 2^{-B}\}$.

*(b)* It fits no better than noise: $R_B(\pi) \geq H(p) - \dfrac{c\Lambda_2 + \epsilon}{c2^m - 1}$.

*(c)* It is short: $\ell(\pi) \leq d + \Delta_\ell$.

*(d)* It holds little information about the error rule: $I(\pi) \leq \Delta_\ell + \Lambda_3$, *so* $L_S - K(S \mid a, \pi) \leq \delta_S + \Delta_\ell + \Lambda_3$.

*(e)* It does not predict the errors: *for every integer* $g \geq 1$, $e(\pi) \leq 1 - u(\pi) + \gamma_g + 2^{-g}$, *where* $\gamma_g := \sqrt{j_g \ln 2/(2s_0)}$ *and* $j_g$ *is the least integer* $j \geq 0$ *with* $j - 2\log(j+1) > \delta_S + \Delta_\ell + \Lambda_3 + 1 + 2\log m + 2\log s_0 + g + 2\log g + \kappa$.

*(iii)* Memoriser regime. *Let* $c\,2^m < 1$, *and let* $J_c^B(\pi) \leq J_b(c) + \epsilon$ *with* $J_b(c) + \epsilon < \Phi_2(c)$. *Then* $\mathrm{lev}(\pi) = 1$ *and* $(1 - c2^m)\min\{R_B(\pi), B_S/2^m\} \leq \varrho := c(\lceil L_S \rceil - L_S + \delta_S + 4 + \Lambda_1 + \kappa_b) + \epsilon$. *If* $\varrho < (1 - c2^m)B_S/2^m$, *then* $R_B(\pi) \leq \varrho/(1 - c2^m)$ *and* $e(\pi) \geq 2^{-R_B(\pi)/p} - 2^{-B}$.

*(iv)* The lower end is sharp. *If* $c < H(p)/(\lceil L_S \rceil + \kappa_b + 4 + \Lambda_1)$, *then* $J_c(\pi_b) < J_c(\pi)$ *for every* $\pi$ *with* $\mathrm{lev}(\pi) = 1$ *and* $R(\pi) \geq H(p)$. *Since* $L_S = 2^m H(p) - O(m)$, *this threshold is* $2^{-m}\big(1 + O((m + \log d + \kappa)/L_S)\big)$.

*Proof idea.* (i), (ii)(b) and (ii)(c) are Lemma 5.3(i),(ii) with the bounds of Proposition 8.6 ($\tau = 2^{-m}$, offset $\Lambda_2/2^m$, length floor $d - 4 - \Lambda_1$, $\eta_{\mathcal{N}} \geq \Phi_2 - J_g$); the lemma's $\Delta(c)$ is $\Delta(c)$. The rest gives the position its meaning: (a) is the risk floor at $R_B \leq R_{\max}$; (d) is list decoding and symmetry of information; in (e), much probability on the erroneous labels would put $S$ in a small set enumerable from the program (Hoeffding, 1963), contradicting (d). (iii) is the side $c < 2^{-m}$ of the same line, (iv) the memoriser's secant (Appendix F.2).

So in the window $2^{-m} < c < c_{\mathrm{hi}}$, for error rules incompressible relative to the model, no product hypothesis beats the good model by more than $\Delta(c)$, and every near-optimal hypothesis knows the parity, is at most $\Delta_\ell$ bits longer than $d$, holds little information about the rule and gives the erroneous labels average probability at most $1 - u + \gamma_g + 2^{-g}$ (at most 0.2 to 0.29 at the example's sizes, against 1 for the memoriser), when the rule is large compared with the information budget. Below $2^{-m}$, by a relative margin, near-optimal hypotheses reproduce the errors. The lower end $2^{-m}$ is the reuse factor: a bit of the rule serves about $n2^{-m}$ of $n$ samples, a bit of noise one.

*Remark* 8.8 (Numbers for the running example). At $d = 100$, $m = 20$, $p = 2^{-10}$, $B = \infty$, $\epsilon = 0$, a direct computation (code in the accompanying repository) gives Table 3, with $J_g(c) + \epsilon$ for $R_{\max}$ in (ii)(a); with $R_{\max}$ the first row has $u \geq 0.988$ and $e \leq 0.195$. The window is $(9.54 \cdot 10^{-7},\ 8.06 \cdot 10^{-3}]$ at $\kappa = 0$ (a ratio of 8453; an illustration) and $(9.54 \cdot 10^{-7},\ 3.65 \cdot 10^{-3}]$ at $\kappa = 30$. For $m = 24$ the bound at $c = 10^{-4}$ is $e \leq 0.065$ ($\kappa = 0$, $\delta_S = 0$). In a band of relative width about 6% ($\kappa = 0$; about 11% at $\kappa = 30$) below $2^{-m}$ nothing useful is proved (Appendix F.2).

| $\kappa$ | $\delta_S$ | $u \geq$ | $\ell \leq$ | $I \leq$ | $L_S - K(S \mid a, \pi) \leq$ | $e \leq$ |
|---|---|---|---|---|---|---|
| 0 | 0 | 0.983 | $d + 20.5$ | 37.8 | 37.8 | 0.200 |
| 0 | 20 | 0.983 | $d + 20.7$ | 38.0 | 58.0 | 0.218 |
| 30 | 0 | 0.981 | $d + 51.1$ | 98.3 | 98.3 | 0.274 |
| 30 | 20 | 0.981 | $d + 51.3$ | 98.5 | 118.5 | 0.287 |

Table 3: Identification at $c = 10^{-4}$ (Theorem 8.7(ii), $g$ optimised) for the running example: $d = 100$, $m = 20$, $p = 2^{-10}$ ($L_S = 11{,}710$ bits), $B = \infty$, $\epsilon = 0$. Rows with $\kappa = 0$ set the machine constant to 0 and are illustrations. Information bounds are in bits, out of $L_S$. Lower bounds are rounded down and upper bounds up.

*Remark* 8.9 (What the theorem relies on). Without one of the first four hypotheses a step of the proof fails as written; the fifth enters every bound. (1) *A list-decodable model class* (Lemma 7.2) gives the left side and the length floor of the right line; majority-like functions do not admit it. (2) *The product form of $E$* gives the right slope $2^{-m}$ instead of $2^{-d}$ (Lemma 7.4). (3) *Uniform inputs* make every slope a rate; the window scales with the input law (Proposition G.4). (4) $G_1 > H(p)$ is the margin of the left wedge; Proposition 8.13 relaxes it to $p \leq 0.1233$, asymptotically. (5) *A small deficiency $\delta_S$* enters the offset of the right line, as the Ville deficiency $\delta^*$ does for noise (Theorem 6.9(i)), with the same tail $2^{-j}$; it is not an artefact of the proof (Proposition 9.1).

*Remark* 8.10 (What is selected). Theorem 8.7 concerns the near-optimal programs, not the minimiser. The natural strengthening that $\pi_g$ is a vertex of the population hull on the whole window fails in explicit rate codes: among the programs "true parity with flip rate $q$", a direct computation (code in the accompanying repository) at the example's sizes, with the machine constant set to 0 or 30 (an illustration), finds that the rate-$p$ program minimises $J_c$ only on the lower 67.7% to 88.6% of the window's logarithmic width; a coarser rate such as $2^{-9}$ takes over above, and satisfies every conclusion of Theorem 8.7(ii). In $\mathcal{H}_\times$ nothing is claimed either way.

### 8.3 Full tables and samples

On the full truth table the reuse factor is exact, and the window becomes one in $\lambda$. Let $N := 2^d$, and write $L_E, B_E, \delta_E$ for $L_S, B_S, \delta_S$ when $m = d$.

**Corollary 8.11** (Full truth tables: the window starts at the reuse factor)**.** *Let $B = \infty$ and let the data be the whole truth table of $y^*$, so $n = N$; a $\lambda$-MAP minimises $\lambda\ell + NR = N\, J_{\lambda/N}$. (i) For every $\lambda > 2^{d-m}$ with $J_g(\lambda/N) < \Phi_2(\lambda/N)$, in particular every $\lambda \in (2^{d-m}, Nc_{\mathrm{hi}})$, every $\lambda$-MAP $\pi$ satisfies Theorem 8.7(ii) with $c = \lambda/N$ and $\epsilon = 0$; so $\ell(\pi) \leq d + \kappa_g + 2^{d-m}\Lambda_2/(\lambda - 2^{d-m})$, $NR(\pi) \geq NH(p) - \lambda 2^{d-m}\Lambda_2/(\lambda - 2^{d-m})$ and $I(\pi) \leq \kappa_g + 2^{d-m}\Lambda_2/(\lambda - 2^{d-m}) + \Lambda_3$. (ii) Let $m = d$. If $G_1 > H(p)$ and $N(G_1 - H(p)) > d + \kappa_g + \log(N + 1)$, then at $\lambda = 1$ the optimum of $\ell + NR$, and the values $\ell(\pi_g) + NR(\pi_g)$ and $\ell(\pi_b) + NR(\pi_b)$, lie within $O(\log N) + \delta_E$ of each other, so $\lambda = 1$ does not decide whether the errors are memorised; lower-order terms and the reference machine do. (iii) For $\lambda < NH(p)/(\lceil L_S \rceil + \kappa_b + 4 + \Lambda_1)$ the memoriser beats every program of level 1 with $R \geq H(p)$.*

*Proof idea.* Theorem 8.7(ii) and (iv) at $c = \lambda/N$, with $c2^m - 1 = (\lambda - 2^{d-m})/2^{d-m}$; (ii) compares the two programs with (D2) and the risk floor (Appendix F.3).

The threshold in (iii) is $2^{d-m}(1 + O((m + \log d + \kappa)/L_S))$: the window in $\lambda$ starts, up to lower-order terms, at the reuse factor $2^{d-m}$, the slope of the right line of Proposition 8.6(b) multiplied by $N$. For $m = d$ it is 1, as for noise: the good model is $b_0$-sufficient, $b_0 = \delta_E + O(\log N)$

(Proposition 7.7), so at $\lambda = 1$ its point lies on the sufficiency line up to $b_0$, like the sampler's in Theorem 6.11, and (ii) is the resulting tie. At $d = m = 20$ and $p = 2^{-10}$ a direct computation (code in the accompanying repository) gives the window $\lambda \in (1, 1.194 \cdot 10^4)$ with the machine constant set to 0 (an illustration) and $(1, 6456)$ at $\kappa = 30$.

Let $D$ be $n \le 2^d$ inputs drawn uniformly without replacement, labelled by $y^*$. For $p < \frac{1}{4}$, by (D1) the memoriser beats $\pi_g$ at every $\lambda$ below $F_g/(\lceil L_S \rceil + \kappa_b + 4 + \Lambda_1) \approx n2^{-m}$, with $F_g := -\log \nu_{\pi_g}(D) \approx nH(p)$. Let $B \ge \log(1/p)$ be an integer, $\delta \in (0,1)$, $\epsilon_1$ as in Proposition 8.3(c), $\hat{L} := (J_c(\pi_g) + \epsilon_1)/c$ and $\epsilon_n := \epsilon_1 + B\sqrt{(\hat{L}\ln 2 + \ln(2/\delta))/(2n)}$.

**Corollary 8.12** (Finite samples)**.** *With probability at least $1-\delta$, every $\lambda$-MAP with $\lambda = cn$ satisfies $J_c^B \le J_g(c) + \epsilon_n$, and hence the conclusions of Theorem 8.7(ii) with $\epsilon = \epsilon_n$, whenever $c2^m > 1$ and $J_g(c) + \epsilon_n < \Phi_2(c)$.*

The proof is uniform convergence over all programs with Kraft weights (Lemma F.3), which is pessimistic: at the example's sizes and $c = 10^{-4}$, a direct computation (code in the accompanying repository) shows that near-population constants need about $10^{11}$ samples (machine constant 0, an illustration; Appendix F.4). The power posterior with $\lambda = cn$ also concentrates on near-optimal programs (Proposition F.5), and Remark 8.1(1) is realised in population form (Example F.4).

## 8.4 Flip rates below 0.1233

The bound $\Phi_2$ charges a program that does not know the parity only through the risk floor. Charging in its length the information it holds about the rule gives a continuous lower bound $\Phi_2^\sharp \ge \Phi_2$ on all programs of level at least 2, and Theorem 8.7(i),(ii) and Proposition 8.3(c) hold with $\Phi_2^\sharp$ in place of $\Phi_2$ for every $c > 2^{-m}$ with $J_g(c) + \epsilon < \Phi_2^\sharp(c)$ (Proposition F.8 in Appendix F.8).

At $c \ge 2^{-m}$ the charge $cI \ge p\,(I/s_0)$ is linear in the information per error cell, while the sharp floor of Section 7.2 lets that information gain only about $2p\gamma_g(I)$, of order $\sqrt{I/s_0}$. Let $\psi_2(p,a) := \inf_{x \ge 0} \left[apx - \log\left(\frac{3}{4} - \frac{p}{2} + 2p\sqrt{x \ln 2/2}\right)\right]$ and $a^*(p) := \inf\{a \ge 1 : \psi_2(p,a) > H(p)\}$.

**Proposition 8.13** (A window for every flip rate below $p_1$, asymptotically)**.** *Fix a dyadic $p \in (0, \frac{1}{4})$, $\kappa$, $B = \infty$ and $a > a^*(p)$. Let $m \to \infty$ with $s_0 = p2^m$, $d \ge m$, $d2^{-m} \to 0$ and $\delta_S = o(2^m)$, and let $c_m \ge a2^{-m}$ satisfy $c_m d \to 0$. Then there is $\epsilon > 0$ such that for all large $m$, $J_g(c) + \epsilon < \Phi_2^\sharp(c)$ for every $c \in [a2^{-m}, c_m]$; so the identification of Theorem 8.7(ii) holds there (Proposition F.8(iii)). Moreover $a^*(p) < \infty$ if and only if $1 - \log(\frac{3}{2} - p) > H(p)$, and $a^*(p) = 1$ if and only if $\psi_2(p,1) \ge H(p)$.*

*Proof idea.* At $c = a2^{-m}$ a budget $I = xs_0$ costs $apx$ per sample, and as $m \to \infty$ the level-2 floor tends to $-\log(\frac{3}{4} - \frac{p}{2} + 2p\sqrt{x \ln 2/2})$; so the level-2 term $\Psi_2$ of $\Phi_2^\sharp$ (Appendix F.8) has $\liminf_m \Psi_2(a2^{-m}) \ge \psi_2(p,a) > H(p)$, while the length terms vanish. As $a \to \infty$, $\psi_2(p,a)$ increases to $1 - \log(\frac{3}{2} - p)$ (Appendix F.8).

The first condition is that of Proposition 7.8(iii), which holds exactly for $p < p_1 = 0.1233072\ldots$ (Section 7.3). $\psi_2(p,1) - H(p)$ is strictly decreasing on $(0, \frac{1}{4}]$, and a direct computation (code in the accompanying repository) puts its root at $p^* = 0.0778658\ldots$ (Appendix F.8). So asymptotically a window reaching far above $2^{-m}$ exists for every dyadic $p \le 0.1233$, reaching down to $a2^{-m}$ for every $a > 1$ when $p \le 0.0778$. At $d = 100$ and $m = 20$, with the machine constant set to 0 (an illustration), windows exist at the flip rates 1/16, 5/64, 3/32, 7/64 and 15/128, where Theorem 8.7 gives none, with $e \le 0.197$ to 0.269 at a value of $c$ inside each (Table 6).

**Question 8.14** (other models, flip rates, samples and schedules)**.** (a) For which classes of simple models other than parities is there a window as in Theorem 8.7? We expect that majority-like functions give none. (b) For $p_1 \le p < \frac{1}{4}$ and $m < d$, with $p_1 = 0.1233072\ldots$ as in Section 7.3, is there a window on which every near-optimal hypothesis has level at most 2? The present floors cannot give level 1 there; the level-3 analogue of $\psi_2$ tends to $-\log(\frac{5}{8} - \frac{p}{4}) > H(p)$, and what is missing is (D2) at level 2. (c) Do near-population constants hold for sampled data far below $10^{11}$ samples? (d) On $n \gg 2^m$ inputs drawn without replacement, with unequal multiplicities of the sampled cells, is the least $\lambda$ at which a parity with flip noise is a $\lambda$-MAP about the mean multiplicity of the sampled error cells, up to $O(\log)$ terms, and does the largest grow like $n$? (e) What do schedules do whose ratio $\lambda_n/n$ leaves the window infinitely often?

# 9 When errors have structure: a second kink

In Theorem 8.7 an incompressible error rule lies to the right of the model's kink, on a line of slope $2^{-m}$ set by its reuse factor (Proposition 8.6). Here the second reason for a steeper penalty fails: a rule with structure has a kink of its own, and steeper selection sorts hull segments by *rate*, risk reduction per bit, not by validity. The setting is Definition 7.1 unless stated otherwise; proofs are in Appendix G.

## 9.1 Which error rules defeat steeper selection

Realistic systematic errors are often low-complexity patterns, while the example of Section 8 fixes only a density and a description length. Split the rule as $S = S_1 \sqcup S_2$, $|S| = s_0 = p2^m$, with $s_i := |S_i|$, $p_1 := s_1/2^m$ and $p_2 := s_2/(2^m - s_1)$, where $S_1$ is *cheap*, $K(S_1 \mid m) \le k_1$; every input of $E_1 := S_1 \times \{0,1\}^{d-m}$ is an error input. The *envelope* $\pi_e$ predicts $f_a \oplus 1$ with probability 1 on $E_1$ and $f_a$ with flip probability $p_2$ elsewhere; it has risk $R_e = (1-p_1)H(p_2) < H(p)$ and length at most $d + k_1 + \kappa_e$, $\kappa_e := 2\log d + 2\log m + 2\log(s_2+1) + \kappa$. Put $J_e(c) := c(d + k_1 + \kappa_e) + R_e$, $\zeta := c(k_1 + \kappa_e + 4 + \Lambda_1) + \epsilon$ and $c_{\mathrm{env}} := (H(p) - R_e)/(k_1 + \kappa_e + 4 + \Lambda_1)$. A program is *$\epsilon$-near-optimal* at $c$ if $J_c(\pi) \le \Phi(c) + \epsilon$.

**Proposition 9.1** (Error envelopes)**.** *Let $B = \infty$, and let $J_c(\pi) \le J_e(c) + \epsilon$ with $J_e(c) + \epsilon < \Phi_2(c)$.*

*(i)* $\mathrm{lev}(\pi) = 1$ *and $R(\pi) \le R_e + \zeta$. So for $c < c_{\mathrm{env}}$ and $\epsilon$ small enough that $\zeta < H(p) - R_e$, no program with $R(\pi) \ge H(p)$, the good model included, is $\epsilon$-near-optimal.*

*(ii) With $B_2 := K(S_2 \mid a, K(a))$, the average probability of reproducing the cheap errors is*

$$e_1(\pi) := \mathbb{E}\big[\nu_\pi(y^*(x) \mid x) \,\big|\, x \in E_1\big] \;\ge\; 2^{-\bar{R}_1},$$
$$\bar{R}_1 := \tfrac{1}{p_1}\big(\zeta + (1-p_1)H(p_2) - \min(c, 2^{-m})\,(B_2 - K(S_1))^+\big).$$

*Proof idea.* (i) is Lemma 5.3(iv) with the envelope's point $(d + k_1 + \kappa_e, R_e)$ as the reference: Theorem 8.7(o) excludes the programs of level at least 2, and (D1) of Lemma 7.5 gives those of level 1 length at least $d - 4 - \Lambda_1$; no right line is needed. (ii) codes $S_2$ given $S_1$ with the hypothesis, as in Lemma 7.4, which bounds its risk off $E_1$ from below.

*A second kink.* The cheap part adds a second drop to the population hull, at length about $d + k_1$. Take a program of level 1 with $R \ge H(p)$; its length is at least $d - 4 - \Lambda_1$, so the envelope either dominates its point or has secant at least $c_{\mathrm{env}}$ from it, which then bounds the right secant there (Definition 5.2) from below. So, under its side conditions, (i) excludes near-optimal fits

no better than noise below $c_{\mathrm{env}}$, whereas for an incompressible rule the identification window of Theorem 8.7 starts just above $2^{-m}$.

At the example's sizes let $S_1 = \{z : z_1 = \cdots = z_j = 0\}$ and $S_2$ be uniformly random. For $j = 13$ (128 cheap and 896 random points) the rule has density $2^{-10}$ and $\log \binom{2^m - s_1}{s_2} = 10{,}418$ bits, of the order of the example's 11,710. With the machine constant set to 0 (an illustration), a direct computation (code in the accompanying repository) gives $c_{\mathrm{env}} = 1.64 \cdot 10^{-5}$: on 31% of the logarithmic width of the window $(9.54 \cdot 10^{-7}, 8.06 \cdot 10^{-3}]$ of Theorem 8.7 (24% at $\kappa = 30$) no hypothesis that fits no better than the good model is near-optimal, and at $c = 1.01 \cdot 2^{-m}$ near-optimal hypotheses reproduce the cheap errors with average probability at least 0.544 for typical $S_2$ ($j = 11$: 47%, 0.862; Appendix G.1). So density and description length alone do not suffice. (Its deficiency is about 1,300 bits; Theorem 8.7 applies to it with this value of $\delta_S$ in its bounds, which are correspondingly weaker, and Proposition 9.1 says nothing about it above $c_{\mathrm{env}}$.) A cheap, dense error region has a high rate and is learned like any valid component.

**Proposition 9.2** (Rate blindness)**.** *Let $B = \infty$ and $\ell(\pi_{\mathrm{coin}}) \le \kappa$.*

*(i) If $H(p)/(\ell(\pi_b) - \ell(\pi_g)) \ge (1 - H(p))/(\ell(\pi_g) - \ell(\pi_{\mathrm{coin}}))$, then at every $c$ the good model is beaten by the memoriser or beaten or tied by the coin.*

*(ii) Suppose that*

$$\frac{H(p)}{\lceil L_S \rceil + \kappa_b + 4 + \Lambda_1} \;\ge\; \frac{1 - H(p)}{d - 4 - \Lambda_1 - \kappa} \quad \text{and} \quad d - 4 - \Lambda_1 - \kappa > 0. \tag{$\star$}$$

*Let $u(\pi) > \frac{3}{4}$ and $R(\pi) \ge H(p)$. Unless $\ell(\pi_b) \le \ell(\pi)$ or $R(\pi) \ge 1$, the memoriser's secant from the point of $\pi$ is at least the coin's secant to it, $R(\pi)/(\ell(\pi_b) - \ell(\pi)) \ge (1 - R(\pi))/(\ell(\pi) - \ell(\pi_{\mathrm{coin}}))$. So at every $c$ every such program is beaten by the memoriser or beaten or tied by the coin.*

*(iii) For $c < 2^{-m}$, near-optimal programs reproduce the errors, under the condition of Theorem 8.7(iii).*

*Proof idea.* (i) compares three points. In (ii), (D1) gives $\pi$ length at least $d - 4 - \Lambda_1$, so the two secants are bounded by the two sides of ($\star$). So the error kink is at least as steep as the model's, and no slope makes such a point the unique optimum. (iii) is Theorem 8.7(iii).

For fixed $m$ and $s_0$, and under a condition on the constants that holds in every computed case, ($\star$) holds for all $d$ beyond a threshold $d_0(\kappa)$ linear in the machine constant (Lemma G.1): once the simple model is long enough, no program that knows the parity and fits no better than noise is ever the unique optimum. For one 3-bit prefix ($m = 3$, $p = 1/8$) a direct computation (code in the accompanying repository) gives $d_0 = 49$ with the machine constant set to 0 (an illustration) and 169 at $\kappa = 30$, while the incompressible rule of the example leaves a window whose ends differ by a factor of about 8500 ($\kappa = 0$) or 3800 ($\kappa = 30$).

## 9.2 Rate, not validity

**Proposition 9.3** (Separable components)**.** *Let the inputs split into finitely many regions $X_j$ and the hypotheses be products of per-region choices with additive code lengths. Each region selects a vertex of its own hull for slope $c$ (Lemma 3.8). If region $j$ offers a default or a component $j$ of extra length $k_j$ and population risk reduction $r_j$, component $j$ is selected if and only if $r_j/k_j > c$ (ties arbitrary). So a valid component whose rate is at most an error component's rate cannot be kept without the error, except at a common tie value. Two worlds with the same teacher channel*

*and input law, one where a component is an error and one where it is a rare valid rule, get the same selection at every $c$.*

*Proof idea.* The objective is a sum over regions, a component lowers it if and only if $c\,k_j < r_j$, and the twin worlds have the same objective.

*Remark* 9.4 (What a hull cannot see). Steeper selection keeps exactly the hull segments whose rate exceeds $c$ (Lemma 3.8(v), Remark 5.8). An incompressible error rule has rate at most about $2^{-m}$ per sample, its reuse factor; a structured rule has the rate of its cheapest dense part, which can lie anywhere. Nothing in the geometry tells the model's kink from an error's (Appendix G.3): a guard with a lower rate than its rule is dropped on an interval of $c$, and the rule is applied where the teacher never applied it (Proposition G.2); the selection, and by the same mechanism the window of Theorem 8.7, moves with the input law, proved here in the separable case (Proposition G.4); and $c$ cannot be chosen from imitation data, since strictly proper scores prefer the teacher's channel, errors included (Gneiting and Raftery, 2007), log-loss validation picks the smallest $c$, and the twin worlds have the same hull (Proposition G.3).

Off the training support the shortest hypotheses that fit the data decide, so steeper SFI extrapolates more reliably than SFI if they extrapolate well and is more confidently wrong if they do not (Proposition G.5).

**Question 9.5** (structured errors and the choice of $c$)**.** (Larger flip rates are the subject of Proposition 8.13 and Question 8.14.) (a) For envelopes whose cheap region has error density $\theta < 1$, only part (i) of Proposition 9.1 carries over (Appendix G.1). Do near-optimal hypotheses then memorise the random part $S_2$, and what is selected above $c_{\mathrm{env}}$? (b) Is there a rule for choosing $c$, using information other than the imitation data, that keeps rare valid rules and drops errors? By Propositions 9.3 and G.3, imitation data alone cannot tell them apart.

## 10 Imitation: a sampler whose coins the environment drives

Imitation motivates both reasons: a learner trained on a person's answers, as in behavioural cloning (Bain and Sammut, 1999; de Haan et al., 2019), should return a copy of the person. In our terms the person is a sampler, a model with fresh coins at each query, and a *copy* is a program whose kernel equals the person's on the support of the inputs. If the person's coins are private, this is the first reason: no predictor gains over the person more than its length plus a Ville deficiency, however it models the environment (Remark H.1), and every fixed $\lambda > 1$ eventually returns a shortest copy, not a fill-in (Theorem 6.3). Systems that infer from the data which person they face are excluded by the choice of the private-randomness class, not by the data (Appendix H.1).

Here the person's coins depend on the environment, so hypotheses that model the environment can predict the person's answers *better* than the person does. The data follow the kernel $P^*$ of Definition 10.1, so a fixed $\lambda$, or any $\lambda_n = o(n)$, eventually returns a shortest program for $P^*$ (Theorem 6.3(a)), not a copy when $P^*$ differs from the person's kernel; in the feature case such a program encodes the coupling. The coins are reused at every input of a cell, like an error rule, and the guarantee below needs the schedule of the second reason, $\lambda/n$ above the reuse factor $2^{-m}$ (Section 8). When the coins are an incompressible function of a variable, the cell, with many more values than the person has bits, we show that for $\lambda/n$ in a window every near-optimal private-randomness hypothesis, whatever it computes, is close to the person in behaviour (Theorem 10.2), and that no procedure reliably copies each of several persons who share one data law (Proposition 10.3). We do not show that a copy itself is selected or that the selected

program computes as the person does, and a coupling with a short description defeats every $\lambda/n$ (Proposition H.4); the guarantee is for the population objective, from samples only asymptotically. Proofs are in Appendix H.

**Definition 10.1** (Environment-driven coins)**.** Let $Y$ and the coin set $\Sigma$ be finite, $m \geq 1$, and $W := \{0,1\}^m$, the *cells*. (a) A *channel* $\mu = (\mu(\cdot \mid w))_{w \in W}$ of finitely supported laws on inputs with rational masses gives the input law $\bar{\mu} := 2^{-m} \sum_w \mu(\cdot \mid w)$, with support $\operatorname{supp} \mu$; in the *feature case* every input in $\operatorname{supp} \mu(\cdot \mid w)$ has $w$ as its first $m$ bits. (b) The *person* has a coin law $\rho$ on $\Sigma$, every $\rho(b)2^m$ a positive integer, and a response function $h : \operatorname{supp} \mu \times \Sigma \to Y$, injective in $b$; its kernel is $\nu_H(y \mid x) := \rho(b)$ if $y = h(x,b)$, and 0 otherwise, and its *model* is $h_0 := (h, \rho)$. A *copy* is a program of $\mathcal{H}_\times$ whose kernel equals $\nu_H$ on $\operatorname{supp} \mu$; $\mathcal{E}$ is the set of copies, $\ell_0 := \min_{\mathcal{E}} \ell$, and $\pi_H$ is a shortest copy. (c) A *coupling* $\beta : W \to \Sigma$ of exact type $\rho$, $|\beta^{-1}(b)| = \rho(b)2^m$, drives the coins: at each query, independently, $w_i$ is uniform on $W$, $x_i \sim \mu(\cdot \mid w_i)$ and $y_i := h(x_i, \beta(w_i))$. The learner sees $(x_i, y_i)$, i.i.d. from $\bar{\mu} \otimes P^*$ with $P^*(y \mid x) := \sum_w \mu(x \mid w)\mathbf{1}[y = h(x, \beta(w))] / \sum_w \mu(x \mid w)$. (d) $R(\pi) := \mathbb{E}[-\log \nu_\pi(y_i \mid x_i)]$, $J_c := c\,\ell + R$ and $\Phi(c) := \inf_{\mathcal{H}_\times} J_c$; every copy has $R = H(\rho)$. $L_\beta := \log\left(2^m! / \prod_b (\rho(b)2^m)!\right)$, $B_\beta := K(\beta \mid h_0, \mu)$, $\delta_\beta := L_\beta - B_\beta$, and $I_\beta(\pi) := B_\beta - K(\beta \mid h_0, \pi, \mu)$, the information of $\pi$ about the coupling. For a probability $P$ and a sub-probability $Q$, $D_{1/2}(P \| Q) := -2 \log(1 - d_{\mathrm{H}}^2(P, Q))$.

The coin used at a query is a fixed function of the cell, which the learner sees in the input (the feature case) or only through the channel (a latent shared cause). The data identify $P^*$ (Theorem 6.3); when the input carries information about the coin $\beta(w)$, that is, when $P^* \neq \nu_H$ on a set of positive $\bar{\mu}$-mass, hypotheses that model $\beta$ predict better than the person, by up to $H(\rho)$ bits per sample in the feature case, where $P^*$ is deterministic. The teacher of Definition 7.1 is the feature case with $h(x,b) = f_a(x) \oplus b$, $\rho = \mathrm{Bern}(p)$ and $\beta = \mathbf{1}_S$: systematic errors are environment-driven coins. As for error rules, $L_\beta \geq 2^m H(\rho) - |\Sigma| \log(2^m + 1)$ (Cover and Thomas, 2006), and $\mathbb{P}[\delta_\beta > j] < 2^{-j}$ for a coupling drawn uniformly from its type class, independently of the person's model and the channel.

**Theorem 10.2** (Robust near-imitation)**.** *In the setting of Definition 10.1 let $c = a2^{-m}$ with $a > 1$, $\epsilon \geq 0$, and $G := \ell_0 + \delta_\beta + |\Sigma| \log(2^m + 1) + \kappa$. Every $\pi \in \mathcal{H}_\times$ with $J_c(\pi) \leq \Phi(c) + \epsilon$, whatever its internal structure, satisfies:*

*(i)* $\ell(\pi) \leq \ell_0 + (G + 2^m\epsilon)/(a-1)$*;*

*(ii)* $-(aG + 2^m\epsilon)/((a-1)2^m) \leq R(\pi) - H(\rho) \leq c\,\ell_0 + \epsilon$*: it predicts neither much better nor much worse than the person;*

*(iii)* $I_\beta(\pi) \leq \ell_0 + (G + 2^m\epsilon)/(a-1) + \kappa$*;*

*(iv) with* $E := 2\log\left(2^{m-1}(2H(\rho) + c\,\ell_0 + \epsilon) + 1\right) \leq 2m + 2\log(2H(\rho) + c\,\ell_0 + \epsilon + 1)$*,*

$$\mathbb{E}_{x \sim \bar{\mu}} D_{1/2}\big(\nu_H(\cdot \mid x) \,\|\, \nu_\pi(\cdot \mid x)\big) \;\leq\; c\,\ell_0 + \frac{2a}{a-1} \cdot \frac{G + E + \kappa}{2^m} + \frac{a+1}{a-1}\,\epsilon,$$

*so its Hellinger gap from the person, $\mathbb{E}_{x \sim \bar{\mu}} d_{\mathrm{H}}^2(\nu_H(\cdot \mid x), \nu_\pi(\cdot \mid x))$, is at most $\frac{\ln 2}{2}$ times the right side.*

*Moreover $J_c(\pi_H) \leq \Phi(c) + G/2^m + c\,\ell_0$. In the feature case, if $|\Sigma| \geq 2$ and the values of $\rho$ are distinct, a program of length at most $\ell_0 + \lceil L_\beta \rceil + a_\rho + \kappa$, $a_\rho := 2\log(m+1) + 2\log(|\Sigma|+1) + |\Sigma|(m+1)$, has kernel $P^*$ and risk 0, and so beats every copy for $c < H(\rho)/(\lceil L_\beta \rceil + a_\rho + \kappa)$, a threshold at least $2^{-m}(1 + (a_\rho + 1 + \kappa)/(2^m H(\rho)))^{-1}$.*

*Proof idea.* A program is a code for the coupling, $K(\beta \mid h_0, \pi, \mu) \le 2^m R(\pi) + \kappa$ (Lemma H.2), which gives the whole class a right line of slope $2^{-m}$ at $(\ell_0, H(\rho))$; Lemma 5.3 gives (i), (ii) and the bound on $\Phi$. For (iv), the divergence is at most that of per-cell geometric means of the predictions (Lemma H.3), which define a semimeasure on couplings; normalised, it would give $\beta$ a code shorter than the coupling's incompressibility allows, unless the divergence is small, the program predicts worse than the person, or it knows more about $\beta$.

The best predictor has $\mathbb{E}_x D_{1/2}(\nu_H \| P^*) = H(\rho)$ in the feature case. So when the coupling is incompressible ($\delta_\beta \ll 2^m H(\rho)$) and the person much simpler than it ($\ell_0 + |\Sigma| m \ll 2^m H(\rho)$), for every $c$ above the reuse factor $2^{-m}$, with $a$ bounded away from 1, and well below $H(\rho)/\ell_0$, every near-optimal hypothesis is close to the person in behaviour, on the scale $H(\rho)$, and holds little of the coupling, although hypotheses that model the environment predict better. The theorem gives near-copies, not copies: nothing excludes a nearby coin law as the minimiser (cf. Remark 8.10); and the hypotheses that predict better gain only by storing an incompressible table. By Proposition D.5(c) with the sampler $P^*$, almost surely every $cn$-MAP satisfies (i)–(iv) with $\epsilon = 0$ for all large $n$. What happens outside this range of $c$, and numerical illustrations, are in Appendix H.3.

**Proposition 10.3** (What no procedure can do)**.** *(a)* One data law, at most one copy. *If worlds $\mathcal{W}_1, \dots, \mathcal{W}_N$ as in Definition 10.1 have one law of the data pairs and persons whose kernels differ pairwise on sets of positive $\bar{\mu}$-mass, then every procedure $\Psi$ (a measurable map from samples and independent randomness to programs) has $\sum_i \mathbb{P}_{\mathcal{W}_i}[\Psi(h_n) \in \mathcal{E}(\nu_i)] \le 1$ at every $n$. In the feature case with $|\Sigma| \ge 2$, one such world has the person $P^*$, who responds to the cell.*

*(b)* A person entangled with its coupling. *Let $X = W$, $\mu(\cdot \mid w) = \delta_w$, $Y = \Sigma = \{0,1\}$, $\rho = \mathrm{Bern}(s_0/2^m)$ with $1 \le s_0 < 2^{m-1}$, and let the person $H_f$ answer $f(w) \oplus b$ for a table $f : W \to \{0,1\}$. Draw $f$ uniformly and $\beta$ uniformly from its type class, independently. Then $\mathbb{P}[\delta_\beta > j] < 2^{-j}$, yet every procedure copies the person with probability at most $2^{-L_\beta}$, at every $n$.*

*Proof idea.* (a) The output has one law in all the worlds, and the sets of copies are disjoint. (b) The data depend on the world only through $f \oplus \mathbf{1}_\beta$, which $2^{L_\beta}$ equally likely worlds with pairwise different persons share.

So imitation data identify behaviour, not mechanism: persons with the same response function and coin law share a data law, however they compute them, and the selected program, a shortest one for its own kernel, may compute it unlike the person and extrapolates off $\operatorname{supp} \bar{\mu}$ as it does. Nor do the data identify the split of $P^*$ into a person and an environment, as a policy does not determine its split into a planner and a reward (Armstrong and Mindermann, 2018): every procedure adopts a convention, the $c$-procedure by rate (Proposition 9.3). An incompressible coupling is not enough: in (b) the person is as complex as its coupling and responds to the same cells, so a hypothesis such as $\ell_0 \ll 2^m$, or the left condition that list decoding gives in Theorem 8.7, is needed. A coupling with a short description, such as a pseudo-random generator with a short seed, defeats every $c$ (Proposition H.4); if it is the person's own deterministic state, the program that models it is arguably the more faithful copy, and the target $\nu_H$ is itself a modelling choice. Data from several environments or from interventions identify more in inverse reinforcement learning and behavioural cloning (Cao et al., 2021; de Haan et al., 2019); whether such data identify which structure belongs to the person here is open (Section 12.1).

# 11 Computable hypothesis classes

Both reasons for a steeper penalty survive resource bounds: the recovery theorems of Section 6 use only countability and the Kraft inequality, and the lower bounds of Sections 7 and 8 only coding properties of the class (proofs in Appendix I).

**Proposition 11.1** (Computable product classes)**.** *Let $\mathcal{H}'$ be a hypothesis class of product hypotheses (Definition 3.5). (a) Lemmas 7.2 to 7.5 and the lower-bound parts of Theorem 8.7 ((o), the bound on $\Phi$ in (i), (ii), and the level and $R_B$ statements of (iii)) hold for every $\pi \in \mathcal{H}'$, with $\kappa$ replaced by a constant that depends on $\mathcal{H}'$. The parts that use reference programs ($\Delta(c)$, $J_g$, $J_b$, (iv)) hold if $\mathcal{H}'$ contains a good model $\pi'_g$ and a memoriser $\pi'_b$, with $\kappa_g := \ell(\pi'_g) - d$ and $\kappa_b := \ell(\pi'_b) - d - \lceil L_S \rceil$. (b) Theorem 6.9 holds in every countable product class with Kraft weights that contains a program for the sampler. (c) Proposition B.5(ii) holds in every product class with a prefix-free code. (d) Theorems 6.3 and 6.7 hold in every countable product class with Kraft weights that contains a program whose kernel equals $\nu$ at $\mu$-almost every input, for example a time-bounded, circuit or finite class; $\ell_0$, $\mathcal{E}_0$ and the margins are then those of the class, in its own code.*

*Proof idea.* Parts (a) and (c) use (F1) and the coding theorem, with constants of the class; (b) and (d) use only countability, the Kraft inequality and Ville's inequality for the class's own mixture. With a finite input set, as for circuits on $\{0,1\}^d$, inputs repeat, so (d) concerns per-query labels; with frozen labels the data eventually form one table (Remark D.4). The classes of Proposition 6.6 are finite classes of this kind.

*Remark* 11.2 (Examples). Examples are probabilistic programs clocked at $t(|x|)$ steps (Hänni, 2026a, Section 9.1), and circuits with private output noise, where a parity costs about $d \log d$ bits. Theorem 4.2 does not transfer: a mixture of all time-$t$ programs is not a time-$t$ program.

For computable $\lambda \geq 1$, in $\mathcal{H}_\times$ and $\mathcal{H}_F$, the steeper mixture $\xi_\lambda$ is lower semicomputable, and its predictive conditionals, raw or normalised, are limit-computable but not computable, as for $M_F$ (Proposition I.1).

*Remark* 11.3 (Neural-network Bayes). With labels independent given the parameters $\theta$, a network's Bayes posterior is a mixture of product laws, of the type of $M_{\text{prod}}$ (Hänni, 2026a, Proposition 10.1); for a countable discretisation with integer code lengths $\ell(\theta)$ and Kraft weights, whose noise model contains the sampler, Theorems 6.3 and 6.9 apply: under the power prior $2^{-\lambda \ell(\theta)}$, $\lambda > 1$, settings that reproduce the noisy labels get posterior mass exponentially small in the sampled randomness, and per query the $\lambda$-MAP is eventually a shortest setting whose kernel equals the sampler $\mu$-almost everywhere. Whether a parameter prior behaves like a description-length prior, as Theorem 8.7 needs, is the bridge question of the companion paper (Hänni, 2026a, Question 10.9 and Remark H.2), which the slides of Vaintrob and Hänni (2026) motivate; nothing here settles it.

**Question 11.4** (resource bounds and computability)**.** (a) Does some time-bounded shared class contain a short member that absorbs everything at $\lambda > 1$, as in Theorem 4.2? If not, steeper simplicity could work in time-bounded Solomonoff induction. (b) For computable $\lambda \geq 1$ and the class $\mathcal{H}_\times$, is there a computable map from data sets to $\lambda$-MAPs?

# 12 Discussion

**Two reasons, one geometry.** Both reasons select at a kink of the hull of (length, fit) and discard what lies to its right (Table 4). Noise enters one likelihood term per unit, so every fixed

$\lambda > 1$ refuses it and per query the split is exact; an error rule enters the term of every input it governs, so $\lambda$ must exceed its reuse factor, and only the near-optimal set, not a single hull vertex, is controlled (Remark 8.10). The Ville deficiency of the noise and the deficiency $\delta_S$ of the rule both offset the right side, with the same tail (Remark 8.9). Imitation needs both: a person with private coins is eventually recovered at every fixed $\lambda > 1$; when the environment drives the coins and the input partly reveals them, every $\lambda_n = o(n)$ eventually returns a program for the data kernel instead, and near-copies are guaranteed for $\lambda/n$ in a window (Section 10).

| | The split into model and noise (first reason) | The teacher's mistakes (second reason) |
|---|---|---|
| What is discarded | sampled randomness: private coins, each used once | a fixed error rule, used at every input of its cell |
| Its reuse factor | one term per unit, per query or frozen and counted once (Prop. F.1) | $2^{d-m}$ on the full table, about $n2^{-m}$ on $n$ samples (Lem. 8.2 and Cor. 8.11) |
| The model's left steepness | linear in $n$ per query; exponential in $d$ on the full parity table (Thm. 6.7 and Prop. 7.7) | $c_{\mathrm{hi}}$ per sample (Prop. 8.6) |
| The $\lambda$ that works | every fixed $\lambda > 1$, with rates that depend on the class only through a margin; in the limit every $1 \le \lambda_n = o(n)$ (Thm. 6.3) | $\lambda/n$ in a compact subset of $(2^{-m}, c_{\mathrm{hi}})$, $\lambda = cn$ the simplest; no $\lambda_n = o(n)$ (Prop. 8.3) |
| What $\lambda = 1$ does | similar posterior weight for sampler and fill-ins on a data set; per query the same limit, possibly exponentially later (Thm. 6.11 and Prop. 6.6) | learns the errors (Prop. F.2) |
| What defeats it | shared randomness (Thm. 4.4); for exact recovery of $\nu$, frozen labels and adaptive inputs (Props. 6.10 and D.5) | structured rules (a second kink); cheap frequent rules (Props. 9.1 and 9.2) |
| What the learner is | for fixed $\lambda$, one Bayesian predictor $\xi_\lambda$ | a family of batch objectives, not a fixed prior (Rem. 3.7 and Prop. F.6) |

Table 4: The two reasons for a steeper complexity penalty, side by side.

**What counts as one hypothesis.** Which kinks exist depends on the class: a universal member flattens every kink far beyond its own length (Proposition 5.7), and the full table of a noisy random parity can have a steep product kink at the parity (Proposition 7.7) but has no steep set kink (Proposition 7.10). So a steeper prior is only as good as the class's notion of one hypothesis; ours is the private-randomness class, with the price and the boundary of Remark 4.6.

**The window and its price.** The lower end $2^{-m}$ of the window of Theorem 8.7 is the reuse factor: a bit about the error rule costs $c$ per sample in length and saves about $2^{-m}$ per sample in risk, so $\lambda$ must grow linearly with $n$. A fixed weight-decay coefficient on an averaged loss has the same form, a Gaussian prior whose precision grows linearly in $n$, and it too can keep a cheaper, worse classifier in the limit (Remark 8.5). The learner with $\lambda = cn$ is deliberately inconsistent for the teacher's labels, at a price (Remark 8.4) that imitation data cannot justify, because strictly proper scores prefer the teacher's channel, errors included (Proposition G.3).

**The gap to realistic errors.** Human labelling errors are often cheap, frequent patterns rather than incompressible sets. In the parity instance steeper selection provably fails for such errors in two ways: a fixed cheap rule defeats every $c$ once the simple model is long enough (Proposition 9.2 and Lemma G.1), and a rule made of a cheap region plus random points, with the original density and description length, makes the good model non-optimal on about a third (with the machine constant set to 0, an illustration) to a quarter ($\kappa = 30$) of the window's logarithmic width (Proposition 9.1). Whether any window survives for structured errors in general is open. Steeper

simplicity sorts components by rate, not by validity (Proposition 9.3), so by itself it does not tell errors from rare valid rules. We do not expect steeper penalisation alone to solve the general problem of learning a norm from imperfect imitation; it works in one toy case and fails in others.

**What the toy cases leave open.** The results concern idealised learners in toy worlds; the problems that motivate them remain far from solved. Six gaps are concrete. (1) Data identify behaviour, not mechanism: the selected program is a shortest program in the class for the learned kernel, which may compute it unlike the person and extrapolates off the support of the inputs as that program does (Section 10). (2) Which structure belongs to the person and which to the environment, and with it the complexity weight, are conventions that imitation data cannot settle (Proposition G.3, above): worlds with one data law cannot all be copied reliably by any procedure (Proposition 10.3(a)), and when a person's apparent randomness is its own deterministic state, returning the person with fresh coins is a modelling choice. (3) Realistic errors and realistic randomness have structure: like the cheap error rules above, cheap couplings defeat every $c$, and in a toy world no procedure reliably copies a person entangled with its coupling (Propositions 10.3 and H.4). (4) Exact copies need the same question answered afresh; with frozen labels no procedure recovers every sampler (Proposition 6.10). (5) People have memory, and excluding hypotheses that learn in context, which absorb what they imitate, is a choice of class (Theorem 4.4, Proposition B.6, and Section 10). (6) The steeper predictors are not computable, and whether the $\lambda$-MAPs are is open (Proposition I.1 and Question 11.4); the margin form of Theorem 6.3 gives no computable sample size in $\mathcal{H}_\times$ (Proposition D.2); and which program counts as the shortest copy, and where every window lies, depend on the hypothesis class and the reference machine (Remark 8.8). We see the results as tools for these problems, with toy cases in which the tools work, not as solutions.

**Neural networks.** For a network whose labels are independent given its parameters, Theorems 6.3 and 6.9 apply to countable discretisations with integer code lengths and Kraft weights whose noise model contains the sampler; whether a parameter prior behaves like a description-length prior, as Theorem 8.7 would need, is the bridge question of the companion paper (Hänni, 2026a, Question 10.9), which the slides of Vaintrob and Hänni (2026) motivate (Remark 11.3).

### 12.1 Open problems

(1) *Recovery.* A computable recovery-time bound in the canonical class; local margins; the joint necessity of $1/(\min(1, \lambda - 1)\gamma)$; frozen labels with repeated inputs; misspecification in countable classes (Question 6.13).

(2) *Set kinks.* Does a product kink at a model with a random parameter always come with a near-sufficient set model of about the complexity of the model without its parameter (Question 7.13)? Does Proposition E.3 hold for lower-semicomputable products (Question E.4)?

(3) *The teacher's mistakes.* Other model classes (we expect none from majority-like functions); a junta window at level at most 2 for $p_1 \leq p < \frac{1}{4}$; near-population constants far below $10^{11}$ samples; unequal cell multiplicities; schedules that leave the window (Question 8.14); non-uniform input laws (separable case only, Proposition G.4).

(4) *Structured error rules.* Envelopes whose cheap region has error density below one, what is selected above $c_{\mathrm{env}}$, and a choice of $c$ that does not use the imitation data (Question 9.5).

(5) *Shared hypotheses.* Is every lower-semicomputable law of a random partial function dominated by $M_F$ up to a constant (Question 4.7)?

(6) *Resource bounds.* Does a time-bounded shared class contain a short member that absorbs everything at $\lambda > 1$? Is there a computable map from data sets to $\lambda$-MAPs (Question 11.4)?

(7) *Imitation* (Section 10). A near-imitation theorem when the coin combines $\beta(w)$ with a private residual coin, or when the cells have unequal masses; Theorem 10.2 for $cn$-MAPs at each finite $n$ with probability $1 - \delta$, for every response function; whether a coupling that is short to describe but slow to compute is private randomness for a time-bounded class; and whether data other than imitation data, from several environments or from interventions, identify which structure belongs to the person.

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## Code and data

Every numerical statement comes from a deterministic or seeded Python script (Python 3.11 with NumPy and SciPy) whose output is saved next to it. The scripts, their outputs and independent re-implementations of the main computations are in the folder `solomonoff-function-induction/` of the repository `https://github.com/kaarelh/AI-works`; its README lists the script behind each number.

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# A Proofs for Section 3 (setting)

## A.1 Standard facts

The following facts are standard (Li and Vitányi, 2008); $\kappa$ is as in Section 3.2.

(F1) $K(\pi) \le \ell(\pi) + \kappa$ for every program $\pi$ of a class $\mathcal{H}$. The identity map on the c.e. prefix-free set $\mathcal{H}$ is a prefix machine, which the universal machine simulates with a fixed prefix; for canonical classes, $\mathcal{H} = \operatorname{dom} V$ and the fact is immediate.

(F2) *Conditional coding theorem.* If a fixed program lower-semicomputes from $z$ a semimeasure $Q_z$ on a countable set, then $K(w \mid z) \le -\log Q_z(w) + \kappa$.

(F3) $K(x) \le K(y) + K(x \mid y) + \kappa$, and $K(x, y \mid z) \le K(x \mid z) + K(y \mid x, z) + \kappa$.

(F4) *Symmetry of information.* $K(x, y) = K(x) + K(y \mid x, K(x)) + O(1)$.

(F5) $K(j) \le 2\log(j+1) + \kappa$ for every integer $j \ge 0$, by a doubling code.

## A.2 Basic properties of the objectives

**Lemma A.1** (Basic properties)**.** *(i) If $J_\lambda(\pi; o) < \infty$ for some $\pi$, the set of $\lambda$-MAPs is finite and nonempty. (ii)* $\arg\min_\pi J_\lambda(\pi; o) = \arg\max_\pi \Pi_\lambda(\pi \mid o) = \arg\max_\pi \Pi^{1/\lambda}(\pi \mid o)$. *(iii) On deterministic hypotheses $P_\pi(D)^\eta = P_\pi(D) \in \{0, 1\}$, so on any set of deterministic hypotheses the tempered posterior has the relative weights of the plain ($\lambda = 1$) posterior.*

*Proof.* (i) $J_\lambda \ge \lambda\ell$, and only finitely many programs have $\ell(\pi) \le J_\lambda(\pi_0; o)/\lambda$. (ii) $2^{-J_\lambda} = (2^{-\ell} P_\pi^{1/\lambda})^\lambda$. □

## A.3 Proof of the hull lemma

We prove Lemma 3.8. Write $\alpha_0 := \min\{\ell(\pi) : L(\pi) < \infty\}$, and call a program *finite* if $L(\pi) < \infty$. All fits are nonnegative.

*Step 1: the vertices, by gift wrapping.* Put $\rho_0 := \min\{L(\pi) : \ell(\pi) = \alpha_0\}$ and $v_0 := (\alpha_0, \rho_0)$. Given $v_i = (\alpha_i, \rho_i)$, let $\sigma_{i+1}$ be the supremum of the slopes $(\rho_i - L(\pi))/(\ell(\pi) - \alpha_i)$ over finite $\pi$ with $\ell(\pi) > \alpha_i$. If there is no such $\pi$ or $\sigma_{i+1} \le 0$, stop, and put $q := i$. Otherwise let $\alpha_{i+1}$ be the largest length of a program attaining the supremum, and $\rho_{i+1} := \rho_i - \sigma_{i+1}(\alpha_{i+1} - \alpha_i)$.

- *The suprema are attained.* If some $\pi$ has slope $s > 0$ from $v_i$, every $\pi'$ with slope at least $s$ has $0 \le L(\pi') \le \rho_i - s(\ell(\pi') - \alpha_i)$, so $\ell(\pi') \le \alpha_i + \rho_i/s$; only finitely many programs qualify.
- $\rho_i = \min\{L(\pi) : \ell(\pi) \le \alpha_i\}$, *attained at length exactly* $\alpha_i$. By induction: a program with $\alpha_i < \ell(\pi) \le \alpha_{i+1}$ has slope at most $\sigma_{i+1}$ from $v_i$, so $L(\pi) \ge \rho_i - \sigma_{i+1}(\ell(\pi) - \alpha_i) \ge \rho_{i+1}$, and the maximising program at length $\alpha_{i+1}$ has fit exactly $\rho_{i+1}$.
- $\sigma_1 > \sigma_2 > \cdots > 0$. If $\sigma_{i+1} \ge \sigma_i$ for some $i \ge 1$, the slope of $v_{i+1}$ seen from $v_{i-1}$ is a weighted mean of $\sigma_i$ and $\sigma_{i+1}$, hence at least $\sigma_i$: strict inequality contradicts the definition of $\sigma_i$, and equality the maximality of $\alpha_i$.
- *Infinitely many vertices.* The $\alpha_i$ are strictly increasing integers, and $\sum_i \sigma_i(\alpha_i - \alpha_{i-1}) = \rho_0 - \lim_i \rho_i \le \rho_0$ with every $\alpha_i - \alpha_{i-1} \ge 1$; so $\alpha_i \to \infty$ and $\sigma_i \to 0$.

*Step 2: the boundary.* Let $P$ be linear between consecutive vertices, equal to $\rho_q$ on $[\alpha_q, \infty)$ if $q < \infty$, and defined on $[\alpha_0, \infty)$ (if there are infinitely many vertices, $\alpha_i \to \infty$). Its slopes $-\sigma_1 < -\sigma_2 < \cdots \le 0$ increase, so $P$ is convex and nonincreasing. Every finite point lies on or above $P$: if $\ell(\pi) \in (\alpha_i, \alpha_{i+1}]$ its slope from $v_i$ is at most $\sigma_{i+1}$; if $\ell(\pi) > \alpha_q$ its slope from $v_q$ is at most 0; if $\ell(\pi) = \alpha_0$, $L(\pi) \ge \rho_0$. So $C$ lies in the epigraph $\{(\alpha, y) : \alpha \ge \alpha_0,\ y \ge P(\alpha)\}$, which is convex and closed under adding $\mathbb{R}^2_{\ge 0}$. Conversely, $(\alpha, P(\alpha))$ is a convex combination of two vertices for $\alpha \in [\alpha_{i-1}, \alpha_i]$, and equals $v_q + (\alpha - \alpha_q, 0)$ for $\alpha \ge \alpha_q$. Hence $C$ is that epigraph and $\underline{C} = P$. Its boundary consists of the vertical ray above $v_0$, the edges $[v_{i-1}, v_i]$ of slope $-\sigma_i$, and, if $q < \infty$, the horizontal ray to the right of $v_q$, the least-$L$ vertex. Its extreme points are exactly the $v_i$, and each is the point of a program of length $\alpha_i$. This proves the first part of (v).

Below, $\langle (c, 1), (\ell, L) \rangle = c\,\ell + L$.

*(i).* Since $(c, 1) \ge 0$, the infimum of $c\,\ell + L$ over $C$ equals its infimum over $\mathcal{A}$. A minimiser therefore minimises a nonzero linear functional over $C$, so the line $\{c\,\ell + L = J_c(\pi)\}$ supports $C$ at its point, which lies on the boundary.

*(ii).* Add $J_c(\pi) \le J_c(\pi')$ and $J_{c'}(\pi') \le J_{c'}(\pi)$ to get $(c' - c)(\ell(\pi') - \ell(\pi)) \le 0$, so $\ell(\pi') \le \ell(\pi)$. Then the first inequality gives $L(\pi) \le L(\pi') + c(\ell(\pi') - \ell(\pi)) \le L(\pi')$.

*(iii).* The two boundary pieces at $v_i$ have slopes $-\sigma_i < -\sigma_{i+1}$. For $c \in (\sigma_{i+1}, \sigma_i)$ the line with normal $(c, 1)$ through $v_i$ meets $C$ only at $v_i$, so $v_i$ is the unique minimising point. Conversely, a point that is the unique minimising point for every $c$ in an open interval is an exposed point of $C$, hence a vertex. A boundary point that is not a vertex lies in the relative interior of an edge, and then both endpoints of the edge minimise whenever it does; or on the vertical ray, where the vertex below it is better; or on the horizontal ray, where the least-$L$ vertex is better. The statement about programs is immediate.

*(iv).* If $\ell(\pi) \le \ell(\pi_c)$, then $L(\pi) \ge L(\pi_c) + c(\ell(\pi_c) - \ell(\pi)) \ge L(\pi_c)$. For the converse: a vertex minimises $J_c$ for $c \in (\sigma_{i+1}, \sigma_i)$, and a point inside an edge of slope $-\sigma_i$ minimises $J_{\sigma_i}$. A point on the open horizontal ray is beaten, for every $c > 0$, by the least-$L$ vertex. A point on the vertical ray above $v_0$ is never a constrained solution, since $v_0$ is feasible whenever it is and fits better. A point off the boundary minimises no $J_c$, by (i).

*(v), the minimisers.* For every finite $\pi$, $J_c(\pi) \ge c\,\ell(\pi) + P(\ell(\pi)) \ge m_c := \inf_{\alpha \ge \alpha_0}(c\,\alpha + P(\alpha))$. The function $\varphi(\alpha) := c\,\alpha + P(\alpha)$ is piecewise linear with slope $c - \sigma_i$ on $(\alpha_{i-1}, \alpha_i)$ and $c$ beyond $\alpha_q$, increasing in $i$. Let $i^* := \min\{i : \sigma_{i+1} \le c\}$, which exists because $\sigma_i \to 0 < c$ or $\sigma_{q+1} = 0$. Then $\varphi$ is nonincreasing up to $\alpha_{i^*}$ and nondecreasing after it, so $m_c = c\,\alpha_{i^*} + \rho_{i^*}$ is attained by

the program at $v_{i^*}$. Hence $\pi$ is a minimiser if and only if $L(\pi) = P(\ell(\pi))$ and $\ell(\pi)$ minimises $\varphi$; the minimising set of $\varphi$ is $\{\alpha_{i^*}\}$ if $\sigma_{i^*+1} < c$, and $[\alpha_{i^*}, \alpha_{i^*+1}]$ if $c = \sigma_{i^*+1}$. This is the description in (v). The set is finite because its lengths are bounded and there are finitely many programs of each length.

*(vi).* By (v), $\Phi(c) = \min_i(c\,\alpha_i + \rho_i)$, a minimum of affine functions, hence concave; on $(\sigma_{i+1}, \sigma_i)$ it equals $c\,\alpha_i + \rho_i$, with derivative $\alpha_i$. For $\alpha \geq \alpha_0$, $\Phi(c) \leq c\,\alpha + P(\alpha)$ for every $c$, so $\sup_c(\Phi(c) - c\,\alpha) \leq P(\alpha)$. Conversely, for $\alpha \in [\alpha_{i-1}, \alpha_i]$ take $c = \sigma_i$, at which every point of the edge minimises, so $\Phi(\sigma_i) = \sigma_i\alpha + P(\alpha)$; for $\alpha \geq \alpha_q$, $\Phi(c) - c\,\alpha = \rho_q - c(\alpha - \alpha_q) \to \rho_q = P(\alpha)$ as $c \downarrow 0$. This is the Legendre–Fenchel duality of convex analysis. □

*Remark* A.2 (The converse in (iv)). The converse in (iv) cannot be weakened to "on the boundary of $C$": in $\mathcal{A} = \{(0,1), (10,0), (20,0)\}$ the point $(20,0)$ lies on the open horizontal ray to the right of the least-$L$ vertex $(10,0)$ and solves $\min\{L : \ell \leq \bar{\ell}\}$ for $\bar{\ell} \geq 20$, since it ties with $(10,0)$ there, but $(10,0)$ is strictly better for every $c > 0$.

*Examples.* In $\{(0,1), (10,0.5), (20,0)\}$ the middle point lies on the segment between the other two, so it is never the unique minimiser; the end points are the unique minimisers on $(0.05, \infty)$ and on $(0, 0.05)$. In $\{(0,1), (5,0.9), (10,0)\}$ the point $(5,0.9)$ lies above the boundary of $C$; it solves the constrained problem for $\bar{\ell} \in [5,10)$ and minimises no penalised objective. A direct computation (code in the accompanying repository) confirms (iii), (v) and (vi) on random point sets with duplicate and collinear points.

### A.4 Proof of the coding bound

We prove Lemma 3.9. Given $\pi$ and the finite set $\operatorname{dom} D$, the map $y \mapsto P_\pi(\operatorname{dom} D \mapsto y)$ on $Y^{\operatorname{dom} D}$ is lower semicomputable, uniformly in $(\pi, \operatorname{dom} D)$, because the class is (Definition 3.5). It is a semimeasure. For a product hypothesis its total mass is $\prod_{x \in \operatorname{dom} D} \sum_y \nu_\pi(y \mid x) \leq 1$. For a shared hypothesis, repeated use of the one-point inequality gives $\sum_y P_\pi(\operatorname{dom} D \mapsto y) \leq P_\pi(\emptyset) \leq 1$. By (F2), applied to one fixed program for all $(\pi, \operatorname{dom} D)$, $K(y_D \mid \pi, \operatorname{dom} D) \leq L(\pi; D) + \kappa$. By (F3) and (F1),

$$K(y_D \mid \operatorname{dom} D) \leq K(\pi) + K(y_D \mid \pi, \operatorname{dom} D) + \kappa \leq \ell(\pi) + L(\pi; D) + \kappa.$$

□

## B Proofs for Section 4 (absorption)

### B.1 Classes with an absorbing member

We prove Remark 4.1. Suppose first that the mixture $\xi_1 = \sum_\pi 2^{-\ell(\pi)} P_\pi$ is itself a member, $\xi_1 = P_{\pi_U}$. Then $P_\pi \leq 2^{\ell(\pi)}\xi_1 = 2^{\ell(\pi)}U$ for every $\pi$, which is (Dom) with $\kappa_0 = 0$, and $\xi_1 = U$. This covers $\mathcal{H}_w$ ($M^w = \sum_i \mathbf{m}(i)\nu_i$ is an lsc weak function semimeasure (Hänni, 2026a, Theorem 3.11), hence occurs in the trimmed enumeration), $\mathcal{H}_{\mathrm{SI}}$ ($M$ is an lsc semimeasure) and $\mathcal{H}_{\mathrm{chr}}$ ($M_H$ is an lsc chronological semimeasure (Hänni, 2026a, Corollary 6.17) and, by Hänni (2026a, Theorem 6.16), the law of a history-oracle function machine at every nonempty history; at the empty history both sides of (Dom) are at most 1 and $P_{\pi_U} = 1$). For $\mathcal{H}_F$: $M_F$ is a machine law (Hänni, 2026a, Theorem 3.8), so it has an index; let $\pi_U$ be a shortest program for it. For $D \neq \emptyset$, $P_\pi(D) = \nu_{T_{V(\pi)}}(D) \leq M_F(D)/\mathbf{m}(V(\pi)) \leq 2^{\ell(\pi)} M_F(D)$, and at $\emptyset$ both sides equal 1. Here $\xi_1 = M_F$ on nonempty data sets and $\xi_1(\emptyset) \leq 1 = M_F(\emptyset)$, so $\xi_1 \leq U$. □

## B.2 Proof of the absorption theorem

We prove Theorem 4.2. Throughout, (Dom) holds and $\lambda > 1$.

*(a).* Let $\hat{\pi}$ be a $\lambda$-MAP for $o$. Since $U(o) > 0$, $J_\lambda(\hat{\pi}; o) \le J_\lambda(\pi_U; o) = \lambda \ell_U - \log U(o) < \infty$, so $L(\hat{\pi}; o) < \infty$. By (Dom), $-\log U(o) \le \ell(\hat{\pi}) + \kappa_0 + L(\hat{\pi}; o)$. Substituting, $\lambda \ell(\hat{\pi}) + L(\hat{\pi}; o) \le \lambda \ell_U + \ell(\hat{\pi}) + \kappa_0 + L(\hat{\pi}; o)$, so $(\lambda - 1)\ell(\hat{\pi}) \le \lambda \ell_U + \kappa_0$, the length bound. The fit bound is $J_\lambda(\hat{\pi}; o) \le J_\lambda(\pi_U; o)$ rearranged. Finally, $\ell^*_\lambda < \bar{\ell}$ if and only if $\lambda(\bar{\ell} - \ell_U) > \bar{\ell} + \kappa_0$.

*(b).* The first inequality keeps the single term of $\pi_U$ in $\xi_\lambda(o) = \sum_\pi 2^{-\lambda \ell(\pi)} P_\pi(o)$; the second holds because $2^{-\lambda \ell} \le 2^{-\ell}$. If $\xi_1 \le 2^{\kappa_1} U$, then $-\log \xi_\lambda(o) \ge -\log \xi_1(o) \ge -\log U(o) - \kappa_1$, and the first inequality gives $-\log \xi_\lambda(o) \le \lambda \ell_U - \log U(o)$. For a data-set class, $\xi_\lambda$ is a weak function semimeasure with $\xi_\lambda(\emptyset) \le \sum_\pi 2^{-\lambda \ell(\pi)} \le 1$, because each $P_\pi$ is one and the weak inequalities are linear; so its sequential log loss on $D_n$ is $-\log \xi_\lambda(D_n) + \log \xi_\lambda(\emptyset)$ in every order (Definition 3.4).

*(c1).* Since $\pi_t$ is deterministic and consistent with $D_n$, $P_{\pi_t}(D_n) = 1$, so $\xi_\lambda(D_n) \ge 2^{-\lambda \ell_t}$. Also, by (Dom), $U(D_n) \ge 2^{-\ell_t - \kappa_0}$, so $\xi_\lambda(D_n) \ge 2^{-\lambda \ell_U} U(D_n) \ge 2^{-\lambda \ell_U - \kappa_0 - \ell_t}$. By telescoping, $\sum_{i \le n} -\log \xi_\lambda(t(x_i) \mid x_i, D_{i-1}) \le -\log \xi_\lambda(D_n)$. Each conditional is at most 1 by the one-point inequality, so each term is nonnegative, and a label with probability below $1/2$ adds more than one bit.

*(c2).* Combine (a) with $-\log U(D_n) \le \ell_t + \kappa_0$.

*(c3).* The set $\mathcal{H}_{\le} := \{\pi : \ell(\pi) \le \ell^*_\lambda\}$ is finite, and by (a) it contains every $\lambda$-MAP for every $D_n$. For $\pi \in \mathcal{H}_{\le}$, $L(\pi; D_n)$ is nondecreasing in $n$, because weak function semimeasures are monotone; let $L(\pi; D_\infty) \in [0, \infty]$ be its limit, and $\mathcal{H}_{\text{fin}} := \{\pi \in \mathcal{H}_{\le} : L(\pi; D_\infty) < \infty\}$. It contains $\pi_U$, because $\ell_U \le \ell^*_\lambda$ and $L(\pi_U; D_\infty) \le \ell_t + \kappa_0$. For $\pi \in \mathcal{H}_{\le} \setminus \mathcal{H}_{\text{fin}}$, $J_\lambda(\pi; D_n) \to \infty$, while $J_\lambda(\pi_U; D_n) \le \lambda \ell_U + \ell_t + \kappa_0$ for every $n$; as there are finitely many such $\pi$, none of them is a $\lambda$-MAP beyond some $n_0$. For $\pi \in \mathcal{H}_{\text{fin}}$, the increments $L(\pi; D_{n+1}) - L(\pi; D_n)$, the losses of $\pi$'s predictions of the next label, sum to at most $L(\pi; D_\infty)$, so $\pi$ gives the next label probability below $1/2$ fewer than $L(\pi; D_\infty) + 1$ times. The union over the finite set $\mathcal{H}_{\text{fin}}$ bounds the number of $n \ge n_0$ at which some $\lambda$-MAP does so.

*(d).* By (Dom), $\xi_\lambda \ge 2^{-\lambda \ell_U} U \ge 2^{-\lambda \ell_U - \kappa_0 - \ell(\pi_{\mathbb{P}})} \mathbb{P}$ on every string. Hence for every $\bar{n}$, with $\omega_{1:\bar{n}}$ drawn from $\mathbb{P}$, $\mathbb{E}_{\mathbb{P}} \log \left(\mathbb{P}(\omega_{1:\bar{n}}) / \xi_\lambda(\omega_{1:\bar{n}})\right) \le \lambda \ell_U + \kappa_0 + \ell(\pi_{\mathbb{P}})$. With $\xi_\lambda(b \mid h) := \xi_\lambda(hb)/\xi_\lambda(h)$, the chain rule over bit positions gives $\log(\mathbb{P}/\xi_\lambda)(\omega_{1:\bar{n}}) = \sum_{j \le \bar{n}} \log \left(\mathbb{P}(\omega_j \mid \omega_{<j}) / \xi_\lambda(\omega_j \mid \omega_{<j})\right) - \log \xi_\lambda(\epsilon)$, and $-\log \xi_\lambda(\epsilon) \ge 0$. At each position the conditional expectation of the summand is $\mathrm{KL}\left(\mathbb{P}(\cdot \mid h) \,\|\, \xi_\lambda(\cdot \mid h)\right) \ge -\log \sum_b \xi_\lambda(b \mid h) \ge 0$, since $\xi_\lambda(\cdot \mid h)$ is a sub-probability. With fixed-length blocks the label positions are fixed, and there $\mathbb{P}(\cdot \mid h)$ is a point mass, because $y_i = t(x_i)$; so the summand is $-\log \xi_\lambda(\omega_j \mid \omega_{<j})$, and these sum over one label block to $-\log \xi_\lambda(y_i \mid x_1 y_1 \cdots x_i)$. Dropping the nonnegative terms of the input positions gives the claim, which is Solomonoff's bound (Solomonoff, 1978) restricted to the label positions. □

## B.3 Proof of the shared-randomness theorem

We prove Theorem 4.4. (i) Since $U(o) > 0$, $J_\lambda(\pi_U; o) < \infty$. By (Dom), $-\log U(o) \le k + \kappa_0 + L(\pi_\nu; o)$, so $J_\lambda(\pi_U; o) \le \lambda \ell_U + \kappa_0 + k + L(\pi_\nu; o)$, which is below $J_\lambda(\pi_\nu; o) = \lambda k + L(\pi_\nu; o)$ exactly when $(\lambda - 1)k > \lambda \ell_U + \kappa_0$; the same condition says $\ell^*_\lambda < k$, and every $\lambda$-MAP has length at most $\ell^*_\lambda$ by Theorem 4.2(a). (ii) $\sum_{\pi \in A} 2^{-\lambda \ell(\pi)} P_\pi(o) \le 2^{-(\lambda - 1) \min_A \ell} \xi_1(o) \le 2^{-(\lambda - 1) \min_A \ell + \kappa_1} U(o)$, and the normaliser $\xi_\lambda(o)$ is at least $2^{-\lambda \ell_U} U(o)$. For SI with adaptive or non-computable inputs the sampler is not an SI hypothesis; in $\mathcal{H}_{\text{chr}}$ it is a chronological semimeasure, and $M_H$ dominates every lsc chronological semimeasure on every input sequence (Hänni, 2026a, Corollary 6.17(a)), so (Dom) holds there with $U = M_H$. □

The product and the shared predictor both still predict well: for fixed $\lambda$ they dominate $\nu$ ($\xi_\lambda^\times \geq 2^{-\lambda k}\nu$; $\xi_\lambda^F \geq 2^{-\lambda\ell_U-\kappa_0-k}\nu$), so by the argument of Hänni (2026a, Theorem 5.3) their summed expected KL divergences to $\nu$'s predictions are at most $\lambda k$ and $\lambda\ell_U+\kappa_0+k$.

**Example B.1** (Noisy parity, full table: an illustration)**.** Let $d = 12$ and let the data be the full table ($N = 4096$ inputs) of a random parity with i.i.d. flips at rate 1/8 (509 flips). A small product class contains the coin, parity-plus-noise samplers ($\ell = 24.2$ bits) and the memoriser (parity plus flip set); a "shared class" adds the finite Bayes mixture $U$ over all parities and a grid of noise rates, with an *assigned* length $\ell_U = 12$. A direct computation (code in the accompanying repository) gives: the product-class MAP is the sampler at $\lambda = 1, 1.1, 1.5, 2, 3, 6$, and at $\lambda = 1$ the memoriser is only 8.2 bits worse, a tie up to $O(\log N)$; the shared-class MAP is the sampler for $\lambda < 1.315$ and $U$ above, the threshold being $(-\log U(D) + \log\nu(D))/(k-\ell_U)$ (the rounded values 2234.0, 2218.0, 24.2, 12 give 1.311). Since $\ell_U = 12$ is assigned and this finite class does not satisfy (Dom), the example illustrates the mechanism of Theorem 4.4; it is not an instance of it, and its numbers are not values for any reference machine.

## B.4 Per-input pricing

For a finite set $F$ of total functions $X \to Y$ and a distribution $Q$ on $F$, let $Q_x$ be the law of $f(x)$ for $f \sim Q$.

**Proposition B.2** (Per-input pricing)**.** *Let $x_1, \ldots, x_n$ be distinct.*

*(i) For every product hypothesis $\nu$ and every $Q$, $\max_{f\in F}\sum_{i\leq n} -\log\nu(f(x_i) \mid x_i) \geq \sum_{i\leq n} H(Q_{x_i})$; the value $\inf_\nu \max_f \sum_i -\log\nu(f(x_i) \mid x_i)$ is $\max_Q \sum_i H(Q_{x_i})$, which can be strictly below $\sum_i \log|\{f(x_i) : f \in F\}|$.*

*(ii) If $F$ is the set of parities on $\{0,1\}^d$ and the $x_i$ are nonzero, every product hypothesis pays at least $n$ bits on some parity, while the Bayes mixture over parities, a shared hypothesis, pays at most $d$ bits on every parity.*

*(iii) A per-input mixture $\nu(y \mid x) = \sum_q w_q\, q(y \mid x)$ is a product hypothesis and pays up to $\log(1/w_q)$ at every input: with the two constant functions and $w = 1/2$ it pays $n$ bits on $n$ ones, where the shared mixture pays 1 bit.*

*Proof.* The bound in (i) is Gibbs' inequality applied to $\mathbb{E}_{f\sim Q}$, and the value follows from Sion's minimax theorem (Sion, 1958).

*(i), the bound.* $\max_f \geq \mathbb{E}_{f\sim Q}$, and for each $i$

$$\mathbb{E}_{f\sim Q}\big[-\log\nu(f(x_i) \mid x_i)\big] = H(Q_{x_i}) + \sum_y Q_{x_i}(y)\log\frac{Q_{x_i}(y)}{\nu(y \mid x_i)},$$

where the last sum is at least $-\log\sum_y \nu(y \mid x_i) \geq 0$, by Gibbs' inequality with a sub-probability.

*(i), the value.* For $\varepsilon \in (0, 1/|Y|)$ let $\Delta_\varepsilon$ be the compact convex set of conditional sub-probabilities on $\{x_1, \ldots, x_n\}$ with all values at least $\varepsilon$. The loss $\mathcal{L}(\nu, Q) := \mathbb{E}_{f\sim Q}\sum_i -\log\nu(f(x_i) \mid x_i)$ is continuous on $\Delta_\varepsilon \times \Delta(F)$, convex in $\nu$ and linear in $Q$, so by Sion's minimax theorem (Sion, 1958) $\min_{\Delta_\varepsilon}\max_Q \mathcal{L} = \max_Q \min_{\Delta_\varepsilon} \mathcal{L}$, and $\max_Q \mathcal{L}(\nu, Q) = \max_{f\in F}\sum_i -\log\nu(f(x_i) \mid x_i)$. For each $Q$, $\min_{\Delta_\varepsilon}\mathcal{L}(\cdot, Q)$ lies between $\sum_i H(Q_{x_i})$ and $\sum_i H(Q_{x_i}) - n\log(1-|Y|\varepsilon)$, the latter attained by $\nu = (1-|Y|\varepsilon)Q_x + \varepsilon$; the error term is uniform in $Q$. Rational conditional probabilities are computable and dense in $\Delta_\varepsilon$, so the infimum over product hypotheses equals that over $\Delta_\varepsilon$. Hence $\max_Q \sum_i H(Q_{x_i}) \leq \inf_\nu \max_f \sum_i -\log\nu(f(x_i) \mid x_i) \leq \max_Q \sum_i H(Q_{x_i}) - n\log(1-|Y|\varepsilon)$;

let $\varepsilon \to 0$. For $F = \{(a,a),(a,b),(b,a),(c,a)\}$ a direct computation (code in the accompanying repository) gives $\max_Q \sum_i H(Q_{x_i}) = 2.5431$, and the same value for $\min_\nu \max_f$ by constrained optimisation over $\nu$, against $\log 3 + \log 2 = 2.5850$.

*(ii).* For $x \neq 0$ exactly half of the parities give each label at $x$, so the uniform $Q$ has $H(Q_x) = 1$, and (i) gives a parity with loss at least $n$. The Bayes mixture with weight $2^{-d}$ on each parity gives every parity's data probability at least $2^{-d}$.

*(iii).* Each term of the per-input mixture assigns $1/2$ to the observed label 1, at every input; the shared mixture gives the constant-1 function weight $1/2$. □

*Remark* B.3 (The universal conditional semimeasure never learns). The universal lsc conditional semimeasure satisfies $\mathbf{m}(y \mid x) \geq 2^{-K(y|x)-\kappa}$. For $Y = \{0,1\}$ this gives a constant $\kappa_{\mathbf{m}}$ with $\mathbf{m}(y \mid x) \geq 2^{-\kappa_{\mathbf{m}}}$, hence $\mathbf{m}(y \mid x) \leq 1 - 2^{-\kappa_{\mathbf{m}}}$, for both labels, so it pays at least $\log \frac{1}{1-2^{-\kappa_{\mathbf{m}}}} > 0$ bits for every label on every data set. It dominates each product hypothesis input by input, $\mathbf{m} \geq 2^{-K(\nu)-\kappa}\nu$, but pays the factor $2^{K(\nu)}$ again at every input; its prediction does not depend on the data.

## B.5 Why private randomness, and its price

*Remark* B.4 (Why private randomness is what matters). At $\lambda = 1$ the decomposition into hypotheses does not change the predictor ($\xi_\times = M_{\text{prod}}$ is one shared hypothesis); at $\lambda > 1$ it does. Over the class $\mathcal{H}_F$ of all machine laws, $M_F$ would be a short member and absorption would return, so steeper SFI is defined over $\mathcal{H}_\times$, whose $\lambda \to 1$ limit is $M_{\text{prod}}$, not $M_F$. At $\lambda = 1$ this costs at most $K(\operatorname{dom} D) + O(1)$ bits against $M_F$, but not a constant: $M_F/M_{\text{prod}}$ is unbounded already against the law $\nu_{\text{len}}$ of $F(x) = \omega_{|x|+1}$, one shared fair coin per input length, although on inputs of length at most $d$ the gap to $\nu_{\text{len}}$ is at most $K^{\text{pad}}(d+1) + O(1)$ bits, $K^{\text{pad}}(N) := \min_{L \geq N} K(L)$ (Hänni, 2026a, Theorem 4.4, Proposition 4.7 and Remark 4.8). At $\lambda > 1$ the cost is a constant factor on shared latent causes (Proposition B.5).

For $s \geq 1$ and $\beta \in \{0,1\}^s$ let $D_\beta := \{(0^i, \beta_i), (1^i, \beta_i) : i \leq s\}$: one fair coin per pair of inputs, shown on both.

**Proposition B.5** (The price of private randomness)**.** *Let $Y = \{0,1\}$. (i) For every $\lambda \geq 1$ and $\beta$: $-\log \xi^\times_\lambda(D_\beta) \geq \min(\lambda, 2)\,(K(\beta \mid s) - \kappa)$. (ii) Hence $J_\lambda(\pi; D_\beta) \geq \min(\lambda,2)\big(K(\beta \mid s) - \kappa\big)$ for every $\pi \in \mathcal{H}_\times$, and in every product class with a prefix-free code, with $\kappa$ depending on the class. (iii) $\nu_{\text{len}}$ is a machine law of complexity $O(1)$ with $\nu_{\text{len}}(D_\beta) = 2^{-s}$, so $-\log M_F(D_\beta) \leq s + \kappa$ and $-\log \xi^F_\lambda(D_\beta) \leq s + \lambda\kappa$. (iv) For uniform $\beta$ and $j \geq 0$, with probability at least $1 - 2^{-j}$ the steeper product predictor pays at least (in the semimeasure convention) $(\min(\lambda,2) - 1)s - 2j - (\lambda+2)\kappa$ bits more than $M_F$ and than $\xi^F_\lambda$.*

*Proof.* Write $a_i := 0^i$, $b_i := 1^i$ and $\bar\lambda := \min(\lambda, 2)$.

*(i).* Fix $\pi \in \mathcal{H}_\times$ with conditional semimeasure $\nu = \nu_\pi$. Then $\nu(D_\beta) = \prod_{i \leq s} \nu(\beta_i \mid a_i)\,\nu(\beta_i \mid b_i) \leq Q_\pi(\beta)^2$ with $Q_\pi(\beta) := \prod_i q_i(\beta_i)$ and $q_i(y) := \frac{1}{2}(\nu(y \mid a_i) + \nu(y \mid b_i))$, by the inequality of arithmetic and geometric means. Each $q_i$ is a sub-probability on $\{0,1\}$, so $Q_\pi$ is a semimeasure on $\{0,1\}^s$, lower semicomputable uniformly in $(\pi, s)$, and so is $\bar Q(\beta) := \sum_{\pi \in \mathcal{H}_\times} 2^{-\ell(\pi)} Q_\pi(\beta)$; by (F2), $\bar Q(\beta) \leq 2^{-K(\beta|s)+\kappa}$. If $1 \leq \lambda \leq 2$, then $2^{-\lambda\ell(\pi)} Q_\pi^2 = (2^{-\ell(\pi)} Q_\pi)^\lambda Q_\pi^{2-\lambda} \leq (2^{-\ell(\pi)} Q_\pi)^\lambda$; if $\lambda \geq 2$, then $2^{-\lambda\ell(\pi)} Q_\pi^2 \leq (2^{-\ell(\pi)} Q_\pi)^2$. Since $\sum_\pi t_\pi^{\bar\lambda} \leq (\sum_\pi t_\pi)^{\bar\lambda}$ for nonnegative $t_\pi$,

$$\xi^\times_\lambda(D_\beta) = \sum_\pi 2^{-\lambda\ell(\pi)} \nu_\pi(D_\beta) \leq \sum_\pi 2^{-\lambda\ell(\pi)} Q_\pi(\beta)^2 \leq \bar Q(\beta)^{\bar\lambda} \leq 2^{-\bar\lambda(K(\beta|s)-\kappa)}.$$

*(ii).* $2^{-J_\lambda(\pi;D_\beta)}$ is one term of $\xi^\times_\lambda(D_\beta)$. For a product class with a prefix-free code, define $\bar{Q}$ over that class; (F2) then holds with a constant for the class.

*(iii).* $\nu_{\text{len}}$ is the law of the function machine $T = \{(x, w, b) : |w| = |x| + 1,\ w_{|x|+1} = b\}$, which outputs the tape bit $\omega_{|x|+1}$. Inputs $a_i, b_i$ have the same length $i$, so they receive the same bit, and distinct $i$ receive independent bits; so $\nu_{\text{len}}(D_\beta) = 2^{-s}$. Universality gives $M_F \geq 2^{-K(\nu_{\text{len}})}\nu_{\text{len}}$ (Hänni, 2026a, Theorem 3.8), so $-\log M_F(D_\beta) \leq s + \kappa$; for $\xi^F_\lambda$ keep the single term of a shortest program for $\nu_{\text{len}}$.

*(iv).* Fewer than $2^{s-j}$ strings $\beta$ have $K(\beta \mid s) < s-j$, so $\Pr[K(\beta \mid s) < s-j] < 2^{-j}$. Otherwise (i) gives $-\log \xi^\times_\lambda(D_\beta) \geq \bar{\lambda}(s - j - \kappa)$, while (iii) gives $-\log M_F(D_\beta) \leq s + \kappa$ and $-\log \xi^F_\lambda(D_\beta) \leq s + \lambda\kappa$. Subtracting, with $\bar{\lambda} \leq 2$ and $\bar{\lambda} \leq \lambda$, the excess is at least $(\bar{\lambda} - 1)s - 2j - (\lambda + 2)\kappa$ in both cases. □

### B.6 Hypotheses that see the history

**Proposition B.6** (Labelled history restores universality)**.** *Suppose a hypothesis's input at position $i$ of a block contains the block's earlier labelled pairs. Let $(\nu_\theta)_{\theta\in\Theta}$, $\Theta$ countable, be uniformly computable history-dependent conditional measures with positive values, and $w$ computable positive weights with a computable sum $\sum_\theta w_\theta \leq 1$. With $Z_i := \sum_\theta w_\theta \prod_{j<i} \nu_\theta(y_j \mid \cdot)$, the in-context Bayes mixture $h_w(y_i \mid \cdot) := Z_{i+1}/Z_i$ is a single product hypothesis on the extended inputs, with a program of length at most $K(w, (\nu_\theta)) + \kappa$, and on every block*

$$\textstyle\sum_i -\log h_w(y_i \mid \cdot) \ \leq\ \min_\theta \big[\log(1/w_\theta) + \sum_i -\log \nu_\theta(y_i \mid \cdot)\big].$$

*So it pays $\log(1/w_\theta)$ once per block, not once per input.*

*Proof.* Each $Z_i$ is a positive real, computable uniformly in the block's history: its terms are computable and positive, and the tail $\sum_{\theta>T} w_\theta \prod_{j<i} \nu_\theta(\cdot)$ is at most $\sum_{\theta>T} w_\theta$, which is computable from $T$ because $w$ and $\sum_\theta w_\theta$ are. So $h_w = Z_{i+1}/Z_i$ is computable, and it sums to 1 over $y_i$ because the $\nu_\theta$ are measures: a computable conditional measure on extended inputs, with a program obtained from a description of $w$ and $(\nu_\theta)$. Over a block of length $n'$ the predictive probabilities telescope to $Z_{n'+1}/Z_1$, and $Z_1 = \sum_\theta w_\theta \leq 1$, so $-\log(Z_{n'+1}/Z_1) \leq \log(1/w_\theta) + \sum_i -\log \nu_\theta(y_i \mid \cdot)$ for every $\theta$. □

Without the computability assumptions, $Z_{i+1}/Z_i$ is a ratio of lsc sums, which need not be lsc. A finite family with rational values and rational weights satisfies them.

**Proposition B.7** (A history of inputs only)**.** *A hypothesis that sees the earlier inputs but not the earlier labels, and draws fresh randomness for each extended input $(x_{<i}, x_i)$, is a product hypothesis on the extended inputs, which are pairwise distinct; so Proposition B.2 and Corollary 4.5 apply and per-input pricing persists. With shared randomness instead, as in* $\text{SFI}^{\text{hist}}$*, $\mathcal{H}_{\text{chr}}$ has the universal member $M_H$ and Theorem 4.2 applies.*

*Proof.* Such a hypothesis is an lsc conditional semimeasure on extended inputs, with private randomness per extended input, and extended inputs at different positions are distinct. In Proposition B.2(ii) take the functions $(x_{<i}, x_i) \mapsto \langle a, x_i\rangle \bmod 2$ with nonzero $x_i$; the arguments are unchanged. The second claim is Remark 4.1 for $\mathcal{H}_{\text{chr}}$. □

# C Proofs for Section 5 (kinks)

## C.1 The hull is the graph of the convex minorant

Let $\mathcal{P}$ be a point family. By Lemma 3.8(v),(vi) and its proof, the lower boundary $\underline{C}$ of $C$ is $+\infty$ left of $\alpha_0$, linear between consecutive vertices and equal to $\rho_q$ right of a last vertex $v_q$; the vertices are points of the family, attained at exactly their lengths, and $\alpha_i \to \infty$ if there are infinitely many, since lengths are integers. $\underline{C}$ is convex and nonincreasing, since its slopes $-\sigma_1 < -\sigma_2 < \cdots \le 0$ increase. Every point $(\ell(\pi), L(\pi))$ lies in $C$, hence on or above $\underline{C}$, so $\underline{C}(\alpha) \le \underline{C}(\ell(\pi)) \le L(\pi)$ whenever $\ell(\pi) \le \alpha$: $\underline{C} \le \rho$, hence $\underline{C} \le \breve{\rho}$; and $\rho(\alpha_i) = \rho_i$, since $\rho(\alpha_i) \ge \underline{C}(\alpha_i)$ and the program at $v_i$ has length $\alpha_i$. Conversely, a convex $f \le \rho$ has $f(\alpha_i) \le \rho_i$, lies below the chord $\underline{C}$ between consecutive vertices, and has $f \le \rho \le \rho_q = \underline{C}$ right of a last vertex. So $\breve{\rho} = \underline{C}$.

## C.2 The kink lemma, and lines against secants

*Proof of Lemma 5.3.* Let $\pi_{\mathrm{r}}$ bound the point. (i) On $\mathcal{N}$, $J'_c \ge J_{\mathrm{r}}(c) + \eta_{\mathcal{N}}(c)$ by definition. For $\pi \in \mathcal{M}$, the right line and $\ell(\pi) \ge \ell_{\mathcal{M}}$ give

$$J'_c(\pi) \ge c\,\ell(\pi) + L_{\mathrm{r}} - g - \tau(\ell(\pi) - \alpha_{\mathrm{r}}) = (c-\tau)\ell(\pi) + \tau\alpha_{\mathrm{r}} + L_{\mathrm{r}} - g \ge J_{\mathrm{r}}(c) - \Delta(c),$$

since $c > \tau$. If $\pi_{\mathrm{r}} \in \mathcal{M}$, the line at $\pi_{\mathrm{r}}$ gives $g \ge L_{\mathrm{r}} - L'(\pi_{\mathrm{r}}) + \tau(\alpha_{\mathrm{r}} - \ell(\pi_{\mathrm{r}})) \ge 0$, and then $\Delta(c) \ge g \ge 0$ because $\ell_{\mathcal{M}} \le \alpha_{\mathrm{r}}$. Finally $\Phi(c) \ge \inf_\pi J'_c(\pi)$ because $L \ge L'$, and $\Phi(c) \le J_c(\pi_{\mathrm{r}}) \le J_{\mathrm{r}}(c)$.

(ii) $J_c(\pi) \le \Phi(c) + \epsilon$ implies $J'_c(\pi) \le J_{\mathrm{r}}(c) + \epsilon$. A program with this property is not in $\mathcal{N}$, so it is in $\mathcal{M}$, and the line gives $(c-\tau)(\ell(\pi) - \alpha_{\mathrm{r}}) \le g + \epsilon$. Since $J'_c(\pi_{\mathrm{r}}) \le J_{\mathrm{r}}(c)$, also $\pi_{\mathrm{r}} \in \mathcal{M}$, so $g \ge 0$ by (i). If $\ell(\pi) \ge \alpha_{\mathrm{r}}$, the line and the length bound give $L'(\pi) \ge L_{\mathrm{r}} - g - \tau(g+\epsilon)/(c-\tau) = L_{\mathrm{r}} - (cg + \tau\epsilon)/(c-\tau)$; if $\ell(\pi) < \alpha_{\mathrm{r}}$, the line gives $L'(\pi) \ge L_{\mathrm{r}} - g$, which is larger, since $(cg+\tau\epsilon)/(c-\tau) - g = \tau(g+\epsilon)/(c-\tau) \ge 0$.

(iii) $\eta_{\mathcal{N}}$ is an infimum of the functions $c \mapsto c(\ell(\pi) - \alpha_{\mathrm{r}}) + L'(\pi) - L_{\mathrm{r}}$, each affine or identically $+\infty$, and $\eta_{\mathcal{N}} > -\infty$ because $J'_c \ge 0$. (iv) Rearrange $J'_c(\pi) \le J_{\mathrm{r}}(c) + \epsilon$ and use $\ell(\pi) \ge \ell_{\mathcal{M}}$. (v) For $\pi \in \mathcal{N}$ put $x := \alpha_{\mathrm{r}} - \ell(\pi)$; then $J'_c(\pi) - J_{\mathrm{r}}(c) = L'(\pi) - L_{\mathrm{r}} - cx$. This is at least $\vartheta$ if $x \le 0$; if $x > 0$ it is at least $\max\{\sigma x, \vartheta\} - cx$, which is at least $(\sigma - c)x \ge (\sigma - c)\vartheta/\sigma$ when $\sigma x \ge \vartheta$ and more than $\vartheta - c\vartheta/\sigma$ otherwise; in all cases it is at least $\vartheta(1 - c/\sigma)$. □

If every program that bounds the point lies in $\mathcal{N}$, then $\eta_{\mathcal{N}}(c) \le J'_c(\pi_{\mathrm{r}}) - J_{\mathrm{r}}(c) \le 0$, and (ii) is vacuous.

Every $\tau \ge 0$ has a right line on every $\mathcal{M}$ with length floor $\ell_{\mathcal{M}}$, with $g = L_{\mathrm{r}} + \tau(\alpha_{\mathrm{r}} - \ell_{\mathcal{M}})$, because $L' \ge 0$; so a slope alone says nothing, and what matters is the offset. The *secant* of $\pi \in \mathcal{M}$ with $\ell(\pi) = \alpha_{\mathrm{r}} + x$, $x > 0$, is $\varsigma(\pi) := (L_{\mathrm{r}} - L'(\pi))/x$. A line bounds how far to the right a near-optimal program can lie; the secant decides whether a program at the point itself survives.

**Lemma C.1** (Lines against secants)**.** *Let $(\alpha_{\mathrm{r}}, L_{\mathrm{r}})$ be bounded and $\mathcal{M} \subseteq \mathcal{P}$. (a) If $(\tau, g)$ is a right line on $\mathcal{M}$, every $\pi \in \mathcal{M}$ that is $x > 0$ bits longer than $\alpha_{\mathrm{r}}$ has $\varsigma(\pi) \le \tau + g/x$. (b) If $\pi_1 \in \mathcal{M}$ is $x_1 > 0$ bits longer than $\alpha_{\mathrm{r}}$, every right line on $\mathcal{M}$ of slope $\tau$ has $g \ge (\varsigma(\pi_1) - \tau)x_1$. (c) Let a program $\pi_0$ have $\ell(\pi_0) = \alpha_{\mathrm{r}}$ and $L'(\pi_0) = L_{\mathrm{r}}$. For $0 \le c < s_+$ some program of $\mathcal{M}$ longer than $\alpha_{\mathrm{r}}$ has $J'_c < J'_c(\pi_0)$; for $c > s_+$ none has $J'_c \le J'_c(\pi_0)$.*

*Proof.* (a) and (b) are the line at $\pi$ and at $\pi_1$, rearranged. (c) $J'_c(\pi) - J'_c(\pi_0) = cx + L'(\pi) - L_{\mathrm{r}} = x(c - \varsigma(\pi))$. If $0 \le c < s_+$, then $s_+ > 0$, so some such $\pi$ has $\varsigma(\pi) > c$ and $J'_c(\pi) < J'_c(\pi_0)$; if $c > s_+$, every such $\pi$ has $\varsigma(\pi) < c$ and $J'_c(\pi) > J'_c(\pi_0)$. □

## C.3 Kinks on one data set

*Proof of Corollary 5.4.* (a) Let $\pi_-$ bound $(\alpha_-, L_-)$, so $J_\lambda(\hat\pi) \le \lambda\alpha_- + L_-$. If $\ell(\hat\pi) < \alpha_- - w$, the kink gives $J_\lambda(\hat\pi) \ge \lambda\alpha_- + L_- + (\sigma - \lambda)(\alpha_- - \ell(\hat\pi)) > \lambda\alpha_- + L_-$ (for $\sigma = \infty$, $L(\hat\pi) = \infty$), a contradiction. The fit bound is $J_\lambda(\hat\pi) \le \lambda\alpha_- + L_-$ rearranged, and for $\lambda \ge 1$ subtracting $(\lambda-1)\ell(\hat\pi)$ gives $\ell(\hat\pi) + L(\hat\pi) \le \alpha_- + L_- + (\lambda-1)(\alpha_- - \ell(\hat\pi)) \le \alpha_- + L_- + (\lambda-1)w$. (b) If $\ell(\hat\pi) > \alpha_+ + w_+$, then $J_\lambda(\hat\pi) \ge \lambda\alpha_+ + L_+ + (\lambda-\tau)(\ell(\hat\pi) - \alpha_+) > J_\lambda(\pi_+)$ for $\pi_+$ bounding $(\alpha_+, L_+)$.

(c) Programs shorter than $\alpha_\mathrm{r}$ have fit above $L_\mathrm{r} \ge \rho(\alpha_\mathrm{r})$, so $\rho(\alpha_\mathrm{r})$ is attained at length $\alpha_\mathrm{r}$. By (a) and (b), every $\lambda$-MAP with $\lambda \in (\tau, \sigma)$ has length $\alpha_\mathrm{r}$ and hence fit $\rho(\alpha_\mathrm{r})$, and $\lambda$-MAPs exist (Lemma A.1). The $\lambda$-MAP point is thus the same on a nonempty open interval, so it is a vertex $v_i$ (Lemma 3.8(iii)). At $\lambda = \sigma_i$ ($i \ge 1$) and at $\lambda = \sigma_{i+1} > 0$ a neighbouring vertex is also a $\lambda$-MAP point (Lemma 3.8(v)), so $(\tau, \sigma) \subseteq (\sigma_{i+1}, \sigma_i)$. Conversely, by convexity $\breve\rho(\ell) \ge \rho_i + \sigma_i(\alpha_i - \ell)$ for $\ell < \alpha_i$ and $\breve\rho(\ell) \ge \rho_i - \sigma_{i+1}(\ell - \alpha_i)$ for $\ell > \alpha_i$, and every point lies on or above $\breve\rho$. □

*The hypotheses are needed.* In the point family $\{(0,10), (1,5), (2,0)\}$ (one program at each point) the middle point is a zero-window $(5,5)$-kink; it is a $\lambda$-MAP point only at $\lambda = 5$, together with both other points, and it is not a vertex, so (c) needs $\tau < \sigma$. In $\{(0,10), (2,0)\}$ the point $(2,0)$ is a zero-window left kink of steepness 5, and at $\lambda = 5$ the point $(0,10)$, of length $0 < 2$, is a $\lambda$-MAP; symmetrically, $(0,10)$ is a zero-window right kink of flatness 5, and at $\lambda = 5$ the point $(2,0)$ is a $\lambda$-MAP. So (a) needs $\lambda < \sigma$, and (b) needs $\lambda > \tau$.

*Proof of Corollary 5.5.* (a) By Lemma 3.9 and $b$-sufficiency, $L(\pi; D) \ge K(y_D \mid \operatorname{dom} D) - \kappa - \ell(\pi) \ge L_\mathrm{r} - (\ell(\pi) - \alpha_\mathrm{r}) - (b + \kappa)$ for every $\pi$: a right line $(1, b+\kappa)$ with length floor 0. At $\pi_s$ it gives $b + \kappa \ge 0$, hence the right kinks (the remark after Definition 5.2). For a $\lambda$-MAP $\hat\pi$, subtract $\ell(\hat\pi) + L(\hat\pi; D) \ge \alpha_\mathrm{r} + L_\mathrm{r} - (b+\kappa)$ from $\lambda\ell(\hat\pi) + L(\hat\pi; D) \le \lambda\alpha_\mathrm{r} + L_\mathrm{r}$ to get $(\lambda-1)(\ell(\hat\pi) - \alpha_\mathrm{r}) \le b + \kappa$, and insert this into the line for the fit bound. (b) is (a) with Corollary 5.4(a), whose two-part bound and $\alpha_\mathrm{r} + L_\mathrm{r} \le K(y_D \mid \operatorname{dom} D) + b$ give the sufficiency. □

At a program attaining the product sophistication, Corollary 5.5(a) gives the following.

**Proposition C.2** (Steeper two-part codes over products)**.** *Let $\lambda > 1$. Every $\lambda$-MAP over $\mathcal{H}_\times$ for $D$ has length at most $\operatorname{soph}_b^\times(D) + (b+\kappa)/(\lambda-1)$ and is $(b + (\lambda-1)\operatorname{soph}_b^\times(D))$-product-sufficient.*

*Proof.* If $\operatorname{soph}_b^\times(D) = \infty$ there is nothing to prove; otherwise let $\pi_s$ attain it. Corollary 5.5(a) in $\mathcal{H}_\times$ gives the length bound, and, since $\ell(\hat\pi) \ge 0$, $\ell(\hat\pi) + L(\hat\pi; D) \le J_\lambda(\hat\pi; D) \le J_\lambda(\pi_s; D) \le K(y_D \mid \operatorname{dom} D) + b + (\lambda-1)\operatorname{soph}_b^\times(D)$. □

*Proof of Theorem 5.6.* The kinks give the lengths by Corollary 5.4(a),(b) with zero windows. Conversely, assume the condition on lengths. The index $i$ exists, since $\sigma_0 = \infty$ and the $\sigma_i$ decrease to 0 or end with $\sigma_{i+1} = 0$; for the same reason $j$ exists when $\tau > 0$ or the vertices are finitely many. On the nonempty interval $(\max\{\tau, \sigma_{i+1}\}, \sigma) \subseteq (\sigma_{i+1}, \sigma_i)$ the $\lambda$-MAPs are the programs at $v_i$, so $\alpha_i \ge \alpha'$. Since $\sigma_{j+1} \le \tau < \sigma$, $j \ge i$, and on $(\tau, \min\{\sigma, \sigma_j\}) \subseteq (\sigma_{j+1}, \sigma_j)$ the $\lambda$-MAPs are the programs at $v_j$, so $\alpha_j \le \alpha''$. By Corollary 5.4(c), $v_i$ is a zero-window left kink of steepness $\sigma_i \ge \sigma$, hence of steepness $\sigma$, and $v_j$ one of flatness $\sigma_{j+1} \le \tau$, hence of flatness $\tau$, at the points $(\alpha_i, \rho(\alpha_i))$ and $(\alpha_j, \rho(\alpha_j))$ (Appendix C.1). The MAP length is nonincreasing in $\lambda$ (Lemma 3.8(ii)), which gives the minimum $\alpha_i$ and the maximum $\alpha_j$. If $\tau = 0$ and there are infinitely many vertices, then $\sigma_i \to 0$ and $\alpha_i \to \infty$: the $\lambda$-MAP lengths are unbounded as $\lambda \downarrow 0$, and no point is a zero-window right kink of flatness 0, since $\rho$ decreases beyond every length; so both conditions fail. □

*Proof of Proposition 5.7.* (i) Let $\pi_\mathrm{r}$ bound $(\alpha_\mathrm{r}, L_\mathrm{r})$. The hypothesis at $\pi_\mathrm{r}$ gives $L(\pi_U) \le L_\mathrm{r} + \alpha_\mathrm{r} + \kappa_0 < \infty$, and the kink gives $L(\pi_U) \ge L_\mathrm{r} + \sigma(\alpha_\mathrm{r} - \ell_U)$, since $\ell_U < \alpha_\mathrm{r} - w$. So $\sigma \le (\alpha_\mathrm{r} + \kappa_0)/(\alpha_\mathrm{r} - \ell_U)$, and $\sigma > 1 + \epsilon$ forces $\alpha_\mathrm{r} - \ell_U < (\ell_U + \kappa_0)/\epsilon$. (ii) A $\lambda$-MAP $\hat\pi$ has $\lambda\ell(\hat\pi) + L(\hat\pi) \le \lambda\ell_U + L(\pi_U) \le \lambda\ell_U + L(\hat\pi) + \ell(\hat\pi) + \kappa_0$, so $(\lambda-1)\ell(\hat\pi) \le \lambda\ell_U + \kappa_0$, as in

Theorem 4.2(a). The hypothesis at $\pi_U$ gives $\ell_U + \kappa_0 \geq 0$; so the bound is at most $\ell_U + (\ell_U + \kappa_0)/\epsilon$ when $\lambda - 1 \geq \epsilon$, and exceeds it when $\ell_U + \kappa_0 > 0$ and $\lambda - 1 < \epsilon$. □

# D Proofs for Section 6 (exact model recovery)

Throughout, $\Lambda_n(\pi) := \nu_\pi(o_n)/\nu(o_n) = 2^{L_n(\nu) - L_n(\pi)}$ is the likelihood ratio of $\pi$ against the sampler; almost surely every label has positive $\nu$-probability, so it is well defined. The prior's tail beyond $\bar{\ell}$ is $t_{\bar{\ell}} := \sum_{\ell(\pi) > \bar{\ell}} 2^{-\ell(\pi)}$, which tends to 0, and $Z_n^{\bar{\ell}} := \sum_{\ell(\pi) > \bar{\ell}} 2^{-\ell(\pi)} \Lambda_n(\pi)$.

## D.1 Likelihood ratios

**Lemma D.1** (Likelihood ratios)**.** *Let the inputs be chosen by any rule of Definition 6.1, indexed by queries for per-query labels and by distinct inputs, in order of first appearance, for frozen labels; let $o_i$ and $\Lambda_i(\pi)$ be the data and the likelihood ratio at the $i$-th step of this indexing, and put $g_i(\pi) := d_{\mathrm{H}}^2(\nu(\cdot \mid x_i), \nu_\pi(\cdot \mid x_i))$ for its $i$-th input, which is fixed before $y_i$ is drawn. (Indexed by queries, $\Lambda$ is constant at a repeated query with frozen labels, so the supermartingale statements of (a) hold in either indexing.)*

*(a) $\mathbb{E}[\Lambda_{n+1}(\pi)/\Lambda_n(\pi) \mid o_n, x_{n+1}] = \sum_{y : \nu(y \mid x_{n+1}) > 0} \nu_\pi(y \mid x_{n+1}) \leq 1$. So $\Lambda_n(\pi)$, $\Lambda_n(\pi)^\eta$ ($0 < \eta \leq 1$) and $Z_n^{\bar{\ell}}$, also with $\Lambda^\eta$ in place of $\Lambda$, are nonnegative supermartingales, and $\mathbb{P}[\sup_n Z_n^{\bar{\ell}} \geq a] \leq t_{\bar{\ell}}/a$ for $a > 0$.*

*(b) On the event $\{\sum_i g_i(\pi) = \infty\}$, almost surely $\Lambda_n(\pi) \to 0$.*

*Now let the data be per query with inputs i.i.d. from $\mu$, so that $\Lambda_n(\pi)$ is a product of $n$ independent copies of $Z := \nu_\pi(y \mid x)/\nu(y \mid x)$, $(x, y) \sim \mu \otimes \nu$.*

*(c) $\mathrm{KL}_\mu(\pi) = -\mathbb{E} \log Z \in [0, \infty]$ is positive iff $\pi \notin \mathcal{E}$; $\mathbb{E}(\log Z)^+ \leq \log e/e$; and $\frac{1}{n} \log \Lambda_n(\pi) \to -\mathrm{KL}_\mu(\pi)$ almost surely, also when $\mathrm{KL}_\mu(\pi) = \infty$.*

*(d) $\mathbb{E}\Lambda_n(\pi)^s = \mathsf{C}_s(\pi)^n$ for $s \in (0, 1]$; for $s < 1$, $\mathsf{C}_s(\pi) \leq (1 - d_{\mathrm{H}\mu}^2(\pi))^{2 \min(s, 1-s)}$, so $\mathsf{C}_s(\pi)^n \leq e^{-2 \min(s, 1-s) d_{\mathrm{H}\mu}^2(\pi) n}$. In particular $\mathbb{E}\sqrt{\Lambda_n(\pi)} = (1 - d_{\mathrm{H}\mu}^2(\pi))^n \leq e^{-n d_{\mathrm{H}\mu}^2(\pi)}$, and $\Lambda_n(\pi) = 1$ for all $n$ and all $\pi \in \mathcal{E}$, almost surely.*

*Proof.* (a) Given $o_n$ and $x_{n+1}$, the new factor is $\nu_\pi(y \mid x_{n+1})/\nu(y \mid x_{n+1})$ with $y \sim \nu(\cdot \mid x_{n+1})$ (per query, and also with frozen labels, where the next distinct input is new and its label is drawn independently of the past). Its mean is the stated sum. Jensen's inequality gives the same for the power $\eta$, and the sums $Z_n^{\bar{\ell}}$ inherit the property termwise; they start at $t_{\bar{\ell}}$. Ville's maximal inequality (Ville, 1939) gives the tail bound.

(b) Given $o_n$ and $x_{n+1}$, $\mathbb{E}\sqrt{\Lambda_{n+1}/\Lambda_n} = \sum_{y : \nu > 0} \sqrt{\nu(y \mid x_{n+1}) \nu_\pi(y \mid x_{n+1})} = 1 - g_{n+1}$. If $g_{n+1} = 1$, then almost surely $\nu_\pi(y_{n+1} \mid x_{n+1}) = 0$ and $\Lambda$ is 0 from then on. Otherwise $M_n := \sqrt{\Lambda_n} / \prod_{i \leq n} (1 - g_i)$ is a nonnegative supermartingale, so it converges almost surely, and on $\{\sum_i g_i = \infty\}$, $\prod_{i \leq n} (1 - g_i) \leq e^{-\sum_{i \leq n} g_i} \to 0$, hence $\Lambda_n \to 0$. This is the argument behind Kakutani's dichotomy (Kakutani, 1948); it needs only that $g_{n+1}$ is fixed before $y_{n+1}$ is drawn.

(c) Conditional Jensen for the strictly concave logarithm gives $\mathbb{E}[\log Z \mid x] \leq \log \mathbb{E}[Z \mid x] \leq 0$, with equality in both iff $\nu_\pi(\cdot \mid x) = c\, \nu(\cdot \mid x)$ on the support of $\nu(\cdot \mid x)$ with $c = 1$, which with $\sum_y \nu_\pi \leq 1$ forces $\nu_\pi(\cdot \mid x) = \nu(\cdot \mid x)$. So $\mathrm{KL}_\mu(\pi) > 0$ iff the kernels differ on a set of positive $\mu$-measure. Since $\ln z \leq z/e$, $\mathbb{E}(\log Z)^+ \leq (\log e/e) \mathbb{E}Z \leq \log e/e$, and the strong law for i.i.d. variables with integrable positive part gives the limit in $[-\infty, 0]$.

(d) Independence gives $\mathbb{E}\Lambda_n^s = (\mathbb{E}Z^s)^n$ and $\mathbb{E}Z^s = \mathsf{C}_s(\pi)$. For $s \in [\frac{1}{2}, 1)$ write $\nu^{1-s}\nu_\pi^s = (\sqrt{\nu\nu_\pi})^{2(1-s)}\nu_\pi^{2s-1}$ and apply Hölder's inequality on $\mu\otimes$(counting measure) with exponents $1/(2(1-s))$ and $1/(2s-1)$; this gives $\mathsf{C}_s \le \mathsf{C}_{1/2}^{2(1-s)}(\mathbb{E}_\mu \sum_y \nu_\pi)^{2s-1} \le \mathsf{C}_{1/2}^{2(1-s)}$. For $s \in (0, \frac{1}{2}]$ write $\nu^{1-s}\nu_\pi^s = (\sqrt{\nu\nu_\pi})^{2s}\nu^{1-2s}$ to get $\mathsf{C}_s \le \mathsf{C}_{1/2}^{2s}$. Then use $\mathsf{C}_{1/2} = 1 - d_{\mathrm{H}\mu}^2$ and $1 - u \le e^{-u}$. For $\pi \in \mathcal{E}$ every $x_i$ falls almost surely where the two kernels agree, and there are countably many programs. □

## D.2 Proof of Theorem 6.3

Work on the almost sure event of Lemma D.1(d) on which $\Lambda_n(\pi) = 1$ for all $n$ and all $\pi \in \mathcal{E}$. A program $\pi_0 \in \mathcal{E}_0$ then has $J_\lambda(\pi_0; h_n) = \lambda\ell_0 + L_n(\nu) < \infty$, so $\lambda$-MAPs exist for every $\lambda > 0$ (Lemma A.1(i)).

*(a).* If $\hat\pi$ is a $\lambda_n$-MAP, then $J_{\lambda_n}(\hat\pi) \le J_{\lambda_n}(\pi_0)$, that is,

$$\log \Lambda_n(\hat\pi) \;\ge\; \lambda_n(\ell(\hat\pi) - \ell_0). \tag{1}$$

*Long programs, uniformly in $n$.* Let $\bar\ell \ge \ell_0$. If a $\lambda_n$-MAP has $\ell(\hat\pi) > \bar\ell$, then (1) and $\lambda_n \ge 1$ give $\log\Lambda_n(\hat\pi) \ge \ell(\hat\pi) - \ell_0$, so $2^{-\ell(\hat\pi)}\Lambda_n(\hat\pi) \ge 2^{-\ell_0}$ and $Z_n^{\bar\ell} \ge 2^{-\ell_0}$. By Lemma D.1(a), $B_{\bar\ell} := \{\exists n : Z_n^{\bar\ell} \ge 2^{-\ell_0}\}$ has probability at most $2^{\ell_0} t_{\bar\ell} \to 0$. Since $Z_n^{\bar\ell}$ is pathwise nonincreasing in $\bar\ell$, the events $B_{\bar\ell}$ decrease, and their intersection is null. So almost surely some finite $\bar\ell_0$ bounds the length of every $\lambda_n$-MAP at every $n$. *Short programs.* By Lemma D.1(c) and countability, almost surely $\frac{1}{n}\log\Lambda_n(\pi) \to -\mathrm{KL}_\mu(\pi) < 0$ for all $\pi \notin \mathcal{E}$ at once. Take $\pi \notin \mathcal{E}$ with $\ell(\pi) \le \bar\ell_0$. If $\ell(\pi) \ge \ell_0$, the left side of (1) tends to $-\infty$ and the right side is nonnegative. If $\ell(\pi) < \ell_0$, the left side is eventually at most $-n\,\mathrm{KL}_\mu(\pi)/2$ (or tends to $-\infty$ if $\mathrm{KL}_\mu(\pi) = \infty$), while the right side is $-\lambda_n(\ell_0 - \ell(\pi)) = -o(n)$. Programs of $\mathcal{E} \setminus \mathcal{E}_0$ have $\Lambda_n = 1$ and $\ell > \ell_0$, so they violate (1) at every $n$. Finitely many programs have length at most $\bar\ell_0$, so from some $n$ on every $\lambda_n$-MAP lies in $\mathcal{E}_0$, and all of $\mathcal{E}_0$ tie.

*(b).* Divide the posterior weights by $\nu(h_n)$; they become $2^{-\lambda\ell(\pi)}\Lambda_n(\pi)$, and the normaliser is at least $W_\lambda$. For every $\bar\ell$, since $2^{-\lambda\ell} \le 2^{-\ell}$,

$$\sum_{\pi\notin\mathcal{E}} 2^{-\lambda\ell(\pi)}\Lambda_n(\pi) \;\le\; \sum_{\pi\notin\mathcal{E},\ \ell(\pi)\le\bar\ell} 2^{-\lambda\ell(\pi)}\Lambda_n(\pi) + Z_n^{\bar\ell}.$$

The finite sum tends to 0 almost surely, by Lemma D.1(c). Choose $\bar\ell_j$ with $t_{\bar\ell_j} \le 4^{-j}$; then $\mathbb{P}[\sup_n Z_n^{\bar\ell_j} \ge 2^{-j}] \le 2^{-j}$, and by the Borel–Cantelli lemma almost surely $\sup_n Z_n^{\bar\ell_j} < 2^{-j}$ for all large $j$. So the mass outside $\mathcal{E}$ tends to 0, the normaliser tends to $W_\lambda$, and the terms of $\mathcal{E}$ are constant. For the tempered posterior use the weights $2^{-\ell(\pi)}\Lambda_n(\pi)^\eta$, $\Lambda_n^\eta \to 0$ off $\mathcal{E}$, and the tail bound for $\Lambda^\eta$; the normaliser tends to $W_1$.

*(c).* A program with $J_\lambda(\pi) \le J_\lambda(\pi_0)$ has $\lambda(\ell(\pi) - \ell_0) \le \log\Lambda_n(\pi) \le \ell(\pi) + \delta_n$ by Lemma 6.2, which is the length bound. A fill-in has $\Lambda_n(\pi) = 2^{L_n(\nu)} \le 2^{\ell(\pi)+\delta^*}$, which is its length bound. Hence, summing over fill-ins,

$$\sum 2^{-\lambda\ell(\pi)}\Lambda_n(\pi) \le 2^{-(\lambda-1)(L_n(\nu)-\delta^*)}\sum 2^{-\ell(\pi)}\Lambda_n(\pi) \le 2^{-(\lambda-1)(L_n(\nu)-\delta^*)+\delta^*},$$

and the normaliser divided by $\nu(h_n)$ is at least $2^{-\lambda\ell_0}$. For $\Pi^{1/\lambda}$ a fill-in has $\Lambda_n^{1/\lambda} = 2^{-(1-1/\lambda)L_n(\nu)}\Lambda_n$, so the fill-ins' weight is at most $2^{-(1-1/\lambda)L_n(\nu)+\delta^*}$, against a normaliser of at least $2^{-\ell_0}$. Finally $L_n(\nu)/n \to H_\mu(\nu)$ by the strong law, since $0 \le \mathbb{E}[-\log\nu(y \mid x)] \le \log|Y|$.

*(d).* Almost surely the programs of $\mathcal{E} \setminus \mathcal{E}_0$ are worse than $\pi_0$ by $\lambda(\ell - \ell_0) > 0$, so the event that the set of $\lambda$-MAPs is not $\mathcal{E}_0$ is contained in $\bigcup_{\pi \notin \mathcal{E}} \{\Lambda_n(\pi) \geq 2^{\lambda(\ell(\pi)-\ell_0)}\}$. For each such $\pi$ and each $s \in (0,1]$, Markov's inequality for $\Lambda_n(\pi)^s$ and Lemma D.1(d) bound its probability by $2^{-s\lambda(\ell(\pi)-\ell_0)}\mathsf{C}_s(\pi)^n$; take the infimum over $s$ and a union bound. For the margin form split the union at $\bar{\ell} := \bar{\ell}_\varepsilon$. A program with $\ell(\pi) > \bar{\ell}$ and $J_\lambda(\pi) \leq J_\lambda(\pi_0)$ forces $\delta^* > (\lambda-1)\bar{\ell} - \lambda\ell_0 \geq \log(2/\varepsilon)$ by (c), which has probability at most $\varepsilon/2$. For $\ell(\pi) \leq \bar{\ell}$ take $s = 1/\lambda$: since $\min(1/\lambda, 1-1/\lambda) = \min(1, \lambda-1)/\lambda$, the sum is at most $\sum_{\ell(\pi)\leq\bar{\ell}} 2^{\ell_0-\ell(\pi)}e^{-an} \leq 2^{\ell_0}e^{-an}$ with $a := 2\min(1,\lambda-1)\gamma/\lambda$, and this is at most $\varepsilon/2$ when $an \geq (\ell_0 + \log(2/\varepsilon))\ln 2$. □

## D.3 The parity classes, and what is necessary

*Parities.* In $\mathcal{C}_k$ and $\mathcal{C}_{k,r_0}$ a parity on $r$ bits is consistent with $h_n$ iff the label vector $y$ lies in the column span over GF(2) of the $n \times r$ matrix $A_r$ of the first $r$ bits of the inputs, and a parity on fewer bits is one on more bits. The labels are fair coins independent of the inputs, so $\mathbb{P}[y \in \mathrm{colspan} A_r] = \mathbb{E}2^{\mathrm{rank}A_r - n} \leq 2^{r-n}$. With i.i.d. uniform rows, $\mathbb{P}[\mathrm{rank}A_r < n] \leq \sum_{i<n} 2^{i-r} < 2^{n-r}$. The coin has $J_\lambda = \lambda k + n$, and a consistent parity has $J_\lambda$ equal to $\lambda$ times its length, which is $\ell(r) := r + 2\lceil\log(r+1)\rceil + 1$ on $r$ bits in $\mathcal{C}_k$; inconsistent parities have $J_\lambda = \infty$. A parity has $\mathsf{C}_s = 2^{s-1}$.

*Proof of Proposition 6.6(a).* At $n = r_0 - 5$, $\mathbb{P}[\mathrm{rank}A_{r_0} < n] < 2^{-5}$, and when the rank is $n$ some parity is consistent, with $J_1 = r_0 + 2 = n + 7 < k + n$, so every 1-MAP is a parity.

*(b), lower end.* For $n$ with $2\lceil\log(n+6)\rceil + 6 < k$ and $r = n+5 \leq d$, $\ell(r) = n+6+2\lceil\log(n+6)\rceil < k+n$, and with probability at least $1 - 2^{-5}$ a parity on $r$ bits is consistent and strictly beats the coin. At $n_b := 2^{(k-8)/2} - 6$ ($k$ even), $2\lceil\log(n_b+6)\rceil + 6 = k - 2 < k$, so the probability is at least $31/32 > \varepsilon$ at $n_b$, and $n_{\mathrm{rec}}(\varepsilon) > n_b$.

*(b), upper end.* Let $r^* := \max\{r \geq 1 : \ell(r) \leq k+n\}$; the set of 1-MAPs differs from {coin} (ties included) iff $y \in \mathrm{colspan} A_{r^*}$, so its probability $P(n)$ is at most $2^{r^*-n}$ (capping $r^*$ at $d$ only lowers it). From $\ell(r^*) \leq k+n$, $r^* - n \leq k - 1 - 2\lceil\log(r^*+1)\rceil$, so $P(n) \leq 2^{k-1}/(r^*+1)^2$. Put $r_1 := n + k - 1 - 2\lceil\log(n+1)\rceil$. If $1 \leq r_1 \leq n$, then $\ell(r_1) \leq n+k$ and $r^* \geq r_1$; if $r_1 > n$, then $\ell(n) < n+k$ and $r^* \geq n$. For $n \geq 32$, $n/2 \geq 2\lceil\log(n+1)\rceil$, so $r_1 \geq n/2$ and $r^* + 1 \geq n/2$. Hence $P(n) \leq 2^{k+1}/n^2$ for $n \geq 32$, a decreasing bound that is at most $\varepsilon$ once $n \geq 2^{(k+1)/2}/\sqrt{\varepsilon}$.

*(c).* Theorem 6.3(d) with $s = 1/\lambda$ for every parity gives at most $2^k 2^{(1/\lambda - 1)n}$ times the total prior weight of the parities, which is at most $1/4$ in either class.

The ingredients of Theorem 6.3(d) are necessary in the following sense.

**Proposition D.2** (What is necessary)**.** *(a)* A margin. *Let $\lambda > 1$, $\ell_0 \geq 2$ and $\lambda(\ell_0 - 1) \geq 9$. For every $N$ some class of two programs, one of them a program of length $\ell_0$ for $\nu$, has $\mathbb{P}[$the set of $\lambda$-MAPs at $n \neq \mathcal{E}_0] \geq 1 - 8.66/(\lambda(\ell_0-1))$ for every $n \leq N$. (b)* The factor $1/\gamma$. *Let $\nu$ be the fair coin with a program of length $\ell_0$, and $\pi'$ a program for* $\mathrm{Bern}(\frac{1}{2}+\varepsilon')$, $0 < \varepsilon' \leq \frac{1}{10}$, *of length $\ell_0 - \Delta \geq 1$, with $\lambda\Delta \geq 9$. Then $d^2_{\mathrm{H}_\mu}(\pi') \in [\varepsilon'^2/2, 2\varepsilon'^2]$, and in the class of these two programs the $\lambda$-MAP is $\pi'$ with probability at least $1 - 8.66/(\lambda\Delta)$ for every $n \leq 0.083\,\lambda\Delta/d^2_{\mathrm{H}_\mu}(\pi')$. (c)* The factor $1/(\lambda-1)$. *In $\mathcal{C}_k$ with $\lambda > 1$, for every $n$ with $(\lambda-1)n < \lambda(k - 6 - 2\lceil\log(n+6)\rceil)$ and $d \geq n+5$, with probability at least $31/32$ the $\lambda$-MAP is a parity and the coin is not a $\lambda$-MAP. (d)* The length bound is attained. *Let $k$ be an integer and $\lambda > 1$ with $n := \lambda(k-2)/(\lambda-1)$ a positive integer. In $\mathcal{C}_{k,n}$, with probability $\prod_{i=1}^n (1 - 2^{-i}) > 0.2887$ the set of $\lambda$-MAPs at $n$ is $\{coin, \varphi\}$ for a parity $\varphi$ with $\ell(\varphi) = (\lambda k + \delta_n)/(\lambda-1) - \log(1 + 2^{2-k})/(\lambda-1)$. (e)* The canonical class. *Let $Y = \{0,1\}$, $\mathcal{C} = \mathcal{H}_\times$, and $\nu$ computable with $\nu(1 \mid x) \in [\beta, 1-\beta]$ for all $x$ and a rational $\beta \in (0, \frac{1}{2}]$. For every computable $g : \mathbb{N} \to (0,1]$, $\gamma_{\bar{\ell}} < g(\bar{\ell})$ for all large $\bar{\ell}$.*

*Proof of Proposition D.2(b).* The $\lambda$-MAP is $\pi'$ iff $\log \Lambda_n(\pi') > -\lambda\Delta$, and $\log \Lambda_n(\pi') = \sum_{i\le n} X_i$ with $X_i \in \{\log(1+2\varepsilon'), \log(1-2\varepsilon')\}$ equally likely. The mean is $\mathbb{E}X_1 = \frac{1}{2}\log(1-4\varepsilon'^2) \ge -2\varepsilon'^2/((1-4\varepsilon'^2)\ln 2)$ and $\operatorname{Var} X_i \le \log^2(1-2\varepsilon') \le 4\varepsilon'^2/((1-2\varepsilon')^2 \ln^2 2)$. For $n \le n_1 := \lambda\Delta(1-4\varepsilon'^2)\ln 2/(4\varepsilon'^2)$ we have $n|\mathbb{E}X_1| \le \lambda\Delta/2$, and Chebyshev's inequality gives

$$\mathbb{P}\Big[\sum_i X_i \le -\lambda\Delta\Big] \le \frac{4n \operatorname{Var} X_1}{(\lambda\Delta)^2} \le \frac{4(1+2\varepsilon')}{(1-2\varepsilon')\ln 2 \cdot \lambda\Delta} \le \frac{8.66}{\lambda\Delta}.$$

Since $\sqrt{1+u}+\sqrt{1-u} \le 2 - u^2/4$, $d^2_{\mathrm{H}\mu}(\pi') \ge \varepsilon'^2/2$; and $d^2_{\mathrm{H}\mu}(\pi') \le \frac{1}{2}\sum_y (P-Q)^2/P = 2\varepsilon'^2$. So $n_1 \ge 0.96 \ln 2 \cdot \lambda\Delta/(8 d^2_{\mathrm{H}\mu}(\pi')) \ge 0.083\lambda\Delta/d^2_{\mathrm{H}\mu}(\pi')$. *(a)* is (b) with $\Delta = \ell_0 - 1$ and $\varepsilon'$ so small that $0.083\lambda(\ell_0-1)/d^2_{\mathrm{H}\mu}(\pi') \ge N$; the Kraft sum is $\frac{1}{2} + 2^{-\ell_0} \le 1$.

*(c).* With $r = n+5$, $\lambda\ell(r) < \lambda k + n$ is the stated condition, and the rank argument applies.

*(d).* If the $n \times n$ matrix $A_n$ is invertible, which has probability $\prod_{i=1}^n (1-2^{-i})$, exactly one parity $\varphi$ is consistent. Then $J_\lambda(\varphi) = \lambda(n+2)$ equals $J_\lambda(\mathrm{coin}) = \lambda k + n$ by the choice of $n$, and all other parities have $J_\lambda = \infty$. The mixture is $\xi_{\mathcal{C}}(h_n) = 2^{-k}2^{-n} + 2^{-(n+2)}$, so $\delta_n = \log(2^{-k} + \frac{1}{4}) = -2 + \log(1 + 2^{2-k})$, and $\ell(\varphi) = n+2 = (\lambda k - 2)/(\lambda - 1)$ is the stated value.

*(e).* Let $\pi_0 \in \mathcal{E}_0$, and let $\tau$ be the flipped kernel, $\tau(y \mid x) := \nu_{\pi_0}(1-y \mid x)$, or $\tau := \mathrm{Bern}(\frac{1}{4})$ if $\nu$ is the fair coin $\mu$-a.e.; either way $\tau$ is lsc and differs from $\nu$ on a set of positive $\mu$-measure (which case holds need not be decided). The kernels $\nu_j := (1-2^{-j})\nu_{\pi_0} + 2^{-j}\tau$ are lsc uniformly in $j$ and outside $\mathcal{E}$, and by closure under prefix-free simulation they have programs of length at most $\ell_0 + K(j) + \kappa$. Since $d^2_{\mathrm{H}}(P,Q) \le \frac{1}{2}\sum_y (P-Q)^2/P$, $|\nu_j - \nu| \le 2^{-j}$ and $\nu \ge \beta$, their gaps are at most $2^{-2j}/\beta$. Approximating $g(\bar\ell)$ until an approximation $a \ge 3 \cdot 2^{-i}$ appears gives a computable rational $g'(\bar\ell) := a - 2^{-i} \in [g(\bar\ell)/2, g(\bar\ell)]$, and $h(m) := \lceil \frac{1}{2}\log(2/(\beta g'(m+\ell_0+\kappa)))\rceil + 1$ is computable with $2^{-2h(m)}/\beta < g(m+\ell_0+\kappa)$. Let $j(\bar\ell)$ be the largest $j$ with $K(j) \le \bar\ell - \ell_0 - \kappa$; since $K(h(m)) \le 2\log(m+1) + \kappa' \le m$ for large $m$, $j(\bar\ell) \ge h(\bar\ell - \ell_0 - \kappa)$, and $\gamma_{\bar\ell} \le 2^{-2j(\bar\ell)}/\beta < g(\bar\ell)$ for all large $\bar\ell$. □

## D.4 Proof of Theorem 6.7 and of Remark D.3

*Remark* D.3 (The $\ell_0$ in the right window). The natural strengthening of Theorem 6.7(iii) that each bit of length buys at most one bit of fit plus $\delta^*$ in total, $\rho(\ell_0) - \rho(\alpha) \le (\alpha - \ell_0) + \delta^*$ for the sample's structure function $\rho$, fails: in $\mathcal{C}_{k,r_0}$ with $r_0 + 2 > k$ and $n = r_0 - 5 \ge 1$, with probability at least $31/32$ a parity is consistent and $\rho(\ell_0) - \rho(r_0+2) - (r_0 + 2 - \ell_0) = k - 7$, so it fails with probability at least $31/32 - 2^{7-k}$. Hence no fixed $\lambda \le 1$ lies in a kink interval guaranteed at finite $n$; only the limit in Theorem 6.7(v) puts $\lambda = 1$ inside.

*Proof of Theorem 6.7(i).* Fix $\pi$ and put $a(\pi) := 2\log e \cdot n d^2_{\mathrm{H}\mu}(\pi) - 2\ell(\pi) - 2\log(1/\varepsilon)$. By Markov's inequality and Lemma D.1(d), $\mathbb{P}[\Lambda_n(\pi) \ge 2^{-a(\pi)}] = \mathbb{P}[\sqrt{\Lambda_n(\pi)} \ge 2^{-a(\pi)/2}] \le 2^{a(\pi)/2} e^{-n d^2_{\mathrm{H}\mu}(\pi)} = \varepsilon 2^{-\ell(\pi)}$. A union bound over $\pi$ costs at most $\varepsilon \sum_\pi 2^{-\ell(\pi)} \le \varepsilon$, and on the complement $L_n(\pi) - L_n(\nu) = -\log \Lambda_n(\pi) > a(\pi)$.

*(ii).* Programs shorter than $\ell_0$ are outside $\mathcal{E}$, so on the event of (i) they have $L_n(\pi) - L_n(\nu) > 2\log e \cdot n\gamma_=(\ell(\pi)) - 2\ell(\pi) - 2\log(1/\varepsilon) \ge \sigma_n(\ell_0 - \ell(\pi))$, and the programs of $\mathcal{E}_0$ have fit $L_n(\nu)$, so they bound the point. For the second form, $(A - 2\ell)/(\ell_0 - \ell) \ge A/\ell_0$ iff $\ell(A - 2\ell_0) \ge 0$.

*(iii).* By Lemma D.1(a), $\mathbb{E}[\xi_{\mathcal{C}}(h_n)/\nu(h_n)] = \sum_\pi 2^{-\ell(\pi)}\mathbb{E}\Lambda_n(\pi) \le 1$, so by Markov's inequality, with probability at least $1-\varepsilon$, $2^{-\ell(\pi)}\Lambda_n(\pi) < 1/\varepsilon$ for every $\pi$, that is, $L_n(\pi) > L_n(\nu) - (\ell(\pi) - \ell_0) - (\ell_0 + \log(1/\varepsilon))$. This is at least $L_n(\nu) - \tau(\ell(\pi) - \ell_0)$ once $(\tau - 1)(\ell(\pi) - \ell_0) \ge \ell_0 + \log(1/\varepsilon)$, for every $\tau > 1$ on the same event. The bound for all $n$ is Lemma 6.2; the offset $\ell_0 + \delta^*$ is nonnegative because $\delta^* \ge \delta_n \ge -\ell_0$.

*(iv).* On the event of (i), a program outside $\mathcal{E}$ of length at most $\bar{\ell}$ has $L_n(\pi) - L_n(\nu) > 2\log e \cdot n\gamma_{\bar{\ell}} - 2\bar{\ell} - 2\log(1/\varepsilon) > 0$ by the hypothesis on $n$; programs of $\mathcal{E}$ have fit $L_n(\nu)$. This is the flat part, and every right secant to a length in $(\ell_0, \bar{\ell}]$ is at most 0. Beyond the cap, on the event of (iii), a secant is below $(\ell + \log(1/\varepsilon))/(\ell - \ell_0) \le \tau_{\bar{\ell}}$ for $\ell > \bar{\ell}$. With (ii), $(\ell_0, L_n(\nu))$ is a zero-window $(\sigma_n, \tau_{\bar{\ell}})$-kink with $\tau_{\bar{\ell}} < \sigma_n$, and Corollary 5.4(c) makes it a vertex whose programs are the $\lambda$-MAPs for $\lambda \in (\tau_{\bar{\ell}}, \sigma_n)$; by the flat part these programs form $\mathcal{E}_0$. The probability is at least $1 - 2\varepsilon$.

*(v).* Apply (i) with $\varepsilon_n := n^{-2}$; by the Borel–Cantelli lemma its event holds for all large $n$, and then $s_-(n) \ge \min_{\ell<\ell_0}(2\log e \cdot n\gamma_=(\ell) - 2\ell - 4\log n)/(\ell_0 - \ell)$; divide by $n$. By Theorem 6.3(a) with $\lambda_n \equiv 1$, almost surely the set of 1-MAPs is eventually $\mathcal{E}_0$, whose programs all sit at the point $(\ell_0, L_n(\nu))$. By Lemma 3.8(v), a set of $\lambda$-MAPs at a single point is the set of programs at a vertex $v_i$ with $\sigma_{i+1} < \lambda < \sigma_i$, since at $\lambda = \sigma_i$ both ends of an edge are minimisers. The largest right secant from $v_i$, floored at 0, is $\sigma_{i+1}$ (it is 0 at a last vertex), and by convexity the left steepness at $v_i$ is at least $\sigma_i$. So $s_+(n) = \sigma_{i+1} < 1 < \sigma_i \le s_-(n)$.

*Remark D.3.* In $\mathcal{C}_{k,r_0}$ with $r_0 + 2 > k$ the coin is the only program of length at most $k = \ell_0$, so $\rho(\ell_0) = L_n(\text{coin}) = n$; a consistent parity has fit 0 and length $r_0+2$, so the excess is $n-(r_0+2-k) = k - 7$. Some parity is consistent with probability at least $\mathbb{P}[\text{rank}A_{r_0} = n] \ge 1 - 2^{n-r_0} = 31/32$, and $\mathbb{P}[\delta^* \ge k - 7] \le 2^{7-k}$. □

## D.5 Proof of Corollary 6.8

(i) $\pi \in \mathcal{E}$ iff $\nu_\pi(\cdot \mid x) = \nu_{f,p}(\cdot \mid x)$, for both labels, at $\mu$-a.e. $x$. For $p < \frac{1}{2}$ the labels have probabilities $1 - p > p$, so such a kernel determines $f(x)$ and $p$; apply Theorem 6.3(a). (ii) is Theorem 6.3(c), with $H(\nu_{f,p}(\cdot \mid x)) = H(p)$ at every $x$. (iii) For $p = \frac{1}{2}$, $\nu_{f,1/2}$ is the coin for every $f$. For $p = 0$, $\nu_{f,0}$ is the point mass on $f$, the programs that compute $f$ $\mu$-a.e. are in $\mathcal{E}$, and $L_n(\nu) = 0$. Finally $\nu_{f,p}(1 \mid x) \in \{p, 1-p\} \subseteq [\beta, 1-\beta]$ for every rational $\beta \in (0, p]$. □

*Remark* D.4 (Frozen labels on a finite domain). Let $\mu$ have finite support $X_0$, with frozen labels. The data set is eventually the full table $F$ on $X_0$, so every $\lambda$-MAP is eventually a minimiser of $J_\lambda(\cdot\,; F)$, a function of $F$ alone, and by Proposition 6.10(a) no procedure identifies every such sampler. Among the programs $(f, q)$ with the true $f$, the one of least $J_\lambda$ has the rate $q$ that minimises $\lambda\ell(f, q) + K\log(1/q) + (|X_0| - K)\log(1/(1-q))$, $K$ the number of flipped inputs (a penalised empirical rate); the $\lambda$-MAP is this program unless another function or the coin has smaller $J_\lambda$. In an explicit class (inputs uniform on $\{0,1\}^6$, parities $a$ with rates $q$ from an eight-element list with a prefix code, and a coin; the truth $(a^*, \frac{1}{8})$ has length 10 and $(a^*, 0)$ length 9), on the event that no input is flipped, of probability $(7/8)^{64}$, the program $(a^*, 0)$ has $J_\lambda = 9\lambda < 10\lambda + 64\log(8/7) = J_\lambda(a^*, \frac{1}{8})$; so for every $\lambda$ the $\lambda$-MAP is eventually wrong with probability at least $(7/8)^{64}$.

A direct computation (code in the accompanying repository) gives the exact failure probabilities 0.255, 0.255, 0.204 and 0.566 in this class at $\lambda = 1$, 1.5, 2 and 4, and 0.136, 0.136, 0.085 and 0.566 in a second class, with another rate code and single-input exception programs; per query Corollary 6.8(i) applies to both classes.

## D.6 Frozen labels, adaptive inputs, misspecification and $\lambda = cn$

*The run with frozen labels.* With frozen labels and i.i.d. inputs on the countable $X$, a program of $\mathcal{E}$ equals $\nu$ at every input of the support of $\mu$, which contains every queried input almost surely, so it can serve as $\pi_\nu$ in Theorem 6.9. If the support is infinite, the distinct inputs eventually

enumerate it, and since the terms are nonnegative, $\nu$-equivalence on the run is $\sum_{x\in\operatorname{supp}\mu} d^2_{\mathrm{H}}(\nu(\cdot \mid x), \nu_\pi(\cdot \mid x)) < \infty$. Per query with i.i.d. inputs, $g_i(\pi)$ are i.i.d. in $[0,1]$ with mean $d^2_{\mathrm{H}\mu}(\pi)$, so for $\pi \notin \mathcal{E}$ their sum is infinite almost surely by the strong law, and for $\pi \in \mathcal{E}$ it is 0.

*Proof of Proposition 6.10(a).* With frozen labels the data process is a measurable function of the input sequence and of the table $F$ on the support of $\mu$, drawn once with independent values $F(x) \sim \nu(\cdot \mid x)$. The input sequence has the same law under both samplers and is independent of $F$. The laws of $F$ under $\nu$ and $\nu'$ are products that differ in the single factor at $x_0$, with density $\nu'(F(x_0) \mid x_0)/\nu(F(x_0) \mid x_0) \in (0,\infty)$ almost surely; so the laws of the data process are mutually absolutely continuous. Since $\mu(x_0) > 0$, $\mathcal{E}_\nu \cap \mathcal{E}_{\nu'} = \emptyset$. If $A := \{\hat\pi_n \in \mathcal{E}_\nu \text{ eventually}\}$ has probability 1 under $\nu$, it has probability 1 under $\nu'$, and on $A$ the estimator is eventually outside $\mathcal{E}_{\nu'}$. In $\mathcal{H}_\times$ take $\nu'(\cdot \mid x_0) := \frac{1}{2}(\nu(\cdot \mid x_0) + \mathbf{1}[\cdot = y_0])$, which is computable, distinct from $\nu(\cdot \mid x_0)$ and has the same support.

*(b).* A deterministic $\varphi$ has $d^2_{\mathrm{H}}(\nu(\cdot \mid x), \mathbf{1}[\cdot = \varphi(x)]) = 1 - \sqrt{\nu(\varphi(x) \mid x)} \geq 1 - \sqrt{1-\beta_x} \geq \beta_x/2$, and gap 1 where it is undefined. So no deterministic program is $\nu$-equivalent on the run, and Theorem 6.9(iv) applies.

*(c).* The table is eventually complete. An $e_j$ with $F(j) = 0$ has likelihood 0. An $e_j$ with $F(j) = 1$ fits $\log 64 = 6$ bits better than $\nu$ and is 4 bits longer, so it beats $\nu$ iff $4\lambda < 6$; all such $e_j$ tie. Some $F(j)$ is 1 with probability $1 - (63/64)^{10}$, and otherwise $\nu$'s program is the $\lambda$-MAP.

For the limits of Theorem 6.3 in directions other than frozen labels, put $\mathrm{KL}_\mu(\pi) := \mathbb{E}_{x\sim\mu}\,\mathrm{KL}(\nu(\cdot \mid x) \,\|\, \nu_\pi(\cdot \mid x)) \in [0,\infty]$, the mean KL divergence, and $R_\nu(\pi) := H_\mu(\nu) + \mathrm{KL}_\mu(\pi) = \mathbb{E}_{\mu\otimes\nu}[-\log \nu_\pi(y \mid x)]$, the expected per-query fit.

**Proposition D.5** (Adaptive inputs, misspecification, $\lambda = cn$)**.** *(a)* Adaptive inputs. *Let the inputs be $x_j = 0^j$, let $\nu(1 \mid 0^j) = \frac{1}{2} + 2^{-j/2-2}$ have a program of length $k$, and let the class consist of it and the coin, of length $k - \Delta \geq 1$ with $\Delta > 0$. The coin is $\nu$-equivalent on the run although it differs from $\nu$ at every query, and for every fixed $\lambda > 0$, with positive probability, it is the $\lambda$-MAP for all large $n$.*

*(b)* Misspecification: no fill-ins per query. *Let $\nu$ be a computable $f$ with independent flips at rate $p \in (0, \frac{1}{2})$, per query with i.i.d. inputs, and let the class consist of deterministic programs and one fair coin, so that $\mathcal{E} = \emptyset$. The coin, the unique minimiser of $\mathrm{KL}_\mu$, is almost surely eventually the unique $\lambda$-MAP for every $\lambda > 0$.*

*(c) $\lambda = cn$. Per query, in a class as in Definition 6.1, let $J_c(\pi) := c\,\ell(\pi) + R_\nu(\pi)$ and suppose $J_c(\pi) < \infty$ for some $\pi$. Then $A_c := \arg\min J_c$ is nonempty and finite, and almost surely every $cn$-MAP is in $A_c$ for all large $n$; if the programs of $A_c$ have one kernel $\mu$-a.e., the set of $cn$-MAPs equals $A_c$ for all large $n$. If $\mathcal{E} \neq \emptyset$ and $c^* := \min\{\mathrm{KL}_\mu(\pi)/(\ell_0 - \ell(\pi)) : \ell(\pi) < \ell_0\}$ ($c^* := \infty$ if no program is shorter than $\ell_0$), then almost surely the set of $cn$-MAPs is eventually $\mathcal{E}_0$ for $c < c^*$, and eventually contains no program of $\mathcal{E}_0$ for $c > c^*$; $c = c^*$ is not covered.*

With adaptive inputs the target is $\nu$ on the run, and the selected program is still a sampler (Theorem 6.9(iii)); with $\lambda = cn$ selection follows the population hull of the points $(\ell(\pi), R_\nu(\pi))$ (Lemma 3.8).

*Proof of Proposition D.5(a).* The factors $\nu_{\mathrm{coin}}(y_j)/\nu(y_j \mid 0^j)$ of $\Lambda_n(\mathrm{coin})$ are independent and positive with mean 1, and $\sum_j d^2_{\mathrm{H}}(\nu(\cdot \mid 0^j), \mathrm{coin}) \leq \sum_j 2 \cdot 2^{-j-4} < \infty$. By Kakutani's theorem (Kakutani, 1948) the product converges almost surely and in $L^1$ to a limit $\Lambda_\infty > 0$ with $\mathbb{E}\Lambda_\infty = 1$, so $\mathbb{P}[\Lambda_\infty \geq 1] > 0$. On $\{\Lambda_\infty > 2^{-\lambda\Delta}\}$, eventually $\log \Lambda_n(\mathrm{coin}) > -\lambda\Delta$, that is, $J_\lambda(\mathrm{coin}) < J_\lambda(\nu)$. The Kraft sum is $2^{-k} + 2^{-(k-\Delta)} \leq 1$.

*(b).* The coin has $\mathrm{KL}_\mu = 1 - H(p) < \infty$, and every deterministic program has $\mathrm{KL}_\mu = \infty$ since $p > 0$. An atom of positive mass is queried infinitely often, and its labels are i.i.d. with both values of probability at least $p$, so almost surely some input eventually receives both labels; from then on every deterministic program has likelihood 0, and the coin, with finite $J_\lambda$, is the unique $\lambda$-MAP.

*(c).* $J_c(\pi) \geq c\ell(\pi)$, so only finitely many programs have $J_c$ below any finite value, and $A_c$ is nonempty and finite. Put $\hat{R}_n(\pi) := L_n(\pi)/n$, so $J_{cn}(\pi; h_n)/n = c\ell(\pi) + \hat{R}_n(\pi)$; by the strong law for nonnegative i.i.d. variables, possibly of infinite mean, $\hat{R}_n(\pi) \to R_\nu(\pi)$ almost surely for all $\pi$ at once. Fix $\pi_c \in A_c$. A $cn$-MAP has $cn\,\ell(\hat{\pi}) \leq J_{cn}(\pi_c)$, so $\ell(\hat{\pi}) \leq \ell(\pi_c) + \sup_n \hat{R}_n(\pi_c)/c$, which is finite almost surely because $\hat{R}_n(\pi_c)$ converges to the finite $R_\nu(\pi_c)$. Each of the finitely many candidates outside $A_c$ has $J_{cn}/n \to J_c > \min J_c$, so it is eventually not a MAP. Programs of $A_c$ with one kernel $\mu$-a.e. have equal $R_\nu$, hence equal lengths, and almost surely equal likelihoods, so they tie. If $\mathcal{E} \neq \emptyset$, then for $\pi_0 \in \mathcal{E}_0$, $J_c(\pi) - J_c(\pi_0) = c(\ell(\pi) - \ell_0) + \mathrm{KL}_\mu(\pi)$: on $\mathcal{E}$ it is nonnegative with equality iff $\pi \in \mathcal{E}_0$; off $\mathcal{E}$ with $\ell(\pi) \geq \ell_0$ it is positive; and for $\ell(\pi) < \ell_0$ it is $\mathrm{KL}_\mu(\pi) - c(\ell_0 - \ell(\pi))$, positive if $c < c^*$. So $A_c = \mathcal{E}_0$ for $c < c^*$. If $c > c^*$, some $\pi'$ with $\ell(\pi') < \ell_0$ has $\frac{1}{n}[J_{cn}(\pi_0) - J_{cn}(\pi')] \to c(\ell_0 - \ell(\pi')) - \mathrm{KL}_\mu(\pi') > 0$. □

## D.7 Proof of the Ville lemma

We prove Lemma 6.2. Let $Z_n := \xi(o_n)/\nu(o_n) = 2^{\delta_n}$. With frozen labels and a repeated query, $Z_{n+1} = Z_n$. Otherwise, given the past and the next input $x_{n+1}$, which the rule fixes before $y_{n+1}$ is drawn, $\mathbb{E}[Z_{n+1}] \leq \sum_y \xi(o_n + (x_{n+1}, y))/\nu(o_n) \leq Z_n$, where $o_n + (x, y)$ is the data after the new query: the first step sums over the labels with $\nu(y \mid x_{n+1}) > 0$, and the second is the one-point inequality, which for a shared hypothesis and a new distinct input is Definition 3.4, and for $\xi_\mathcal{C}$ holds termwise because $\sum_y \nu_\pi(y \mid x) \leq 1$, per query or on a new distinct input (Lemma D.1(a)). So $(Z_n)$ is a nonnegative supermartingale with $Z_0 = \xi(\emptyset) \leq 1$, whatever the rule, and $\nu$ enters only in the denominator. Ville's maximal inequality (Ville, 1939) gives $\mathbb{P}[\sup_n Z_n \geq 2^j] \leq 2^{-j}$. Finally $\xi_\mathcal{C} \geq 2^{-\ell(\pi)}\nu_\pi$, so $\nu_\pi(o_n) \leq 2^{\ell(\pi)+\delta_n}\nu(o_n)$. □

## D.8 Fill-ins from stored stream prefixes

**Lemma D.6** (Fill-ins from stored stream prefixes)**.** *Let the class be $\mathcal{H}_\times$, let $\nu$ be a sampler with a program $\pi_\nu$ of length $k$, and let $D$ be a data set with $n$ inputs, all of length at most $d$, with $\beta_D := \min_{(x,y)\in D} \nu(y \mid x) > 0$. Put $i_D := \lceil \log(6n/\beta_D) \rceil$ and $t_D := \lceil \log(2d + 2 + 2\log(i_D + 1)) \rceil$. Then*

$$W_{\det}(D) \;\geq\; 2^{-k-2\log(t_D+1)-\kappa}\,\nu(D).$$

*Since $\beta_D \geq \nu(D)$, $i_D \leq \log(6n) + \log(1/\nu(D)) + 1$; so the cost $2\log(t_D + 1)$ is doubly logarithmic in $d$ and triply logarithmic in $n$ and in the sampled randomness $-\log \nu(D)$.*

*Proof.* Order $Y = \{y^{(1)}, \ldots, y^{(|Y|)}\}$.

*Private streams.* Let $j(x)$ be the length-lexicographic index of $x$, so $j(x) \leq 2^{d+1} - 2$ for $|x| \leq d$, and let $\langle a, b \rangle := (a+b)(a+b+1)/2 + b$ be the Cantor pairing, injective and increasing in $b$. For a finite string $w$ (bits indexed from 0), the *stream prefix* of $x$ in $w$ is $w_{\langle j(x),0\rangle} w_{\langle j(x),1\rangle} \cdots$, continued while the positions are below $|w|$. Distinct inputs read disjoint positions.

*Certification.* For an input $x$ put $F_h := \sum_{h' \leq h} \nu(y^{(h')} \mid x)$, so $F_0 = 0$ and $F_{|Y|} = 1$. At depth $i \geq 1$ compute, by a fixed algorithm from the program for $\nu$, rationals $q_{i,h}$ with $|q_{i,h} - F_h| \leq 2^{-i-2}$ ($q_{i,0} = 0$, $q_{i,|Y|} = 1$). A stream prefix $b_1 \cdots b_i$ names the dyadic interval $I = [0.b_1 \cdots b_i,\ 0.b_1 \cdots b_i +$

$2^{-i}$). It *certifies* $y^{(h)}$ at depth $i$ if $q_{i,h-1} + 2^{-i-2} \le \min I$ and $\sup I \le q_{i,h} - 2^{-i-2}$; then $I \subseteq [F_{h-1}, F_h]$.

*Consistency.* If the stream of $x$ certifies $y^{(h)}$ at depth $i$ and $y^{(h')}$ at depth $i'$ with $h < h'$, the two intervals are nested, since one stream prefix extends the other; the inner one has positive length and lies in $[F_{h-1}, F_h] \cap [F_{h'-1}, F_{h'}]$, which is empty or the single point $F_h$, a contradiction. So $\varphi_w(x)$, the label certified at some depth within the stream prefix of $x$ in $w$ (undefined if none), is a well-defined partial computable function, and an index of it is computable from $(V(\pi_\nu), w)$.

*Probability.* Every depth-$i$ dyadic interval inside $[F_{h-1} + 2^{-i-1}, F_h - 2^{-i-1}]$ certifies $y^{(h)}$, because $q_{i,h-1} + 2^{-i-2} \le F_{h-1} + 2^{-i-1}$ and $q_{i,h} - 2^{-i-2} \ge F_h - 2^{-i-1}$. These intervals cover that segment up to at most $2 \cdot 2^{-i}$, and the segment has length $\nu(y^{(h)} \mid x) - 2^{-i}$. So a uniform stream prefix of length at least $i$ gives $\varphi_w(x) = y^{(h)}$ with probability at least $\nu(y^{(h)} \mid x) - 3 \cdot 2^{-i}$.

*Independence.* For $w$ uniform on $\{0,1\}^L$ the stream prefixes of distinct inputs are independent and uniform. If $L \ge (2^{d+1} + i)^2$, every input of length at most $d$ has a stream prefix of length at least $i$, because $\langle j(x), i-1\rangle < (j(x)+i)^2 \le (2^{d+1}+i)^2$. Take $i = i_D$; then $3 \cdot 2^{-i_D} \le \beta_D/(2n)$, and

$$\Pr_w[\varphi_w \supseteq D] \;\ge\; \prod_{(x,y)\in D} (\nu(y \mid x) - 3 \cdot 2^{-i_D}) \;\ge\; \nu(D)\Big(1 - \frac{1}{2n}\Big)^n \;\ge\; \frac{\nu(D)}{2}.$$

*Programs.* Let $V'$ be the prefix machine that reads $\pi_\nu$ (by simulating $V$), then a self-delimiting code of an integer $t$ of length at most $2\log(t+1)+1$, then $L_t := 2^{2^t}$ bits $w$, and outputs an index of the deterministic conditional semimeasure $\varphi_w$. By closure under prefix-free simulation (Definition 3.5) there is $\sigma$ with $V(\sigma u) = V'(u)$, and the programs $\pi_w := \sigma\, \pi_\nu\, \mathrm{code}(t)\, w$ are distinct for distinct $w$. With $t = t_D$, $2^{t_D} \ge 2d + 2 + 2\log(i_D+1)$, so $L_{t_D} \ge (2^{d+1}(i_D+1))^2 \ge (2^{d+1}+i_D)^2$.

*Sum.*

$$W_{\det}(D) \ge \sum_w 2^{-|\pi_w|} \mathbf{1}[\varphi_w \supseteq D] = 2^{-|\sigma|-k-|\,\mathrm{code}(t_D)|} \Pr_w[\varphi_w \supseteq D] \ge 2^{-k-2\log(t_D+1)-|\sigma|-2}\, \nu(D). \;\square$$

A direct computation (code in the accompanying repository) gives the cost $2\log(t_D+1)$ as 5.6 bits at $d = 20$, $n = 10^3$ ($\beta_D = 1/3$); 6.3 bits at $d = 100$, $n = 10^6$ ($\beta_D = 1/3$ or $2^{-10}$); 7.2 bits at $d = 1000$, $n = 10^{12}$ ($\beta_D = 10^{-3}$); and 8.9 bits at $d = 10^6$, $n = 10^{15}$ ($\beta_D = 1/3$).

## D.9 Solomonoff induction at $\lambda = 1$

For Theorem 6.11: a sampler that gives every label probability at least 1/3, such as the Bern(1/3) threshold hypotheses named after the theorem, has $\beta_n \ge 1/3$ and so $i_n \le \lceil \log(18|\operatorname{dom} D_n|)\rceil$. In SI the analogous bounds hold, and in its representation by deterministic programs every hypothesis is a fill-in:

**Proposition D.7** (SI at $\lambda = 1$)**.** *Let a string $z_{1:n}$ be drawn from a computable measure $\nu$ with a program $\pi_\nu \in \mathcal{H}_{\mathrm{SI}}$ of length $k$, and let $\delta^* := \sup_n \log(M(z_{1:n})/\nu(z_{1:n}))$, where $M = \xi_1$ is the mixture of $\mathcal{H}_{\mathrm{SI}}$. (i) $\Pi_1(\pi_\nu \mid z_{1:n}) \ge 2^{-k-\delta^*}$ for all $n$, and $\mathbb{P}[\delta^* \ge j] \le 2^{-j}$. (ii) The deterministic hypotheses consistent with $z_{1:n}$ (point masses on computable infinite extensions) have total weight at least $2^{-k-K(n)-\kappa}\nu(z_{1:n})$. (iii) In the deterministic-program representation of $M$ every hypothesis is a fill-in; the sampler appears only as the programs "decoder for $\nu$, then a code of $z$", of total weight at least $2^{-k-\kappa}\nu(z)$.*

*Proof.* (i) Under $\nu$, $M(z_{1:n})/\nu(z_{1:n})$ is a nonnegative supermartingale with initial value at most 1, because $M(z0) + M(z1) \le M(z)$; Ville's inequality gives the tail, and $\Pi_1(\pi_\nu \mid z_{1:n}) = 2^{-k}\nu(z_{1:n})/M(z_{1:n})$. (ii) Let $w := z_{1:n}0^\infty$ and $\mathbf{1}_w$ the point mass on its prefixes. Given $n$ and

$\pi_\nu$, the restriction of $\nu$ to $\{0,1\}^n$ is a computable measure, so (F2) gives $K(z_{1:n} \mid n, \pi_\nu) \le -\log \nu(z_{1:n}) + \kappa$, and (F1), (F3) give $K(z_{1:n}) \le K(n) + k - \log \nu(z_{1:n}) + \kappa$. An index of $\mathbf{1}_w$ is computable from $z_{1:n}$, so the class contains a program for $\mathbf{1}_w$ of length at most $K(z_{1:n}) + \kappa$, consistent with $z_{1:n}$. (iii) The programs "a fixed decoder, a shortest program for $\nu$, then a Shannon–Fano code of $z$ under $\nu$" have total weight at least $2^{-k-\kappa}\nu(z)$; this is the usual proof of $M \ge 2^{-K(\nu)-O(1)}\nu$ (Li and Vitányi, 2008). □

*Remark* D.8 (No split in SI's program representation). Solomonoff's $M$ is a mixture over deterministic programs. Tempering does not change deterministic hypotheses (Lemma A.1(iii)), and a power prior $2^{-\lambda\ell(\pi)}$ charges sampled bits and model bits alike, so it is indifferent between "decoder for $\nu$ plus a code of the randomness" and a literal fill-in of the same length. The split needs a representation in which the randomness is not part of the program: a product hypothesis with private coins.

## D.10 Proof of the private-randomness theorem

We prove Theorem 6.9. Write $o = o_n$ and $\Lambda_n(\pi) := \nu_\pi(o_n)/\nu(o_n)$, so that $\Lambda_n(\pi_\nu) = 1$ for all $n$ almost surely.

*(i).* $J_\lambda(\hat\pi; o) \le J_\lambda(\pi_\nu; o) = \lambda k + L_n(\nu)$, and by Lemma 6.2, $L_n(\hat\pi) \ge L_n(\nu) - \ell(\hat\pi) - \delta^*$. Substituting gives $(\lambda-1)\ell(\hat\pi) \le \lambda k + \delta^*$. The upper bound on the fit is $J_\lambda(\hat\pi; o) \le J_\lambda(\pi_\nu; o)$ with $\lambda\ell(\hat\pi) \ge 0$.

*(ii).* $\sum_{\ell(\pi)>\bar\ell} 2^{-\lambda\ell(\pi)}\nu_\pi(o) \le 2^{-(\lambda-1)\bar\ell}\xi_{\mathcal{C}}(o) \le 2^{-(\lambda-1)\bar\ell+\delta^*}\nu(o)$, and the normaliser $\xi_\lambda(o)$ is at least the sampler's term $2^{-\lambda k}\nu(o)$.

*(iii).* A fill-in $\pi$ has $1 = \nu_\pi(o) \le 2^{\ell(\pi)+\delta^*}\nu(o)$, which is the length bound. Summing over fill-ins, $\sum 2^{-\lambda\ell(\pi)} = \sum 2^{-\ell(\pi)}2^{-(\lambda-1)\ell(\pi)} \le 2^{-(\lambda-1)(L_n(\nu)-\delta^*)}\sum 2^{-\ell(\pi)}$, with $\sum 2^{-\ell(\pi)} \le \xi_{\mathcal{C}}(o) \le 2^{\delta^*}\nu(o)$ and a normaliser of at least $2^{-\lambda k}\nu(o)$. For the tempered posterior the fill-ins' weight is at most $2^{\delta^*}\nu(o)$ and the normaliser at least $2^{-k}\nu(o)^\eta$.

*(iv), bounded length, every $\lambda \ge 1$.* A $\lambda$-MAP has $J_\lambda(\hat\pi; o) \le J_\lambda(\pi_\nu; o)$, so $\log\Lambda_n(\hat\pi) \ge \lambda(\ell(\hat\pi) - k)$. For $\bar\ell \ge k$ and $\ell(\hat\pi) > \bar\ell$ this and $\lambda \ge 1$ give $\log\Lambda_n(\hat\pi) \ge \ell(\hat\pi) - k$, hence $Z_n^{\bar\ell} \ge 2^{-k}$. By Lemma D.1(a), which holds for every input rule and for both kinds of labels, $\mathbb{P}[\exists n : Z_n^{\bar\ell} \ge 2^{-k}] \le 2^k t_{\bar\ell} \to 0$, and these events decrease in $\bar\ell$. So almost surely some finite $\bar\ell_0$ bounds every $\lambda$-MAP at every $n$. (For $\lambda > 1$ this also follows from (i).)

*(iv), MAPs.* By Lemma D.1(b) and countability (with frozen labels along the distinct inputs; $\Lambda$ is constant at a repeated query), almost surely $\Lambda_n(\pi) \to 0$ simultaneously for every $\pi$ that is not $\nu$-equivalent on the run. A $\lambda$-MAP has $\log\Lambda_n(\hat\pi) \ge -\lambda k$, and each of the finitely many such programs of length at most $\bar\ell_0$ eventually has $\Lambda_n < 2^{-\lambda k}$, so eventually none is a $\lambda$-MAP.

*(iv), posterior.* Dividing by $\nu(o)$, the normaliser is at least $2^{-\lambda k}$, so for every $\bar\ell$

$$\Pi_\lambda(\text{not } \nu\text{-equivalent on the run} \mid o_n) \;\le\; 2^{\lambda k}\Big[\sum_{\ell(\pi)\le\bar\ell,\ \text{not equivalent}} 2^{-\lambda\ell(\pi)}\Lambda_n(\pi) + Z_n^{\bar\ell}\Big].$$

The finite sum tends to 0. Choose $\bar\ell_j$ with $t_{\bar\ell_j} \le 4^{-j}$; by Lemma D.1(a) and the Borel–Cantelli lemma, almost surely $\sup_n Z_n^{\bar\ell_j} < 2^{-j}$ for all large $j$, so the lim sup is at most $2^{\lambda k - j}$ for every large $j$. □

# E Proofs for Section 7 (the noisy random parity)

Throughout, the setting is Definition 7.1, and (F1)–(F5) are the standard facts of Appendix A.1. In Appendices E.1 and E.2, for $z \in \{0,1\}^m$ and $w \in \{0,1\}^{d-m}$ we write $(z,w)$ for the input with prefix $z$, and $w$ is uniform on $\{0,1\}^{d-m}$ in every expectation $\mathbb{E}_w$; from Appendix E.3 on, $z$ is the table, as in Section 7.3. Since $a \in \{0,1\}^d$, the length $d$ can be computed from $a$. We write $\tau_B := 2^{1-B}$ and $\beta_B := 2^m \log(1+\tau_B)$, both 0 if $B = \infty$, so that $G_j = 1 - \log(1 + 2^{-j} + 2p + \tau_B)$ and $\Lambda_1 = 2\log d + 2\log m + 2\log B \cdot \mathbf{1}[B < \infty] + \beta_B + \kappa$.

## E.1 List decoding

*Proof of Lemma 7.2.* Write $h_y(x) := \nu_\pi(y \mid x)$, $s_h := h_0 - h_1 \in [-1,1]$, $\chi_{a'}(x) := (-1)^{\langle a', x\rangle}$ and $\hat{s}(a') := \mathbb{E}_x[s_h \chi_{a'}]$. Since $\nu_\pi(f_{a'}(x) \mid x) = \frac{1}{2}(h_0 + h_1) + \frac{1}{2}\chi_{a'}(x) s_h(x)$, the agreement with $f_{a'}$ is

$$u(\pi; a') := \mathbb{E}_x \nu_\pi(f_{a'}(x) \mid x) = \tfrac{1}{2}\big(\mathbb{E}_x[h_0 + h_1] + \hat{s}(a')\big) \le \tfrac{1}{2}\big(1 + \hat{s}(a')\big).$$

The characters are orthonormal under the uniform law on $\{0,1\}^d$, so Parseval's identity gives $\sum_{a'} \hat{s}(a')^2 = \mathbb{E}_x s_h^2 \le 1$. Hence $A_t := \{a' : u(\pi; a') > \theta_t\} \subseteq \{a' : \hat{s}(a') > 2^{-t}\}$ has fewer than $2^{2t}$ elements, the list-size bound for the Hadamard code (Goldreich and Levin, 1989; Kushilevitz and Mansour, 1993). The value $u(\pi; a')$ is a finite average of lower semicomputable values, so it is lower semicomputable uniformly in $(\pi, a', d)$, and $A_t$ is computably enumerable given $(\pi, d, t)$. If $\mathrm{lev}(\pi) = t$, then $a \in A_t$. So $a$ is described, given $\pi$, by $d$, by $t$, and by its index in the enumeration of $A_t$, which takes $2t$ bits given $t$. By (F5), $K(a \mid \pi) \le 2t + 2\log(t+1) + 2\log d + \kappa = s(t) + 2\log d + \kappa$. □

The argument uses the lower-semicomputable agreement $u$ rather than the correlation $\hat{s}$, which is not computable for semimeasures; this extends the lemma of the earlier report (Claude (Anthropic), 2026), for computable hypotheses, to all product hypotheses.

## E.2 The risk floors, the coding lemma and the two-part bounds

*Proof of Lemma 7.3.* Since $\min\{B, -\log v\} = -\log\max\{v, 2^{-B}\}$ and $-\log$ is convex, Jensen's inequality gives

$$R_B(\pi) \ge -\log \mathbb{E}_x \max\{\nu_\pi(y^*(x) \mid x), 2^{-B}\} \ge -\log\big(\mathbb{E}_x \nu_\pi(y^*(x) \mid x) + 2^{-B}\big).$$

Off $E$, $y^* = f_a$; on $E$, $\nu_\pi(y^* \mid x) \le 1$, and $\Pr[E] = p$. So $\mathbb{E}_x \nu_\pi(y^*(x) \mid x) \le u(\pi) + p$. If $\mathrm{lev}(\pi) = t \ge 2$, then $u(\pi) \le \theta_{t-1} = \frac{1}{2} + 2^{-t}$, and $\frac{1}{2} + 2^{-t} + p + 2^{-B} = \frac{1}{2}\big(1 + 2^{-(t-1)} + 2p + \tau_B\big)$, which gives $R_B \ge G_{t-1}$. If $\mathrm{lev}(\pi) = \infty$, then $u \le \frac{1}{2}$ and $R_B \ge G_\infty$. □

The floor is tight as a function of $u$ alone: a hypothesis that gives the teacher's label probability 1 on $E$ and is uniform elsewhere has $u = \frac{1-p}{2}$ and $\mathbb{E}_x \nu(y^* \mid x) = u + p$; it knows $S$. Lemma 7.6 is the semimeasure version of a lemma of the earlier report (Claude (Anthropic), 2026) for computable hypotheses.

*Proof of Lemma 7.6.* For $z \in \{0,1\}^m$ let $A(z) := \mathbb{E}_w \nu_\pi(f_a(z,w) \mid (z,w))$ and $V(z) := \mathbb{E}_w \nu_\pi(f_a(z,w) \oplus 1 \mid (z,w))$. Then $A + V \le 1$, $u = 2^{-m}\sum_z A(z)$, and $\bar{v} := 2^{-m}\sum_z V(z) \le 1 - u$. Off $E$ the teacher's label is $f_a$ and on $E$ it is $f_a \oplus 1$, so

$$\mathbb{E}_x \nu_\pi(y^*(x) \mid x) = 2^{-m}\Big[\sum_{z \notin S} A(z) + \sum_{z \in S} V(z)\Big] = u - p\,\mathrm{avg}_S A + p\,\mathrm{avg}_S V. \tag{2}$$

*Information budget.* By definition $K(S \mid a,\pi) = B_S - I(\pi) = L_S - \delta_S - I(\pi)$. By (F3), $B_S = K(S \mid a, K(a)) \le K(\pi \mid a, K(a)) + K(S \mid a,\pi) + \kappa$, so $I(\pi) \le K(\pi \mid a, K(a)) + \kappa \le \ell(\pi) + \kappa$ by (F1).

*Coding step.* Let $j$ satisfy $j - 2\log(j+1) > \delta_S + I + 1 + 2\log m + 2\log s_0 + g + 2\log g + \kappa$ for some $I \ge I(\pi)$; the least such $j$ for $I = I(\pi)$ is the one of the statement. Put $\gamma := \sqrt{j \ln 2/(2s_0)}$. Suppose $\mathcal{B}$ is a set of $s_0$-subsets of $\{0,1\}^m$ with $|\mathcal{B}| \le 2^{L_S - j}$ that contains $S$ and is computably enumerable given $(\pi, a, m, s_0)$ and a parameter $\theta$ with $K(\theta) \le g + 2\log g + \kappa$. Then $S$ is described by $j$, $m$, $s_0$, $\theta$ and its index among the first $2^{\lceil L_S \rceil - j}$ enumerated elements, so $K(S \mid a,\pi) \le \lceil L_S \rceil - j + 2\log(j+1) + 2\log m + 2\log s_0 + g + 2\log g + \kappa$. With the budget this gives $j - 2\log(j+1) \le \delta_S + I(\pi) + 1 + 2\log m + 2\log s_0 + g + 2\log g + \kappa$, which the choice of $j$ excludes. So no such $\mathcal{B}$ exists.

*The $V$ side.* If $1 - u + \gamma \ge 1$, then $\operatorname{avg}_S V \le 1 \le 1 - u + \gamma$. Otherwise let $\theta \le 1$ be the least multiple of $2^{-g}$ with $\theta \ge 1 - u + \gamma$; then $K(\theta) \le g + 2\log g + \kappa$ and $\theta \ge \bar{v} + \gamma$. If $\operatorname{avg}_S V > \theta$, the set $\mathcal{B} := \{S' : |S'| = s_0,\ \operatorname{avg}_{S'} V > \theta\}$ contains $S$ and is computably enumerable, because $V$ is lower semicomputable. For $S'$ uniform among the $s_0$-subsets, Hoeffding's inequality for sampling without replacement (Hoeffding, 1963) gives $\Pr[\operatorname{avg}_{S'} V - \bar{v} \ge \gamma] \le e^{-2s_0\gamma^2} = 2^{-j}$, so $|\mathcal{B}| \le 2^{L_S - j}$, which the coding step excludes. Hence $\operatorname{avg}_S V \le 1 - u + \gamma + 2^{-g}$.

*The $A$ side.* $A$ is only lower semicomputable, so a set of the form "$\operatorname{avg}_{S'} A$ is small" need not be enumerable; we use a finite stage. Let $\nu_s \uparrow \nu_\pi$ be the stage-$s$ lower approximations, $A_s(z) := \mathbb{E}_w \nu_s(f_a(z,w) \mid (z,w))$ and $u_s := 2^{-m}\sum_z A_s(z) \uparrow u$. If $u \le 2^{-g}$, then $\operatorname{avg}_S A \ge 0 > u - 2^{-g} - \gamma$. Otherwise let $\theta'$ be the largest multiple of $2^{-g}$ strictly below $u$, so $\theta' \ge u - 2^{-g}$ and $K(\theta') \le g + 2\log g + \kappa$, and let $s^*$ be the first stage with $u_{s^*} > \theta'$. Then $A_{s^*}$ is computable from $(\pi, a, m, \theta')$. If $\operatorname{avg}_S A_{s^*} < u_{s^*} - \gamma$, the set $\mathcal{B} := \{S' : |S'| = s_0,\ \operatorname{avg}_{S'} A_{s^*} < u_{s^*} - \gamma\}$ is decidable from these data and contains $S$; the lower-tail form of the same inequality gives $|\mathcal{B}| \le 2^{L_S - j}$, which the coding step excludes. Hence $\operatorname{avg}_S A \ge \operatorname{avg}_S A_{s^*} \ge u_{s^*} - \gamma > u - 2^{-g} - \gamma$.

*Combination.* By (2), $\mathbb{E}_x \nu_\pi(y^* \mid x) \le u - p(u - 2^{-g} - \gamma) + p(1 - u + \gamma + 2^{-g}) = p + (1-2p)u + p(2\gamma + 2^{1-g})$. Conclude with Jensen's inequality as in Lemma 7.3. □

*Monotonicity of the sharp floor.* The function $f(j) := j - 2\log(j+1)$ is at most 0 for $j \le 5$ and increasing for $j \ge 2$. So the least integer $j$ with $f(j) > T$ is 0 for $T < 0$ and the least $j \ge 6$ with $f(j) > T$ for $T \ge 0$; it is nondecreasing in $T$. The threshold in $\gamma_g(I)$ increases with $I$, so $\gamma_g(I)$ is nondecreasing in $I$, each bracket in the definition of $F_t^\sharp(I)$ is nonincreasing in $I$, and so is their supremum. Each bracket is nondecreasing in $t$, because $p < \frac{1}{2}$. Since the coding step allows every $I \ge I(\pi)$, a program with $\operatorname{lev}(\pi) > t$ and $I(\pi) \le I$ has $R(\pi) \ge F_t^\sharp(I)$ at $B = \infty$, by the lemma with $u(\pi) \le \theta_t$ and $p + (1-2p)\theta_t = \frac{1}{2}(1 + (1-2p)2^{-t})$.

*Proof of Lemma 7.4.* For $z \in \{0,1\}^m$ and $b \in \{0,1\}$ let

$$q_z(b) := 2^{-\mathbb{E}_w \min\{B,\, -\log \nu_\pi(f_a(z,w) \oplus b \mid (z,w))\}}.$$

By Jensen's inequality for the convex function $t \mapsto 2^{-t}$, $q_z(b) \le \mathbb{E}_w \max\{\nu_\pi(f_a(z,w) \oplus b \mid (z,w)), 2^{-B}\}$, so $\sum_b q_z(b) \le 1 + \tau_B$. Each $q_z(b)$ is lower semicomputable from $(\pi, a, m, B)$, because the exponent is an average of upper semicomputable values. Define, on subsets $S' \subseteq \{0,1\}^m$, $Q(S') := \prod_{z \in \{0,1\}^m} q_z(\mathbf{1}_{S'}(z))/(1+\tau_B)$, normalised factor by factor. Its total mass is $\prod_z \sum_b q_z(b)/(1+\tau_B) \le 1$, so $Q$ is a lower-semicomputable semimeasure given $(\pi, a, m, B)$. Since $y^*(z,w) = f_a(z,w) \oplus \mathbf{1}_S(z)$,

$$-\log Q(S) = \sum_z \mathbb{E}_w \min\{B, -\log \nu_\pi(y^*(z,w) \mid (z,w))\} + 2^m \log(1+\tau_B) = 2^m R_B(\pi) + \beta_B.$$

By (F2), $K(S \mid \pi, a, m, B) \le 2^m R_B(\pi) + \beta_B + \kappa$, and by (F3) and (F5) the descriptions of $m$ and of the integer $B$ cost at most $2\log m + 2\log B + \kappa$ more. For $B = \infty$ the terms in $B$ vanish. □

Normalising once by $1 + \tau_B$, not per factor, does not in general give a semimeasure.

*Proof of Lemma 7.5.* (D1). By (F3), (F1) and Lemma 7.2, $d \le K(a) \le K(\pi) + K(a \mid \pi) + \kappa \le \ell(\pi) + s(t) + 2\log d + \kappa$. All terms of $\Lambda_1$ other than $2\log d + \kappa$ are nonnegative. (D2). By symmetry of information (F4), $K(a, S) = K(a) + K(S \mid a, K(a)) + O(1) \ge d + B_S - \kappa$. By (F3), $K(a, S) \le K(\pi) + K(a \mid \pi) + K(S \mid a, \pi) + \kappa$. Insert (F1), Lemma 7.2 and Lemma 7.4: $d + B_S \le \ell(\pi) + s(t) + 2\log d + 2^m R_B(\pi) + \beta_B + 2\log m + 2\log B \cdot \mathbf{1}[B < \infty] + \kappa$. □

### E.3 The full table

Here $m = d$, $B = \infty$ and $p < 1/4$. *Minimum-distance decoding.* Two distinct parities differ on exactly $N/2$ inputs and $|E| = pN < N/4$, so $f_a$ is the unique parity within Hamming distance $N/4$ of $z$, and $E = \{x : z(x) \ne f_a(x)\}$; $d$ is determined by $|z| = N$. So $K(a, E) \le K(z) + \kappa$, and by (F4) $K(a, E) = K(a) + K(E \mid a, K(a)) + O(1) \ge d + B_E - \kappa$. Hence

$$K(z) \;\ge\; d + L_E - \delta_E - \kappa. \tag{3}$$

Also $0 \le NH(p) - L_E \le \log(N+1)$, by Definition 7.1(b), and $\operatorname{dom} D = \{0,1\}^d$ and $d$ are computable from each other, so by (F3) and (F5)

$$K(y_D \mid \operatorname{dom} D) \;\ge\; K(z \mid d) - \kappa \;\ge\; K(z) - 2\log(d+1) - \kappa \;\ge\; d + L_E - \delta_E - 2\log(d+1) - \kappa. \tag{4}$$

*Proof of Proposition 7.7.* (i) Since $|E| = pN$ exactly, $L(\pi_g; D) = NH(p)$. By (4), $\ell(\pi_g) + L(\pi_g; D) \le d + \kappa_g + NH(p) \le K(y_D \mid \operatorname{dom} D) + b_0$.

(ii) The point $(\ell(\pi_g), NH(p))$ is bounded by $\pi_g$. Since $u(\pi_g) = 1 - p > \frac{3}{4}$, $\pi_g$ has level 1, and (D1) with $s(1) = 4 \le s(t_0)$ gives $w \ge 0$; $w \le \kappa_g + s(t_0) + \Lambda_1$ because $\ell(\pi_g) \le d + \kappa_g$. Let $\ell(\pi) < d - s(t_0) - \Lambda_1$. A program of level $t' \le t_0$ has $\ell(\pi) \ge d - s(t') - \Lambda_1 \ge d - s(t_0) - \Lambda_1$ by (D1), since $s$ is increasing; so $\operatorname{lev}(\pi) = t + 1$ for an integer $t \ge t_0$, or $\operatorname{lev}(\pi) = \infty$. In the first case (D1) gives $\ell(\pi) \ge (d - s(t+1) - \Lambda_1)^+$, so $0 < \ell(\pi_g) - \ell(\pi) \le d + \kappa_g - (d - s(t+1) - \Lambda_1)^+ = \operatorname{den}_t$; Lemma 7.3 gives $R(\pi) \ge G_t$; and since $\operatorname{lev}(\pi) > t$ and $I(\pi) \le \ell(\pi) + \kappa < d + \kappa$, the sharp floor (Appendix E.2) gives $R(\pi) \ge F_t^\sharp(d + \kappa)$. Hence

$$L(\pi; D) - NH(p) \;\ge\; N(\Gamma_t^\circ - H(p)) \;\ge\; \sigma^{(t_0)} \operatorname{den}_t \;\ge\; \sigma^{(t_0)}(\ell(\pi_g) - \ell(\pi)).$$

In the second case $R(\pi) \ge \max\{G_\infty, F_\infty^\sharp(d+\kappa)\} = \Gamma_\infty^\circ \ge \Gamma_t^\circ$ for every $t$, because both floors are nondecreasing in $t$; with any integer $t \ge \max(t_0, T)$, where $\operatorname{den}_t = \kappa_g + d \ge \ell(\pi_g) - \ell(\pi)$, the same chain holds. This is the left kink.

(iii) is Corollary 5.5(b) with $\pi_s = \pi_g$, $b = b_0$, $(\alpha_{\mathrm{r}}, L_{\mathrm{r}}) = (\ell(\pi_g), NH(p))$, $\sigma = \sigma^{(t_0)}$ and window $w$. □

*Proof of Proposition 7.8.* Let $\hat\pi$ be a $\lambda$-MAP; it exists by Lemma A.1(i), and $J_\lambda(\hat\pi; D) \le J_\lambda(\pi_g; D) \le \lambda(d + \kappa_g) + NH(p)$.

*(b), upper.* Corollary 5.5(a) with $\pi_s = \pi_g$ and $b = b_0$ (Proposition 7.7(i)) gives $\ell(\hat\pi) \le \ell(\pi_g) + (b_0 + \kappa)/(\lambda - 1) \le \bar\ell_\lambda$.

*(a).* Suppose $\operatorname{lev}(\hat\pi) = t + 1$ with an integer $t \ge t_0$, or $\operatorname{lev}(\hat\pi) = \infty$, and then put $t := \infty$. By (D1), $\ell(\hat\pi) \ge (d - s(t+1) - \Lambda_1)^+$ (read as 0 for $t = \infty$). By (b), $I(\hat\pi) \le \ell(\hat\pi) + \kappa \le \bar\ell_\lambda + \kappa$, so Lemma 7.3 and the sharp floor give $R(\hat\pi) \ge \Gamma_t(\lambda)$. Since $(d - x)^+ + \min\{d, x\} = d$ for $x \ge 0$,

$$J_\lambda(\hat\pi; D) \;\ge\; \lambda(d - s(t+1) - \Lambda_1)^+ + N\Gamma_t(\lambda) \;=\; \lambda(d + \kappa_g) - \lambda \operatorname{den}_t + N\Gamma_t(\lambda).$$

If $t \le \max(t_0, T)$, then $C_{t_0}(\lambda)$ makes the right side exceed $\lambda(d+\kappa_g) + NH(p) \ge J_\lambda(\hat\pi; D)$, a contradiction. If $t > t' := \max(t_0, T)$, then $s(t'+1) + \Lambda_1 \ge d$, so $\mathrm{den}_t = \mathrm{den}_{t'} = \kappa_g + d$, and $\Gamma_t(\lambda) \ge \Gamma_{t'}(\lambda)$ because both floors are nondecreasing in $t$; the condition at $t'$ gives the same contradiction. So $\mathrm{lev}(\hat\pi) \le t_0$ and $u(\hat\pi) > \theta_{t_0}$, and Lemma 7.2, with $s$ increasing, bounds $K(a \mid \hat\pi)$.

*(b), lower.* (D1) at the level $\mathrm{lev}(\hat\pi) \le t_0$.

*(c).* By (b), $L(\hat\pi; D) \le J_\lambda(\pi_g; D) - \lambda\ell(\hat\pi) \le NH(p) + \lambda(d + \kappa_g - d + s(t_0) + \Lambda_1)$. The lower bound is Corollary 5.5(a).

*(d).* By (b), $\ell(\hat\pi) + L(\hat\pi; D) = J_\lambda(\hat\pi; D) - (\lambda-1)\ell(\hat\pi) \le \ell(\pi_g) + L(\pi_g; D) + (\lambda-1)(\ell(\pi_g) - d + s(t_0) + \Lambda_1)$; apply Proposition 7.7(i) and $\ell(\pi_g) \le d + \kappa_g$.

*(e).* By (F3), $B_E \le K(\hat\pi \mid a, K(a)) + K(E \mid a, \hat\pi) + \kappa$, so $I(\hat\pi) \le K(\hat\pi \mid a, K(a)) + \kappa$. By (F4) and (F3), $K(\hat\pi \mid a, K(a)) = K(\hat\pi, a) - K(a) + O(1) \le K(\hat\pi) + K(a \mid \hat\pi) - K(a) + \kappa$. Insert (F1), the bound of (a), $K(a) \ge d$ and (b).

*(f).* Given $x \in E$, $x$ is uniform on $E$ and $y^*(x) = f_a(x) \oplus 1$, so $e(\hat\pi) = \mathrm{avg}_E V$ with $V$ as in the proof of Lemma 7.6 ($m = d$). The $V$ side of that proof, with the budget $I_e \ge I(\hat\pi)$ of (e), gives $\mathrm{avg}_E V \le 1 - u(\hat\pi) + \gamma_g(I_e) + 2^{-g}$.

*(g).* By (c), $R(\hat\pi) = L(\hat\pi; D)/N \le R_{\max}$. Lemma 7.3 at $B = \infty$ gives $R(\hat\pi) \ge -\log(u(\hat\pi) + p)$, so $u(\hat\pi) \ge 2^{-R_{\max}} - p$. Lemma 7.6 at $B = \infty$, with the budget $I_e$, gives $R(\hat\pi) \ge -\log\big(p + (1-2p)u(\hat\pi) + p(2\gamma_g(I_e) + 2^{1-g})\big)$ for every $g \ge 1$; solve for $u(\hat\pi)$, using $1 - 2p > 0$. The term $\theta_{t_0}$ is (a).

*(ii).* Three facts. ($\alpha$) $\bar\ell_\lambda$ is nonincreasing in $\lambda$, because $b_0 + \kappa \ge 0$: Lemma 3.9 at $\pi_g$ gives $\ell(\pi_g) + L(\pi_g; D) \ge K(y_D \mid \mathrm{dom}\, D) - \kappa$, and (i) of Proposition 7.7 gives $\le K(y_D \mid \mathrm{dom}\, D) + b_0$. ($\beta$) $F_t^\sharp(I)$ is nonincreasing in $I$ (Appendix E.2). ($\gamma$) $\mathrm{den}_t \ge 0$: in $\mathcal{H}_\times$, $\kappa_g \ge 0$; in another class $\mathrm{den}_t = \ell(\pi_g) \ge 0$ if the minimum is $d$, and otherwise (D1) for $\pi_g$ gives $\mathrm{den}_t \ge s(t+1) - 4 > 0$. By ($\alpha$) and ($\beta$), $\Gamma_t(\lambda)$ is nondecreasing in $\lambda$; so for $\lambda \in [\lambda_{\mathrm{lo}}, \lambda_{\mathrm{hi}}]$ and $t = t_0, \dots, \max(t_0, T)$, by ($\gamma$), $\lambda\,\mathrm{den}_t \le \lambda_{\mathrm{hi}}\mathrm{den}_t < N(\Gamma_t(\lambda_{\mathrm{lo}}) - H(p)) \le N(\Gamma_t(\lambda) - H(p))$.

*(iii).* The flip rate $p$ is fixed, so $K(p)$ is a constant, and $pN$ is an integer once $2^d$ is a multiple of the denominator of $p$. *The budget is $O(d)$:* $\kappa_g = O(\log d)$, and $b_0 = O(d)$ by $NH(p) - L_E \le \log(N+1)$ and $\delta_E = O(d)$, so $\bar\ell_\lambda = O(d)$. In $F_t^\sharp(\bar\ell_\lambda + \kappa)$ the threshold is then $O(d+g)$, since $\log s_0 \le d$, so $j = O(d+g)$ and $\gamma_g = O(\sqrt{(d+g)/(p2^d)})$. *The floors converge:* with $g := d$, $\epsilon_d := 2\gamma_g(\bar\ell_\lambda + \kappa) + 2^{1-g} \to 0$, and since $\log(y+x) \le \log y + x \log e$ for $y \ge 1$, $x \ge 0$, every $t$ has $F_t^\sharp(\bar\ell_\lambda + \kappa) \ge f_t(p) - 2p\,\epsilon_d \log e$, where $f_t(p) := 1 - \log(1 + (1-2p)2^{-t})$ is increasing in $t$. *Conclusion:* for $t_0 \in \{1, 2\}$ and $t \ge t_0$, $N(\Gamma_t(\lambda) - H(p)) \ge N\big(f_{t_0}(p) - H(p) - 2p\,\epsilon_d \log e\big)$, while $\lambda\,\mathrm{den}_t \le \lambda(\kappa_g + d) = O(d)$. So $C_{t_0}(\lambda)$ holds for all large $d$ whenever $f_{t_0}(p) > H(p)$. For $t_0 = 2$ this holds on all of $(0, \frac14]$ (the computation reported after the proposition). For $t_0 = 1$, $f_1(p) = 1 - \log(\frac32 - p)$, and $f_1(p) > H(p)$ is the hypothesis. □

By Proposition 7.8(ii), finitely many cells certify $C_{t_0}(\lambda)$ on an interval of $\lambda$; a direct computation (code in the accompanying repository) of such covers, in high-precision arithmetic, gives the intervals of Table 5, with $\delta_E = 0$. For $p = 2^{-10}$, where the floors $G_t$ alone exceed $H(p)$, the range reaches $\lambda \to 1$, and at $d = 20$ it agrees to the printed digits with the window of Corollary 8.11. For $p \ge 1/16$, at the levels $t_0$ of the table, the condition fails near $\lambda = 1$, where the budget $\bar\ell_\lambda$ grows like $(b_0 + \kappa)/(\lambda - 1)$; there the proposition says nothing. With the machine constant set to 0 (an illustration), at $d = 20$, $p = 2^{-10}$ and $\delta_E = 0$, the same computation gives the identification bounds $u \ge 0.9928$, $\ell \le d + 45.7$, $I \le 58.4$, $e \le 0.209$ at $\lambda = 2$, and $u \ge 0.9903$, $\ell \le d + 15.6$, $I \le 28.3$, $e \le 0.184$ at $\lambda = 105$.

The same computation covers all 40 cases $p \in \{\frac{1}{16}, \frac18, \frac{3}{16}, \frac{7}{32}, \frac{15}{64}\}$, $d \in \{20, 30, 60, 100\}$, $\kappa \in \{0, 30\}$: in each a certified range exists with $t_0 \le 2$, and for $d \ge 30$ its lower end is below $1 + 10^{-3}$.

| $p$ | $d$ | $\kappa$ | $t_0$ | floors $G_t$ alone | with the sharp floor |
|---|---|---|---|---|---|
| $2^{-10}$ | 20 | 0 | 1 | (1, 11944.9] | (1, 12008.5] |
| $2^{-10}$ | 20 | 30 | 1 | (1, 6456.2] | (1, 6486.2] |
| $2^{-10}$ | 30 | 0 | 1 | $(1, 9.978 \cdot 10^6]$ | $(1, 1.0047 \cdot 10^7]$ |
| 1/16 | 20 | 0 | 1 | – | [1.00049, 4276] |
| 1/8 | 20 | 0 | 2 | – | [1.00059, 6529.7] |
| 15/64 | 20 | 0 | 2 | – | [1.08656, 465.39] |
| 15/64 | 20 | 30 | 2 | – | [1.3422, 199.8] |

Table 5: Certified ranges of $\lambda$ on which every $\lambda$-MAP of the full table satisfies Proposition 7.8(i); $\delta_E = 0$, and $\kappa_g = 2\log d + 2\log r + \kappa$ for $p = 2^{-r}$, $2\log d + 2\log(h+1) + 2\log(e+1) + \kappa$ for $p = h/2^e$ with $h > 1$ odd. Rows with $\kappa = 0$ set the machine constant to 0 (an illustration). Lower ends are rounded up, upper ends down. A dash: the floors $G_t$ alone certify nothing.

*The crossings.* On $(0, \frac{1}{4}]$, $(f_1 - H)'(p) = \log e/(\frac{3}{2} - p) - \log\frac{1-p}{p} \le \log e/\frac{5}{4} - \log 3 < 0$, and $(G_1 - H)'(p) = -2\log e/(\frac{3}{2} + 2p) - \log\frac{1-p}{p} < 0$ at $B = \infty$. Both differences are positive near 0 and negative at $\frac{1}{4}$, so each has exactly one root in $(0, \frac{1}{4})$, the values $p_1$, $p_0$ of Section 7.3.

### E.4 Set models

**Proposition E.1** (Steeper two-part codes over sets)**.** *Let $\lambda > 1$ and let $A_\lambda$ minimise $\lambda K(A) + \log|A|$ over $A \ni z$. Then, for every $b \ge 0$,*

$$K(A_\lambda) \le \operatorname{soph}_b^{\mathrm{set}}(z) + \frac{b+\kappa}{\lambda - 1},$$

*and $A_\lambda$ is $\big(b + (\lambda-1)\operatorname{soph}_b^{\mathrm{set}}(z)\big)$-sufficient.*

*Proof.* Let $A_b$ attain $\operatorname{soph}_b^{\mathrm{set}}(z)$. Then $\lambda K(A_\lambda) + \log|A_\lambda| \le \lambda K(A_b) + \log|A_b| \le (\lambda-1)\operatorname{soph}_b^{\mathrm{set}}(z) + K(z) + b$. Subtract the sufficiency line for the first claim; use $K(A_\lambda) \le \lambda K(A_\lambda)$ for the second. A minimiser exists because only finitely many $A$ have $K(A) \le L$. □

*Remark* E.2 ($\lambda$ near 1). The bounds do not show that $\lambda$ slightly above 1 selects an approximately minimal sufficient statistic: as $\lambda \to 1$ the length bound becomes vacuous, and as $\lambda$ grows the sufficiency deficiency grows. The proposition is useful at $\lambda - 1 = \Theta(1)$ when the sophistication is small. The product version (Proposition C.2) has the same trade-off.

**Proposition E.3** (Sets from products)**.** *Let $\pi$ be product-$b$-sufficient for $D$, and let $\nu_\pi$ be a computable conditional measure with rational values. Put $r_D := \lceil -\log \nu_\pi(D) \rceil$ and $A := \{y' \in Y^{\operatorname{dom} D} : \nu_\pi(\operatorname{dom} D \mapsto y') \ge 2^{-r_D}\}$. Then $y_D \in A$, $K(A \mid \operatorname{dom} D) \le \ell(\pi) + 2\log(r_D+1) + \kappa$, and $K(A \mid \operatorname{dom} D) + \log|A| \le K(y_D \mid \operatorname{dom} D) + b + 2\log(r_D+1) + \kappa$. Consequently $\operatorname{soph}_{b'}^{\mathrm{set}}(y_D \mid \operatorname{dom} D) \le \operatorname{soph}_b^{\times}(D) + 2\log(r_D+1) + \kappa$ with $b' = b + 2\log(r_D+1) + \kappa$, provided a computable rational product hypothesis attains $\operatorname{soph}_b^{\times}(D)$. Against the unconditional set sophistication of the string $y_D$, add $K(\operatorname{dom} D)$.*

*Proof.* Since $Y$ is finite and $\nu_\pi$ is computable with rational values, the test $\nu_\pi(\operatorname{dom} D \mapsto y') \ge 2^{-r_D}$ is decidable, so $A$ is computable from $(\pi, \operatorname{dom} D, r_D)$. By (F1), (F3) and (F5), $K(A \mid \operatorname{dom} D) \le \ell(\pi) + 2\log(r_D+1) + \kappa$. It contains $y_D$ because $\nu_\pi(D) \ge 2^{-r_D}$, and $|A| \le 2^{r_D}$ because $\sum_{y'} \nu_\pi(\operatorname{dom} D \mapsto y') \le 1$. Hence

$$K(A \mid \operatorname{dom} D) + \log|A| \le \ell(\pi) + 2\log(r_D+1) + \kappa + r_D \le \ell(\pi) - \log\nu_\pi(D) + 1 + 2\log(r_D+1) + \kappa,$$

and product-$b$-sufficiency of $\pi$ gives the claim. The consequence follows by taking $\pi$ to attain $\mathrm{soph}_b^{\times}(D)$; against the unconditional set sophistication of $y_D$, describe $\mathrm{dom}\, D$ first, at cost $K(\mathrm{dom}\, D)$. □

**Question E.4** (lower-semicomputable product models)**.** Does Proposition E.3 hold for lower-semicomputable product hypotheses, for which the set $A$ is only computably enumerable?

### E.5 Sets against products

**Proposition E.5** (Sets against products)**.** *In the setting of Section 7.3, in $\mathcal{H}_{\times}$, let $c^* := 2\log d + K(p) + \kappa$. (i)* Sets: the parity is noise. *$A_0 := \{f_{a'} \oplus \mathbf{1}_{E'} : a' \in \{0,1\}^d,\ |E'| = pN\}$ has $|A_0| = 2^{d+L_E}$ and $K(A_0) \le c^*$, and $K(z) \ge d + L_E - \delta_E - \kappa$; so $A_0$ is $(\delta_E + c^*)$-sufficient and $\mathrm{soph}_b^{\mathrm{set}}(z) \le c^*$ for every $b \ge \delta_E + c^*$. The structure function $h_z$ lies within $\delta_E + O(\log d)$ of the sufficiency line for $\alpha$ between $O(\log d)$ and $d + O(\log d)$, and every $\lambda$-optimal set model with $\lambda > 1$ holds at most $O(\log d) + (\delta_E + O(\log d))/(\lambda - 1)$ bits of information about $a$. (ii)* Products: the parity is model. *For every integer $t \ge 1$ and every $b$ with $\delta_E + 2\log d + \kappa_g + \log(N+1) + \kappa \le b < N\Gamma_t^{\circ} - d - \lceil L_E \rceil - \kappa_b - 2\kappa$, one has $d - s(t) - \Lambda_1 \le \mathrm{soph}_b^{\times}(D) \le d + \kappa_g$. (iii)* The gap. *For such $b$, $\mathrm{soph}_b^{\times}(D) - \mathrm{soph}_b^{\mathrm{set}}(z) \ge d - s(t) - \Lambda_1 - c^*$. For a fixed dyadic $p \in (0, \frac{1}{4})$ and $\delta_E = O(d)$, the range of $b$ in (ii) with $t = 2$ is nonempty, of length of order $N$, for all large $d$.*

*Proof of Proposition E.5. (i), injectivity and size.* If $f_{a'} \oplus \mathbf{1}_{E'} = f_{a''} \oplus \mathbf{1}_{E''}$ with $a' \ne a''$, then the nonzero parity $f_{a'} \oplus f_{a''}$, which has $N/2$ ones, equals $\mathbf{1}_{E'} \oplus \mathbf{1}_{E''}$, which has at most $2pN < N/2$ ones. So $a' = a''$, and then $E' = E''$. Hence $|A_0| = 2^d \binom{N}{pN} = 2^{d+L_E}$.

*(i), complexity.* $A_0$ is computable from $d$ and $p$, so $K(A_0) \le c^*$ by (F3) and (F5). With (3), $K(A_0) + \log|A_0| \le K(z) + \delta_E + c^*$ (the constants merged into $\kappa$).

*(i), flatness.* For an integer $0 \le v \le d$ let $A_v$ be the subset of $A_0$ whose parities $a'$ agree with $a$ on the first $d - v$ coordinates. Then $\log|A_v| = v + L_E$ and $K(A_v) \le d - v + 2\log(v+1) + c^*$, since $A_v$ is computable from $d$, $p$, $v$ and those $d - v$ bits. So every $A_v$ has two-part length $d + L_E + O(\log d)$, within $\delta_E + O(\log d)$ of the sufficiency line, which is itself a lower bound; so $h_z$ lies within $\delta_E + O(\log d)$ of the sufficiency line for every $\alpha$ from $O(\log d)$ up to $d + O(\log d)$.

*(i), $\lambda$-optimal set models.* By Proposition E.1 with $b = \delta_E + c^*$, $K(A_\lambda) \le c^* + (b + \kappa)/(\lambda - 1)$. By (F3), $K(a) - K(a \mid A_\lambda) \le K(A_\lambda) + \kappa$, which is the claim.

*(ii), upper bound.* By Proposition 7.7(i), $\pi_g$ is $b_0$-sufficient, and $b_0 \le \delta_E + 2\log d + \kappa_g + \log(N+1) + \kappa$ since $NH(p) - L_E \le \log(N+1)$. So $\pi_g$ is product-$b$-sufficient for every $b$ in the range, and $\mathrm{soph}_b^{\times}(D) \le \ell(\pi_g) \le d + \kappa_g$.

*(ii), lower bound.* The memoriser computes $z$ from $d$, so by (F1) $K(y_D \mid \mathrm{dom}\, D) \le K(z \mid d) + \kappa \le \ell(\pi_b) + 2\kappa \le d + \lceil L_E \rceil + \kappa_b + 2\kappa$. Let $\pi$ be product-$b$-sufficient; then $\ell(\pi) + NR(\pi) \le K(y_D \mid \mathrm{dom}\, D) + b < N\Gamma_t^{\circ}$ by the upper end of the range of $b$. If $\ell(\pi) < d - s(t) - \Lambda_1$, then $\mathrm{lev}(\pi) = t' + 1$ with $t' \ge t$, or $\infty$, by (D1), and as in the proof of Proposition 7.7, $R(\pi) \ge \Gamma_{t'}^{\circ} \ge \Gamma_t^{\circ}$, a contradiction. So $\ell(\pi) \ge d - s(t) - \Lambda_1$.

*(iii).* The lower end of the range of $b$ is at least $\delta_E + c^*$, so $\mathrm{soph}_b^{\mathrm{set}}(z) \le c^*$ by (i), and (ii) gives the gap. With $\lceil L_E \rceil \le NH(p) + 1$, $\kappa_b = O(d)$ and $\delta_E = O(d)$, the range of $b$ at $t = 2$ has length at least $N(\Gamma_2^{\circ} - H(p)) - O(d)$. The argument for Proposition 7.8(iii), with the budget $d + \kappa$ in place of $\bar{\ell}_\lambda + \kappa$, gives $\Gamma_2^{\circ} \ge F_2^{\sharp}(d + \kappa) \ge f_2(p) - o(1)$, and $f_2(p) - H(p) \ge 0.0187$; so the length is of order $N$ for all large $d$. □

*Proof of Proposition 7.10.* The kink point is bounded: some $A \ni z$ has $K(A) \le \alpha_{\mathrm{r}}$ and $\log|A| \le L_{\mathrm{r}}$. By the sufficiency line and (3), $L_{\mathrm{r}} \ge K(z) - \kappa - \alpha_{\mathrm{r}} \ge d + L_E - \delta_E - 2\kappa - \alpha_{\mathrm{r}}$. The set $A_0$ has

$K(A_0) \le c^* < \alpha_{\mathrm{r}} - w$, so the kink gives $d + L_E = \log|A_0| \ge L_{\mathrm{r}} + \sigma(\alpha_{\mathrm{r}} - K(A_0)) \ge L_{\mathrm{r}} + \sigma(\alpha_{\mathrm{r}} - c^*)$. Combining, $\sigma(\alpha_{\mathrm{r}} - c^*) \le \alpha_{\mathrm{r}} + \delta_E + 2\kappa$, which is the claim. □

Within $O((\log d + \delta_E)/d)$ of $\lambda = 1$ the bound says nothing, and the $O(\log d)$ terms of the code decide. In explicit classes of the sets $A_v$ of the proof of Proposition E.5, with code lengths assigned to $v$, a direct computation (code in the accompanying repository) gives: with a fixed-length field for $v$, the set $\lambda$-MAP knows nothing about $a$ for all $\lambda \in (1, 27.5)$ at $d = 8$; with an Elias code for $v$, the set that knows $a$ is a vertex of the set hull whose $\lambda$-interval reaches 3.0 at $d = 8$ and 1.36 at $d = 30$.

# F Proofs for Section 8 (the teacher's mistakes)

Throughout, the setting is Definition 7.1, and (F1)–(F5) are the facts of Appendix A.1. For $z \in \{0,1\}^m$ and $w \in \{0,1\}^{d-m}$, $(z, w)$ is the input with prefix $z$, and $w$ is uniform in every $\mathbb{E}_w$. We write $\tau_B := 2^{1-B}$ (0 if $B = \infty$), so that $G_j = 1 - \log(1 + 2^{-j} + 2p + \tau_B)$; $T_g := \delta_S + \Delta_\ell + \Lambda_3 + 1 + 2\log m + 2\log s_0 + g + 2\log g + \kappa$, the threshold that defines $j_g$ in Theorem 8.7(ii)(e); $\mathrm{loss}_\pi(x) := -\log \nu_\pi(y^*(x) \mid x) \in [0, \infty]$; and $N_n(x)$ for the number of queries at $x$ among the first $n$.

## F.1 Noise and errors: frozen noise, plain Bayes and schedules

**Proposition F.1** (Frozen noise, counted once or per query)**.** *Let the inputs be i.i.d. from a law $\mu$ on $X$, let $\nu$ be a sampler with a program $\pi_0$ of length $k$, in a class $\mathcal{C}$ as in Definition 6.1, whose kernel equals $\nu$ on* $\operatorname{supp}\mu$*, and let the table $F$ have independent values $F(x) \sim \nu(\cdot \mid x)$, drawn once. (a)* FR. *For fixed $\lambda > 1$, at every $n$, repeats or not, every $\lambda$-MAP has length at most $k + (k + \delta^*)/(\lambda - 1)$, where $\delta^*$ is the Ville deficiency of $\xi_{\mathcal{C}}$ along the distinct inputs (Lemma 6.2), so $\mathbb{P}[\delta^* \ge j] \le 2^{-j}$. A program that agrees with $\nu$ off a finite set $A \subseteq \operatorname{supp}\mu$ fits $D_n$ better than $\pi_0$ by at most $\sum_{x \in A} \log(1/\nu(F(x) \mid x))$ bits, however often the inputs of $A$ are queried. (b)* FQ. *Let $A \subseteq \operatorname{supp}\mu$ be finite with $\nu(F(x) \mid x) < 1$ for some $x \in A$, and let $\pi_A \in \mathcal{C}$ agree with $\nu$ off $A$ and give $F(x)$ probability 1 on $A$. The FQ fits of $\pi_0$ and $\pi_A$ differ by $\sum_{x \in A} N_n(x) \log(1/\nu(F(x) \mid x))$. So for every schedule $\lambda_n = o(n)$, almost surely each program whose kernel equals $\nu$ on* $\operatorname{supp}\mu$ *is a $\lambda_n$-MAP for only finitely many $n$. (c) If $\mu$ is uniform on a finite set $X_0$ and every input of $X_0$ has been queried $\bar{n}$ times, the FQ $\lambda$-MAPs are the $(\lambda/\bar{n})$-MAPs of the table $F|_{X_0}$, so every $\lambda$-threshold of that table is multiplied by $\bar{n}$.*

By (c) and Corollary 8.11, in FQ frozen noise needs $\lambda$ of order $n/|X_0|$, like errors with $m = d$.

*Proof.* (a) The next new input is fixed by the i.i.d. input stream, independent of $F$, and $F(x)$ at a new $x$ is independent of the earlier labels and inputs; almost surely all queried inputs lie in $\operatorname{supp}\mu$, so $\nu_{\pi_0}(D_n) = \nu(D_n)$ for all $n$. So Theorem 6.9(i) applies with $\pi_\nu = \pi_0$, and Lemma 6.2 gives the tail of $\delta^*$. If $\pi$ agrees with $\nu$ off $A$, then $\nu_\pi(D_n)/\nu_{\pi_0}(D_n) = \prod_{x \in A \cap \operatorname{dom} D_n} \nu_\pi(F(x) \mid x)/\nu(F(x) \mid x) \le \prod_{x \in A} 1/\nu(F(x) \mid x)$. (b) Off $A$ the per-query losses agree; on $A$, $\pi_A$ has loss 0 and $\pi_0$ loss $\log(1/\nu(F(x) \mid x))$ at each query. As $X$ is countable, $\operatorname{supp}\mu$ is the set of atoms, and $N_n(x)/n \to \mu(x) > 0$ almost surely; so the difference divided by $n$ tends to a positive limit. Every program $\pi'$ whose kernel equals $\nu$ on $\operatorname{supp}\mu$ has the FQ fit of $\pi_0$, so $J_{\lambda_n}(\pi') - J_{\lambda_n}(\pi_A) \ge -\lambda_n \ell(\pi_A) + \sum_{x \in A} N_n(x) \log(1/\nu(F(x) \mid x)) \to +\infty$ when $\lambda_n = o(n)$. (c) The FQ fit is $\bar{n}\, L(\pi; F|_{X_0})$; apply Lemma 8.2. □

**Proposition F.2** (Plain Bayes, MDL and SI learn the errors)**.** *(a) Let $P$ be a shared hypothesis with $P \geq 2^{-\ell(\pi)-\kappa}\nu_\pi$ for every $\pi \in \mathcal{H}_\times$, such as $M_{\mathrm{prod}}$ or $M_F$. For every sequence of distinct inputs and every $n$, $\sum_{i\leq n} -\log P(y^*(x_i) \mid x_i, D_{i-1}) \leq \ell(\pi_b) + \kappa \leq d + \lceil L_S \rceil + \kappa_b + \kappa$. So at most $d + \lceil L_S \rceil + \kappa_b + \kappa$ inputs in total ($d + L_S + O(\log d + \log m)$ if $p = 2^{-r}$) get probability below $\frac{1}{2}$ for the teacher's label (semimeasure convention, hence also normalised), and every other revealed error input gets the erroneous label with probability at least $\frac{1}{2}$. (b) Let $B < \infty$, $\delta \in (0,1)$, let $\lambda = \lambda_n \geq 1$ be fixed or depend on $n$, and draw $n \leq 2^d$ inputs uniformly without replacement. With probability at least $1-\delta$, every $\lambda$-MAP $\hat{\pi}$ has $R_B(\hat{\pi}) \leq \lambda\hat{L}/n + B\sqrt{(\hat{L}\ln 2 + \ln(2/\delta))/(2n)}$ with $\hat{L} := d + \lceil L_S \rceil + \kappa_b$, hence $e(\hat{\pi}) \geq 2^{-R_B(\hat{\pi})/p} - 2^{-B}$. In particular $e(\hat{\pi}) \geq 1 - 2^{-B} - o(1)$ whenever $(\lambda_n/n)\hat{L} \to 0$ and $\hat{L}/n \to 0$, along any sequence of instances $(d_n)$ with fixed $m$, $p$ and error rule and $n \leq 2^{d_n}$. On a fixed instance the bound is informative only once $(\lambda_n/n)(d + L_S)$ is small compared with $p$; for slowly decaying schedules such as $\lambda_n = n^{0.9}$ at the example's sizes this never happens for $n \leq 2^d$. (c) On the interleaved string with i.i.d. uniform inputs, Solomonoff's $M$ has expected total label loss at most $d + \lceil L_S \rceil + \kappa_b + \kappa$, and for fixed $\lambda$ the steeper SI predictor pays at most $\lambda\ell_U$ more (Theorem 4.2(d)).*

For instance, at the example's sizes with all inputs revealed and the machine constant set to 0 (an illustration), a direct computation (code in the accompanying repository) shows that the exceptions in (a) are at most 11,853 of the $1.238 \cdot 10^{27}$ error inputs, a fraction at most $9.6 \cdot 10^{-24}$. With $B = 30$ and $\delta = 0.01$ (machine constant 0, an illustration), the same computation gives $e \geq 0.649$ at $n = 10^{13}$ for $\lambda = 1$, and $e \geq 0.734$ at $n = 10^{15}$ for $\lambda_n = \sqrt{n}$.

*Proof.* (a) The memoriser is consistent with every $D_n$, so $P(D_n) \geq 2^{-\ell(\pi_b)-\kappa}$, and the sequential loss of a shared hypothesis telescopes to $-\log P(D_n) + \log P(\emptyset) \leq \ell(\pi_b) + \kappa$. Each term is nonnegative, a probability below $\frac{1}{2}$ costs more than a bit, and normalised conditionals are at least the raw ones. For $p = 2^{-r}$ the memoriser can name $s_0 = 2^{m-r}$ through $r$. For $M_F$, $P \geq 2^{-K(\nu_\pi)-O(1)}\nu_\pi$ as product laws are machine laws, and $K(\nu_\pi) \leq \ell(\pi) + \kappa$; this is Hänni (2026a, Corollary 5.4). (b) Apply Lemma F.3 with $\pi_{\mathrm{ref}} = \pi_b$, so $A = 0$, $\epsilon_1 = 0$, $c = \lambda_n/n$ and $\hat{L} = \ell(\pi_b)$; it holds for each $n$, so $\lambda$ may depend on $n$. The error-label bound is the last step of the proof of Theorem 8.7(iii). The bound on $R_B$ is at least $(\lambda_n/n)(d + L_S)$; for $\lambda_n = n^{0.9}$ at the example's sizes, $\lambda_n/n \geq 2^{-10} = p$ for every $n \leq 2^{100}$. (c) The law $\mathbb{P}$ of the interleaved sequence (fixed-length blocks) is computable from $a$, $S$, $d$, $m$ and $s_0$, so it has a program in $\mathcal{H}_{\mathrm{SI}}$ of length at most $d + \lceil L_S \rceil + \kappa_b + \kappa$, and $M$ dominates it; the chain-rule argument in the proof of Theorem 4.2(d) bounds the expected label loss by that length. For $\lambda > 1$ apply Theorem 4.2(d), with $\kappa_0 = 0$ in $\mathcal{H}_{\mathrm{SI}}$. □

*Proof of Proposition 8.3.* Let $\hat{R}_n(\pi) := \sum_{x\in\{0,1\}^d} (N_n(x)/n)\,\mathrm{loss}_\pi(x)$. As $N_n(x)/n \to 2^{-d}$ for each of the finitely many $x$, almost surely $\hat{R}_n(\pi) \to R(\pi)$ for every $\pi$ (if $R(\pi) = \infty$, $\hat{R}_n(\pi) = \infty$ once an input of infinite loss is queried). (a) The memoriser has fit 0, so a $\lambda_n$-MAP $\hat{\pi}$ has $\lambda_n\ell(\hat{\pi}) \leq \lambda_n\ell(\pi_b)$; as $\lambda_n > 0$ and by Kraft's inequality, finitely many candidates remain. For each with $R(\pi) > 0$, $\frac{1}{n}[J_{\lambda_n}(\pi) - J_{\lambda_n}(\pi_b)] = \frac{\lambda_n}{n}(\ell(\pi) - \ell(\pi_b)) + \hat{R}_n(\pi) \to R(\pi) > 0$, so it is a $\lambda_n$-MAP for only finitely many $n$. Under uniform inputs $R(\pi) = 0$ means $\nu_\pi(y^*(x) \mid x) = 1$ at every $x$, so $e(\pi) = 1$. (b) The coin has $R = 1$, so $\ell_{\min}$ exists; fix $\pi_0$ with $\ell(\pi_0) = \ell_{\min}$ and $R(\pi_0) < \infty$. Lengths are integers, so every $\pi$ with $\ell(\pi) > \ell_{\min}$ has $J_{\lambda_n}(\pi) \geq \lambda_n(\ell_{\min} + 1) > J_{\lambda_n}(\pi_0)$ once $\lambda_n > n\hat{R}_n(\pi_0)$, which holds eventually. In $\mathcal{H}_\times$, $\ell_{\min} \leq \ell(\pi_{\mathrm{coin}}) \leq \kappa$, while level-1 programs have length at least $d - 4 - \Lambda_1$ by (D1). (c) Fix $n$ and $c := \lambda_n/n \in [c_1, c_2]$; a $\lambda_n$-MAP minimises $\hat{J}_c := J_{\lambda_n}/n$. Lemma F.3 with $\pi_{\mathrm{ref}} = \pi_g$ (losses in $[0, \log(1/p)]$) holds for every $c$, because its two events do not involve $c$; for PQ data, Hoeffding's inequality for independent bounded variables gives the same events. Its excess

has $\hat{L}(c) = (J_c(\pi_g) + \epsilon_1)/c \le \bar{L}$, as $J_g$ is increasing and $c \ge c_1$. So with probability at least $1-\delta$, $J_c^B(\hat{\pi}) \le J_g(c) + \bar{\epsilon}_n$, and Theorem 8.7(ii) applies. $\Phi_2 - J_g$ is a minimum of finitely many strictly decreasing affine functions, positive on $[c_1, c_2] \subset (0, c_{\rm hi})$, so it has a positive minimum there, while $\bar{\epsilon}_n \to 0$, also with $\delta = n^{-2}$; then the failure probabilities are summable, and the Borel–Cantelli lemma gives the almost-sure form. □

## F.2 The kink bounds and the theorem

Write $\psi_c(R) := c\max\{0, B_S - 2^m R\} + R = \max\{R,\ cB_S - (c2^m - 1)R\}$. We prove (o), then Proposition 8.6, then the rest.

*Proof of Theorem 8.7(o).* If $\mathrm{lev}(\pi) = 1$, (D1) and (D2) with $s(1) = 4$ give $\ell(\pi) \ge d - 4 - \Lambda_1 + \max\{0, B_S - 2^m R_B(\pi)\}$; multiply by $c$ and add $R_B(\pi)$. If $\mathrm{lev}(\pi) = t \in [2, \infty)$, (D1), $\ell \ge 0$ and Lemma 7.3 give $J_c^B(\pi) \ge c(d - s(t) - \Lambda_1)^+ + G_{t-1} \ge \varphi(t) := c(d - 4t - \Lambda_1)^+ + G_{t-1}$, as $s(t) \le 4t$. Extend $\varphi$ to real $t \ge 2$. If $t^* \ge 2$: on $[2, t^*]$ the first term is linear and $G_{t-1}$ concave, so $\varphi \ge \min\{\varphi(2), \varphi(t^*)\}$; for $t > t^*$, $\varphi(t) \ge G_{t-1} \ge G_{t^*-1}$. If $t^* < 2$: $\varphi(t) \ge G_1$. At level $\infty$, $J_c^B \ge R_B \ge G_\infty \ge G_j$ for every $j$. ($G_j$ is increasing and concave in $j$: with $v = 2^{-j}$, $\mathrm{d}G/\mathrm{d}j = v/(1+v+2p+\tau_B) > 0$ and $\mathrm{d}^2G/\mathrm{d}j^2 = -\ln 2 \cdot v(1+2p+\tau_B)/(1+v+2p+\tau_B)^2 < 0$.) □

*Proof of Proposition 8.6.* (a) By (o), every $\pi \in \mathcal{N}$ has $J_c^B(\pi) \ge \Phi_2(c)$, and the reference value is $J_g(c)$; $J_g - \Phi_2$ is a maximum of strictly increasing affine functions, hence $\le 0$ exactly on $(0, c_{\rm hi}]$ and $< 0$ on $(0, c_{\rm hi})$. At $c = c_{\rm hi} > 0$, $c_{\rm hi}\ell(\pi) + R_B(\pi) \ge \Phi_2(c_{\rm hi}) \ge J_g(c_{\rm hi})$, the slanted side of the wedge; the flat side is $R_B(\pi) \ge G_{\mathrm{lev}(\pi)-1} \ge G_1$ (Lemma 7.3). (b) (D1) with $s(1) = 4$ gives $\ell(\pi) \ge d - 4 - \Lambda_1$, and (D2) gives $R_B(\pi) \ge B_S/2^m - 2^{-m}(\ell(\pi) - d + 4 + \Lambda_1) = H(p) - \Lambda_2/2^m - 2^{-m}(\ell(\pi) - d - \kappa_g)$. If $p < \frac{1}{4}$, then $u(\pi_g) = 1 - p > \frac{3}{4}$, so $\pi_g \in \mathcal{M}$, and the line at $\pi_g$ ($\ell(\pi_g) \le d + \kappa_g$, $R_B(\pi_g) = H(p)$) gives $\Lambda_2 \ge 0$. □

*Proof of Theorem 8.7(i)–(iv).* Apply Lemma 5.3 in the population reading, $L = R \ge L' = R_B$, at the reference point $(d + \kappa_g, H(p))$, bounded by $\pi_g$, so that $J_{\rm r} = J_g$, with $\mathcal{N}, \mathcal{M}$ and the right line of Proposition 8.6. For $c > 2^{-m}$ the lemma's $\Delta(c)$ is $\Lambda_2/2^m + (c - 2^{-m})(4 + \Lambda_1 + \kappa_g) = c(4 + \Lambda_1 + \kappa_g) + H(p) - B_S/2^m$.

*(i).* $c_{\rm hi} \ge c > 0$ forces $G_1 > H(p)$, hence $p < p_0 < \frac{1}{4}$ ($G_1$ is largest at $B = \infty$), $\Lambda_2 \ge 0$, and the lemma's $\Delta(c)$ is $\Delta(c) \ge 0$; also $\eta_{\mathcal{N}}(c) \ge 0$. Lemma 5.3(i) gives $\Phi^B(c) \ge \min\{\Phi_2(c), J_g(c) - \Delta(c)\}$, with $J_g(c) - \Delta(c) = c(d - 4 - \Lambda_1) + B_S/2^m$, and $J_c(\pi_g) \le J_g(c) \le \Phi(c) + \Delta(c)$; $\Phi \ge \Phi^B$ as $R \ge R_B$. Finally $H(p) - B_S/2^m = (2^m H(p) - L_S + \delta_S)/2^m \le (\log(2^m + 1) + \delta_S)/2^m$.

*(ii).* Every $\pi \in \mathcal{N}$ has $J_c^B(\pi) \ge \Phi_2(c) > J_g(c) + \epsilon$, the pointwise margin of Lemma 5.3(ii); and $J_c(\pi) \le \Phi(c) + \epsilon$ implies $J_c^B(\pi) \le J_g(c) + \epsilon$, as $\Phi(c) \le J_c(\pi_g) \le J_g(c)$. So the lemma gives $\mathrm{lev}(\pi) = 1$,

$$\ell(\pi) \le d + \kappa_g + \frac{\Lambda_2/2^m + \epsilon}{c - 2^{-m}} = d + \Delta_\ell, \qquad R_B(\pi) \ge H(p) - \frac{c\Lambda_2/2^m + 2^{-m}\epsilon}{c - 2^{-m}} = H(p) - \frac{c\Lambda_2 + \epsilon}{c2^m - 1},$$

using $\frac{\epsilon}{c} + \frac{\epsilon/c}{c2^m - 1} = \frac{\epsilon 2^m}{c2^m - 1}$: these are (c) and (b). (a) By (D1), $\ell(\pi) \ge (d - 4 - \Lambda_1)^+$, so $R_B(\pi) = J_c^B(\pi) - c\ell(\pi) \le R_{\max}$; then $-\log(u + p + 2^{-B}) \le R_B$ (Lemma 7.3), and $\mathrm{lev}(\pi) = 1$ gives $u > \frac{3}{4}$. (d) By (F3), $B_S \le K(\pi \mid a, K(a)) + K(S \mid a, \pi) + \kappa$, and by (F4), (F3), $K(\pi \mid a, K(a)) \le K(\pi) + K(a \mid \pi) - K(a) + O(1)$. By (F1), Lemma 7.2 with $t = 1$, $K(a) \ge d$ and (c), this is at most $\ell(\pi) - d + 4 + 2\log d + \kappa \le \Delta_\ell + \Lambda_3$. (e) Given $x \in E$, $z$ is uniform on $S$ and $y^* = f_a \oplus 1$, so $e(\pi) = \mathrm{avg}_S V$ with $V$ as in the proof of Lemma 7.6. By (d), $I(\pi) \le \Delta_\ell + \Lambda_3$, and for this budget the least admissible $j$ in that proof is $j_g$ (its threshold is $T_g$); so the $V$ side of that proof gives

$\mathrm{avg}_S V \le 1-u(\pi)+\gamma_g+2^{-g}$. (The argument: if $\mathrm{avg}_S V$ exceeded the least multiple $\theta$ of $2^{-g}$ above $1-u+\gamma_g$, then $S$ would lie in the set of $s_0$-subsets with average above $\theta$, which is enumerable from $(\pi, a, m, s_0, \theta)$ and, by Hoeffding's inequality for sampling without replacement (Hoeffding, 1963), has at most $2^{L_S - j_g}$ elements; this would make $K(S \mid a, \pi)$ smaller than (d) allows.)

*(iii).* $\mathrm{lev}(\pi) = 1$ as in (ii). For $c2^m < 1$ and $R \ge 0$, $\psi_c(R) \ge cB_S + (1-c2^m)\min\{R, B_S/2^m\}$ (equality for $R \le B_S/2^m$; for $R \ge B_S/2^m$, $\psi_c(R) = R$). Compare with $J_c^B \le J_b(c) + \epsilon$ and (o), and use $L_S = B_S + \delta_S$: $(1-c2^m)\min\{R_B, B_S/2^m\} \le c(\lceil L_S \rceil + \kappa_b - B_S + 4 + \Lambda_1) + \epsilon = \varrho$. If $\varrho < (1-c2^m)B_S/2^m$, the minimum is $R_B \le \varrho/(1-c2^m)$. For the error label, $p\,\mathbb{E}[\min\{B, -\log \nu_\pi(y^* \mid x)\} \mid x \in E] \le R_B$, and by Jensen's inequality $\mathbb{E}[\nu_\pi(y^* \mid x) \mid E] \ge \mathbb{E}[\max\{\nu_\pi(y^* \mid x), 2^{-B}\} \mid E] - 2^{-B} \ge 2^{-R_B/p} - 2^{-B}$.

*(iv).* By (D1) and $R(\pi) \ge H(p)$, $J_c(\pi) \ge c(d-4-\Lambda_1) + H(p) > c(d + \lceil L_S \rceil + \kappa_b) \ge J_c(\pi_b)$. □

**Numbers.** All numbers of Section 8 and this appendix come from deterministic or seeded scripts in the accompanying repository. The script for Table 3 computes the windows in closed form and by bisection on $J_g \le \Phi_2$ (the two agree to $10^{-15}$), the identification bounds with $g \le 40$, and the memoriser regime. With $\delta_S = 0$, (iii) applies only for $c < 0.99418 \cdot 2^{-m}$ at $\kappa = 0$ (an illustration) and $c < 0.98914 \cdot 2^{-m}$ at $\kappa = 30$; it gives $e \ge 0.954$, 0.830, 0.658, 0.483, 0.222 at $c2^m = 0.5$, 0.8, 0.9, 0.94, 0.97 ($\kappa = 0$) and $e \ge 0.916$, 0.705, 0.456, 0.255, 0.059 ($\kappa = 30$), and $e \ge \frac{1}{2}$ needs $c \le 0.937 \cdot 2^{-m}$ ($0.888 \cdot 2^{-m}$ at $\kappa = 30$).

### F.3 Full truth tables

*Proof of Corollary 8.11.* On the whole truth table the empirical loss is $N$ times the population risk, so $J_\lambda(\pi; D) = N\, J_{\lambda/N}(\pi)$. (i) Put $c = \lambda/N$, so $c2^m - 1 = (\lambda - 2^{d-m})/2^{d-m} > 0$; $J_g(c) < \Phi_2(c)$ for $c < c_{\mathrm{hi}}$, as $J_g - \Phi_2$ is strictly increasing. A $\lambda$-MAP has $J_c = \Phi(c)$, so Theorem 8.7(ii) applies with $\epsilon = 0$, and $1/(c2^m - 1) = 2^{d-m}/(\lambda - 2^{d-m})$ gives the stated forms. (ii) $\ell(\pi_g) + NR(\pi_g) \le d + \kappa_g + L_E + \log(N+1)$, as $NH(p) \le L_E + \log(N+1)$, and $\ell(\pi_b) + NR(\pi_b) \le d + \lceil L_E \rceil + \kappa_b$. Level-1 programs have $\ell + NR \ge d + L_E - \delta_E - 4 - \Lambda_1$ by (D2). Programs of level at least 2 have $\ell + NR \ge NG_1$ (Lemma 7.3), which by the hypotheses exceeds $\ell(\pi_g) + NR(\pi_g)$, since $L_E \le NH(p)$. As $d = \log N$ and $\kappa_g$, $\kappa_b$, $\Lambda_1$ are $O(\log N)$, the claim follows. (iii) is Theorem 8.7(iv) at $c = \lambda/N$. □

At $d = m = 20$, $p = 2^{-10}$ and $\lambda = 2$, with $g$ optimised, every $\lambda$-MAP has $u \ge 0.991$, $\ell \le d + 58.2$, $I \le 70.8$ and $e \le 0.220$ with the machine constant set to 0 (an illustration) and $\delta_E = 0$, and $\ell \le d + 148.2$, $I \le 190.8$, $e \le 0.320$ at $\kappa = 30$ (a direct computation, code in the accompanying repository); $\delta_E = 20$ adds 20 bits to $\ell$ and $I$. Near $\lambda = 1$ the deficiency enters at full weight, through $\Lambda_2/(\lambda - 1)$.

### F.4 Finite samples

**Lemma F.3** (Finite sample)**.** *Let $m \le d$ and $B < \infty$. Draw $n \le 2^d$ inputs uniformly without replacement and label them by $y^*$; let $\hat{J}_c(\pi) := c\,\ell(\pi) + \frac{1}{n}\sum_{i \le n} -\log \nu_\pi(y^*(x_i) \mid x_i)$. Let $\pi_{\mathrm{ref}}$ be a fixed program whose per-input losses lie in $[0, A]$. With probability at least $1-\delta$, every minimiser $\hat{\pi}$ of $\hat{J}_c$ satisfies*

$$J_c^B(\hat{\pi}) \le J_c(\pi_{\mathrm{ref}}) + \epsilon_1 + B\sqrt{\frac{\hat{L}\ln 2 + \ln(2/\delta)}{2n}}, \qquad \epsilon_1 := A\sqrt{\frac{\ln(2/\delta)}{2n}}, \quad \hat{L} := \frac{J_c(\pi_{\mathrm{ref}}) + \epsilon_1}{c}.$$

*Proof.* Each sampled input is marginally uniform. By Hoeffding's inequality for sampling without replacement (Hoeffding, 1963), with probability at least $1-\delta/2$, $\hat{J}_c(\pi_{\mathrm{ref}}) \le J_c(\pi_{\mathrm{ref}}) + \epsilon_1$. The truncated losses of each $\pi$ lie in $[0, B]$; the same inequality with confidence $(\delta/2)2^{-\ell(\pi)}$ and a union bound, which costs at most $\delta/2$ by Kraft's inequality, give, with probability at least $1-\delta/2$, simultaneously for all $\pi$,

$$\hat{J}_c(\pi) \ge c\,\ell(\pi) + \frac{1}{n}\sum_{i\le n} \min\{B, -\log \nu_\pi(y^*(x_i) \mid x_i)\} \ge J_c^B(\pi) - B\sqrt{\frac{\ell(\pi)\ln 2 + \ln(2/\delta)}{2n}}.$$

On both events a minimiser has $c\,\ell(\hat{\pi}) \le \hat{J}_c(\hat{\pi}) \le \hat{J}_c(\pi_{\mathrm{ref}}) \le J_c(\pi_{\mathrm{ref}}) + \epsilon_1$, so $\ell(\hat{\pi}) \le \hat{L}$, and the displays combine. □

The bound is one-sided and truncated, so no noise floor is needed; as $c$ is fixed for the given $n$, it applies to any schedule. It adapts the empirical-selection lemma of the earlier report.

*Proof of Corollary 8.12.* A $\lambda$-MAP with $\lambda = cn$ minimises $\hat{J}_c$. Apply Lemma F.3 with $\pi_{\mathrm{ref}} = \pi_g$, whose losses $-\log(1-p)$ and $-\log p$ lie in $[0, \log(1/p)]$: $J_c^B(\hat{\pi}) \le J_c(\pi_g) + \epsilon_n \le J_g(c) + \epsilon_n$. □

At the example's sizes, with $B = 30$, $\delta = 0.01$, $\delta_S = 0$, $g = 7$ and the machine constant set to 0 (an illustration), a direct computation (code in the accompanying repository) gives, at $c = 10^{-4}$, an excess $\epsilon_n/c$ of 92.5 bits at $n = 10^9$ ($\ell \le d + 113.9$, $e \le 0.284$), 9.2 bits at $n = 10^{11}$ ($u \ge 0.9825$, $\ell \le d + 29.8$, $e \le 0.212$) and 0.9 bits at $n = 10^{13}$ ($u \ge 0.9830$, $\ell \le d + 21.5$, $e \le 0.204$).

## F.5 The total-bits version of the example

**Example F.4** (The total-bits version of the example)**.** Take $d = 100$, $m = 10$ and $p = 2^{-6}$, so that $s_0 = 16$ and $L_S = 115.6$ bits. At $n = 10^4$ the model saves $n(1 - H(p)) = 8839$ bits for its 100 bits (88 bits per bit) and the error rule saves $nH(p) = 1161$ bits for its 116 bits (10.0 bits per bit): the sizes and rates of Remark 8.1(1). A direct computation (code in the accompanying repository) gives the window $c \in (2^{-10}, c_{\mathrm{hi}}]$ with $c_{\mathrm{hi}} = 5.80 \cdot 10^{-3}$ with the machine constant set to 0 (an illustration) and $2.53 \cdot 10^{-3}$ at $\kappa = 30$; at $n = 10^4$ and $\kappa = 0$ this is $\lambda \in (9.77, 58.0]$, against $(10, 100)$ for the three hypotheses. Only coarse identification is proved at this size: near-optimal hypotheses have $u > \frac{3}{4}$, and at $\kappa = 0$, $\delta_S = 0$, $c = 5.5 \cdot 10^{-3}$ they have $\ell \le d + 28.3$ and $I \le 45.6$ bits out of 115.6; (e) is vacuous because $s_0 = 16$ is too small. At $n = 10^4$ samples Corollary 8.12 gives nothing: every admissible $B$ makes its excess at least 0.321, more than the largest margin $\Phi_2 - J_g$ in the window, 0.224.

## F.6 Concentration of the power posterior

Let $\hat{J}_c(\pi) := c\,\ell(\pi) + \frac{1}{n}\sum_{i\le n} -\log \nu_\pi(y_i \mid x_i)$ (Remark 3.7) and $\hat{J}_g := \hat{J}_c(\pi_g)$.

**Proposition F.5** (The power posterior concentrates on near-minimisers)**.** *Let $\lambda = cn$ with $cn \ge 1$, and let $D$ be any data set of size $n$. For every $\epsilon > 0$,*

$$\Pi_\lambda(\{\pi : \hat{J}_c(\pi) \ge \hat{J}_g + \epsilon\} \mid D)/\Pi_\lambda(\pi_g \mid D) \;\le\; 2^{-n\epsilon + (\hat{J}_g + \epsilon)/c}.$$

*Consequently, for data sampled as in Corollary 8.12, with probability at least $1-\delta$ all but this fraction of the posterior mass lies on programs that satisfy Theorem 8.7(ii) with $\epsilon' := \epsilon + \epsilon_1 + B\sqrt{(\hat{L}'\ln 2 + \ln(2/\delta))/(2n)}$ (with $B$ as in Theorem 8.7) in place of $\epsilon$, where $\hat{L}' := (J_g(c) + \epsilon_1 + \epsilon)/c$, provided $J_g(c) + \epsilon' < \Phi_2(c)$.*

*Proof.* $2^{-\lambda\ell(\pi)}\nu_\pi(D) = 2^{-n\hat{J}_c(\pi)}$, and with the empirical risk $\hat{R} \geq 0$, $n\hat{J}_c - \ell = (\lambda-1)\ell + n\hat{R} \geq (1 - 1/\lambda)\, n\hat{J}_c$. So on the set $\{\hat{J}_c \geq \hat{J}_g + \epsilon\}$, $2^{-n\hat{J}_c(\pi)} \leq 2^{-\ell(\pi)} 2^{-(1-1/\lambda)n(\hat{J}_g+\epsilon)}$, and by Kraft's inequality the set has unnormalised weight at most $2^{-(1-1/\lambda)n(\hat{J}_g+\epsilon)}$; divide by $2^{-n\hat{J}_g}$ and use $n/\lambda = 1/c$. For the consequence, on the event of Lemma F.3 with $\pi_{\mathrm{ref}} = \pi_g$, $\hat{J}_g \leq J_g(c) + \epsilon_1$, so a program with $\hat{J}_c(\pi) < \hat{J}_g + \epsilon$ has $\ell(\pi) < \hat{L}'$ and $J_c^B(\pi) \leq \hat{J}_c(\pi) + B\sqrt{(\hat{L}' \ln 2 + \ln(2/\delta))/(2n)} \leq J_g(c) + \epsilon'$. □

Near-minimisers can be $\epsilon/c$ bits longer than the minimiser, hence the budget $\hat{L}'$. With $J_g = 0.0232$ in place of $\hat{J}_g$, $c = 10^{-4}$, $\epsilon = 10^{-3}$, $n = 10^6$ and the machine constant set to 0 (an illustration), the ratio is at most $2^{-1000+241.7} = 2^{-758.3}$. So the steeper predictive, averaged over the unseen error inputs, gives the erroneous label probability at most $\left(\sup e(\pi) + 2^{-n\epsilon+(\hat{J}_g+\epsilon)/c}\right) |E|/(|E| - n)$, the supremum over the programs just described.

## F.7 $\lambda = cn$ and weight decay

A *fixed prior* is a map $w$ from programs to $[0, \infty)$ chosen before the data; its MAPs on a per-query sample $h$ maximise $w(\pi)\nu_\pi(h)$. For $\lambda \geq 1$ the $\lambda$-MAPs are the MAPs of the fixed prior $2^{-\lambda\ell}$. Write $h^r$ for the sample $h$ repeated $r$ times.

**Proposition F.6** (No fixed prior gives the $cn$-MAPs)**.** *Let $c > 0$, and let the class be countable, with Kraft weights and kernels, as in Definition 6.1.*

*(i) If samples $h$ and $h'$ of sizes $n$ and $n'$ have the same empirical distribution, the $cn$-MAPs of $h$ are the $cn'$-MAPs of $h'$.*

*(ii) Let $h$ have size $n_0$, let $\pi^*$ attain $P^* := \max_\pi \nu_\pi(h) > 0$, and let no $cn_0$-MAP of $h$ attain $P^*$. If $w(\pi^*) > 0$, then for every $r > \max_{\hat{\pi}} \log \frac{w(\hat{\pi})}{w(\pi^*)} / \log \frac{P^*}{\nu_{\hat{\pi}}(h)}$, the maximum over the $cn_0$-MAPs $\hat{\pi}$ of $h$ with $w(\hat{\pi}) > 0$ (and every $r \geq 1$ if there are none), no $crn_0$-MAP of $h^r$ is a MAP of $w$ on $h^r$. So no fixed prior with full support gives the $cn$-MAPs on all samples; nor does any fixed prior if $\pi^*$ is a $cn$-MAP of some sample whose $cn$-MAPs are not all programs. In $\mathcal{H}_\times$ samples as in (ii) exist for every $c > 0$.*

*(iii) Per query, let $\mathcal{E} \neq \emptyset$ and let $w$ be summable with $w(\pi) > 0$ for some $\pi \in \mathcal{E}$. Almost surely, for all large $n$, the MAPs of $w$ are the programs of $\mathcal{E}$ of largest weight. If $c > c^*$ (Proposition D.5), almost surely for all large $n$ no $cn$-MAP lies in $\mathcal{E}$, and every minimiser $\pi$ of $c\,\ell + R_\nu$ has $\ell(\pi) < \ell_0$ and $R_\nu(\pi) > R_\nu(\pi_0)$ for $\pi_0 \in \mathcal{E}_0$. If $c < c^*$, the $cn$-MAPs and the MAPs of $2^{-\ell}$ are both eventually $\mathcal{E}_0$.*

*(iv) In the instance of Definition 7.1, with PQ data $\nu = \delta_{y^*}$, inputs uniform on $\{0,1\}^d$ and $B = \infty$, let $c_+ := H(p)/(B_S - 4 - \Lambda_1 - \kappa_g)$, the denominator being positive. Every $c \in (\max\{2^{-m}, c_+\}, c_{\mathrm{hi}})$ has $c > c^*$.*

*Proof.* (i) $J_{cn}(\pi; h)/n = c\,\ell(\pi) + \mathbb{E}_{\hat{P}_h}[-\log \nu_\pi(y \mid x)]$ depends on $h$ only through its empirical distribution $\hat{P}_h$. (ii) By (i) the $crn_0$-MAPs of $h^r$ are the $cn_0$-MAPs $\hat{\pi}$ of $h$; they are finitely many (Lemma A.1), with $0 < \nu_{\hat{\pi}}(h) < P^*$, and $w(\hat{\pi})\nu_{\hat{\pi}}(h)^r < w(\pi^*)(P^*)^r$ for the stated $r$. A prior $w$ whose MAPs on the sample where $\pi^*$ is a $cn$-MAP are these $cn$-MAPs has $w(\pi^*) > 0$: otherwise every program would have score 0 there and be a MAP. In $\mathcal{H}_\times$ take $n_0$ distinct inputs whose labels $y$ have $K(y \mid \mathrm{dom}\, D) \geq n_0 \log |Y|$, which exist by Kraft's inequality, with $n_0 \log |Y| - \kappa - \ell_{\mathrm{u}} > \log |Y|/c$, where $\ell_{\mathrm{u}}$ is the length of a program for the uniform kernel: a lookup program attains $P^* = 1$, and by Lemma 3.9 every program of likelihood 1 has $J_{cn_0} \geq cn_0(n_0 \log |Y| - \kappa) > cn_0\ell_{\mathrm{u}} + n_0 \log |Y|$, the uniform kernel's value. (iii) Let $W := \max_{\mathcal{E}} w$.

Almost surely $\nu_\pi(h_n) = \nu(h_n) > 0$ for all $\pi \in \mathcal{E}$ and all $n$, so a MAP $\hat{\pi}$ of $w$ has $w(\hat{\pi})\Lambda_n(\hat{\pi}) \geq W$. For a cofinite set $T$ of programs, $\sum_{\pi\in T} w(\pi)\Lambda_n(\pi)$ is a nonnegative supermartingale started at $w(T)$ (Lemma D.1(a)), so by Ville's inequality some MAP lies in $T$ at some $n$ with probability at most $w(T)/W$, which tends to 0 along cofinite sets decreasing to $\emptyset$. Of the finitely many other programs, those outside $\mathcal{E}$ have $\Lambda_n \to 0$ almost surely (Lemma D.1(c)), and those of $\mathcal{E}$ with $w < W$ are never MAPs. For $c > c^*$ some $\pi'$ with $\ell(\pi') < \ell_0$ has $J_c(\pi') < J_c(\pi_0) \leq J_c(\pi)$ for all $\pi \in \mathcal{E}$, and $J_c(\pi) > J_c(\pi_0)$ for $\pi \notin \mathcal{E}$ with $\ell(\pi) \geq \ell_0$; so the minimisers of $J_c$ are shorter than $\ell_0$ and lie outside $\mathcal{E}$, where $\mathrm{KL}_\mu > 0$, and Proposition D.5(c) puts every $cn$-MAP among them for all large $n$. For $c < c^*$ use Proposition D.5(c) and the first part with $w = 2^{-\ell}$. (iv) Here $\mathcal{E}$ is the set of programs with $R = 0$; it contains the memoriser. Let $\pi$ minimise $J_c$, so $J_c(\pi) = \Phi(c)$. As $c2^m > 1$ and $J_g(c) < \Phi_2(c)$ on $(0, c_{\mathrm{hi}})$, Theorem 8.7(ii)(b) with $\epsilon = 0$ gives $R(\pi) \geq H(p) - c\Lambda_2/(c2^m - 1)$, which is positive for $c > c_+$ because $2^m H(p) - \Lambda_2 = B_S - 4 - \Lambda_1 - \kappa_g$. So no minimiser of $J_c$ lies in $\mathcal{E}$. For $c < c^*$ the minimisers are $\mathcal{E}_0$ (Proposition D.5(c)); at $c = c^*$ the programs of $\mathcal{E}_0$ are minimisers, since $J_c(\pi) - J_c(\pi_0) = \mathrm{KL}_\mu(\pi) - c^*(\ell_0 - \ell(\pi)) \geq 0$ when $\ell(\pi) < \ell_0$. Hence $c > c^*$. □

Full support cannot be dropped from (ii) in general: in the class of a coin of length 1 and a deterministic $f$ of length $1 + k$, for $ck > 1$ the coin is the only $cn$-MAP on every sample, and it is also the only MAP of the prior concentrated on it. With two programs that give the label 1 probability $\frac{1}{2}$ and $\frac{1}{4}$ at every input, of lengths 1 and $1 + k$, the last clause of (ii) applies for every $0 < ck < \log\frac{3}{2}$, and a direct computation (code in the accompanying repository) finds at $ck = 0.3$ no fixed prior whose MAPs are the $cn$-MAPs even on the samples of size at most 4. At the example's sizes, with the machine constant set to 0 or 30 (an illustration) and $\delta_S \in \{0, 20\}$, a direct computation (code in the accompanying repository) puts $c_+$ between $1.0044 \cdot 2^{-m}$ and $1.0115 \cdot 2^{-m}$.

*Weight decay.* For a per-sample loss $f(\theta; z)$ let $\hat{L}_n := \frac{1}{n}\sum_{i\leq n} f(\cdot\,; z_i)$ and $L := \mathbb{E}f(\cdot\,; z)$. A step $\theta \leftarrow \theta - \eta(\nabla\hat{L}_B(\theta) + \omega\theta)$ on a uniformly drawn minibatch $B$ is a stochastic gradient step on $F_n := \hat{L}_n + \frac{\omega}{2}\|\theta\|^2$; with full batches its fixed points are the stationary points of $F_n$, for every $\eta$. If $f = -\ln p_\theta(y \mid x)$, then $nF_n(\theta) = -\ln\left(p_\theta(y_{1:n} \mid x_{1:n})\, e^{-n\omega\|\theta\|^2/2}\right)$. As in (i), the minimisers of $F_n$ depend on the sample only through its empirical distribution. If $\theta^*$ minimises $F := L + \frac{\omega}{2}\|\theta\|^2$ and $L$ is differentiable, then $\nabla L(\theta^*) = -\omega\theta^*$, so $\theta^*$ is a stationary point of $L$ only if $\theta^* = 0$; and $L(\theta^*) - L(\theta) \leq \frac{\omega}{2}(\|\theta\|^2 - \|\theta^*\|^2)$ for every $\theta$, the analogue of $R_\nu(\pi_c) - R_\nu(\pi) \leq c(\ell(\pi) - \ell(\pi_c))$ for a minimiser $\pi_c$ of $J_c$. Unlike a vertex of the hull, a point $\theta \neq 0$ is stationary for $L + \frac{\omega}{2}\|\theta\|^2$ at one value of $\omega$ at most when $L$ is differentiable; without differentiability it can be selected on an interval ($L(\theta) = |\theta - 1|$ selects $\theta = 1$ for every $\omega \in (0, 1]$). If the decay is not multiplied by the learning rate $\alpha$ of the loss, $\theta \leftarrow \theta - \eta_t(\alpha\nabla L(\theta) + \omega_{\mathrm{d}}\theta)$, the fixed points are the stationary points of $L + \frac{\omega_{\mathrm{d}}}{2\alpha}\|\theta\|^2$: the effective coefficient then depends on $\alpha$.

**Example F.7** (A worse classifier in the limit)**.** Let $y$ be uniform on $\{\pm 1\}$, $\xi = 1$ with probability $1-q$ and $\xi = -1$ otherwise, independently, $0 < q < \frac{1}{2}$, $\varepsilon > 0$, $x = (\varepsilon y, \xi y)$ and $p_\theta(y \mid x) = \sigma(y\,\theta\cdot x)$. The first feature separates the classes, at scale $\varepsilon$; $\inf L = 0$ is approached only by classifiers of error rate 0. If $\varepsilon^2 < 1 - 2q$, the minimiser of $F$ has error rate 0 for $\omega < \omega^*$ and $q$ for $\omega \geq \omega^*$ (a tie counted as an error), where

$$\omega^* = \frac{2q\varepsilon^2}{(1-\varepsilon^2)\,a^*}, \qquad a^* = \ln\frac{2(1-q)(1-\varepsilon^2) - q(1+\varepsilon^2)}{q(1+\varepsilon^2)};$$

if $\varepsilon^2 \geq 1 - 2q$, it has error rate 0 for every $\omega > 0$. At $\varepsilon = 0.1$ and $q = 0.2$, $\omega^* = 2.101 \cdot 10^{-3}$. A direct computation (code in the accompanying repository) finds the empirical minimisers with

$\omega = 0.01$ at error rate 0.2 in all of 20 samples for each $n \in \{10^2, 10^3, \dots, 10^6\}$, and with $\omega = 1/n$, the fixed prior $\mathcal{N}(0, I)$, at error rate 0 in all of 20 for each $n \geq 10^3$; minibatch gradient descent reaches the same regimes. If full-batch AdamW with decoupled coefficient $\omega_{\mathrm{d}}$ converges, under a non-increasing learning rate with divergent partial sums, its limit is a KKT point of $L$ under $\|\theta\|_\infty \leq 1/\omega_{\mathrm{d}}$ (Xie and Li, 2024), here the minimiser of $L$ on this box, as $L$ is convex; in the same computation AdamW ends near it. That minimiser has error rate $q$ exactly when $\omega_{\mathrm{d}} \geq 2\varepsilon/\ln((2-3q)/q)$, for every $\varepsilon \leq 1$: at $\varepsilon = 0.1$, $q = 0.2$ the threshold is 0.1028, and at $\varepsilon = 0.8$, $q = 0.2$, where $L_2$ decay never switches, it is 0.8222 (both to four digits).

*Proof idea.* With $a := \varepsilon\theta_1 + \theta_2$, $b := \varepsilon\theta_1 - \theta_2$, the error rate is $(1-q)[a \leq 0] + q[b \leq 0]$. The stationarity equations of the strictly convex $F$ give $\theta_1, \theta_2 > 0$, so the error rate is $q[b \leq 0]$. At $b = 0$ they force $(1-q)(1-\varepsilon^2)\sigma(-a) = \frac{q}{2}(1+\varepsilon^2)$. If $\varepsilon^2 \geq 1 - 2q$ this is impossible: for $\varepsilon \geq 1$ the left side is not positive, and otherwise it needs $\sigma(-a) \geq \frac{1}{2}$, against $a = 2\theta_2 > 0$. If $\varepsilon^2 < 1 - 2q$ it fixes $a = a^*$ and $\omega = \omega^*$. As $b$ is continuous in $\omega$ and $b > 0$ for $\omega \leq \varepsilon^2 q/\ln((2-3q)/q)$, and $\omega\, b \to (\varepsilon^2 - (1-2q))/2$ as $\omega \to \infty$, the claim follows. Under the box, $\theta_1 = 1/\omega_{\mathrm{d}}$ because $\partial L/\partial\theta_1 < 0$, and for $\varepsilon \leq 1$ the error rate is $q$ iff $\partial L/\partial\theta_2 \leq 0$ at $\theta_2 = \varepsilon/\omega_{\mathrm{d}}$, that is iff $(1-q)\sigma(-2\varepsilon/\omega_{\mathrm{d}}) \geq q/2$.

### F.8 Charging the information about the rule

For $t \in \{2, 3, \dots\} \cup \{\infty\}$ ($s(\infty) := \infty$, $\theta_{\infty-1} := \frac{1}{2}$, $G_{\infty-1} := G_\infty$), integers $I \geq 0$ and $c > 0$ let

$$\mathrm{Floor}_t(I) := \max\Big\{0,\ \sup_{g \geq 1} -\log\big(p + (1-2p)\theta_{t-1} + p(2\gamma_g(I) + 2^{1-g}) + 2^{-B}\big)\Big\},$$

$$\Psi_t(c) := \min_{I \in \{0,1,\dots\}} \big[cI + \mathrm{Floor}_t(I)\big], \qquad \Phi_2^\sharp(c) := \inf_t LB_t(c),$$

$$LB_t(c) := \max\big\{c(d - s(t) - \Lambda_1)^+ + G_{t-1},\ c\big(d - \min\{s(t) + 2\log d, d\} - 2\log(2d) - \kappa\big) + \Psi_t(c)\big\},$$

with $\gamma_g(I)$ as in Section 7.2; at $B = \infty$, $\mathrm{Floor}_t = \max\{0, F_{t-1}^\sharp\}$.

**Proposition F.8** (Charging the information about the rule)**.** *In the instance of Definition 7.1, let $d \geq 2\log(d+1) + \kappa + 1$. (i) Every program $\pi$ of level $t \in \{2, 3, \dots\} \cup \{\infty\}$ has $\ell(\pi) \geq d - \min\{s(t) + 2\log d,\ d\} + I(\pi) - \kappa$, and $I(\pi) \geq -2\log(2d) - \kappa$ is an integer. (ii) Every such $\pi$ has $J_c^B(\pi) \geq LB_t(c)$ for every $c > 0$. So $\Phi_2^\sharp \geq \Phi_2$ bounds $J_c^B$ on all programs of level at least 2, and $\Phi_2^\sharp$ is continuous on $(0, \infty)$. (iii) Theorem 8.7(i) and (ii), with all their conclusions, hold with $\Phi_2^\sharp$ in place of $\Phi_2$ (with $B$ as in Theorem 8.7) for every $c$ in the window $\{c > 2^{-m} : J_g(c) + \epsilon < \Phi_2^\sharp(c)\}$; there, and under the hypotheses of Theorem 8.7(ii), (ii)(a) holds in the sharper form $u(\pi) \geq (2^{-R_{\max}} - p(1 + 2\gamma_g + 2^{1-g}) - 2^{-B})/(1-2p)$ for every integer $g \geq 1$, with $\gamma_g$ as in (ii)(e). (iv) Proposition 8.3(c) holds with $\Phi_2^\sharp$, computed at the same $B$, in place of $\Phi_2$, for every compact $[c_1, c_2] \subset \{c > 2^{-m} : J_g(c) < \Phi_2^\sharp(c)\}$.*

*Proof of Proposition F.8.* (i) By (F3), $I(\pi) \leq K(\pi \mid a, K(a)) + \kappa$, and by (F4), (F3), $K(\pi \mid a, K(a)) \leq K(\pi) + K(a \mid \pi) - K(a) + \kappa$. Insert $K(\pi) \leq \ell(\pi) + \kappa$ and $K(a \mid \pi) \leq \min\{s(t) + 2\log d,\ K(a)\} + \kappa$ (Lemma 7.2, or trivially), and use $\min\{s, K(a)\} - K(a) \leq \min\{s, d\} - d$, as $K(a) \geq d$. Also $K(S \mid a, \pi) \leq K(S \mid a) + \kappa \leq B_S + K(K(a)) + \kappa$ by (F3), $K(K(a)) \leq 2\log(K(a)+1) + \kappa$ by (F5), and $K(a) \leq d + 2\log(d+1) + \kappa \leq 2d - 1$; so $I(\pi) \geq -2\log(2d) - \kappa$. $B_S$ and $K(S \mid a, \pi)$ are integers. (ii) The first term of $LB_t$ is the bound in the proof of (o) before $s(t) \leq 4t$ is used, so $\Phi_2^\sharp \geq \Phi_2$ by that proof ($LB_\infty \geq G_\infty \geq \Phi_2$). For the second, with $I := I(\pi)$, (i) gives $c\,\ell(\pi) \geq c(d - \min\{s(t) + 2\log d, d\} - 2\log(2d) - \kappa) + cI^+$ (if $I < 0$, use

$cI \geq -c(2\log(2d)+\kappa)$). Lemma 7.6 with the budget $I^+ \geq I$, $u(\pi) \leq \theta_{t-1}$ and $1-2p>0$ gives $R_B(\pi) \geq \mathrm{Floor}_t(I^+)$; add and minimise over $I^+$. *Continuity.* Let $t_c$ be the least integer with $s(t_c) \geq d-\Lambda_1$ and $s(t_c)+2\log d \geq d$. For $t \geq t_c$ the first term of $LB_t$ is $G_{t-1}$, the coefficient of $c$ in the second is $-2\log(2d)-\kappa$, and $\Psi_t$ is nondecreasing in $t$ as $\theta_{t-1}$ decreases; also $LB_\infty \geq LB_{t_c}$. So $\Phi_2^\sharp = \min_{2\leq t\leq t_c} LB_t$. As $p+(1-2p)\theta_{t-1} \geq \frac{1}{2}$, $0 \leq \mathrm{Floor}_t \leq 1$, so on a compact subset of $(0,\infty)$ only finitely many $I$ compete in $\Psi_t$, and each $LB_t$ is continuous. (iii) The proofs of Theorem 8.7(i),(ii) use (o) only through "level at least 2 implies $J_c^B \geq \Phi_2(c)$", that is, through $\eta_{\mathcal{N}} \geq \Phi_2 - J_g$, and (i) uses $c \leq c_{\mathrm{hi}}$ only for $J_g \leq \Phi_2$; by (ii) both hold with $\Phi_2^\sharp$ on the window. Parts (a)–(e) use only level 1, (D1), (D2) and the bounds on $\ell$ and $I$. For the sharper (a): by (ii)(d), $I(\pi) \leq \Delta_\ell + \Lambda_3$, so Lemma 7.6 with this budget, whose $j$ is $j_g$, gives $2^{-R_B} \leq p+(1-2p)u+p(2\gamma_g+2^{1-g})+2^{-B}$; with $R_B \leq R_{\max}$, solve for $u$. On this window $\Lambda_2 \geq 0$: if $\pi_g$ had level at least 2, then by (ii) $J_g(c) \geq J_c(\pi_g) \geq J_c^B(\pi_g) \geq \Phi_2^\sharp(c)$, against $J_g(c)+\epsilon < \Phi_2^\sharp(c)$; so $u(\pi_g) = 1-p > \frac{3}{4}$, hence $p < \frac{1}{4}$, and $\Lambda_2 \geq 0$ by Proposition 8.6(b). (iv) The proof of Proposition 8.3(c) uses $\Phi_2$ only through Theorem 8.7(ii) and the continuity and positivity of $\Phi_2 - J_g$ on $[c_1, c_2]$. □

| $p$ | window, $\kappa=0$ | window, $\kappa=30$ | $u \geq$ | $e \leq$ |
|---|---|---|---|---|
| $2^{-10}$ | $(1,\ 8.198\cdot10^{-3})$ | $(1,\ 3.686\cdot10^{-3})$ | 0.989 | 0.193 |
| 1/16 | $(1,\ 2.529\cdot10^{-3})$ | $(1,\ 6.559\cdot10^{-4})$ | 0.827 | 0.197 |
| 5/64 | $(1.014,\ 1.521\cdot10^{-3})$ | $(1.016,\ 4.285\cdot10^{-4})$ | 0.802 | 0.221 |
| 3/32 | $(2.141,\ 9.097\cdot10^{-4})$ | $(2.155,\ 2.487\cdot10^{-4})$ | 0.780 | 0.241 |
| 7/64 | $(6.223,\ 3.184\cdot10^{-4})$ | $(6.532,\ 8.558\cdot10^{-5})$ | 0.759 | 0.260 |
| 15/128 | $(21.433,\ 6.225\cdot10^{-5})$ | none | 0.750 | 0.269 |
| 1/8 | none | none | – | – |

Table 6: The windows $\{c > 2^{-m} : J_g(c) < \Phi_2^\sharp(c)\}$ of Proposition F.8 at $d=100$, $m=20$, $B=\infty$, $\delta_S=0$; lower ends in units of $2^{-20}$, rounded up, and upper ends rounded down. Rows with $\kappa=0$ set the machine constant to 0 and are illustrations; $\kappa_g = 2\log d + K(p) + \kappa$, with $K(p)$ taken as $2\log r$ for $p=2^{-r}$ and $2\log(h+1)+2\log(e+1)$ for $p=h/2^e$, $h>1$ odd. The last columns give the sharper (ii)(a) of Proposition F.8(iii) and the resulting (ii)(e) at $\kappa=0$, $\epsilon=0$, $g$ optimised, at $c=10^{-4}$ ($p=2^{-10}$) and $c=6.745, 5.282, 4.278, 4.337, 3.586\cdot10^{-5}$ (the other rates in order); $u$ is rounded down and $e$ up. At $p=2^{-10}$ and $c=10^{-4}$ this is the sharpest of three forms of (ii)(a): Table 3 uses $J_g(c)+\epsilon$ for $R_{\max}$ ($u \geq 0.983$, $e \leq 0.200$), and Remark 8.8 uses $R_{\max}$ ($u \geq 0.988$, $e \leq 0.195$).

*Proof of Proposition 8.13.* Let $f(x) := -\log(\frac{3}{4} - \frac{p}{2} + 2p\sqrt{x\ln 2/2})$, so $\psi_2(p,a) = \inf_{x\geq 0}[apx+f(x)]$; at $c = a2^{-m}$ a budget $I = xs_0$ costs $cI = apx$. *The floor converges.* Let $r_g := \delta_S + 1 + 2\log m + 2\log s_0 + g + 2\log g + \kappa$; in $\gamma_g(I)$, $j$ is the least integer with $j - 2\log(j+1) > I + r_g$, so $I + r_g < j \leq I + r_g + O(\log(I+|r_g|+2))$. Take $g = g_m \to \infty$ with $g_m = o(s_0)$; as $\delta_S = o(2^m)$, $|r_{g_m}| = o(s_0)$, so $\gamma_{g_m}(xs_0)^2 = x\ln 2/2 + o(1)$ uniformly on compacts, and $p2^{1-g_m} \to 0$. $\mathrm{Floor}_2$ is a supremum over $g$ and $\theta_1 = \frac{3}{4}$, so by uniform continuity $\mathrm{Floor}_2(xs_0) \geq f(x) - o(1)$ uniformly on $[0,X]$; take $X$ with $apX > \psi_2(p,a)+1$, and note $cI + \mathrm{Floor}_2 \geq apX$ for $x > X$. So $\liminf_m \Psi_2(a2^{-m}) \geq \psi_2(p,a)$. *All levels.* $\Psi_t \geq \Psi_2$ as $\theta_{t-1} \leq \theta_1$, and the coefficient of $c$ in the second term of $LB_t$ is at least $-2\log(2d)-\kappa$, so uniformly in $t$, $J_g(c) - LB_t(c) \leq c(d + \kappa_g + 2\log(2d) + \kappa) + H(p) - \Psi_2(c)$. For $c \in [a2^{-m}, c_m]$ the first term is at most $c_m(d + O(\log d)) \to 0$, and $\Psi_2(c) \geq \Psi_2(a2^{-m})$. As $a > a^*(p)$ and $\psi_2(p,\cdot)$ is nondecreasing, $\psi_2(p,a) > H(p)$, and $\epsilon := \frac{1}{2}(\psi_2(p,a) - H(p))$ works for all large $m$ (where also $d \geq 2\log(d+1) + \kappa + 1$). *The readings.* $\psi_2(p,\cdot)$ is an infimum

of affine functions, hence concave, continuous on $(0,\infty)$ and nondecreasing. The infimum is attained at some $x^* > 0$, as $f'(x) \to -\infty$ when $x \downarrow 0$; so for $a' > a$, with $x^*$ the minimiser at $a'$, $\psi_2(p,a') = a'px^* + f(x^*) > apx^* + f(x^*) \geq \psi_2(p,a)$: $\psi_2(p,\cdot)$ is strictly increasing. Also $\psi_2(p,a) \leq f(0) = 1 - \log(\frac{3}{2} - p)$, and $\psi_2(p,a) \to f(0)$ as $a \to \infty$, since $f$ is continuous at 0 and $apx \to \infty$ for $x$ bounded away from 0. So $\psi_2(p,a) < f(0)$ for every $a$, and some $a$ has $\psi_2(p,a) > H(p)$ if and only if $f(0) > H(p)$. If $\psi_2(p,1) \geq H(p)$, strict increase gives $a^*(p) = 1$; if $\psi_2(p,1) < H(p)$, continuity gives $a^*(p) > 1$. □

*Monotonicity of $h(p) := \psi_2(p,1) - H(p)$ on $(0, \frac{1}{4}]$.* With $y := \sqrt{x}$, $A := \frac{3}{4} - \frac{p}{2}$ and $k := 2p\sqrt{\ln 2/2}$, $\psi_2(p,1) = \min_{y\geq 0}[py^2 - \log(A + ky)]$; the function is strictly convex with negative slope at 0, so its minimiser $y^*$ is unique, positive, and solves $y^*(A + ky^*) = q := 1/\sqrt{2\ln 2} = 0.8493\ldots$. For $p \leq \frac{1}{4}$, $A \geq \frac{5}{8}$ and $k \leq 0.2944$, so $y^* \leq q/A \leq 1.359$, and $y^* \geq 0.7385$ as $0.7385(\frac{3}{4} + 0.2944 \cdot 0.7385) < q$. By Danskin's theorem

$$h'(p) = y^{*2} - \frac{2\sqrt{\ln 2/2}\, y^* - \frac{1}{2}}{(A + ky^*)\ln 2} - \log\frac{1-p}{p},$$

where, with $A + ky^* = q/y^*$, the middle term is $(2\sqrt{\ln 2/2}\, y^* - \frac{1}{2})y^*/(q \ln 2) \geq 0.463$, increasing in $y^* \geq 0.7385$; and $\log\frac{1-p}{p} \geq \log 3$. So $h'(p) \leq 1.359^2 - 0.463 - \log 3 < -0.2$. As $h(p) \geq -\log(A + ky^*) - H(p) \to \log\frac{4}{3}$ for $p \to 0$ and $h(\frac{1}{4}) \leq 1 - \log\frac{5}{4} - H(\frac{1}{4}) < 0$, $h$ has exactly one root $p^*$ in $(0, \frac{1}{4})$.

At $d = 100$, $m = 20$, $B = \infty$ and $\delta_S = 0$, a direct computation (code in the accompanying repository), with the minimum over integer budgets taken exactly, gives the windows of Table 6; each computed window is an interval. Of these rates only $2^{-10}$ has $G_1 > H(p)$, so Theorem 8.7 gives no window for the others.

# G Proofs for Section 9 (structured errors)

The setting is Definition 7.1 with $B = \infty$ unless stated otherwise, so $\Lambda_1 = 2\log d + 2\log m + \kappa$; (F1)–(F5) are the facts of Appendix A.1, and (D1), (D2) the bounds of Lemma 7.5.

## G.1 Error envelopes

*Proof of Proposition 9.1. The envelope.* Off $E_1$ the error inputs are those with prefix in $S_2$, a fraction $p_2$, so $R(\pi_e) = (1 - p_1)H(p_2)$, and $p = p_1 + (1 - p_1)p_2$ with concavity of $H$ gives $H(p) \geq R_e$. The program $\pi_e$ needs $a$, $d$, $m$, a description of $S_1$ given $m$, and $s_2$, which gives the length bound.

(i) Apply Lemma 5.3 in the population reading ($L = L' = R$) with the reference point $(d + k_1 + \kappa_e, R_e)$, bounded by $\pi_e$, so that $J_{\mathrm{r}} = J_e$; let $\mathcal{N}$ be the programs of level at least 2 (level $\infty$ included) and $\mathcal{M}$ those of level 1. By Theorem 8.7(o) every program of $\mathcal{N}$ has $J_c \geq \Phi_2(c) > J_e(c) + \epsilon$, so $\mathrm{lev}(\pi) = 1$; by (D1) with $s(1) = 4$ every program of $\mathcal{M}$ has length at least $d - 4 - \Lambda_1$. Lemma 5.3(iv) gives $R(\pi) \leq R_e + \epsilon + c(k_1 + \kappa_e + 4 + \Lambda_1) = R_e + \zeta$. Every $\epsilon$-near-optimal program has $J_c \leq \Phi(c) + \epsilon \leq J_e(c) + \epsilon$; so if $\zeta < H(p) - R_e$, which forces $c < c_{\mathrm{env}}$, it has $R < H(p)$.

(ii) Write $R(\pi) = p_1 R^{(1)} + (1 - p_1)R^{(2)}$, with $R^{(1)}$ and $R^{(2)}$ the risks conditional on $E_1$ and off $E_1$, and put $x := (1 - p_1)R^{(2)}$. *Coding $S_2$ given $S_1$.* For $z \notin S_1$ define $q_z(b)$ as in the proof of Lemma 7.4 with $B = \infty$. Then $Q(S') := \prod_{z\notin S_1} q_z(\mathbf{1}_{S'}(z))$ is a lower-semicomputable semimeasure on subsets of $\{0,1\}^m \setminus S_1$, given $(\pi, a, m, S_1)$. For $z \notin S_1$ the teacher's label is $f_a \oplus \mathbf{1}_{S_2}(z)$, so

$-\log Q(S_2) = (2^m - s_1)R^{(2)} = 2^m x$, and (F2) gives $K(S_2 \mid a, \pi, S_1) \le 2^m x + 2\log m + \kappa$. *Chain.* By (F3), (F1) and Lemma 7.2 with $t = 1$, $K(a, S_2 \mid S_1) \le \ell(\pi) + 4 + 2\log d + 2^m x + 2\log m + \kappa$; by (F3) and (F4), $K(a, S_2 \mid S_1) \ge K(a, S_2) - K(S_1) - \kappa \ge d + B_2 - K(S_1) - \kappa$. With (D1),

$$\ell(\pi) \;\ge\; d - 4 - \Lambda_1 + \max\{0,\; B_2' - 2^m x\}, \qquad B_2' := \big(B_2 - K(S_1)\big)^+.$$

*Combination.* $J_c(\pi) = c\,\ell(\pi) + p_1 R^{(1)} + x \le J_e(c) + \epsilon$ gives

$$p_1 R^{(1)} \le \zeta + (1 - p_1)H(p_2) - \big[x + c\max\{0, B_2' - 2^m x\}\big].$$

Over $x \ge 0$ the bracket is piecewise linear: $cB_2' + (1 - c2^m)x$ for $x \le B_2'/2^m$, and $x$ beyond. Its minimum is $B_2'/2^m$ if $c2^m \ge 1$ and $cB_2'$ if $c2^m < 1$, that is, $\min(c, 2^{-m})B_2'$. Hence $R^{(1)} \le \bar{R}_1$, and Jensen's inequality gives $e_1(\pi) \ge 2^{-R^{(1)}} \ge 2^{-\bar{R}_1}$.

For $c \ge 2^{-m}$, $(2^m - s_1)H(p_2) \le \log \binom{2^m - s_1}{s_2} + \log(2^m + 1)$ gives $\bar{R}_1 \le (\zeta + (\delta_2 + \log(2^m + 1) + K(S_1))/2^m)/p_1$ with $\delta_2 := \log \binom{2^m - s_1}{s_2} - B_2$. □

*Numbers.* Take $d = 100$, $m = 20$, $p = 2^{-10}$, $S_1 = \{z : z_1 = \cdots = z_j = 0\}$, so that $K(S_1 \mid m) \le 2\log(j + 1) + \kappa$, and $S_2$ uniformly random, so that $\delta_2$ is small with high probability. With the machine constant set to 0 (an illustration) unless stated, a direct computation (code in the accompanying repository) gives the following. For $j = 11$ (512 cheap and 512 random points), $\log \binom{2^m - s_1}{s_2} = 6{,}364$, $H(p) - R_e = 0.00510$ and $c_{\text{env}} = 6.98 \cdot 10^{-5}$ ($3.12 \cdot 10^{-5}$ at $\kappa = 30$); for $j = 13$ ($128 + 896$ points), 10,418, 0.00123 and $1.64 \cdot 10^{-5}$ ($0.746 \cdot 10^{-5}$). The share of the logarithmic width of the window $(2^{-m}, c_{\text{hi}}]$ of Theorem 8.7 below $c_{\text{env}}$, $\log(c_{\text{env}}2^m)/\log(c_{\text{hi}}2^m)$, is 0.47 and 0.31 (0.42 and 0.24 at $\kappa = 30$). In both instances $J_e < \Phi_2$ on $(0, c_{\text{env}}]$, as part (i) requires. From the relaxed bound, with $K(S_1) \le k_1 + 2\log m + \kappa$, the lower bound on $e_1$ at $c = 1.01 \cdot 2^{-m}$ is 0.862 and 0.544 for $\delta_2 = 0$ (0.839 and 0.489 for $\delta_2 = 20$); it falls to 0.042 and 0.000 at $c = 3 \cdot 10^{-5}$. Against $L_S = 11{,}710$ the deficiencies of the two rules are about 5,300 and 1,300 bits.

*Envelopes with density $\theta < 1$.* If the cheap region contains errors with density $\theta$, the envelope predicts the error there with probability $\theta$, and the proof of (i) goes through with $R_e = p_1 H(\theta) + (1 - p_1)H(p_2)$ and the corresponding length. For $\theta > \frac{1}{2}$ the envelope is a learned systematic error. Part (ii) is proved only for $\theta = 1$.

## G.2 Rate blindness: the criterion in the machine constant

*Proof of Proposition 9.2.* (i) $J_c(\pi_b) < J_c(\pi_g)$ if and only if $c < H(p)/(\ell(\pi_b) - \ell(\pi_g))$, and $J_c(\pi_{\text{coin}}) \le J_c(\pi_g)$ if and only if $c \ge (1 - H(p))/(\ell(\pi_g) - \ell(\pi_{\text{coin}}))$; under the hypothesis these ranges cover all $c > 0$.

(ii) Here $\text{lev}(\pi) = 1$, so $\ell(\pi) \ge d - 4 - \Lambda_1 > \kappa \ge \ell(\pi_{\text{coin}})$ by (D1) and ($\star$). If $\ell(\pi_b) \le \ell(\pi)$, the memoriser beats $\pi$ at every $c$, since $R(\pi_b) = 0 < R(\pi)$; if $R(\pi) \ge 1$, the coin beats or ties it at every $c$. Otherwise $0 < \ell(\pi_b) - \ell(\pi) \le \lceil L_S \rceil + \kappa_b + 4 + \Lambda_1$, so the memoriser's secant $R(\pi)/(\ell(\pi_b) - \ell(\pi))$ is at least the left side of ($\star$), and $\ell(\pi) - \ell(\pi_{\text{coin}}) \ge d - 4 - \Lambda_1 - \kappa$, so the coin's secant $(1 - R(\pi))/(\ell(\pi) - \ell(\pi_{\text{coin}}))$ is at most the right side. Since $J_c(\pi_b) < J_c(\pi)$ for $c$ below the first secant and $J_c(\pi_{\text{coin}}) \le J_c(\pi)$ for $c$ at or above the second, these ranges cover all $c > 0$. (iii) is Theorem 8.7(iii). □

**Lemma G.1** (The criterion is robust in the machine constant)**.** *Fix $m$ and $s_0$, let $\kappa$ bound the machine constants in $\Lambda_1$, $\kappa_b$ and $\ell(\pi_{\text{coin}})$, and put $H := H(p)$, $q := (1 - H)/H$, $\alpha_0 := 2 + 4q$, $\beta_0 := 4 + 2\log m + 2\kappa + q(\lceil L_S \rceil + 4\log m + 2\log s_0 + 2\kappa + 4)$ and $F(d) := d - \alpha_0 \log d - \beta_0$. (i) For $d \ge 1$, ($\star$) holds if and only if $F(d) \ge 0$. (ii) If $\beta_0 \ge \alpha_0/\ln 2$, then $\{d \ge 1 : F(d) \ge 0\} = [d_0, \infty)$ for a unique real $d_0 = d_0(\kappa)$. (iii) $d_0(\kappa)$ grows linearly in $\kappa$, with slope $(2 + 2q)(1 + o(1))$.*

*Proof.* (i) With $B = \infty$, $\kappa_b = 2\log d + 2\log m + 2\log s_0 + \kappa$. The first denominator of $(\star)$ is positive. When the second is positive, $(\star)$ is equivalent to $H\,(d - 4 - \Lambda_1 - \kappa) \geq (1 - H)(\lceil L_S \rceil + \kappa_b + 4 + \Lambda_1)$; dividing by $H$, the difference of the two sides is

$$d - 4 - 2\log d - 2\log m - 2\kappa - q\big(\lceil L_S \rceil + 4\log d + 4\log m + 2\log s_0 + 2\kappa + 4\big) = F(d).$$

Conversely, $F(d) \geq 0$ implies $d - 4 - \Lambda_1 - \kappa \geq F(d) + 4q\log d + q(\lceil L_S \rceil + 4) > 0$, so the side condition holds as well. (ii) $F'(d) = 1 - \alpha_0/(d \ln 2)$, so $F$ is strictly increasing on $[\alpha_0/\ln 2, \infty)$ and tends to $\infty$. On $[1, \alpha_0/\ln 2]$, $F(1) = 1 - \beta_0 < 0$ because $\beta_0 \geq 4$, and for $d > 1$, $F(d) < d - \beta_0 \leq \alpha_0/\ln 2 - \beta_0 \leq 0$. So $\{d \geq 1 : F(d) \geq 0\} = [d_0, \infty)$ for the unique root $d_0 > \alpha_0/\ln 2$. (iii) $\beta_0$ is affine in $\kappa$ with slope $2 + 2q$, $\alpha_0$ does not depend on $\kappa$, and $d_0 = \beta_0 + \alpha_0 \log d_0$. Taking the three constants equal to a common upper bound is legitimate because $(\star)$ only becomes harder to satisfy when any of them grows. □

A direct computation (code in the accompanying repository) gives $d_0(\kappa)$, the least integer $d$ with $F(d) \geq 0$, at $\kappa = 0$ (an illustration), 10, 20, 30, 50 and 100: 49, 90, 130, 169, 245 and 434 for one 3-bit prefix ($m = 3$, $p = 1/8$), and 450, 639, 823, 1005, 1364 and 2248 for one 6-bit prefix ($m = 6$, $p = 1/64$); in each case $\beta_0 \geq \alpha_0/\ln 2$, and $(\star)$ fails at $d_0 - 1$ and holds at $d_0$.

### G.3 Separable components, guard erosion, non-identifiability, input law

*Proof of Proposition 9.3.* The objective $c\,\ell + R$ is a sum over regions of $c$ times the extra length plus the risk on the region, with independent choices, so it is minimised region by region, by Lemma 3.8 in each; with two options, component $j$ lowers it if and only if $c\,k_j < r_j$. If a valid component with rate $\sigma_v$ is kept, then $c < \sigma_v \leq \sigma_e$ (up to ties), so an error component with rate $\sigma_e$ is kept too. $J_c$ depends on the data only through the teacher's channel and the input law, which coincide in the two worlds. □

**Proposition G.2** (Guard erosion)**.** *In a region of contexts a schema $\varphi$ is one of $n_{\mathrm{st}}$ possible steps. A guard $G$ splits the region into $G$-contexts (input mass $\mu(G)$), where the teacher plays $\varphi$ with probability $1 - \epsilon_G + \epsilon_G/n_{\mathrm{st}}$ and every other step with probability $\epsilon_G/n_{\mathrm{st}}$, and $\neg G$-contexts (mass $\mu(\neg G)$), where it plays one fixed step other than $\varphi$. Over a common base, three options have extra lengths $0$, $k_\varphi$ and $k_\varphi + k_G$: "none" (uniform), "unguarded $\varphi$" (the $G$-context law everywhere) and "guarded $\varphi$" (the $G$-context law in $G$-contexts, uniform over the other $n_{\mathrm{st}} - 1$ steps elsewhere). With $H_G$ the entropy of the $G$-context law, put $\sigma_\varphi := (\mu(G)(\log n_{\mathrm{st}} - H_G) - \mu(\neg G)\log(1/\epsilon_G))/k_\varphi$ and $\sigma_G := \mu(\neg G)(\log(n_{\mathrm{st}}/\epsilon_G) - \log(n_{\mathrm{st}} - 1))/k_G$. If $\sigma_G < \sigma_\varphi$, then for every $c \in (\sigma_G, \sigma_\varphi)$ the unguarded rule is the unique selection, and it applies $\varphi$ where the teacher never did. Otherwise the selection goes from guarded to none.*

*Proof.* Per context, the expected log losses are $\log n_{\mathrm{st}}$ for "none"; $H_G$ in $G$-contexts and $\log(n_{\mathrm{st}}/\epsilon_G)$ in $\neg G$-contexts for "unguarded $\varphi$", since the teacher's fixed step has probability $\epsilon_G/n_{\mathrm{st}}$ under the $G$-context law; and $H_G$ and $\log(n_{\mathrm{st}} - 1)$ for "guarded $\varphi$". Relative to "none", the options are the points $(0, 0)$, $(k_\varphi, -k_\varphi\sigma_\varphi)$ and $(k_\varphi + k_G, -k_\varphi\sigma_\varphi - k_G\sigma_G)$ in the (length, risk) plane, and adding back $\log n_{\mathrm{st}}$ makes Lemma 3.8 applicable. The middle point is a vertex if and only if $\sigma_\varphi > \sigma_G$, and it is then the unique minimiser for $c \in (\sigma_G, \sigma_\varphi)$ by Lemma 3.8(iii); otherwise the selection switches between the outer points at $c = (k_\varphi\sigma_\varphi + k_G\sigma_G)/(k_\varphi + k_G)$. □

For $n_{\mathrm{st}} = 64$, $\epsilon_G = 0.05$, $\mu(G) = 0.05$, $\mu(\neg G) = 5 \cdot 10^{-4}$, $k_\varphi = 22$ and $k_G = 12$, a direct computation (code in the accompanying repository) gives $\sigma_\varphi = 1.2226 \cdot 10^{-2}$ and $\sigma_G = 1.8103 \cdot$

$10^{-4}$, so the unguarded rule is the unique selection on an interval spanning a factor of 67 in $c$; for $\mu(G) = \mu(\neg G) = 0.002$, $\sigma_\varphi < \sigma_G$ and the guard is never dropped. The teacher has noise $\epsilon_G$ in $G$-contexts; what it never does is apply $\varphi$ outside $G$.

**Proposition G.3** ($\lambda$ cannot be chosen from imitation data)**.** *(i) Every strictly proper scoring rule has expected score on the teacher's data uniquely maximised by the teacher's channel $P^*(\cdot \mid x)$, errors included (Gneiting and Raftery, 2007), so validation by such a score prefers the teacher's channel to the intended model. (ii) Along the population selection path the held-out log loss $R(\pi_c)$ of a minimiser $\pi_c$ of $J_c$ is nonincreasing as $c$ decreases, so log-loss validation over a grid of values of $c$ is minimised at the smallest, error-memorising, value. (iii) In the twin worlds of Proposition 9.3, every rule that chooses $c$ from imitation data alone is wrong in one of them.*

For the three hypotheses of Remark 8.1, with the machine constant set to 0 (an illustration), a direct computation (code in the accompanying repository) gives the held-out losses 1.0, 0.0112, 0.0112 and 0 at $c = 10^{-1}, 10^{-3}, 10^{-5}, 10^{-7}$.

*Proof.* (i) A scoring rule is strictly proper if, for each $x$, the expected score of a forecast $Q$ under $P^*(\cdot \mid x)$ is uniquely maximised at $Q = P^*(\cdot \mid x)$ (Gneiting and Raftery, 2007); average over $x$. (ii) By Lemma 3.8(ii), $c < c'$ implies $R(\pi_c) \le R(\pi_{c'})$. (iii) An imitation-only rule sees the same data distribution in both worlds, and Proposition 9.3 gives the same selection at every $c$; the selection that is right in one world keeps an error or drops a valid rule in the other. □

**Proposition G.4** (The selection depends on the input law)**.** *In the setting of Proposition 9.3, let the inputs have law $\mu$ and let $\Delta_j$ be component $j$'s expected log-loss reduction per input of $X_j$. Then component $j$ is selected if and only if $\mu(X_j)\Delta_j > c\,k_j$. So the window that keeps a valid component $j$ and drops an error component $j'$ is $c \in (\mu(X_{j'})\Delta_{j'}/k_{j'},\ \mu(X_j)\Delta_j/k_j)$; it moves with $\mu$ and is empty once $\mu(X_{j'})/\mu(X_j) \ge \Delta_j k_{j'}/(\Delta_{j'} k_j)$, and after a shift to $\mu'$ a $c$ chosen in $\mu$'s window keeps the error if and only if $\mu'(X_{j'})\Delta_{j'} > c\,k_{j'}$.*

*Proof.* The population risk is $\sum_j \mu(X_j)\mathbb{E}[\text{loss} \mid x \in X_j]$, so choosing component $j$ changes $J_c$ by $c\,k_j - \mu(X_j)\Delta_j$; the other claims are rearrangements. □

## G.4 Unseen properties

For $M_F$, a behaviour $b$ on a decidable set $Q$ absent from the training data gets probability at least $2^{-K(Q,b)-O(1)}$, where $K(Q,b)$ is the length of a shortest code for a decider of $Q$ and a program for $b$ (Hänni, 2026a, Proposition 8.1). In $\mathcal{H}_\times$ the bound becomes $2^{-\lambda(K(Q,b)+\kappa)}$, and programs that fit the training data equally well have posterior odds $2^{-\lambda(\ell(\pi_1)-\ell(\pi_2))}$ (Proposition G.5). With $\lambda = cn$ the bound degrades to $2^{-cn(K(Q,b)+\kappa)}$, and the shortest hypotheses that fit the training data decide the behaviour on $Q$: steeper SFI extrapolates more reliably than SFI if they extrapolate well, and is more confidently wrong if they do not.

**Proposition G.5** (Steeper simplicity and unseen properties)**.** *In $\mathcal{H}_\times$, let $Q \subseteq X$ be decidable, $b$ partial computable, $D$ a data set with $\operatorname{dom} D \cap Q = \emptyset$, $Z \subseteq Q \cap \operatorname{dom} b$ finite, and $\lambda \ge 1$. (i) A family of programs that answer $b(x)$ on $Q$ and run an arbitrary program of the class elsewhere (one for each such program) carries posterior mass exactly $2^{-\lambda(K(Q,b)+\kappa)}$, so all programs with this behaviour together carry at least this mass, and $\xi_\lambda(D \cup b|_Z) \ge 2^{-\lambda(K(Q,b)+\kappa)}\xi_\lambda(D)$. (ii) If $\nu_{\pi_1}(D) = \nu_{\pi_2}(D) > 0$, then $\Pi_\lambda(\pi_1 \mid D)/\Pi_\lambda(\pi_2 \mid D) = 2^{-\lambda(\ell(\pi_1)-\ell(\pi_2))}$.*

*Proof.* (i) Let $V'$ be the prefix machine that reads a self-delimiting code $z$ of a pair (decider $q$ for $Q$, program $q'$ for $b$), then a program $\pi' \in \operatorname{dom} V$, and outputs an index of the conditional semimeasure that is the point mass at $b(x)$ (or 0 while $b(x)$ is undefined) if $q$ accepts $x$, and $\nu_{\pi'}(\cdot \mid x)$ otherwise; it is lower semicomputable. By closure under prefix-free simulation (Definition 3.5) there is $\sigma$ with $V(\sigma z \pi') = V'(z\pi')$; let $\kappa := |\sigma|$ and $|z| = K(Q, b)$. Distinct $\pi'$ give distinct programs $\sigma z \pi'$; since $\operatorname{dom} D \cap Q = \emptyset$ and $Z \subseteq Q \cap \operatorname{dom} b$, $\nu_{\sigma z \pi'}(D) = \nu_{\sigma z \pi'}(D \cup b|_Z) = \nu_{\pi'}(D)$. So $\sum_{\pi'} 2^{-\lambda|\sigma z \pi'|} \nu_{\sigma z \pi'}(D) = 2^{-\lambda(K(Q,b)+\kappa)} \xi_\lambda(D)$, which is the posterior mass claim, and the same sum with $D \cup b|_Z$ in place of $D$ is a lower bound for $\xi_\lambda(D \cup b|_Z)$. (ii) Equal likelihoods cancel in the ratio of posteriors, which leaves the ratio of the prior weights. □

At $\lambda = 1$, (i) is Hänni (2026a, Proposition 8.1(a)) with $M_{\text{prod}}$ in place of $W$ and $M_F$. On a conjunction ladder (Hänni, 2026a, Theorem 9.10 and Remark G.11), selecting a least-size consistent formula, a $\lambda$-MAP rule, makes exponentially many mistakes on some ordering for every tie rule, while the $\lambda \to \infty$ power posterior, uniform over those formulas, makes $O(d^2)$ Gibbs mistakes; the danger in Proposition G.5(ii) is sharpest for single-hypothesis selection.

# H Proofs for Section 10 (imitation)

The setting is Definition 10.1, and (F1)–(F5) are the facts of Appendix A.1; we use them relative to a condition such as $(h_0, \mu)$, whose proofs are the unconditional ones, uniformly in the condition (Li and Vitányi, 2008). The channel $\mu$ is a finite object from which $m$, $W$ and $\operatorname{supp} \mu$ are computable.

## H.1 Private coins

*Remark* H.1 (Private coins). If, given its input, the person's answer is independent of everything a hypothesis may use (the earlier data and any variables of the environment, earlier or later), the argument of Lemma 6.2 with this larger filtration shows that no predictor, however it models the environment, gains over the person more than its length plus a Ville deficiency $\delta^*$ with $\mathbb{P}[\delta^* \geq j] \leq 2^{-j}$ (proved below). For product hypotheses that also see an environment variable in a countable set, with i.i.d. pairs, none then predicts better in population, and Theorem 6.3 returns the person. This is the degenerate case: a hypothesis can only be lucky on a sample, at one bit of length per bit of luck.

Let $\mathcal{A}_1 \subseteq \mathcal{A}_2 \subseteq \cdots$ be $\sigma$-algebras, $x_i$ being $\mathcal{A}_i$-measurable and $y_i$ being $\mathcal{A}_{i+1}$-measurable, that contain whatever a hypothesis may use to predict $y_i$, and let the coins be private: $\mathbb{P}[y_i = y \mid \mathcal{A}_i] = \nu_H(y \mid x_i)$ almost surely. A *predictor* $\pi$ assigns to each $i$ an $\mathcal{A}_i$-measurable sub-probability $q_{\pi,i}$ on $Y$, with fit $L_n(\pi) := -\sum_{i \leq n} \log q_{\pi,i}(y_i)$. The claim of Remark H.1 is: for every countable family of predictors with $\sum_\pi 2^{-\ell(\pi)} \leq 1$, $\delta^* := \sup_n \log \sum_\pi 2^{-\ell(\pi) + L_n(\nu_H) - L_n(\pi)}$ has $\mathbb{P}[\delta^* \geq j] \leq 2^{-j}$, and $L_n(\pi) \geq L_n(\nu_H) - \ell(\pi) - \delta^*$ for all $\pi$ and $n$.

*Proof of the claim.* Almost surely $\nu_H(y_i \mid x_i) > 0$ for all $i$, so $M_n := \sum_\pi 2^{-\ell(\pi)} \prod_{i \leq n} q_{\pi,i}(y_i) / \nu_H(y_i \mid x_i)$ is well defined and $\mathcal{A}_{n+1}$-measurable, with $M_0 \leq 1$. By privacy and monotone convergence, $\mathbb{E}[M_{n+1} \mid \mathcal{A}_{n+1}] = \sum_\pi 2^{-\ell(\pi)} \prod_{i \leq n} \frac{q_{\pi,i}(y_i)}{\nu_H(y_i|x_i)} \sum_{y : \nu_H(y|x_{n+1}) > 0} q_{\pi,n+1}(y) \leq M_n$. So $(M_n)$ is a nonnegative supermartingale, and Ville's inequality (Ville, 1939) gives $\mathbb{P}[\sup_n M_n \geq 2^j] \leq 2^{-j}$; put $\delta^* := \sup_n \log M_n$. Keeping one term, $2^{-\ell(\pi)} 2^{L_n(\nu_H) - L_n(\pi)} \leq M_n \leq 2^{\delta^*}$. For product hypotheses on i.i.d. pairs $(x_i, z_i)$, privacy gives $R(\pi) - R(\pi_H) = \mathbb{E}_{x,z} \operatorname{KL}(\nu_H(\cdot \mid x) \| \nu_\pi(\cdot \mid x, z)) \geq 0$, and the triples are per-query data on the input set of pairs with the sampler $\nu_H(y \mid x)$, to which Theorem 6.3 applies. □

*Hypotheses that learn in context.* A system that predicts the answers by inferring, from the data seen so far, which person it faces is a single hypothesis only in a class whose hypotheses carry information across queries. There a Bayes mixture $P_U$ over persons that gives the person weight $w$ has $P_U \geq wP_H$ on every data set, private coins or not, so it loses at most $\log(1/w)$ bits of fit to the person's own program and beats it at every $n$ once $\lambda$ times the length it saves exceeds $\log(1/w)$ (as in Theorem 4.4). The private-randomness class excludes such hypotheses by construction (Corollary 4.5), not because of anything in the data, and a person with memory is not a product sampler.

## H.2 The coding and closeness lemmas

**Lemma H.2** (Coding the coupling with a hypothesis)**.** *For every $\pi \in \mathcal{H}_\times$, $K(\beta \mid h_0, \pi, \mu) \leq 2^m R(\pi) + \kappa$.*

*Proof.* For $w \in W$ and $b \in \Sigma$ put $q_w(b) := \sum_x \mu(x \mid w)\, \nu_\pi(h(x,b) \mid x)$, a finite sum, lower semicomputable from $(\pi, h_0, \mu)$. Since $h(x,\cdot)$ is injective, $\sum_b q_w(b) \leq \sum_x \mu(x \mid w) \sum_y \nu_\pi(y \mid x) \leq 1$, so $Q(\beta') := \prod_w q_w(\beta'(w))$ is a semimeasure on $\Sigma^W$ for every well-formed $(h_0, \mu)$, which is decidable from this finite object (put $Q := 0$ otherwise), and (F2) gives $K(\beta \mid h_0, \pi, \mu) \leq -\log Q(\beta) + \kappa$. By Jensen's inequality for the convex function $-\log$,

$$\begin{aligned} -\log Q(\beta) &= \sum_w -\log \mathbb{E}_{x\sim\mu(\cdot|w)} \nu_\pi(h(x,\beta(w)) \mid x) \\ &\leq \sum_w \mathbb{E}_{x\sim\mu(\cdot|w)} \big[ -\log \nu_\pi(h(x,\beta(w)) \mid x) \big] = 2^m R(\pi). \end{aligned}$$ □

For a program $\pi$ put

$$\tilde{q}_w(b) := \prod_x \nu_\pi(h(x,b) \mid x)^{\mu(x|w)}, \qquad \bar{D}(\pi) := 2^{-m} \sum_w D_{1/2}(\rho \| \tilde{q}_w)$$

and $E(\pi) := 2\log\big(2^{m-1}(H(\rho) + R(\pi)) + 1\big)$. Here $\tilde{q}_w(b)$ is the geometric mean, over the inputs of cell $w$, of the probability that $\pi$ gives to the answer of coin $b$.

**Lemma H.3** (Closeness)**.** *For every $\pi \in \mathcal{H}_\times$ with $R(\pi) < \infty$,*

$$\begin{aligned} \mathbb{E}_{x\sim\bar{\mu}} D_{1/2}\big(\nu_H(\cdot \mid x) \| \nu_\pi(\cdot \mid x)\big) &\leq \bar{D}(\pi) \\ &\leq R(\pi) - H(\rho) + \frac{2\big(I_\beta(\pi) + \delta_\beta + |\Sigma| \log(2^m + 1) + E(\pi) + \kappa\big)}{2^m}. \end{aligned}$$

*Proof. Hölder.* Fix $w$. The generalised Hölder inequality $\sum_b \prod_x f_x(b)^{\theta_x} \leq \prod_x \big(\sum_b f_x(b)\big)^{\theta_x}$, for $f_x \geq 0$, $\theta_x \geq 0$ and $\sum_x \theta_x = 1$, with $\theta_x := \mu(x \mid w)$ and $f_x(b) := \sqrt{\rho(b)\nu_\pi(h(x,b) \mid x)}$, gives $\sum_b \sqrt{\rho(b)\tilde{q}_w(b)} \leq \prod_x \big(\sum_b \sqrt{\rho(b)\nu_\pi(h(x,b) \mid x)}\big)^{\mu(x|w)}$. By injectivity, $\sum_b \sqrt{\rho(b)\nu_\pi(h(x,b) \mid x)} = \sum_y \sqrt{\nu_H(y \mid x)\nu_\pi(y \mid x)}$. Taking $-2\log$ gives $D_{1/2}(\rho\|\tilde{q}_w) \geq \mathbb{E}_{x\sim\mu(\cdot|w)} D_{1/2}(\nu_H(\cdot \mid x)\|\nu_\pi(\cdot \mid x))$; average over $w$.

*A semimeasure.* $\tilde{q}_w(b) \leq q_w(b)$ (geometric against arithmetic mean), so $\sum_b \tilde{q}_w(b) \leq 1$ and, by the Cauchy–Schwarz inequality, $Z_w := \sum_b \sqrt{\rho(b)\tilde{q}_w(b)} \leq 1$. So $Q(\beta') := \prod_w \sqrt{\rho(\beta'(w))\, \tilde{q}_w(\beta'(w))}$ has total mass $Z := \prod_w Z_w \leq 1$ on $\Sigma^W$, and it is lower semicomputable from $(\pi, h_0, \mu)$. As $R(\pi) < \infty$, $Z \geq Q(\beta) > 0$.

*Normalising.* Let $k := \lfloor -\log Z \rfloor \geq 0$. Then $2^k Q$ is a semimeasure, lower semicomputable from $(\pi, h_0, \mu, k)$ (trimmed as usual, so that every value of the condition gives a semimeasure; for the true $k$ nothing is trimmed), and (F2), (F3), (F5) give

$$K(\beta \mid h_0, \pi, \mu) \leq -\log Q(\beta) - k + K(k) + \kappa \leq -\log Q(\beta) + \log Z + 1 + 2\log(k+1) + \kappa.$$

*Evaluating.* Exact type gives $\sum_w \log(1/\rho(\beta(w))) = 2^m H(\rho)$, and the definition of $R$ gives $\sum_w -\log \tilde{q}_w(\beta(w)) = 2^m R(\pi)$. So $-\log Q(\beta) = 2^{m-1}(H(\rho) + R(\pi))$ and $\log Z = -2^{m-1}\bar{D}(\pi)$; and $k \leq -\log Z \leq -\log Q(\beta)$, so $2\log(k+1) \leq E(\pi)$.

*Lower bound.* $K(\beta \mid h_0, \pi, \mu) = B_\beta - I_\beta(\pi) = L_\beta - \delta_\beta - I_\beta(\pi) \geq 2^m H(\rho) - |\Sigma| \log(2^m + 1) - \delta_\beta - I_\beta(\pi)$. Combine with the two previous steps and divide by $2^{m-1}$; the remaining $2/2^m$ is absorbed in $\kappa$. □

## H.3 Proof of Theorem 10.2

*The right line.* For every $\pi \in \mathcal{H}_\times$, (F3), (F1) and Lemma H.2 give $B_\beta \leq \ell(\pi) + K(\beta \mid h_0, \pi, \mu) + \kappa \leq \ell(\pi) + 2^m R(\pi) + \kappa$. With $B_\beta = L_\beta - \delta_\beta \geq 2^m H(\rho) - |\Sigma| \log(2^m + 1) - \delta_\beta$ and $G = \ell_0 + \delta_\beta + |\Sigma| \log(2^m + 1) + \kappa$,

$$R(\pi) \;\geq\; H(\rho) - \frac{G}{2^m} - 2^{-m}(\ell(\pi) - \ell_0) \qquad \text{for every } \pi \in \mathcal{H}_\times :$$

in the population reading of Definition 5.2, with the reference point $(\ell_0, H(\rho))$, bounded by $\pi_H$, the whole class has the right line $(2^{-m}, G/2^m)$ with length floor 0. Similarly $I_\beta(\pi) = B_\beta - K(\beta \mid h_0, \pi, \mu) \leq \ell(\pi) + \kappa$.

*(i)–(iv) and the bound on $\Phi$.* Apply Lemma 5.3 with $L = L' = R$, $\mathcal{M}$ the whole class, $\mathcal{N} = \emptyset$ (so $\eta_\mathcal{N} = +\infty$), $\tau = 2^{-m} < c$, $g = G/2^m \geq 0$ and $\ell_\mathcal{M} = 0$; then $c - \tau = (a-1)2^{-m}$ and $\Delta(c) = G/2^m + (c - 2^{-m})\ell_0$. A program with $J_c(\pi) \leq \Phi(c) + \epsilon$ has $J_c(\pi) \leq J_c(\pi_H) + \epsilon$. Lemma 5.3(ii) gives $\ell(\pi) \leq \ell_0 + (g + \epsilon)/(c - \tau)$, which is (i), and $R(\pi) \geq H(\rho) - (cg + \tau\epsilon)/(c - \tau)$, the left side of (ii); Lemma 5.3(iv) gives $R(\pi) \leq H(\rho) + \epsilon + c\,\ell_0$, the right side; and Lemma 5.3(i) gives $\Phi(c) \geq J_c(\pi_H) - \Delta(c) \geq J_c(\pi_H) - G/2^m - c\,\ell_0$. (iii) is $I_\beta(\pi) \leq \ell(\pi) + \kappa$ with (i). For (iv), Lemma H.3 applies, as $R(\pi) < \infty$; insert the right side of (ii), which also gives $E(\pi) \leq E$, and (iii):

$$\bar{D}(\pi) \leq c\,\ell_0 + \epsilon + \frac{2}{2^m}\Big[\ell_0 + \frac{G + 2^m\epsilon}{a-1} + \delta_\beta + |\Sigma| \log(2^m + 1) + E + 2\kappa\Big].$$

Since $\ell_0 + \delta_\beta + |\Sigma| \log(2^m + 1) + \kappa = G$, $2 + \frac{2}{a-1} = \frac{2a}{a-1}$ and $1 + \frac{2}{a-1} = \frac{a+1}{a-1}$, this is at most the bound in (iv). Finally $D_{1/2} = -2\log(1 - d_\mathrm{H}^2) \geq 2d_\mathrm{H}^2/\ln 2$.

*The program with kernel $P^*$.* It reads a self-delimiting description of $m$, $|\Sigma|$ and the numerators $\rho(b)2^m \leq 2^m$ ($a_\rho$ bits), then the index of $\beta$ among the couplings of type $\rho$, in $\lceil L_\beta \rceil$ bits, a length computable from what it has read, then $\pi_H$. On an input $x$ with first $m$ bits $w$ it computes $\nu_{\pi_H}(\cdot \mid x)$ exactly: the values are multiples of $2^{-m}$ that sum to 1 for $x \in \operatorname{supp}\mu$, so the program waits until the lower approximations sum to more than $1 - 2^{-m}$ and rounds each up to a multiple of $2^{-m}$. The label of mass $\rho(\beta(w))$ is $h(x, \beta(w))$, since the values of $\rho$ are distinct and $h(x, \cdot)$ is injective, and the program gives it probability 1; where the search does not end it gives nothing, which keeps a semimeasure. So its kernel is $P^*$ on $\operatorname{supp}\mu$, and its risk is 0. It has smaller $J_c$ than $\pi_H$, hence than every copy, if $c(\lceil L_\beta \rceil + a_\rho + \kappa) < H(\rho)$; and $\lceil L_\beta \rceil \leq 2^m H(\rho) + 1$. □

*Outside the range of $c$ discussed after the theorem.* If the values of $\rho$ are distinct, the program that models the coupling wins below about $2^{-m}$ (the last part of the theorem); with ties it must also describe which label each tied coin gives, and it can lose to the copies far below $2^{-m}$. Above

about $H(\rho)/\ell_0$ the bound (iv) exceeds $H(\rho)$, and above about $(\log|Y| - H(\rho))/\ell_0$ a simpler, worse program can win (the comparison with $\pi_{\mathrm{u}}$ in the proof of Proposition H.4), as above $c_{\mathrm{hi}}$ in Section 8.

*Numbers.* With $m = 20$, $\epsilon = \delta_\beta = 0$ and the machine constant set to 0 (an illustration, not a value for a fixed reference machine), a direct computation (code in the accompanying repository) gives at $c = 2^{-19}$ the bound (iv) of $6.2 \cdot 10^{-4}$ bits for a person who answers with a fair coin ($h(x,b) = b$, $\ell_0 \le a_\rho < 54$), against $H(\rho) = 1$, the divergence of the best predictor, and $9.5 \cdot 10^{-4}$ for the parity person of Section 8 ($\ell_0 < 120$), against $H(\rho) = 0.0112$; at $c = 10^{-4}$ the latter bound exceeds $H(\rho)$, and only Theorem 8.7 says more.

## H.4 Proof of Proposition 10.3

*(a).* In every $\mathcal{W}_i$ the sample $h_n$ has the same law, so $\Psi(h_n)$ has the same law. A program in $\mathcal{E}(\nu_i) \cap \mathcal{E}(\nu_j)$ would have $\nu_i = \nu_j$ $\bar{\mu}$-almost everywhere, so the events are disjoint. In the feature case, the world with $\Sigma = \{*\}$, $\rho(*) = 1$ and the response $h'(x,*) := h(x, \beta(w))$, $w$ the first $m$ bits of $x$, has the same channel and the data law $\bar{\mu} \otimes P^*$, and its person's kernel $P^*$ is deterministic, while $\nu_H$ is not at any input when $|\Sigma| \ge 2$.

*(b).* Conditional on $f$, hence on $h_0$ and $\mu$, $\beta$ is uniform on its type class, and fewer than $2^{L_\beta - j}$ programs are shorter than $L_\beta - j$; so $\mathbb{P}[\delta_\beta > j] < 2^{-j}$. The map $(f, \beta) \mapsto (T, \beta)$, $T := f \oplus \mathbf{1}_\beta$, is a bijection, so $T$ is uniform and independent of $\beta$, and the data, $(w_i, T(w_i))$ with $w_i$ uniform, depend on the world only through $T$. Fix $T$, the $w_i$ and the randomness of $\Psi$; then $\Psi$ outputs one program $\psi$. The kernels of the persons $H_{T \oplus \mathbf{1}_{\beta'}}$ for distinct $\beta'$ differ at every cell of the symmetric difference, each of mass $2^{-m}$, because $s_0 \ne 2^{m-1}$; so $\psi$ copies the person for at most one $\beta'$, and given $T$ each $\beta'$ has probability $\binom{2^m}{s_0}^{-1}$. Average over the conditioning.

*No window in (b).* Moreover, with probability at least $1 - 2^{-j}$ over $f$, if $H(p)\,(2^m - j - 2\log(m+1) - 2\kappa) > (1 - H(p))\,(j + 4\log(m+1) + 2\kappa)$ with $p := s_0/2^m$, then at every $c > 0$ the deterministic table $f \oplus \mathbf{1}_\beta$ or the fair coin has smaller $J_c$ than every copy. Indeed, from a copy $\pi$ the table $f$ is computable given $m$ ($f(w)$ is the label whose lower approximations of $\nu_\pi(\cdot \mid w)$ eventually exceed $\frac{1}{2}$, as $p < \frac{1}{2}$), so $K(f) \le \ell(\pi) + 2\log(m+1) + \kappa$; and $\mathbb{P}_f[K(f) < 2^m - j] < 2^{-j}$. The deterministic program for $T$ reads $m$ self-delimitingly and then $2^m$ bits, and has risk 0; the fair coin has length at most $\kappa$ and risk 1. So, with probability at least $1 - 2^{-j}$, every copy has $J_c$ larger than the table's for $c < H(p)/(j + 4\log(m+1) + 2\kappa)$ and larger than the coin's for $c > (1 - H(p))/(2^m - j - 2\log(m+1) - 2\kappa)$, and the stated condition makes these two ranges cover every $c > 0$. □

## H.5 Cheap couplings

A coupling with a short description given a shortest copy of the person defeats every $c$, and in the feature case a window for copies needs many cells.

**Proposition H.4** (Cheap couplings)**.** *In the feature case of Definition 10.1, with $|\Sigma| \ge 2$ and the values of $\rho$ distinct, let $\pi_{\mathrm{u}}$, shorter than $\ell_0$, compute the uniform kernel on $Y$.*

*(i) If $H(\rho)/(K(\beta \mid \pi_H) + \kappa) \ge (\log|Y| - H(\rho))/(\ell_0 - \ell(\pi_{\mathrm{u}}))$, then at every $c > 0$ some program that is not a copy has $J_c$ at most that of every copy.*

*(ii) If some $c > 0$ makes the copies the only minimisers of $J_c$, then $2^m > (\ell_0 - \ell(\pi_{\mathrm{u}}))/(\log|Y| - H(\rho)) - (a_\rho + 1 + \kappa)/H(\rho)$, with $a_\rho$ as in Theorem 10.2, whatever the coupling.*

*Proof.* (i) As in the last part of the proof of Theorem 10.2, a program that reads $\pi_H$ and then a shortest description of $\beta$ given $\pi_H$ computes $m$ and $\rho$ from $\beta$ and has kernel $P^*$ on $\operatorname{supp}\mu$, risk 0 and length $\ell_A \le \ell_0 + K(\beta \mid \pi_H) + \kappa$. If $c \le H(\rho)/(K(\beta \mid \pi_H)+\kappa)$, it has $J_c \le c\,\ell_0 + H(\rho) = J_c(\pi_H)$. Otherwise $c > (\log|Y| - H(\rho))/(\ell_0 - \ell(\pi_{\mathrm{u}}))$ by the hypothesis, and $J_c(\pi_{\mathrm{u}}) = c\,\ell(\pi_{\mathrm{u}}) + \log|Y| < J_c(\pi_H)$. Neither program is a copy: the first is deterministic, and the second is shorter than $\ell_0$. Every copy has $J_c \ge J_c(\pi_H)$. The bound $K(\beta \mid \pi_H) \le K(\beta) + \kappa$ holds because a description of $\beta$ need not read $\pi_H$.

(ii) By the last part of Theorem 10.2 the copies are not minimisers for $c < H(\rho)/(\lceil L_\beta \rceil + a_\rho + \kappa)$, and by the comparison with $\pi_{\mathrm{u}}$ in (i) they are not the only minimisers for $c \ge (\log|Y| - H(\rho))/(\ell_0 - \ell(\pi_{\mathrm{u}}))$. So $H(\rho)(\ell_0 - \ell(\pi_{\mathrm{u}})) < (\log|Y| - H(\rho))(\lceil L_\beta \rceil + a_\rho + \kappa)$. Insert $\lceil L_\beta \rceil \le 2^m H(\rho) + 1$ and divide by $H(\rho)(\log|Y| - H(\rho))$, which is positive because distinct values of $\rho$ give $H(\rho) < \log|\Sigma| \le \log|Y|$. □

Part (i) is a form of rate blindness (Proposition 9.2): if the person's apparent randomness has a short description, such as a pseudo-random generator with a short seed, no $c$ returns the person with fresh coins. Part (ii) counts cells and holds for every coupling; a compressible coupling only makes the program with kernel $P^*$ cheaper. With $\ell(\pi_{\mathrm{u}}) = 0$ and the machine constant set to 0 (an illustration), a direct computation (code in the accompanying repository) shows that by (ii) a person of $10^6$ bits with $H(\rho) \approx \frac{1}{2}$ and $|Y| = 2$ needs more than $1.99 \cdot 10^6$ cells, whatever the coupling.

# I Proofs for Section 11 and the correspondence with the earlier report

*Proof of Proposition 11.1.* We go through the proofs and list, for each, the property of the class it uses and the constant that changes. Since $\mathcal{H}'$ is c.e. and prefix-free, (F1) holds for it with a constant $\kappa_2$ that depends on $\mathcal{H}'$: $K(\pi) \le \ell(\pi) + \kappa_2$.

- Lemma 7.2: the agreement $u(\pi; a')$ is lower semicomputable uniformly in $(\pi, a', d)$, as in every class with uniformly lower-semicomputable likelihoods; the constant of the enumeration of the list depends on the class.
- Lemma 7.4: $q_z$ is lower semicomputable uniformly in $(\pi, a, m, B)$, so (F2) applies with a constant that depends on the class.
- Lemma 7.3: Jensen's inequality only; no constant changes.
- Lemma 7.5: the two lemmas above and (F1) with $\kappa_2$.
- Theorem 8.7: the lemmas above, and the reference programs only through $J_g$ and $J_b$, whose constants are $\kappa_g$ and $\kappa_b$ of $\pi'_g$ and $\pi'_b$.
- Theorem 6.9: Ville's inequality for the class's mixture $\xi_{\mathcal{C}}$ (Lemma 6.2), Kraft's inequality and countability; no constant enters.
- Proposition B.5(ii): the coding theorem for the semimeasure $\tilde{R}$ built from the class, which holds with a constant that depends on the class.
- Theorems 6.3 and 6.7: Definition 6.1 and the proofs in Appendix D take an arbitrary countable class $\mathcal{C}$ with Kraft weights and sub-probability kernels that contains a program for $\nu$, and use nothing else: bounds for single programs are combined by Ville's inequality for $\xi_{\mathcal{C}}$ and by union bounds weighted by $2^{-\ell(\pi)}$, which Kraft's inequality makes summable.

No machine constant enters, and $\ell_0$, $\mathcal{E}_0$ and the margins are computed from the class's own lengths and kernels; so a time-bounded class may select a longer but faster program than $\mathcal{H}_\times$ would. □

**Proposition I.1** (Computability of the steeper predictor)**.** *Let $\lambda \geq 1$ be computable, the class $\mathcal{H}_\times$ or $\mathcal{H}_F$, $X$ infinite and $\{0,1\} \subseteq Y$. (a) $\xi_\lambda(D)$ is lower semicomputable uniformly in $D$. (b) Neither the raw conditional $\xi_\lambda(D + (x,y))/\xi_\lambda(D)$ nor its normalised version is computable. (c) Both are limit-computable.*

*Proof of Proposition I.1.* (a) Each term $2^{-\lambda \ell(\pi)} P_\pi(D)$ is lower semicomputable uniformly in $(\pi, D)$, because $\lambda$ is computable, and the sum of uniformly lower-semicomputable nonnegative terms is lower semicomputable.

(b) This is the diagonal argument for $M_F$ (Hänni, 2026a, Proposition A.10(c)). Identify $\mathbb{N}$ with infinitely many inputs. Suppose the raw conditional were computable. Define $f(t)$, for $t \in \mathbb{N}$, as the first $y \in \{0,1\}$ that a dovetailed search certifies to satisfy $\xi_\lambda(y \mid D_t, t) < 0.6$, where $D_t := \{(s, f(s)) : s < t\}$. The two conditionals sum to at most 1, so one of them is at most $\frac{1}{2}$ and the search halts. Then $f$ is total computable, and by telescoping $\xi_\lambda(D_n) \leq \xi_\lambda(\emptyset) \cdot 0.6^n \leq 0.6^n$. But the class contains a deterministic program for $f$ of length at most $K(f) + \kappa$, so $\xi_\lambda(D_n) \geq 2^{-\lambda(K(f)+\kappa)}$ for every $n$, a contradiction for large $n$. For the normalised conditional, run the same construction with it in place of the raw one; then its cumulative loss on $D_n$ is at least $n \log(1/0.6)$. Each normaliser $\sum_{y'} \xi_\lambda(D_t + (t, y'))$ is at most $\xi_\lambda(D_t)$, so the normalised conditional is at least the raw one, and the normalised cumulative loss is at most the raw one, $-\log \xi_\lambda(D_n) + \log \xi_\lambda(\emptyset) \leq \lambda(K(f) + \kappa)$. This is again a contradiction for large $n$.

(c) Lower-semicomputable reals are computable in the limit, and the denominators are positive, because every finite data set has a consistent deterministic program. □

## I.1 Correspondence with the earlier report

Claude (Anthropic) (2026, Section 8) formalised the proposal for a general imitation channel, mostly by comparing a few hypotheses by rate. We re-derive the results we use from this earlier report: per-input and per-block pricing, the sequence-case threshold, the lemmas behind its kink theorem (here for semimeasures), the hull lemma, rate thresholds, rate blindness, non-identifiability, guard erosion and error envelopes. The table names them as they are named there. Two of its statements change here. Its full-table window at $d = 20$ has upper end about $(1 - H(p))/d \approx 0.049$ when logarithmic terms are ignored; with the $2 \log d$ terms, as large as $d$ there, a direct computation gives $c_{\mathrm{hi}} = 0.01139$ with the machine constant set to 0, an illustration (Corollary 8.11). And its lemma that the randomness of the error set removes the slack $2p$, stated for computable hypotheses, holds for semimeasures (Lemma 7.6).

| earlier result | here | change |
|---|---|---|
| per-input versus once-only pricing | Proposition B.2 | exact minimax value kept |
| the sequence pathology | Theorem 4.2(a) | same threshold; adds (b)–(d) |
| universal conditional semimeasure | Remark B.3 | – |
| per-block pricing | Proposition B.6 | countable families with computable weights |
| kink: list-size and floor lemmas | Lemmas 7.2 and 7.3 | semimeasures, via the lower-semicomputable agreement; truncation |
| kink: error segment | Lemma 7.5(D2) | – |
| kink: shared randomness | Proposition E.5(i) | every $\lambda$-optimal set model |
| randomness of $E$ removes the $2p$ slack | Lemma 7.6 | semimeasures; used for the full-table kink at every $p < 1/4$ (Propositions 7.7 and 7.8) |
| identification in the parity example | Corollary 8.11 | every rule length $m$; adds information and error-label bounds; rigorous $d = 20$ window |
| empirical selection follows population selection | Lemma F.3 | one-sided, truncated, without replacement |
| selection traces the lower hull | Lemma 3.8 | adds the data reading (v) and duality (vi) |
| rate threshold, separable classes; rate blindness | Proposition 9.3 | – |
| steepness not identifiable | Proposition G.3 | – |
| guard erosion | Proposition G.2 | teacher model stated; exact crossing |
| error envelopes | Proposition 9.1 | all hypotheses; cheap errors reproduced |
| input-law dependence | Proposition G.4 | separable case only |

Not reused: the idealised rate threshold for arbitrary hypotheses and the realisability of every frontier shape, because Theorem 8.7 covers one model with one error rule over all hypotheses and Propositions 9.3 and G.4 cover separable components; the treatment of inference rules, which is not about function induction; and the pathologies of parasitic errors and language relativity, which hold for steeper SFI by the same arguments but add nothing specific to it.